Text Command Reference Manual

BASE24®

Sep-2011, Release 6.0 Version 10 Publication Number: BA-AE000-05

Contents

BASE24 Base and Shared Process Commands *continued*

Process-Generated Commands *continued*

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What’s New

The following table highlights the major changes that have been made in the most recent update to the ***BASE24 Text Command Reference Manual.*** The first column of the table lists the sections and appendixes in which major changes have been made. The second column of the table describes the major changes for each section or appendix. These changes update the manual to release 6.0, version 10.

Section/ Appendix	Major Changes
3	Adds new commands LOADPRIPUB (LPRIPUB), LOADMSTRCOM (LMSTRCOM), and DELPUBLIC (DPUBLIC) and modifies existing commands LOADALLKEYS, LOADMASTER, and LOADPUBLIC. Updates Device Handler Process Command charts.

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Preface

The ***BASE24 Text Command Reference Manual*** is designed to give readers a thorough understanding of the BASE24 DELIVER PROCESS and interprocess commands. This manual provides detailed descriptions of DELIVER PROCESS text commands originating from an XPNET network control facility, along with BASE24 network commands sent between processes. It provides information on topics important to gaining a basic understanding of these commands, such as command descriptions and syntax. However, this manual is also helpful to readers who require a detailed explanation of these BASE24 commands and their processing steps.

Audience

This manual is intended for three types of readers. One type of reader reads this manual to obtain a general overview of the DELIVER PROCESS and interprocess commands used by a specific process. Another type of reader is interested in the processing steps a specific process takes when it receives one of these commands. A third type of reader of this manual is an operator who is interested in using the manual as a reference guide to explain the commands they use in network management and control.

Prerequisites

Although a knowledge of BASE24 network management and control is encouraged for the customer who is reading this manual, the manual is designed to be a stand-alone manual. Operators who seek an overview of the DELIVER PROCESS and interprocess commands should be able to find as much information about these commands as they will need to perform their functions. However, if more information about network management and control is desired, readers should refer to the ***NET24-XPNET Network Control Operations Guide***.

Additional Documentation

The BASE24 documentation set is arranged so that each BASE24 manual presents a topic or group of related topics in detail. When one BASE24 manual presents a topic that has already been covered in detail in another BASE24 manual, the topic is summarized and the reader is directed to the other manual for additional information. Information has been arranged in this manner to be more efficient for readers who do not need the additional detail and at the same time provide the source for readers who require the additional information. This manual contains references to the following BASE24 publications:

- The ***BASE24 BIC ANSI Interface Manual*** describes in detail the BIC ANSI Interface process, which is used to establish a communications link between a BASE24 network and a co-network. The manual provides a comprehensive description of all the functions performed by the BIC ANSI Interface process involving incoming and outgoing message traffic for both acquirer and issuer processors.
- The ***BASE24 BIC ISO Interface Manual*** describes in detail the BIC ISO Interface process, which is used to establish a communications link between a BASE24 network and a co-network. The manual provides a comprehensive description of all the functions performed by the BIC ISO Interface process involving incoming and outgoing message traffic for both acquirer and issuer processors.
- The ***BASE24 BIC ISO Standards Manual*** contains specifications for the BIC ISO External Message. It includes a general overview of the message, explanations of the external message types used by the BIC ISO Interface, a set of message defaults for each BASE24 product, a description of each data element contained in the message, and descriptions of the message flows between BASE24 and a co-network.
- The ***BASE24 Device Control Manual*** describes in detail the EMT Control Command screen.
- The ***BASE24 ITS Kernel Files Maintenance Manual*** describes the screens used to send commands to BASE24-telebanking processes.
- The ***BASE24 Log Message Reference Manual*** discusses in detail how to access event messages generated by BASE24 processes.
- The ***BASE24 Logical Network Configuration File Manual*** discusses in detail the assigns and params used in configuring all BASE24 processes.

- The ***BASE24 Refresh and Extract Operators Manual*** discusses in detail the operator procedures for performing refreshes and extracts using the BASE24 Refresh and Super Extract processes. It also provides an overview of processing for these procedures.
- The ***BASE24 Remote Banking Transaction Processing Manual*** describes the ASSIGN WARMBOOT, LCONF WARMBOOT, and PARAM WARMBOOT commands, and server control commands that can be sent to core objects.
- The ***BASE24 Remote Banking End-of-Period Processing and Reporting Manual*** describes the server control commands that can be sent to the End-of-Period Processing object.
- The ***BASE24 Transaction Security Manual*** discusses in detail how to set up transaction security for procedural BASE24 processes.
- The ***BASE24-atm Settlement and Reporting Manual*** discusses in detail BASE24-atm settlement, cutover, balancing, and reporting.
- The ***BASE24-atm Transaction Processing Manual*** discusses in detail transaction processing used in configuring BASE24-atm.
- The ***BASE24-pos Settlement and Reporting Manual*** discusses in detail BASE24-pos settlement, cutover, balancing, and reporting.
- The ***BASE24-pos Transaction Processing Manual*** discusses in detail transaction processing used in configuring BASE24-pos.
- The ***BASE24-teller Settlement and Reporting Manual*** discusses in detail BASE24-teller settlement, cutover, balancing, and reporting.
- The ***BASE24-teller Transaction Processing Manual*** discusses in detail transaction processing used in configuring BASE24-teller.

This manual also contains references to the following NET24 publications:

- The ***NET24-XPNET Network Control Operations Guide*** discusses in detail network management and control used to maintain a BASE24 system.
- The ***NET24-XPNET Network Control Command Reference Manual*** discusses in detail the process, station, line and device declarations used in configuring all BASE24 processes that run under control of the XPNET process.

This manual also contains references to the following vendor publications:

- The ***IBM 3624 Programmer's Guide*** provides information on ATM configuration version data.
- The Diebold ***Enhanced Monochrome Graphics (EMG) Programming Manual*** provides information for Diebold 10XX/478X ATMs with Enhanced Monochrome Graphics (EMG).

Software Release

This manual documents standard processing as of its publication date. Software that is not current and custom software modifications (CSMs) may result in processing that differs from the material presented in this manual. The customer is responsible for identifying and noting these changes.

Manual Summary

The following is a summary of the contents of this manual.

"Conventions Used in This Manual" follows this preface and describes notation and documentation conventions necessary to understand the information in the manual.

Section 1, "Introduction," provides an overview of BASE24 text commands. This section includes information on how to use the manual.

Section 2, "BASE24 Base or Shared Process Commands," explains the text commands used by BASE24 Base or shared processes. This includes a description of each command, the command syntax, the processing steps the process takes when it receives the command, and an example of the command when necessary.

Section 3, "BASE24-atm Process Commands," explains the text commands used by BASE24-atm processes. This includes a description of each command, the command syntax, the processing steps the process takes when it receives the command, and an example of the command when necessary.

Section 4, "BASE24-billpay Process Commands," explains the text commands used by BASE24-billpay processes. This includes a description of each command, the command syntax, the processing steps the process takes when it receives the command, and an example of the command when necessary.

Section 5, "BASE24-pos Process Commands," explains the text commands used by BASE24-pos processes. This includes a description of each command, the command syntax, the processing steps the process takes when it receives the command, and an example of the command when necessary.

Section 6, "BASE24-teller Process Commands," explains the text commands used by BASE24-teller processes. This includes a description of each command, the command syntax, the processing steps the process takes when it receives the command, and an example of the command when necessary.

Appendix A, "Restricted Use Operator Commands," explains the restricted use operator commands used universally by several BASE24 processes across product lines. This appendix contains only those restricted use operator commands common to more than one BASE24 process.

Appendix B, "Process-Generated Commands," explains the process-generated commands used by BASE24/Shared, BASE24-atm, BASE24-pos, and BASE24-teller processes. This includes a description of each command, the command syntax, and the processing steps the process takes when it receives the command.

Publication Identification

Three entries appearing at the bottom of each page uniquely identify this BASE24 publication. The publication number (for example, BA-AE000-05 for the ***BASE24 Text Command Reference Manual***) appears on every page to assist readers in identifying the manual from which a page of information was printed. The publication date (for example, Sept-2011 for September, 2011) indicates the issue of the manual. The software release information (for example, R6.0v10 for release 6.0, version 10) specifies the software that the manual describes. This information matches the document information on the copyright page of the manual.

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Conventions Used in This Manual

This section explains how syntax notation is documented in this manual.

Syntax Notation

Syntax descriptions in this manual use several symbols and conventions to denote how commands must be entered. These conventions are described in the table below.

Notation	Meaning
UPPERCASE LETTERS	Indicate key words and literals that you must enter exactly as shown.
lowercase italics	Identify variables that you must provide.
Brackets [ ]	Enclose optional syntax items that you can enter in the command.
Braces { }	Enclose a group of items from which you are required to choose one item. The items are separated within the braces by a vertical bar (|).
Vertical bar |	Separates alternatives in a list of items.
Ellipsis …	Indicates the immediately preceding syntactical item can occur one or more times.
Punctuation	Must be entered as shown. This includes parentheses, commas, semicolons, and other symbols not previously described.
Spaces	Are required as delimiters wherever it would be otherwise impossible to discriminate between adjacent words or symbols. You can insert additional spaces between any adjacent words or symbols, but not within a word or multiple character symbol.

Notation	Meaning
Indented lines	Indicates a continuation of the preceding line of syntax. If the syntax of a command is too long to fit on a single line, each continuation line is indented.

1: Introduction

BASE24 processes can receive a number of commands that cause them to perform specialized processing. Some of these commands are entered by operators from an XPNET network control facility, while others are sent between processes. These commands are used to control or change the way in which a particular BASE24 process is operating. They are also used to receive information about particular BASE24 processes. This section provides an overview of the following topics:

- Using XPNET network control facility commands
- Types of BASE24 commands
- Using this manual
- Locating commands

Using Network Control Commands

Network control interfaces allow the network operator to monitor and control the network. Operators enter network control commands from an XPNET network control facility using defined syntax. In most cases, network control commands are processed by the XPNET process. However, one network control command allows text strings to be sent to a process. Typically, these text strings are commands that cause the application process to perform specialized processing.

Command Syntax

Like all network control commands, DELIVER PROCESS commands must be entered using defined syntax. In addition, the text strings that are sent using these commands have a defined syntax. This manual provides an overview of the DELIVER PROCESS commands and defines the text strings that are sent to BASE24 application processes using these commands.

Security

Access to network control commands can be secured by user ID, password, and command profile. If the ENABLE-SECURITY parameter is included in the declaration for the NCPI Server process in the Pathway Configuration File (PATHCONF), the network operator must be allowed access to the DELIVER PROCESS command in a command profile defined in the Network Control Supervisor Profile File (NCSP) before text commands can be used.

Text String Case Sensitivity

Some BASE24 processes require text commands to be entered in either all uppercase or all lowercase characters. For most processes, command text entered in lowercase characters is automatically upshifted to uppercase. If a command entered in lowercase fails, you may need to reenter the command in all uppercase text.

Note: Network operator user ID passwords are always case sensitive.

Entering XPNET DELIVER PROCESS Commands

XPNET commands can be entered from any XPNET network control facility, such as the Network Control Point Communications (NCPCOM) utility or the Network Control Supervisor (NCS) screen. The NCPCOM utility is a subsystem that provides an interactive prompt for entering commands. The NCS screen is a formatted screen that contains seven fields. The NCS screen is shown in the illustration below:

```
 NET24-NCS   OPEN NETWORK CONTROL     LLLL        YY/MM/DD  HH:MM   01 OF 01
   USER:                    PASSWORD:                  SYSNAME:
   NODE: *                    OBJECT:                  NAME:
COMMAND:















*******************************************************************************
                 DESTINATION:   ____       LOGICAL NETWORK ID:  ____
 F1-ENTER  F3-PREV-PAGE  F4-NEXT-PAGE  F5-NEXT-OBJECT  F12-HELP
```

Whatever XPNET network control facility is used, applications running under XPNET are sent text strings using the DELIVER PROCESS command. The general syntax for this command is given below.

Syntax

DELIVER [PROCESS] [*object-name*] , *text* [, *filter*] [, WAIT] [, QUIET]
[, REFRESH] [, ONEATATIME | , ONEONLY]

object-name — Identifies the specific process or processes affected by the command. You can enter the fully qualified symbolic name of a specific process, a symbolic name that identifies one or more processes with the same name, or an object name that uses wildcard characters to identify one or more processes.

text — The text to be sent to the process. The text can be up to 500 characters in length, and must be enclosed in one of the following sets of characters:

Double quotation marks (“ ”)
Single quotation marks (‘ ’)
Slashes (/ /)
Brackets ([ ])
Parentheses (( ))
Angle brackets (< >)

For simplicity, double quotation marks are used throughout the syntax examples provided in this manual.

filter — Specifies a process state or a process relationship that limits the processes affected by the command. For more information on filters, see the ***NET24-XPNET Network Control Command Reference Manual***.

WAIT — Indicates a reply is requested. The command does not complete until a response is received or the command times out.

Note: This option should not be used. There are currently no BASE24 application processes that return a reply to this command. If this parameter is included in a command that is sent to BASE24 application process, the XPNET network control facility will not be available until the command times out.

QUIET — Indicates the command is to be executed against the specified processes without returning a positive response for each object. Error and warning messages are returned.

REFRESH — Instructs the NCP Server process to update the table it uses to determine which objects meet wildcard criteria when wildcard characters are used in the object name. The NCP Server process updates this table at intervals according to an NCP Server process parameter in the PATHCONF, or when you include the REFRESH modifier in an XPNET network control command.

ONEATATIME — (ONE AT A TIME) Indicates the command is to be issued to a group of objects one at a time. When this modifier is used, the command is executed for the first object and a MORE prompt is returned. When you press the **Enter** key, the command is executed for the next object.

ONEONLY — (ONE ONLY) Indicates the command is to be issued to only one of a group of objects. When this modifier is used, the command is executed for the first matching object found.

Filters and the WAIT, QUIET, REFRESH, ONEATATIME, or ONEONLY modifiers can be added after the text commands documented in this manual, but are not shown in the documentation. For example, the documentation for the LOGON command for the ANSI Host Interface shows the following syntax when logging on all DPCs:

DELIVER PROCESS *object-name,* "LOGONALL"

This command could be entered using the DELIVER PROCESS command as shown below:

```
DELIVER PROCESS P1A^HISF, "LOGONALL"
```

If a filter and the QUIET modifier are added to the command, the command could be entered as shown below:

```
DELIVER PROCESS P1A^HISF, "LOGONALL", STARTED, QUIET
```

Note: There are a number of variations on how XPNET network control commands can be entered. The sample commands above show the most complete command entry possible, and could be entered from any XPNET network control facility.

For more information on the NCS screen and NCPCOM utility, see the ***NET24-XPNET Network Control Operations Guide***. For more information on the DELIVER PROCESS command, see the ***NET24-XPNET Network Control Command Reference Manual.***

Types of BASE24 Commands

The BASE24 commands documented in this manual can be separated into the following three categories: process-generated commands, general use operator commands, and restricted use operator commands. These commands are described below.

Process-Generated Commands

Process-generated commands are generated by processes running in the system and are sent to other processes running in the system. These commands are also referred to as *interprocess commands*. Interprocess commands are not manually entered from an XPNET network control facility, but are automatically generated when certain processing conditions are met or when certain events occur. For example, when a Settlement Initiator process determines that cutover is to occur, it sends interprocess cutover messages to notify specific processes that they should begin cutover procedures. These commands are generally used to notify the process receiving the command of actions occurring, or that have already occurred, elsewhere in the system.

General Use Operator Commands

General use operator commands are those commands entered by an operator in day-to-day use of the system. These commands enable the operator to control or change the way a specific process is operating.

Restricted Use Operator Commands

Restricted use operator commands should be entered by an operator only at the request of ACI personnel. These commands allow programmers to test functionality and determine how a process is running when changes are made to the code. It is not recommended to use these commands except in a test environment.

Using this Manual

The ***BASE24 Text Command Reference Manual*** provides information on all operator-entered and process-generated commands used by BASE24. This manual is organized based on the type commands described and the products that use them. This organizational hierarchy is explained below.

First, general and restricted use operator commands are organized by BASE24 product. The general and restricted use operator commands for the following BASE24 products are documented in separate sections:

- BASE24 Base and Shared
- BASE24-atm
- BASE24-billpay
- BASE24-pos
- BASE24-teller

Within each of the above product sections, the general and restricted use operator commands are organized by the process to which they are sent. All processes associated with a particular BASE24 product are documented in alphabetical order within that BASE24 product section. In addition, all commands for a particular process are documented in alphabetical order.

For convenience, restricted use operator commands common to several processes across BASE24 product lines and process-generated commands have been included in appendices to separate them from the general use commands an operator can issue. Restricted use operator commands are documented in alphabetical order. Process-generated commands are documented using the same hierarchy described above: by product, by process, and then in alphabetical order. Restricted use operator commands are documented in appendix A and process-generated commands are documented in appendix B.

Each command is organized using four main elements. The first element is a description of the command. The description identifies the type of command (i.e., process-generated, general use operator, or restricted use operator) and explains the purpose of the command, including its intended use. The description is followed by the syntax of the command. The command syntax indicates how an operator must enter the command from an XPNET network control facility or how the command is received by a process for process-generated commands. Next, the processing steps describe the actions the process takes when it receives the command. Finally, the processing steps are followed by an example of the command, if an example is necessary to further explain the command.

Limits of this Manual

The scope of the ***BASE24 Text Command Reference Manual*** is limited to the documentation of BASE24 commands generated using the DELIVER PROCESS syntax issued from an XPNET network control facility. In addition, this manual also includes documentation on the interprocess commands generated and sent from one running process in the system to another.

Related BASE24 Documentation

DELIVER PROCESS and interprocess commands are only one aspect of network management and control. Other commands and network management functions are necessary for keeping a BASE24 network operating efficiently according to an institution's policies and operating procedures. Since the network operator is responsible for the overall well-being of a BASE24 system, it is important that he or she has an understanding of all aspects of network management and control, such as the various processes in the system and how these processes interact with each other and the various files in the network. Along with this manual, a network operator will find it necessary to refer to the ***NET24-XPNET Network Control Operations Guide*** for more information on network management and control. This manual provides comprehensive descriptions of the various functions used in monitoring and controlling the network.

BASE24-telebanking Process Commands

Due to the object-oriented nature of some BASE24-telebanking processes, BASE24 commands are sent to these processes using integrated transaction services (ITS) kernel screens rather than using the DELIVER PROCESS syntax. These BASE24-telebanking commands are described in the following publications:

- The ***BASE24 Remote Banking Transaction Processing Manual*** describes the ASSIGN WARMBOOT, LCONF WARMBOOT, and PARAM WARMBOOT commands, and server control commands that can be sent to core objects.
- The ***BASE24 Remote Banking End-of-Period Processing and Reporting Manual*** describes the server control commands that can be sent to the End-of-Period Processing object.
- The ***NET24 ITS Kernel Files Maintenance Manual*** describes the screens used to send commands to BASE24-telebanking processes.

Note: Commands sent to shared procedural processes on behalf of BASE24-telebanking (i.e., ISO Host Interface process, Enscribe Refresh process, and Super Extract process) are sent using the DELIVER PROCESS syntax described in this manual.

Locating Commands

This manual contains a comprehensive index designed to assist operators in locating commands. The index is located at the end of the manual and is organized in two ways. First, it is organized by process. This means that all commands used by a specific process are listed under that process name in the index. Therefore, an operator interested in commands for a specific process can look up the process name and locate all the commands used by the process.

The index is also organized by command name. This means that an operator interested in a particular command can look up the command name in the index and locate where in the manual the command is documented. As a supplement to the index, the beginning of each process subsection includes a summary of the commands, by command type, used by the specified process.

2: BASE24 Base and Shared Process Commands

This section contains the process-generated, general operator use, and restricted operator use commands received by the following Base or shared processes:

- ANSI Host Interface process
- BIC ANSI Interface process
- BIC ISO Interface process
- From Host Maintenance process
- Refresh processes
  - Base National Negative Refresh process
  - Enscribe Refresh process
  - INAS National Negative Refresh process
  - SQL Refresh process
- ISO Host Interface process
- Super Extract process

The processes are presented in alphabetical order. Furthermore, specific commands are documented in alphabetical order within the command categories. The documentation for each command includes the following information:

- A description of the command
- The syntax of the command
- The processing steps performed when the command is received
- An example of the command (when necessary)

ANSI Host Interface Process Commands

The ANSI Host Interface process can receive a number of BASE24 commands that cause it to perform specialized processing. These commands can be separated into three categories: process-generated commands, general use operator commands, and restricted use operator commands. All BASE24 general and restricted use operator commands that can be received by the ANSI Host Interface process are documented in this subsection. Process-generated commands are documented in appendix B.

Note: The ANSI Host Interface process supports only the ANSI-based external message format. For information on the support of the ISO-based external message, refer to the "ISO Host Interface Commands" topic provided later in this section. For additional information on the ANSI Host Interface process, refer to the ***BASE24 ANSI Host Interface Manual***.

All of the commands that can be received by the ANSI Host Interface process are listed below by category. All of these commands must be entered in all uppercase or all lowercase characters.

Process-Generated Commands	General Use Operator Commands	Restricted Use Operator Commands
CUTOVER DEVICE STATE SAFEXTCOMP SAFEXTRQST	KEY CHANGE KEY REPEAT KEY VERIFY LOGOFF LOGON PERFORMANCE STARTSAF STATUS STOPSAF WARMBOOT	DEBUG DUMP TRACE OFF TRACE ON

DEBUG Command

A restricted use operator command that places the ANSI Host Interface process into the debug state. The debug message places the ANSI Host Interface process under control of the Hewlett-Packard (HP) NonStop Inspect product and displays an Inspect prompt on the home terminal (the terminal specified for the process by the HOMETERM attribute or the home terminal of the XPNET process). Inspect is an HP NonStop program debugging tool. For more information on the DEBUG command, refer to appendix A.

DUMP Command

A restricted use operator command that directs the ANSI Host Interface process to request a programmatic dump. A programmatic dump allows a programmer to automatically capture an unusual situation for evaluation. The ANSI Host Interface process can continue running after the dump is generated. For more information on the DUMP command, refer to appendix A.

KEY CHANGE Command

A general use command that causes the ANSI Host Interface process to either request or generate and send a new PIN key or Message Authentication Code (MAC) key.

Note: The KEY CHANGE command can only be used when the ANSI Host Interface process is set up to use Dynamic Key Management (DKM).

Syntax

DELIVER PROCESS *object-name, “identifier* KEY CHANGE *option dpc”*

object-name — The symbolic name assigned to the ANSI Host Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

identifier — Identifies whether the change key request is for PIN keys or MAC keys. Valid values are as follows:

MAC = Change key request is to change the MAC keys. This value is not currently used with the ANSI Host Interface process.
PIN = Change key request is to change the PIN keys.

option — Identifies the working keys to be changed by the KEY CHANGE command. Valid values are as follows:

B = Change both the inbound and the outbound working keys.
I = Change only the inbound working key.
O = Change only the outbound working key.

dpc — Identifies the data processing center (DPC) for which the KEY CHANGE command is being issued.

Processing

The ANSI Host Interface process either requests a new key from the host or generates and sends a new key depending on which entity generates the key specified in the command.

If the ANSI Host Interface process is to generate the specified key, it generates a new key and sends that new key to the host. If the host is to generate the specified key, the ANSI Host Interface process generates a change key request and sends it to the host.

For more information on the steps required to process KEY CHANGE commands, along with the associated message flows, refer to the ***BASE24 ANSI Host Interface Manual***.

Example

The following is an example of a KEY CHANGE command to change the inbound PIN key for DPC 0001. In this example, the ANSI Host Interface process is P1A^HISF1.

DELIVER PROCESS P1A^HISF1, “PIN KEY CHANGE I 0001”

KEY REPEAT Command

A general use command that causes the ANSI Host Interface process to send a repeat key message to the host for a specific PIN key or Message Authentication Code (MAC) key. In turn, the host will respond to the repeat key message with a copy of the requested key.

Note: The KEY REPEAT command can only be used when the ANSI Host Interface process is set up to use Dynamic Key Management (DKM).

Syntax

DELIVER PROCESS *object-name, "identifier* KEY REPEAT *option dpc"*

object-name — The symbolic name assigned to the ANSI Host Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

identifier — Identifies whether the repeat key request is for PIN keys or MAC keys. Valid values are as follows:

MAC = Repeat key request is to repeat the MAC keys. This value is not used with the ANSI Host Interface process.
PIN = Repeat key request is to repeat the PIN keys.

option — Identifies the working keys to be repeated by the PIN KEY REPEAT command. Valid values are as follows:

B = Repeat both the inbound and the outbound working keys.
I = Repeat only the inbound working key.
O = Repeat only the outbound working key.

dpc — Identifies the data processing center (DPC) for which a repeat key message is being generated.

Processing

The KEY REPEAT command is sent to the process that originally received the key in question. The command itself is not a repeat key message. The command instructs a process to generate the repeat key request, so it is directed to the process that is to generate the repeat key request rather than the process that is to receive the repeat key request. The process that originally generated the key in question receives this message and responds with the key previously generated.

For more information on the steps required to process KEY REPEAT commands, along with the associated message flows, refer to the ***BASE24 ANSI Host Interface Manual***.

Example

The following is an example of a KEY REPEAT command for the outbound PIN key for DPC 0001. In this example, the ANSI Host Interface process is P1A^HISF5.

```
DELIVER PROCESS P1A^HISF5, “PIN KEY REPEAT O 0001”
```

KEY VERIFY Command

A general use command that causes the ANSI Host Interface process to send a verify key message to the host for the specified PIN key or Message Authentication Code (MAC) key. The host then verifies the key by comparing check digits.

Note: The KEY VERIFY command can only be used when the ANSI Host Interface process is set up for Dynamic Key Management (DKM).

Syntax

DELIVER PROCESS *object-name, “identifier* KEY VERIFY *option dpc”*

object-name — The symbolic name assigned to the ANSI Host Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

identifier — Identifies whether the verify key request is for PIN keys or MAC keys. Valid values are as follows:

MAC = Verify key request is to verify the MAC keys. This value is not currently used with the ANSI Host Interface process.
PIN = Verify key request is to verify the PIN keys.

option — Identifies the working keys to be verified by the KEY VERIFY command. Valid values are as follows:

B = Verify both the inbound and the outbound working keys.
I = Verify only the inbound working key.
O = Verify only the outbound working key.

dpc — Identifies the data processing center (DPC) for which a verify key message is being generated.

Processing

The ANSI Host Interface process generates a verify key message containing the necessary check digits and sends it to the host.

For more information on the steps required to process KEY VERIFY commands, along with the associated message flows, refer to the ***BASE24 ANSI Host Interface Manual***.

Example

The following is an example of a KEY VERIFY command to verify the inbound PIN key for DPC 0001. In this example, the ANSI Host Interface process is P1A^HISF3.

DELIVER PROCESS P1A^HISF3, “PIN KEY VERIFY I 0001”

LOGOFF Command

A general use operator command that initiates a logoff sequence for one or more data processing centers (DPCs).

Syntax

DELIVER PROCESS *object-name,* “LOGOFFALL”

DELIVER PROCESS *object-name*, “LOGOFF *dpc”*

object-name — The symbolic name assigned to the ANSI Host Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

dpc — Identifies the DPC for which the logoff is being initiated.

Processing

The LOGOFF command causes the ANSI Host Interface process to generate a logoff message to the specified DPCs. If ALL is specified in the message, action is taken for all DPCs processed by the ANSI Host Interface process. If a DPC number is specified, action is taken for that DPC only.

Logoff messages are network management messages used to halt communications between BASE24 and a DPC. They are sent as 0800 messages and require no responses from the ANSI Host Interface process.

Logoff messages are distinguished by the value of 002 in the NM-INFO-CDE field of the network management message. Logoff messages can be sent by either BASE24 or a host. In either case, after a DPC is logged off, the ANSI Host Interface process does not continue to echo-test any of the DPC’s stations, nor does it send or respond to non-network management messages. Any non-network management messages that must be forwarded to the DPC are placed in the Store-and-Forward File (SAF) instead.

The ANSI Host Interface process only attempts to reopen the link to a DPC that is logged off if a LOGON command is received from an XPNET network control facility. The DPC, on the other hand, can reopen the link at any time by sending an echo-test or logon message. The ANSI Host Interface process responds to either type of message.

Note: When a DPC is logged off, any transactions in progress are allowed to complete.

The ANSI Host Interface process only sends logoff messages to a DPC as a result of a LOGOFF command issued from an XPNET network control facility.

To log a DPC off, the ANSI Host Interface process sends a logoff message to an available send/receive station and marks the DPC as down and logged off. A response can be returned by the host, but is not required.

When a host logs off because of a scheduled down or unavailable period, it can transmit a logoff message from any of its stations. Upon receipt of a logoff message, the ANSI Host Interface process marks the DPC down and logged off. It does not return a response to the DPC.

For more information on logoff and echo-test message processing and message flows, refer to the ***BASE24 ANSI Host Interface Manual.***

LOGON Command

A general use operator command that initiates a logon sequence for one or more data processing centers (DPCs).

Syntax

DELIVER PROCESS *object-name,* “LOGONALL”

DELIVER PROCESS *object-name*, “LOGON *dpc”*

object-name — The symbolic name assigned to the ANSI Host Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

dpc — Identifies the DPC for which the logon is being initiated.

Processing

This command causes the ANSI Host Interface process to generate an echo-test message to each of a DPC’s send/receive stations. If the DPC is currently logged off, it also causes the ANSI Host Interface process to set a new performance timer for the DPC.

If ALL is specified in the message, action is taken for all DPCs processed by the ANSI Host Interface process. If a DPC number is specified, action is taken for that DPC only.

Logon messages are network management messages used to manage communications access between BASE24 and a DPC’s stations. They are sent as 0800 messages and in some cases require 0810 messages in response.

Logon messages are distinguished by the value of 001 in the NM-INFO-CDE field of the network management message.

The ANSI Host Interface process sends logon messages to DPCs in response to echo-test messages. This is done regardless of whether the DPC initiated the echo-test message or was simply responding to a prior BASE24-initiated echo-test message.

DPCs send logon messages to start a logon sequence after a period of being logged off. They also send logon messages in response to BASE24-initiated echo-test or logon messages.

When the ANSI Host Interface process receives a logon message after the DPC has been logged off for a period of time, the ANSI Host Interface process marks the DPC as logged on and begins sending echo-test messages to that DPC's stations.

Note: If the ANSI Host Interface process receives a logon message when the DPC is not logged off, it simply drops the message.

When the ANSI Host Interface process receives a logon message in response to a BASE24-initiated message, the ANSI Host Interface process marks the station up and begins processing.

For more information on logon and echo-test message processing and message flows, refer to the ***BASE24 ANSI Host Interface Manual***.

PERFORMANCE Command

A general use operator command that requests performance statistics from the ANSI Host Interface process.

Syntax

DELIVER PROCESS *object-name,* “PERFORMANCEALL”

DELIVER PROCESS *object-name*, “PERFORMANCEB[ASE]24”

DELIVER PROCESS *object-name,* “PERFORMANCE *dpc”*

object-name — The symbolic name assigned to the ANSI Host Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

dpc — Identifies the data processing center (DPC) for which performance statistics are being generated.

Processing

The ANSI Host Interface process tracks performance statistics for each DPC it handles.

If ALL is specified in the command, the ANSI Host Interface process retrieves statistics for all DPCs. If B[ASE]24 is specified in the command, the process retrieves statistics for the first DPC only. If a particular DPC is specified in the command, the process retrieves statistics for the specified DPC only.

These performance statistics are gathered on an interval basis—the interval length established by the TIMER-LMTS.PERFORMANCE field in the Host Configuration File (HCF)—and statistics are available for the five most current intervals. The ANSI Host Interface process performs the following steps when it receives a PERFORMANCE command:

1. Retrieves performance statistics for BASE24-atm and BASE24-pos, which include the number of minutes in the performance period, the outbound request limits, the inbound request limits, the number of requests in the period, the number of timeouts in the period, the percentage of timeouts per request in the period, and the average time (in seconds) it took for the ANSI Host Interface process to receive a response.
2. Sends performance statistics in a series of event messages to the EMS collector process. These event messages are described in the BAHISF log message documentation file. If ALL is specified in the command, the ANSI

Host Interface process logs performance statistics for all of the DPCs it processes. If a DPC number is specified in the command, the ANSI Host Interface process logs performance statistics for that DPC only.

a. If there have been no requests in either direction for a DPC, then the ANSI Host Interface process logs event message 7901.

b. For each period during which there has been activity for the current and four previous periods for the DPC, the ANSI Host Interface process logs event messages 7902 and 7903.

STARTSAF Command

A general use operator command that starts a store-and-forward cycle.

Syntax

DELIVER PROCESS *object-name,* "STARTSAF *dpc*"

object-name — The symbolic name assigned to the ANSI Host Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

dpc — Identifies the data processing center (DPC) for which the store-and-forward cycle is being started.

Processing

The STARTSAF command is intended to restart a store-and-forward cycle stopped by a previous entry of a STOPSAF command. Note that since the STOPSAF command only stops the store-and-forward cycle temporarily, this command has limited value.

When the ANSI Host Interface process receives a STARTSAF command, it performs the following steps:

1. Checks to ensure the DPC is not logged off or marked down.

   If the DPC is logged off or marked down, the ANSI Host Interface process drops the message.
2. Checks to ensure that a Store-and-Forward File (SAF) extract is not in progress. If a SAF extract is in progress, the ANSI Host Interface process drops the message.
3. Checks to ensure that a store-and-forward cycle is not already in progress, which could have been initiated by another station. If a store-and-forward cycle is in progress, the ANSI Host Interface process drops the message.
4. Starts the store-and-forward cycle and updates the DPC status to indicate that a store-and-forward cycle is in progress.

STATUS Command

A general use operator command that requests status information for one or all of the data processing centers (DPCs) that the ANSI Host Interface process handles. The requested information is displayed in a series of event messages sent to the EMS collector process.

Syntax

DELIVER PROCESS *object-name,* “STATUS *dpc*”

object-name — The symbolic name assigned to the ANSI Host Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

dpc — Identifies the data processing center (DPC) for which status information is requested.

Processing

The STATUS command generates a series of event messages. These event messages are numbered in the range of 7910 through 7913 and are described in the BAHISF log message documentation file. The ANSI Host Interface process performs the following steps when it receives a STATUS command:

1. Determines the current state of the ANSI Host Interface process. Based on the current state of the ANSI Host Interface process, different event messages are logged for the requested DPCs.
2. Logs the appropriate event messages to describe the current state of the ANSI Host Interface process for the requested DPCs.

STOPSAF Command

A general use operator command that requests the temporary termination of a store-and-forward cycle.

Syntax

DELIVER PROCESS *object-name,* "STOPSAF *dpc"*

object-name — The symbolic name assigned to the ANSI Host Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

dpc — Identifies the data processing center (DPC) for which the store-and-forward cycle is being terminated.

Processing

The STOPSAF command can be used to stop store-and-forward processing if a Store-and-Forward File (SAF) extract is to be performed. Since a SAF extract cannot be started while a store-and-forward cycle is in progress, an operator must stop the cycle before starting the extract.

The STOPSAF command only stops store-and-forward processing temporarily. It will start again automatically under the normal conditions for starting a store-and-forward cycle. Refer to the STARTSAF command description provided earlier in this section for information on manually starting the store-and-forward cycle.

TRACE ON and TRACE OFF Commands

Restricted use operator commands that activate or deactivate a trace function within the ANSI Host Interface process. When the trace function is activated, it generates an event message each time the ANSI Host Interface process enters a new procedure. The event message displays the name of the procedure from which it was written. For more information on the TRACE ON and TRACE OFF commands, refer to appendix A.

WARMBOOT Command

A general use operator command that causes the ANSI Host Interface process to reinitialize.

Syntax

DELIVER PROCESS *object-name,* “9501”

DELIVER PROCESS *object-name*, “WARMBOOT”

object-name — The symbolic name assigned to the ANSI Host Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

This command causes the ANSI Host Interface process to reinitialize file information. Warmboot processing is similar to process initialization, except that a warmboot only includes those steps required to update the file information carried by the ANSI Host Interface process in memory.

During warmboot processing, the ANSI Host Interface process opens the following files:

- Logical Network Configuration File (LCONF)
- Log Message Configuration File (LMCF)
- Host Configuration File (HCF)
- Institution Definition File (IDF)
- Key File (KEYF)

In addition, the ANSI Host Interface process updates its internal tables with current information from these files. It also opens the Store-and-Forward File (SAF) and counts those messages pending for each of its data processing centers (DPCs).

This command does not cause the ANSI Host Interface process to reallocate extended memory or initiate new echo-tests to any of its DPCs. It does, however, cause the process to retain all of its current timers, processing flags, status indicators, and performance accumulators as they are so that current processing is not affected.

Following successful warmboot processing, the ANSI Host Interface process sends event message 7011 to the EMS collector process. This message is described in the BAHISF log message documentation file.

Example

The following are examples of the WARMBOOT command. In these examples, the ANSI Host Interface process is P1A^HISF7.

DELIVER PROCESS P1A^HISF7, “9501”

DELIVER PROCESS P1A^HISF7, “WARMBOOT”

BIC ANSI Interface Process Commands

This section describes the processing performed by the BIC ANSI Interface process. The BIC ANSI Interface process can receive a number of BASE24 commands that cause it to perform specialized types of processing. These commands can be separated into three categories: process-generated commands, general use operator commands, and restricted use operator commands. All BASE24 general and restricted use operator commands that can be received by the BIC ANSI Interface process are documented in this subsection. Process-generated commands are documented in appendix B.

The BIC ANSI Interface process maintains the relationship between BASE24 and a co-network. In order to maintain that relationship, the BIC ANSI Interface process must perform a variety of functions. These include enabling message exchange between BASE24 and the co-network, providing store-and-forward capabilities, enabling settlement between BASE24 and the co-network, and managing communications between BASE24 and the co-network.

All of the commands that can be received by the BIC ANSI Interface process are listed below by category.

Process-Generated Commands	General Use Operator Commands	Restricted Use Operator Commands
CUTOVER DEVICE STATE	KEY CHANGE KEY REPEAT KEY VERIFY LOGOFF LOGON PERFORMANCE STATUS WARMBOOT	DEBUG DUMP ILF TRACE TRACE OFF TRACE ON

DEBUG Command

A restricted use operator command that places the BIC ANSI Interface process into the debug state. The debug message places the BIC ANSI Interface process under control of the HP NonStop Inspect product and displays an Inspect prompt on the home terminal (the terminal specified for the process by the HOMETERM attribute or the home terminal of the XPNET process). Inspect is an HP NonStop program debugging tool. For more information on the DEBUG command, refer to appendix A.

DUMP Command

A restricted use operator command that directs the BIC ANSI Interface process to request a programmatic dump. A programmatic dump allows a programmer to automatically capture an unusual situation for evaluation. The BIC ANSI Interface process can continue running after the dump is generated. For more information on the DUMP command, refer to appendix A.

ILF TRACE Command

A general use command that causes the BIC ANSI Interface process to reset the current trace number to a specified number.

Note: This command must be entered in all uppercase characters.

Syntax

DELIVER PROCESS *object-name,* "ILF TRACE *curr-trc-num"*

object-name — The symbolic name assigned to the BIC ANSI Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

curr-trc-num — The number to which the current trace number is to be reset. Valid values are 1–99999.

Processing

The BIC ANSI Interface process sets the current trace number to the specified value. Subsequent trace processing is based on this value.

Example

The following is an example of an ILF TRACE command to change the current trace number to 5. In this example, the BIC ANSI Interface process is P1A^BCAN5.

DELIVER PROCESS P1A^BCAN5, "ILF TRACE 5"

KEY CHANGE Command

A general use command that causes the BIC ANSI Interface process to either request or generate and send a new PIN key or message authentication code (MAC) key.

Note: The KEY CHANGE command can only be used when the BIC ANSI Interface process is set up for Dynamic Key Management (DKM).

Syntax

DELIVER PROCESS *object-name,* "*identifier* KEY CHANGE *option"*

object-name — The symbolic name assigned to the BIC ANSI Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

identifier — Identifies whether the change key request is for PIN keys or MAC keys. Valid values are as follows:

MAC = Change key request is to change the MAC key. This value is not currently used with the BIC ANSI Interface process.
PIN = Change key request is to change the PIN key.

option — Identifies the working keys to be changed by the KEY CHANGE command. Valid values are as follows:

B = Change both the inbound and the outbound working keys.
I = Change only the inbound working key.
O = Change only the outbound working key.

Processing

The BIC ANSI Interface process either requests a new key from the co-network or generates and sends a new key depending on which entity generates the key specified in the command.

If the BIC ANSI Interface process generates the specified key, it generates a new key and sends that new key to the co-network. If the co-network generates the specified key, the BIC ANSI Interface process generates a change key request and sends it to the co-network.

For more information on the steps required to process KEY CHANGE commands, along with the associated message flows, refer to the ***BASE24 BIC ANSI Interface Manual***.

Example

The following is an example of a KEY CHANGE command to change the inbound PIN key. In this example, the BIC ANSI Interface process is P1A^BCAN5.

DELIVER PROCESS P1A^BCAN5, “PIN KEY CHANGE I”

KEY REPEAT Command

A general use command that causes the BIC ANSI Interface process to send a repeat key message to the co-network for a specific PIN key or message authentication code (MAC) key. In turn, the co-network will respond to the repeat key message with a copy of the requested key.

Note: The KEY REPEAT command can only be used when the BIC ANSI Interface process is set up to use Dynamic Key Management (DKM).

Syntax

DELIVER PROCESS *object-name,* "*identifier* KEY REPEAT *option"*

object-name — The symbolic name assigned to the BIC ANSI Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

identifier — Identifies whether the repeat key request is for PIN keys or MAC keys. Valid values are as follows:

MAC = Repeat key request is to change the MAC keys. This value is not currently used with the BIC ANSI Interface process.
PIN = Repeat key request is to change the PIN keys.

option — Identifies the working keys to be repeated by the KEY REPEAT command. Valid values are as follows:

B = Repeat both the inbound and the outbound working keys.
I = Repeat only the inbound working key.
O = Repeat only the outbound working key.

Processing

The BIC ANSI Interface process generates a repeat key message and sends it to the co-network.

For more information on the steps required to process KEY REPEAT commands, along with the associated message flows, refer to the ***BASE24 BIC ANSI Interface Manual***.

Example

The following is an example of a KEY REPEAT command for the outbound PIN key. In this example, the BIC ANSI Interface process is P1A^BCAN5.

DELIVER PROCESS P1A^BCAN5, "PIN KEY REPEAT O"

KEY VERIFY Command

A general use command that causes the BIC ANSI Interface process to send a verify key message to the co-network for the specified PIN key or message authentication code (MAC) key. The co-network then verifies the key by comparing check digits.

Note: The KEY VERIFY command can only be used when the BIC ANSI Interface process is set up to use Dynamic Key Management (DKM).

Syntax

DELIVER PROCESS *object-name,* "*identifier* KEY VERIFY *option"*

object-name — The symbolic name assigned to the BIC ANSI Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

identifier — Identifies whether the verify key request is for PIN keys or MAC keys. Valid values are as follows:

MAC = Verify key request is to change the MAC keys. This value is not currently used with the BIC ANSI Interface process.
PIN = Change key request is to change the PIN keys.

option — Identifies the working keys to be verified by the KEY VERIFY command. Valid values are as follows:

B = Verify both the inbound and the outbound working keys.
I = Verify only the inbound working key.
O = Verify only the outbound working key.

Processing

The BIC ANSI Interface process generates a verify key message containing the necessary check digits and sends it to the co-network.

For more information on the steps required to process KEY VERIFY commands, along with the associated message flows, refer to the ***BASE24 BIC ANSI Interface Manual***.

Example

The following is an example of a KEY VERIFY command to verify the inbound PIN key. In this example, the BIC ANSI Interface process is P1A^BCAN3.

DELIVER PROCESS P1A^BCAN3, "PIN KEY VERIFY I"

LOGOFF Command

A general use command that initiates a logoff sequence for one or more stations. It causes the BIC ANSI Interface process to send a logoff message to each station indicated in the message.

Syntax

DELIVER PROCESS *object-name,* "LOGOFF *station-name"*

object-name — The symbolic name assigned to the BIC ANSI Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

station-name — The symbolic station name defined for the BIC ANSI Interface process in the Interchange Configuration File (ICF). The BIC ANSI Interface process builds a Process Control Table (PCT) at initialization that contains the station names from the information in the ICF. The PCT contains configuration and control information for the process.

If ALL is specified in this variable in the command, a logoff sequence is initiated for all stations defined to the process in the ICF. If an individual station is specified, the logoff sequence is initiated for that station only.

Processing

The BIC ANSI Interface process performs the following processing steps when it receives a LOGOFF command. A logoff can be initiated either by BASE24 or by a co-network.

Logoff Initiated by BASE24. The BIC ANSI Interface process performs the following steps when a logoff is initiated by BASE24:

1. Receives the command to log off.
2. Sends an 0800 network management message to the BIC Interchange indicating that a LOGOFF command has been received.
3. Deletes any timers associated with the station named in the LOGOFF command.
4. Disables the logons for the stations named in the LOGOFF command.

   The BIC ANSI Interface process does not require a response to the logoff message.

Logoff Initiated by Co-Network. The BIC ANSI Interface process performs the following steps when a logoff is initiated by a co-network:

1. Receives an 0800 network management message from the co-network indicating a LOGOFF command.
2. Deletes any timers associated with the station named in the LOGOFF command.
3. Disables the station named in the LOGOFF command from receiving any outgoing messages.

   The BIC ANSI Interface process does not send any response to the logoff message.

LOGON Command

A general use command that initiates a logon sequence for one or more stations. It causes the BIC ANSI Interface process to send a logon message to each station indicated in the message.

Syntax

DELIVER PROCESS *object-name,* "LOGON *station-name*"

object-name — The symbolic name assigned to the BIC ANSI Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

station-name — The symbolic station name defined for the BIC ANSI Interface process in the Interchange Configuration File (ICF). The BIC ANSI Interface process builds a Process Control Table (PCT) at initialization that contains the station names from the information in the ICF. The PCT contains configuration and control information for the process.

If ALL is specified in this variable in the command, a logon sequence is initiated for all stations defined to the process in the ICF. If an individual station is specified, the a logon sequence is initiated for that station only.

Processing

The BIC ANSI Interface process performs the following processing steps when it receives a LOGON command. A logon can be initiated either by BASE24 or by a co-network.

Logon Initiated by BASE24. The BIC ANSI Interface process performs the following steps when a logon is initiated by BASE24:

1. Receives the command to log on.
2. Sends an 0800 network management message to the co-network indicating that a LOGON command has been received.
3. Receives an 0800 message response from the co-network that either accepts or denies the logon.

   If the co-network accepts the logon, the process enables the station named in the LOGON command.

   If the co-network denies the logon, the process does not enable the station named in the LOGON command and logs an event message indicating why the logon was denied.

Logon Initiated by Co-Network. The BIC ANSI Interface process performs the following steps when a logon is initiated by a co-network:

1. Receives an 0800 network management message from the co-network indicating a LOGON command.
2. Accepts or denies the logon and returns an 0800 network management message to the co-network indicating the success or failure of the logon. The BIC ANSI Interface process performs the following steps to determine the outcome of the logon:
   a. Determines whether BASE24 currently accepts logons. Logons are disabled by a LOGOFF command sent from an XPNET network control facility and are enabled by another network control command or by reinitializing the BIC ANSI Interface process. If logons are disabled, information in the 0800 message sent by the BIC ANSI Interface process to the co-network indicates that the logon was denied because BASE24 is not currently accepting logons.
   b. Determines whether the version number and options sent in the logon message are compatible with the version number and logon options set for the BIC ANSI Interface process in the ICF. If the version number or options are incompatible, the information in the 0800 message sent by the BIC ANSI Interface process to the co-network indicates that the logon failed due to incompatibility with the co-network.

   If logons are enabled and logon options are compatible, the information in the 0800 message sent by the BIC ANSI Interface process to the co-network indicates a successful logon.

PERFORMANCE Command

A general use command that causes the performance statistics kept in memory by the BIC ANSI Interface process be sent to the EMS collector process. The BIC ANSI Interface process tracks performance statistics for each station assigned to it. These statistics are gathered on an interval basis—the interval length being established in the ICF—and statistics are available for the last five intervals.

Syntax

DELIVER PROCESS *object-name,* "PERFORMANCE"

object-name — The symbolic name assigned to the BIC ANSI Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

The PERFORMANCE command generates a series of event messages for the BASE24-atm and BASE24-pos products showing the statistics for the past five intervals. These event messages are numbered in the range 0095 through 0160 and are described in the BICLOGM log message documentation file. The BIC ANSI Interface process performs the following steps when it receives a PERFORMANCE command:

1. Retrieves the following performance statistics for BASE24-atm and BASE24-pos:
   - The number of minutes in the performance period
   - The outbound and inbound request limits
   - The number of requests in the period
   - The number of timeouts in the period
   - The percentage of timeouts per request in the period
   - The average time (in seconds) it took for the BIC ANSI Interface process to receive a response
2. Logs a series of event messages showing the statistics gathered for the past five intervals. The interval length is configured in the ICF. The event messages generated for BASE24-atm and BASE24-pos are identical except for the event message numbers. For BASE24-atm, the event messages generated for performance statistics are 0095, 0100, 0105, 0110, 0115, 0120, and 0125. For BASE24-pos, the event message numbers are 0130, 0135, 0140, 0145, 0150, 0155, and 0160.

STATUS Command

A general use command that requests the status of the BIC ANSI Interface process. The requested information is displayed in a series of event messages sent to the EMS collector process.

Syntax

DELIVER PROCESS *object-name,* “STATUS”

object-name — The symbolic name assigned to the BIC ANSI Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

The STATUS command generates a series of event messages. These event messages are numbered in the range of 0165 through 0220 and are described in the BICLOGM log message documentation file. The BIC ANSI Interface process performs the following steps when it receives a STATUS command:

1. Determines the current state of the BIC ANSI Interface process. Based on the current state of the BIC ANSI Interface process, different event messages are logged.
2. Logs the appropriate event messages to describe the current state of the BIC ANSI Interface process.

TRACE ON and TRACE OFF Commands

Restricted use operator commands that activate or deactivate a trace function within the BIC ANSI Interface process. When activated, the trace function generates an event message each time the BIC ANSI Interface process enters a new procedure. The event message displays the name of the procedure from which it was written. For more information on the TRACE ON and TRACE OFF commands, refer to appendix A.

WARMBOOT Command

A general use operator command that causes the BIC ANSI Interface process to reinitialize.

Syntax

DELIVER PROCESS *object-name,* "9501"

DELIVER PROCESS *object-name*, "WARMBOOT"

object-name — The symbolic name assigned to the BIC ANSI Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

The BIC ANSI Interface process performs the following processing steps when it receives a WARMBOOT command. Warmboot processing is similar to process initialization, except that the BIC ANSI Interface process closes all open files prior to initializing.

1. Saves the current dynamic key management information in temporary storage.
2. Reopens the Logical Network Configuration File (LCONF), reads the following assigns and params, and closes the LCONF:

   Assigns:

   ICF
   ILF
   KEYF
   LMCF
   RPT
   RPT-ERROR-FILE
   SAF
   SW-RPT-OBEY-FILE
   SW-RPT01-PROG-NAME
   SW-RPT09-PROG-NAME
   TKN

Params:

ACCT-NUM-TYP
ATM-LN-CUTOVER
BIC-ILF-NOTIFY-INTERVAL
BIC-ILF-NOTIFY-PERCENTAGE
BIC-SAF-NOTIFY-INTERVAL
BIC-SAF-NOTIFY-PERCENTAGE
BIC-SEM-LOGGING
BIC-VERSION-NUM
DATE-DISPLAY-TYPE
POS-LN-CUTOVER
SPAN-FLAG
SW-ILF-FILE-RETENTION
SW-RPT-STARTUP

3. Reads the Log Message Configuration File (LMCF) for its modified event messages. The LMCF contains a record for each modified event message in the system. The BIC ANSI Interface process opens the LMCF, retrieves its modified event messages, and closes the LMCF. Whenever the BIC ANSI Interface process generates an event message, it checks the LMCF records in memory to determine whether the message needs to be modified.
4. Reopens the ICF. If an error occurs, the BIC ANSI Interface process logs an error message and abends.
5. Using its process name, reads its own ICF record and then closes the ICF.
6. Reopens the KEYF. If an error occurs, the BIC ANSI Interface process logs an error message and abends.
7. Using its process name, reads its own KEYF record and then closes the KEYF.
8. Reinitializes the Process Control Table (PCT). The PCT contains configuration and control information for the process. This information is read from the ICF and KEYF, and includes all of the controls needed to process transactions.
9. Determines the posting dates for BASE24-atm, BASE24-pos, and the BIC interchange by comparing the current time to the BASE24-atm and BASE24-pos cutover times taken from the LCONF.
10. Reopens the current and previous day's Interchange Log File (ILF).
11. Reopens the Store-and-Forward File (SAF).
12. Opens the Token File (TKN) and rebuilds the token extended memory table.
13. If reports are required, starts a timer to expire when the reports are to be run.

14. If the automatic logon option was chosen, sends a logon message to the co-network.
15. Restores the process's dynamic key management information after warmboot processing is complete, including performance statistics, station status, and key threshold information.
16. Waits for a message from the co-network or a BASE24 process.

Example

The following are examples of WARMBOOT commands. In this example, the BIC ANSI Interface process is P1A^BCAN7.

DELIVER PROCESS P1A^BCAN7, "9501"

DELIVER PROCESS P1A^BCAN7, "WARMBOOT"

BIC ISO Interface Process Commands

This section describes the processing performed by the BIC ISO Interface process. The BIC ISO Interface process can receive a number of BASE24 commands that cause it to perform specialized processing. These commands can be separated into three categories: process-generated commands, general use operator commands, and restricted use operator commands. All BASE24 general and restricted use operator commands that can be received by the BIC ISO Interface process are documented in this subsection. Process-generated commands are documented in appendix B.

The BIC ISO Interface process maintains the relationship between BASE24 and a co-network. In order to maintain that relationship, the BIC ISO Interface process must perform a variety of functions. These include enabling message exchange between BASE24 and the co-network, providing store-and-forward capabilities, enabling settlement between BASE24 and the co-network, and managing communications between BASE24 and the co-network.

All of the commands that can be received by the BIC ISO Interface process are listed below by category.

Process-Generated Commands	General Use Operator Commands	Restricted Use Operator Commands
CUTOVER DEVICE STATE	ALTERATTRIBUTE KEY CHANGE KEY REPEAT KEY VERIFY LOGOFF LOGON PERFORMANCE PMONON PMONOFF PMONRESET STATUS SWL TOTALS WARMBOOT	DEBUG DUMP TRACE OFF TRACE ON

ALTERATTRIBUTE Command

A general use operator command that causes the BIC ISO Interface process to warmboot (reallocate) a specified shared extended memory table after it has been rebuilt. Whenever a shared extended memory table is modified, all processes that read the table must reallocate the table before the modified data is used in transaction processing. The Process Control Terminal (PCT) Server process automatically sends this command to each process specified in an EMT warmboot notify list param in the Logical Network Configuration File (LCONF) when a network operator enters an ALTERATTRIBUTE command from the EMT Control Command screen (screen 5) in the Device Control Terminal (DCT). You can also enter the ALTERATTRIBUTE command manually from a network control facility to warmboot an extended memory table for an individual process. The EMT warmboot notify list params are only used when issuing the command from the DCT. For more information on the EMT Control Command screen, refer to the ***BASE24 Device Control Manual***.

Note: This command is not available in releases prior to release 6.0.

Syntax

DELIVER PROCESS *object-name,* "ALTERATTR[IBUTE]*file-id*EMT*"*

object-name — The symbolic name assigned to the BIC ISO Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

file-id — The acronym of the file from which the shared extended memory table is built. Valid values are as follows:

APCF = Acquirer Processing Code File
IPCF = Issuer Processing Code File

Processing

This command causes the BIC ISO Interface process to reallocate the specified shared extended memory table. If the command completes successfully, a message stating that the EMT warmboot completed successfully is displayed on the network log.

Examples

The following are examples of ALTERATTRIBUTE commands sent to the BIC ISO Interface process from an XPNET network control facility. In these examples, the symbolic name of the BIC ISO Interface process is P1A^BCISO5 and the shared extended memory table to be warmbooted is the Acquirer Processing Code File (APCF) extended memory table.

DELIVER PROCESS P1A^BCISO5, “ALTERATTRIBUTEAPCFEMT”

DELIVER PROCESS P1A^BCISO5, “ALTERATTRAPCFEMT”

DEBUG Command

A restricted use operator command that places the BIC ISO Interface process into the debug state. The debug message places the BIC ISO Interface process under control of the HP NonStop Inspect utility and displays an Inspect prompt on the home terminal (the terminal specified for the process by the HOMETERM attribute or the home terminal of the XPNET process). Inspect is an HP NonStop program debugging tool. For more information on the DEBUG command, refer to appendix A.

DUMP Command

A restricted use operator command that directs the BASE24 BIC ISO Interface process to request a programmatic dump. A programmatic dump allows a programmer to automatically capture an unusual situation for evaluation. The BIC ISO Interface process can continue running after the dump is generated. For more information on the DUMP command, refer to appendix A.

KEY CHANGE Command

A general use command that causes the BIC ISO Interface to either request or generate and send a new PIN key or Message Authentication Code (MAC) key.

Note: The KEY CHANGE command can only be used when the BIC ISO Interface process is set up to use Dynamic Key Management (DKM).

Syntax:

DELIVER PROCESS *object-name,* “*identifier* KEY CHANGE *option”*

object-name — The symbolic name assigned to the BIC ISO Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

identifier — Identifies whether the change key request is for PIN keys or MAC keys. Valid values are as follows:

MAC = Change key request is to change the MAC key.
PIN = Change key request is to change the PIN key.

option — Identifies the working keys to be changed by the KEY CHANGE command. Valid values are as follows:

B = Change both the inbound and the outbound working keys.
I = Change only the inbound working key.
O = Change only the outbound working key.

Processing

The BIC ISO Interface either requests a new key from the co-network or generates and sends a new key depending on which entity is to generate the key specified in the command.

If the BIC ISO Interface process is to generate the specified key, it generates a new key and sends that new key to the co-network. If the co-network is to generate the specified key, the BIC ISO Interface process generates a change key request and sends it to the co-network.

For more information on the steps required to process KEY CHANGE commands, along with the associated message flows, refer to the ***BASE24 BIC ISO Interface Manual***.

Example

The following is an example of a command to change the inbound PIN key. In this example, the BIC ISO Interface process is P1A^BCISO5.

DELIVER PROCESS P1A^BCISO5, “PIN KEY CHANGE I”

KEY REPEAT Command

A general use command that causes the BIC ISO Interface process to send a repeat key message to the co-network for a specific PIN key or Message Authentication Code (MAC) key. In turn, the co-network will respond to the repeat key message with a copy of the requested key.

Note: The KEY REPEAT command can only be used when the BIC ISO Interface is set up to use Dynamic Key Management (DKM).

Syntax

DELIVER PROCESS *object-name,* "*identifier* KEY REPEAT *option"*

object-name — The symbolic name assigned to the BIC ISO Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

identifier — Identifies whether the repeat key request is for a PIN key or a MAC key. Valid values are as follows:

MAC = Repeat key request is for a MAC key.
PIN = Repeat key request is for a PIN key.

option — Identifies the working keys to be repeated by the KEY REPEAT command. Valid values are as follows:

B = Repeat both the inbound and the outbound working keys.
I = Repeat only the inbound working key.
O = Repeat only the outbound working key.

Processing

The BIC ISO Interface process generates a repeat key message and sends it to the co-network.

For more information on the steps required to process KEY REPEAT commands, along with the associated message flows, refer to the ***BASE24 BIC ISO Interface Manual***.

Example

The following is an example of a KEY REPEAT command for the outbound MAC key. In this example, the BIC ISO Interface process is P1A^BCISO5.

DELIVER PROCESS P1A^BCISO5, "MAC KEY REPEAT O"

KEY VERIFY Command

A general use command that causes the BIC ISO Interface process to send a verify key message to the co-network for the specified PIN key or Message Authentication Code (MAC) key. The co-network then verifies the key by comparing check digits.

Note: The KEY VERIFY command can only be used when the BIC ISO Interface process is set up to use Dynamic Key Management (DKM).

Syntax

DELIVER PROCESS *object-name,* "*identifier* KEY VERIFY *option*"

object-name — The symbolic name assigned to the BIC ISO Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

identifier — Identifies whether the verify key request is for a PIN key or a MAC key. Valid values are as follows:

MAC = Verify key request is for a MAC key.
PIN = Verify key request is for a PIN key.

option — Identifies the working keys to be verified by the KEY VERIFY command. Valid values are as follows:

B = Verify both the inbound and the outbound working keys.
I = Verify only the inbound working key.
O = Verify only the outbound working key.

Processing

The BIC ISO Interface generates a verify key message containing the necessary check digits and sends it to the co-network.

For more information on the steps required to process KEY VERIFY commands, along with the associated message flows, refer to the ***BASE24 BIC ISO Interface Manual***.

Example

The following is an example of a KEY VERIFY command for the PIN key. In this example, the BIC ISO Interface process is P1A^BCISO3 and the both the inbound and outbound working keys are being verified.

DELIVER PROCESS P1A^BCISO3, "PIN KEY VERIFY B"

LOGOFF Command

A general use command that initiates a logoff sequence for one or more stations. It causes the BIC ISO Interface process to send a logoff message to each station indicated in the message.

Syntax

DELIVER PROCESS *object-name,* "LOGOFF *station-name"*

DELIVER PROCESS *object-name,* "LOGOUT *station-name"*

object-name — The symbolic name assigned to the BIC ISO Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

station-name — The symbolic station name defined for the BIC ISO Interface process in the Interchange Configuration File (ICF) or Enhanced Interchange Configuration File (ICFE). The BIC ISO Interface process builds a Process Control Table (PCT) at initialization that contains the station names from the information in the ICF or ICFE. The PCT contains configuration and control information for the process.

If ALL is specified in this variable instead of a specific symbolic station name, a logoff sequence is initiated for all stations defined to the process in the ICF or ICFE. If an individual station is specified, the logoff sequence is initiated for that station only.

Processing

The BIC ISO Interface process performs the following processing steps when it receives a LOGOFF command. A logoff can be initiated either by BASE24 or by a co-network.

Logoff Initiated by BASE24. The BIC ISO Interface process performs the following steps when a logoff is initiated by BASE24:

1. Receives the command to logoff.
2. Sends an 0800 network management message to the BIC Interchange indicating that a LOGOFF command has been received.

   The BIC ISO Interface does not require a response to this logoff message.
3. Deletes any timers associated with the station named in the LOGOFF command.

Logoff Initiated by Co-Network. The BIC ISO Interface process performs the following steps when a logoff is initiated by a co-network.

1. Receives an 0800 network management message from the co-network indicating that a LOGOFF command has been received.
2. Deletes any timers associated with the station named in the LOGOFF command.
3. Disables the station named in the LOGOFF command from receiving any outgoing messages.

   The BIC ISO Interface does not send any response to the logoff message.

LOGON Command

A general use command that initiates a logon sequence for one or more stations. It causes the BIC ISO Interface process to send a logon message to each station indicated in the message.

Syntax

DELIVER PROCESS *object-name,* "LOGON *station-name"*

DELIVER PROCESS *object-name*, "LOGIN *station-name"*

object-name — The symbolic name assigned to the BIC ISO Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

station-name — The symbolic station name defined for the BIC ISO Interface process in the Interchange Configuration File (ICF) or Enhanced Interchange Configuration File (ICFE). The BIC ISO Interface process builds a Process Control Table (PCT) at initialization that contains the station names from the information in the ICF or ICFE. The PCT contains configuration and control information for the process.

If ALL is specified in this variable instead of a specific symbolic station name, a logon sequence is initiated for all stations defined to the process in the ICF or ICFE. If an individual station is specified, a logon sequence is initiated for that station only.

Processing

The BIC ISO Interface process performs the following processing steps when it receives a LOGON command. A logon can be initiated either by BASE24 or by a co-network.

Logon Initiated by BASE24. The BIC ISO Interface process performs the following steps when a logon is initiated by BASE24.

1. Receives the command to logon.
2. Sends an 0800 network management message to the co-network indicating that a LOGON command has been received.

3. Receives an 0810 message from the co-network that either accepts or denies the logon.

   If the co-network accepts the logon, the process enables the station named in the LOGON command.

   If the co-network denies the logon, the process does not enable the station named in the LOGON command and logs an event message indicating why the logon was denied.

Logon Initiated by Co-Network. The BIC ISO Interface process performs the following steps when a logon is initiated by a co-network.

1. Receives an 0810 network management message from the co-network indicating a LOGON command.
2. Determines whether the logon should be accepted or denied and returns an 0810 network management message to the co-network indicating the success or failure of the logon. The BIC ANSI Interface process performs the following steps to determine the outcome of the logon:
   a. Determines whether the version number and options sent in the logon message are compatible with the version number and logon options set for the BIC ISO Interface process in the ICF or ICFE. If the version number or options are incompatible, the information in the 0810 message sent by the BIC ISO Interface process to the co-network indicates that the logon failed due to incompatibility with the co-network.
   b. If the logon options are compatible, the information in the 0810 message sent by the BIC ISO Interface process to the co-network indicates a successful logon.

PERFORMANCE Command

A general use command that causes the performance statistics kept in memory by the BIC ISO Interface process be sent to the EMS collector process. The BIC ISO Interface process tracks performance statistics for each station assigned to it. These statistics are gathered on an interval basis—the interval length being established in the Interchange Configuration File (ICF) or Enhanced Interchange Configuration File (ICFE)—and statistics are available for the last five intervals.

Syntax

DELIVER PROCESS *object-name,* "PERFORMANCE"

object-name — The symbolic name assigned to the BIC ISO Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

The PERFORMANCE command generates a series of event messages for the BASE24-atm and BASE24-pos products showing the statistics for the past five intervals. These event messages are numbered in the range 0900 through 0930 for BASE24-atm and 1050 through 1080 for BASE24-pos, and are described in the BABILOGM log message documentation file. The BIC ISO Interface process performs the following steps when it receives a PERFORMANCE command:

1. Retrieves performance statistics for BASE24-atm and BASE24-pos which include the number of minutes in the performance period, the outbound request limits, the inbound request limits, the number of requests in the period, the number of timeouts in the period, the percentage of timeouts per request in the period, and the average time (in seconds) it took for the BIC ISO Interface process to receive a response.
2. Logs a series of event messages showing the statistics gathered for the past five intervals. The interval length is configured in the ICF or ICFE. The event messages generated for BASE24-atm and BASE24-pos are identical except for the event message numbers. For BASE24-atm, the event messages generated for performance statistics are 0900, 0905, 0910, 0915, 0920, 0925, and 0930. For BASE24-pos, the event message numbers are 1050, 1055, 1060, 1065, 1070, 1075, and 1080.

PMONOFF Command

A general use command that deactivates performance monitoring in the BIC ISO Interface process. Once performance monitoring has been deactivated with this command, it remains turned off until it is turned back on using the PMONON command.

Syntax

DELIVER PROCESS *object-name,* "PMONOFF"

object-name — The symbolic name assigned to the BIC ISO Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

The PMONOFF command deactivates performance monitoring in the BIC ISO Interface process. Refer to the informal ***BASE24 Switch Performance Monitoring Manual*** provided on the SW*xx*SWPM subvolume, where *xx* is the number of the current release, for detailed information on the PMONOFF command.

PMONON Command

A general use command that activates performance monitoring in the BIC ISO Interface process. Once performance monitoring has been activated with this command, it continues until it is turned off using the PMONOFF command. While performance monitoring is running, you can reset performance monitoring counters to zero at any time using the PMONRESET command. You can obtain performance monitoring statistics in a series of event messages sent to the EMS collector process by entering the PERFORMANCE or STATUS commands.

Syntax

DELIVER PROCESS *object-name,* "PMONON"

object-name — The symbolic name assigned to the BIC ISO Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

The PMONON command activates performance monitoring in the BIC ISO Interface process. Refer to the informal ***BASE24 Switch Performance Monitoring Manual*** provided on the SW*xx*SWPM subvolume, where *xx* is the number of the current release, for detailed information on the PMONON command and the counters maintained by the Switch Performance Monitoring module.

PMONRESET Command

A general use command that resets all performance monitoring counters in the BIC ISO Interface process to zero. Refer to the informal ***BASE24 Switch Performance Monitoring Manual*** provided on the SW*xx*SWPM subvolume, where *xx* is the number of the current release, for detailed information on the counters maintained by the Switch Performance Monitoring module.

Syntax

DELIVER PROCESS *object-name,* "PMONRESET"

object-name — The symbolic name assigned to the BIC ISO Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

The PMONRESET command resets all performance monitoring counters in the BIC ISO Interface process to zero.

STATUS Command

A general use command that requests the status of the BIC ISO Interface process. The requested information is displayed in a series of event messages sent to the EMS collector process.

Syntax

DELIVER PROCESS *object-name,* "STATUS"

object-name — The symbolic name assigned to the BIC ISO Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

The STATUS command generates a series of event messages. These event messages are numbered in the range of 0601 through 0619 and are described in the BABILOGM log message documentation file. The BIC ISO Interface process performs the following steps when it receives a STATUS command:

1. Determines the current state of the BIC ISO Interface process. Based on the current state of the BIC ISO Interface process, different event messages are logged.
2. Logs the appropriate event messages to describe the current state of the BIC ISO Interface process.

Event messages 0617, 0618, and 0619 are repeated for each station defined in the Interchange Configuration File (ICF) or Enhanced Interchange Configuration File (ICFE).

SWL Command

A general use operator command that changes the event message logging facility from the current facility to the indicated facility.

Syntax

DELIVER PROCESS *object-name,* "SWL *logging-facility"*

object-name — The symbolic name assigned to the BIC ISO Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

logging-facility — The facility to which logging is being changed.

Processing

The BIC ISO Interface begins sending event messages to the logging facility specified in the SWL command.

Note: This command is not the same as the command used to switch log files for the XPNET process. For more on the command used to switch log files for the XPNET process, refer to the ***NET24-XPNET Network Control Command Reference Manual.***

Example

The following is an example of a SWL command. In this example, the BIC ISO Interface process is P1A^BCISO3 and the logging facility is being switched from the current facility to the HP NonStop primary Event Management Service (EMS) collector process $0.

DELIVER PROCESS P1A^BCISO3, "SWL $0"

TOTALS Command

A general use command that requests current information regarding totals for transaction activity. The requested information is displayed in a series of event messages sent to the EMS collector process.

Syntax

DELIVER PROCESS *object-name,* "TOTALS ACQ"

DELIVER PROCESS *object-name*, "TOTALS ISS"

object-name — The symbolic name assigned to the BIC ISO Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

The TOTALS command generates a series of event messages sent to the EMS collector process. These event messages include event messages numbered in the range 0230 through 0245 and 0260 through 0285. Detailed descriptions of these event messages can be found in the BABILOGM log message documentation file. The BIC ISO Interface process only checks for ACQ or ISS in the command. The operator does not need to enter the entire word. If ACQ is specified in the command, only totals for the acquirer are displayed. If ISS is specified in the command, only totals for the issuer are displayed.

The BIC ISO Interface process logs the appropriate messages to describe the current transaction totals for the BIC ISO Interface process.

TRACE ON and TRACE OFF Commands

Restricted use operator commands that activate or deactivate a trace function within the BIC ISO Interface process. When activated, the trace function generates an event message each time the BIC ISO Interface process enters a new procedure. For more information on the TRACE ON and TRACE OFF commands, refer to appendix A.

WARMBOOT Command

A general use operator command that causes the BIC ISO Interface process to reinitialize.

Syntax

DELIVER PROCESS *object-name,* “9501”

DELIVER PROCESS *object-name,* “WARMBOOT”

object-name — The symbolic name assigned to the BIC ISO Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

Warmboot processing is similar to process initialization, except that the BIC ISO Interface closes all open files prior to initializing. The BIC ISO Interface process performs the following steps when it receives a WARMBOOT command:

1. Stores all current information so if an error occurs, it can be restored.
2. Reopens the Logical Network Configuration File (LCONF), reads the following assigns and params, and closes the LCONF. If an error occurs, the BIC ISO Interface process logs an error message and abends.

   Assigns:

   EMF
   ICF or ICFE
   ILF
   ITF
   KEYF
   LMCF
   SAF
   RPT
   RPT-ERROR-FILE
   SW-RPT-OBEY-FILE
   SW-RPT01-PROG-NAME
   SW-RPT09-PROG-NAME
   TKN

   Params:

   ACCT-NUM-TYP
   ATM-LN-CUTOVER
   BASE24-REL

DATE-DISPLAY-TYPE
ILF-NOTIFY-INTERVAL
ILF-NOTIFY-PERCENTAGE
POS-LN-CUTOVER
RSM-TERM-MASTER-KEY
SAF-NOTIFY-INTERVAL
SAF-NOTIFY-PERCENTAGE
SW-ILF-FILE-RETENTION
SW-RPT-STARTUP

3. Rereads the LMCF for its modified event messages. The LMCF contains a record for each modified event message in the system. The BIC ISO Interface process reopens the LMCF, retrieves its modified event messages, and closes the LMCF. Whenever the BIC ISO Interface process generates an event message, it checks the LMCF records in memory to determine whether the message requires modification.
4. Opens the Interchange Configuration File (ICF) or Enhanced Interchange Configuration File (ICFE). If an error occurs, the BIC ISO Interface process logs an error message and abends.
5. Using its process name, reads its own ICF or ICFE record.
6. Reopens the Key File (KEYF). If an error occurs, the BIC ISO Interface process logs an error message and abends.
7. Using its process name, reads its own KEYF record and then closes the KEYF.
8. Reinitializes the extended memory required for this process.
9. Reopens the External Message File (EMF), reads the appropriate records, and then closes the EMF. If an error occurs, the BIC ISO Interface process logs an error message and abends.
10. Rebuilds the Process Control Table (PCT). The PCT contains configuration and control information for the process. This information is read from the ICF or ICFE and the KEYF, and includes all of the controls needed to process transactions. If any errors occur, the old information is restored.
11. Opens the Interchange Log Files (ILFs) for the previous day, the current day, and the next day. If an error occurs, the BIC ISO Interface process logs an error message and abends.
12. Opens the Store-and-Forward File (SAF). If an error occurs, the BIC ISO Interface process logs an error message and abends.
13. Determines the posting dates for BASE24-atm, BASE24-pos, and the co-network by comparing the current time to the BASE24-atm and BASE24-pos cutover times taken from the LCONF.

14. Updates the ICF or ICFE with the current posting dates and then closes the ICF or ICFE.
15. If processing is successful to this point, the old files are closed.
16. Reinitializes a security device if applicable.
17. If reports are required, starts a timer to expire when the reports are to be run. If an error occurs, the BIC ISO Interface process logs an error message and abends.
18. Restarts a performance timer. If an error occurs, the BIC ISO Interface process logs an error message and abends.
19. Restarts the Interchange Totals File (ITF) interval timer. If an error occurs, the BIC ISO Interface process logs an error message and abends.
20. Deletes the Store-and-Forward File (SAF) timer, if one exists.
21. Resets the Dynamic Key Management (DKM) timers.
22. Closes the LCONF and waits for a message from the co-network or a local BASE24 Authorization process.

Example

The following are examples of WARMBOOT commands. In these examples, the BIC ISO Interface process is P1A^BCISO7.

```
DELIVER PROCESS P1A^BCISO7, “9501”
DELIVER PROCESS P1A^BCISO7, “WARMBOOT”
```

From Host Maintenance Process Commands

This section describes the processing performed by the From Host Maintenance process. The From Host Maintenance process can receive a number of BASE24 commands that cause it to perform specialized processing. These commands can be separated into three categories: process-generated commands, general use operator commands, and restricted use operator commands. All BASE24 general and restricted use operator commands that can be received by the From Host Maintenance process are documented in this subsection. Process-generated commands are documented in appendix B.

The From Host Maintenance process updates the BASE24 application files. When an update message is received, the From Host Maintenance process edits the information in the message, applies the requested change to the database, and logs a message to the Update Log File (ULF).

All of the commands that can be received by the From Host Maintenance process are listed below by category.

Process-Generated Commands	General Use Operator Commands	Restricted Use Operator Commands
CUTOVER REFRESH	WARMBOOT ALL WARMBOOT FIID	DEBUG DUMP TRACE OFF TRACE ON

DEBUG Command

A restricted use operator command that places the From Host Maintenance process under control of the HP NonStop Inspect product and displays an Inspect prompt on the home terminal (the terminal specified for the process by the HOMETERM attribute or the home terminal of the XPNET process). Inspect is an HP NonStop program debugging tool. For more information on the DEBUG command, refer to appendix A.

DUMP Command

A restricted use operator command that directs the From Host Maintenance process to request a programmatic dump. A programmatic dump allows a programmer to automatically capture an unusual situation for evaluation. The From Host Maintenance process can continue running after the dump is generated. For more information on the DUMP command, refer to appendix A.

TRACE ON and TRACE OFF Commands

Restricted use operator commands that activate or deactivate a trace function within the From Host Maintenance process. When activated, the trace function generates an event message each time the From Host Maintenance process enters a new procedure. The event message displays the name of the procedure from which it was written.

Syntax

DELIVER PROCESS *object-name,* "TRACE { ON | OFF } *bit*"

object-name — The symbolic name assigned to the From Host Maintenance process in the XPNET node. For more information on entering the object name, refer to section 1.

{ ON | OFF } — An identifier specifying whether trace is activated or deactivated. Valid values are as follows:

ON = Trace has been activated for the specified process.
OFF = Trace has been deactivated for the specified process.

bit — Activates or deactivates the trace function for only one module. If this parameter is omitted from the TRACE command syntax, all modules are activated or deactivated. The table below shows the module identified by each bit. Valid values for *bit* are as follows:

0 = Main
1 = Cardholder Authorization File (CAF)
2 = No Book File (NBF)
3 = Negative Card File (NEG)
4 = Positive Balance File (PBF)
5 = Stop Payment File (SPF)
6 = Warning/Hold/Float File (WHFF)
7 = Corporate Check File (CCF)
8 = Check Status File (CSF)
9 = From Host Maintenance (FHM) utilities
10 = Customer Table (CSTT)
11 = Customer/Account Relation Table (CACT)
12 = Personal Information Table (PIT)
13 = Customer/Cardholder Information File (CCIF)
14 = Customer/Cardholder Memo File (CCMF)
15 = Customer/Personal ID Relation Table (CPIT)

Processing

The TRACE command is intended for use in trouble-shooting the From Host Maintenance process. It can be activated for all modules of the From Host Maintenance process or for a specific module. The TRACE ON form of the command is used to activate the trace function for all modules. To activate the trace function for only one module, the TRACE ON *bit* form is used. When the trace function is set to on, event messages are sent to the current logging facility.

The TRACE OFF command deactivates trace for all modules. The TRACE OFF *bit* command deactivates the trace function for a specific module within the From Host Maintenance process.

The From Host Maintenance process performs the following processing steps when it receives a TRACE command:

1. Reads the command.
2. Determines whether the TRACE command activates or deactivates all modules of the From Host Maintenance process or a specific module by checking for the presence of the *bit* parameter

   If the *bit* parameter is present in the command, a specific module is activated or deactivated.

   If the *bit* parameter is absent from the command, all modules within the From Host Maintenance process are activated or deactivated.
3. Activates or deactivates the trace function within the From Host Maintenance process.

Example

The following are examples of TRACE commands. The From Host Maintenance process for which the command is being entered is P1A^FHFHM2. In the first example, the No Book File (NBF) module is activated (i.e., the value 2 is present in the *bit* field). In the second example, all modules are deactivated (i.e., no value is present in the *bit* field).

```
DELIVER PROCESS P1A^FHFHM2, “TRACE ON 2”
DELIVER PROCESS P1A^FHFHM2, “TRACE OFF”
```

WARMBOOT ALL Command

A general use operator command that causes the From Host Maintenance process to reinitialize.

Syntax

DELIVER PROCESS *object-name,* "9501 ****"

DELIVER PROCESS *object-name*, "WARMBOOT ****"

object-name — The symbolic name assigned to the From Host Maintenance process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

When the From Host Maintenance process receives a WARMBOOT ALL command, it goes through its startup procedures. The startup procedures are the same as the initialization procedures, except that the From Host Maintenance process closes all open files prior to initializing.

WARMBOOT FIID Command

A general use operator command that causes the From Host Maintenance process to update the current information that it has in memory for an institution. A WARMBOOT FIID command reinitializes the specified From Host Maintenance process.

Syntax

DELIVER PROCESS *object-name,* "9501 *fiid"*

DELIVER PROCESS *object-name*, "WARMBOOT *fiid"*

object-name — The symbolic name assigned to the From Host Maintenance process in the XPNET node. For more information on entering the object name, refer to section 1.

fiid — The FIID of the financial institution being warmbooted.

Processing

When the From Host Maintenance process receives this command, it performs the following steps:

1. Locates the IDF record for the FIID in extended memory. If an IDF record cannot be found in extended memory for this FIID, a message is logged indicating that the WARMBOOT FIID command can not be used to add new FIIDs. Adding a new FIID is accomplished using the WARMBOOT ALL command.
2. Reads the corresponding IDF record from the disk file.
3. Closes any Negative Card Files (NEGs), Cardholder Authorization Files (CAFs), Positive Balance Files (PBFs), No Book Files (NBFs), Stop Payment Files (SPFs), or Warning/Hold/Float Files (WHFFs) open for the institution.
4. Overlays the IDF record in extended memory with the new IDF record from disk.
5. Reopens all NEGs, CAFs, PBFs, NBFs, SPFs, and WHFFs for that FIID and updates the indexes in the IDF table.

If a fatal error occurs during this processing, the From Host Maintenance process logs an error message indicating that no transactions will be processed for this FIID.

Example

The following are examples of WARMBOOT FIID commands. In these examples, the From Host Maintenance process is P1A^FHFHM1. The FIID being warmbooted is BNK3.

DELIVER PROCESS P1A^FHFHM1, “9501 BNK3”

DELIVER PROCESS P1A^FHFHM1, “WARMBOOT BNK3”

ISO Host Interface Process Commands

The ISO Host Interface process translates ISO-based external messages between the host and BASE24-atm, BASE24-pos, and BASE24-teller. This section describes the BASE24 commands recognized by the ISO Host Interface process.

Note: The ISO Host Interface process supports only the ISO-based external message format. For information on support of the ANSI-based external message format, refer to the ANSI Host Interface process commands documentation provided earlier in this section.

The ISO Host Interface process can receive a number of BASE24 commands that cause it to perform specialized processing. These commands can be separated into three categories: process-generated commands, general use operator commands, and restricted use operator commands. All BASE24 general and restricted use operator commands that can be received by the ISO Host Interface process are documented in this subsection. Process-generated commands are documented in appendix B.

All of the commands that the ISO Host Interface process can receive are listed below by category.

Process-Generated Commands	General Use Operator Commands	Restricted Use Operator Commands
CUTOVER DEVICE STATE SAFEXTCOMP SAFEXTRQST	KEY CHANGE KEY REPEAT KEY VERIFY LOGOFF LOGON PACINGSTAT PERFORMANCE PERFSTAT SAFPACING STA STATUS STARTSAF STOPSAF SWL WARMBOOT	DEBUG DUMP TRACE ON TRACE OFF

DEBUG Command

A restricted use operator command that places the ISO Host Interface process into the debug state. The debug message places the ISO Host Interface process under control of the HP NonStop Inspect product and displays an Inspect prompt on the home terminal (the terminal specified for the process by the HOMETERM attribute or the home terminal of the XPNET process). Inspect is an HP NonStop program debugging tool. For more information on the DEBUG command, refer to appendix A.

DUMP Command

A restricted use operator command that directs the ISO Host Interface process to request a programmatic dump. A programmatic dump allows a programmer to automatically capture an unusual situation for evaluation. The ISO Host Interface process can continue running after the dump is generated. For more information on the DUMP command, refer to appendix A.

KEY CHANGE Command

A general use command that causes the ISO Host Interface process to either request or generate and send a new PIN key or message authentication code (MAC) key.

Note: The KEY CHANGE command can only be used when the ISO Host Interface process is set up to use Dynamic Key Management (DKM).

Syntax

DELIVER PROCESS *object-name,* "*identifier* KEY CHANGE *option dpc"*

object-name — The symbolic name assigned to the ISO Host Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

identifier — Identifies whether the change key request is for PIN keys or MAC keys. Valid values are as follows:

MAC = Change key request is to change MAC keys.
PIN = Change key request is to change PIN keys.

option — Identifies the working keys to be changed by the KEY CHANGE command. Valid values are as follows:

B = Change both the inbound and the outbound working keys.
I = Change only the inbound working key.
O = Change only the outbound working key.

dpc — Identifies the data processing center (DPC) for which the KEY CHANGE command is being issued.

Processing

The ISO Host Interface process either requests a new key from the host or generates and sends a new key depending on which entity is to generate the key specified in the command.

If the ISO Host Interface process is to generate the specified key, it generates a new key and sends that new key to the host. If the host is to generate the specified key, the ISO Host Interface process generates a change key request and sends it to the host.

For more information on the steps required to process KEY CHANGE commands, along with the associated message flows, refer to the ***BASE24 ISO Host Interface Manual***.

Example

The following is an example of a KEY CHANGE command to change the inbound PIN key for DPC 0001. In this example, the ISO Host Interface process is P1A^HISO1.

```
DELIVER PROCESS P1A^HISO1, “PIN KEY CHANGE I 0001”
```

KEY REPEAT Command

A general use command that causes the ISO Host Interface process to send a repeat key message to the host for a specific PIN key or message authentication code (MAC) key. In turn, the host will respond to the repeat key message with a copy of the requested key.

Note: The KEY REPEAT command can only be used when the ISO Host Interface process is set up to use Dynamic Key Management (DKM).

Syntax

DELIVER PROCESS *object-name,* "*identifier* KEY REPEAT *option dpc-num*"

object-name — The symbolic name assigned to the ISO Host Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

identifier — Identifies whether the repeat key request is for PIN keys or MAC keys. Valid values are as follows:

MAC = Repeat key request is for the MAC keys.
PIN = Repeat key request is for the PIN keys.

option — Identifies the working keys to be repeated by the KEY REPEAT command. Valid values are as follows:

B = Repeat both the inbound and the outbound working keys.
I = Repeat only the inbound working key.
O = Repeat only the outbound working key.

dpc-num — Identifies the data processing center (DPC) number for which the KEY REPEAT command is being performed.

Processing

The ISO Host Interface process generates a repeat key message and sends it to the host.

For more information on the steps required to process KEY REPEAT commands, along with the associated message flows, refer to the ***BASE24 ISO Host Interface Manual***.

Example

The following is an example of a KEY REPEAT command for the outbound PIN key for DPC 0001. In this example, the ISO Host Interface process is P1A^ HISO5.

```
DELIVER PROCESS P1A^HISO5, “PIN KEY REPEAT O 0001”
```

KEY VERIFY Command

A general use command that causes the ISO Host Interface process to send a verify key message to the host for the specified PIN key or message authentication code (MAC) key. The host then verifies the key by comparing check digits.

Note: The KEY VERIFY command can only be used when the ISO Host Interface process is set up to use Dynamic Key Management (DKM).

Syntax

DELIVER PROCESS *object-name,* “*identifier* KEY VERIFY *option dpc-num”*

object-name — The symbolic name assigned to the ISO Host Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

identifier — Identifies whether the verify key request is for PIN keys or MAC keys. Valid values are as follows:

MAC = Verify key request is for the MAC keys.
PIN = Verify key request is for the PIN keys.

option — Identifies the working keys to be verified by the KEY VERIFY command. Valid values are as follows:

B = Verify both the inbound and the outbound working keys.
I = Verify only the inbound working key.
O = Verify only the outbound working key.

dpc-num — Indicates the number of the data processing center (DPC) for which the KEY VERIFY command is being performed.

Processing

The ISO Host Interface process generates a verify key message containing the necessary check digits and sends it to the host.

For more information on the steps required to process KEY VERIFY commands, along with the associated message flows, refer to the ***BASE24 ISO Host Interface Manual***.

Example

The following is an example of a KEY VERIFY command to verify the inbound MAC key for DPC 0001. In this example, the ISO Host Interface process is P1A^HISO3.

DELIVER PROCESS P1A^HISO3, “MAC KEY VERIFY I 0001”

LOGOFF Command

A general use operator command that initiates a logoff sequence for one or more data processing centers (DPCs).

Syntax

DELIVER PROCESS *object-name,* "LOGOFFALL"

DELIVER PROCESS *object-name*, "LOGOFF *dpc"*

DELIVER PROCESS *object-name*, "LOGOUTALL"

DELIVER PROCESS *object-name,* "LOGOUT *dpc"*

object-name — The symbolic name assigned to the ISO Host Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

dpc — Identifies the DPC for which the logoff is being initiated.

Processing

The LOGOFF command causes the ISO Host Interface process to generate a logoff message and send it to each send/receive station supported by a specified DPC. If ALL is specified in the message, the stations associated with all DPCs the ISO Host Interface supports are logged off. If a DPC number is specified, only the stations supported by the specified DPC are logged off.

Logoff messages are network management messages used to halt communications between BASE24 and a DPC. They are sent as 0800 messages and require 0810 messages in response.

Logoff messages are distinguished by the value of 002 in the Network Management Information Code (S-70) field of the BASE24 External Message. Logoff messages can be sent by either BASE24 or a host. In either case, after a DPC is logged off, BASE24 does not continue to echo-test any of the DPC's stations, nor does it send or respond to non-network management messages. Any non-network management messages that must be forwarded to the DPC are placed in the Store-and-Forward File (SAF) instead.

BASE24 only attempts to reopen the link to a DPC that is logged off if a LOGON command is received from an XPNET network control facility. The DPC, on the other hand, can reopen the link at any time by sending an echo-test or logon message. BASE24 responds to either type of message.

Note: When a DPC is logged off, any transactions in progress are allowed to complete.

BASE24 only sends logoff messages to a DPC as a result of a LOGOFF command issued from an XPNET network control facility. BASE24 then requires a logoff response from the DPC.

If no logoff response is received, The ISO Host Interface process still marks the DPC as logged off, but continues to send logoff messages to the DPC until a response is received.

When a host logs off because of a scheduled down or unavailable period, it can transmit a logoff message from any of its stations. Upon receipt of a logoff message, the ISO Host Interface process marks the DPC down and logged off and returns a logoff response to the DPC.

For more information on logoff and echo-test message processing and message flows, refer to the ***BASE24 ISO Host Interface Manual***.

LOGON Command

A general use operator command that initiates a logon sequence for one or more data processing centers (DPCs).

Syntax

DELIVER PROCESS *object-name,* "LOGONALL"

DELIVER PROCESS *object-name*, "LOGON *dpc"*

DELIVER PROCESS *object-name*, "LOGINALL"

DELIVER PROCESS *object-name*, "LOGIN *dpc"*

object-name — The symbolic name assigned to the ISO Host Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

dpc — Identifies the DPC for which the logon is being initiated.

Processing

The LOGON command causes the ISO Host Interface process to generate an echo-test message and send it to each of a DPC's send/receive stations. If the DPC is currently logged off, it also causes the ISO Host Interface process to set a new performance timer.

If ALL is specified in the message, the above actions are taken for all DPCs processed by the ISO Host Interface process. If a DPC number is specified, the above actions are taken for that DPC only. Logon messages are network management messages used to manage communications access between BASE24 and a DPC's stations. They are sent as 0800 messages and in some cases require 0810 messages in response.

Logon messages are distinguished by the value of 001 in the Network Management Information Code (S-70) field of the BASE24 External Message.

The ISO Host Interface process initiates logon messages at initialization and upon receipt of a LOGON command from an XPNET network control facility.

At initialization, the ISO Host Interface process sends a logon message to each send/receive station.

Upon receiving a LOGON command from an XPNET network control facility, the ISO Host Interface process sends a logon message to each send/receive station associated with the DPC indicated in the command.

After a period of being logged off, DPCs can send logon messages to log on to BASE24.

If the ISO Host Interface process receives a logon message after the DPC has been logged off for a period, the ISO Host Interface process marks the DPC as logged on and begins sending echo-test messages to that DPC's stations. The ISO Host Interface process then returns a logon response to the DPC. If the ISO Host Interface process receives a logon message when the DPC is not logged off, it simply returns a logon response.

Note: In its responses, the ISO Host Interface process always sets the Response Code (P-39) field in the BASE24 External Message to a value of 00 to indicate an approval.

For more information on logon and echo-test message processing and message flows, refer to the ***BASE24 ISO Host Interface Manual***.

PACINGSTAT Command

A general use operator command that requests the status of store-and-forward (SAF) pacing from the ISO Host Interface process.

Syntax

DELIVER PROCESS *object-name,* “PACINGSTAT <dpc>”

object-name — The symbolic name assigned to the ISO Host Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

dpc — Identifies the data processing center (DPC) for which the SAF pacing status command is generated.

Processing

The PACINGSTAT command displays the status of SAF pacing.

1. If SAF pacing is enabled, an event message is generated displaying “SAF Pacing turned ON - set to $$$$” where $$$$ is the SAF pacing interval.
2. If SAF pacing is not enabled, an event message is generated displaying “SAF Pacing turned OFF.”

PERFORMANCE Command

A general use operator command that requests performance statistics from the ISO Host Interface process. Statistics are sent in a series of event messages to the EMS collector process.

Syntax

DELIVER PROCESS *object-name,* "PERFORMANCEALL"

DELIVER PROCESS *object-name*, "PERFORMANCE *dpc*"

object-name — The symbolic name assigned to the ISO Host Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

dpc — Identifies the data processing center (DPC) for which performance statistics are being generated.

Processing

The ISO Host Interface process tracks performance statistics for each DPC it supports. These performance statistics are gathered on an interval basis—the interval length being established by the TIMER-LMTS.PERFORMANCE field in the HCF. Statistics are available for the four most current intervals. The ISO Host Interface process performs the following steps when it receives a PERFORMANCE command:

1. Retrieves statistics for BASE24-atm, BASE24-pos, BASE24-telebanking, and BASE24-teller. Performance statistics include the number of minutes in the performance period, the outbound request limits, the inbound request limits, the number of requests in the period, the number of timeouts in the period, the percentage of timeouts per request in the period, the average time (in seconds) it took for the ISO Host Interface process to receive a response, and the number of rejects in the period. The rejects counter is used to count transactions not sent to the host or not sent to BASE24. These could be due to format errors, station down, etc.
2. Sends performance statistics in a series of event messages numbered in the range of 0430 through 0435 to the EMS collector process. If ALL is specified in the command, the ISO Host Interface process logs performance statistics for all of the DPCs it supports. If a DPC number is specified in the message, the ISO Host Interface process logs performance statistics for that DPC only. If no requests have been made in either direction for a DPC, the ISO Host

Interface process logs a single message indicating that performance data is not available for the DPC. All of these event messages are described in the BAHISO log message documentation file.

PERFSTAT Command

A general use operator command that requests performance statistics from the ISO Host Interface process to display data for the current performance period, as well as the status information for the data processing center (DPC).

Syntax

DELIVER PROCESS *object-name,* “PERFSTAT <dpc>”

object-name — The symbolic name assigned to the ISO Host Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

dpc — Identifies the data processing center (DPC) for which enhanced performance status information is being generated.

Processing

The ISO Host Interface process tracks performance statistics for each DPC it supports. These performance statistics are gathered on an interval basis—the interval length being established by the TIMER-LMTS.PERFORMANCE field in the HCF. Statistics are available for the four most current intervals. The ISO Host Interface process performs the following steps when it receives a PERFSTAT command:

1. Retrieves statistics for the current performance period for BASE24-atm and BASE24-pos. Performance statistics include the number of minutes in the performance period, the outbound request limits, the inbound request limits, the number of requests in the period, the number of timeouts in the period, the percentage of timeouts per request in the period, the average time (in seconds) it took for the ISO Host Interface process to receive a response and the number of rejects in the period.
2. Sends performance statistics in a series of event messages to the EMS collector process. The ISO Host Interface process logs performance statistics for the specified DPC only. If no requests have been made in either direction for a DPC, the ISO Host Interface process logs a single message indicating that performance data is not available for the DPC.
3. If the link to the host DPC is down, the ISO Host Interface process sends an event message indicating that the link is down. If the link to the host is up, but the host is not logged on, the process sends an event message indicating the host is logged off. However, if the link is up and logged on, an event message is sent indicating how many of the stations are up. If there are records in the Store-and-Forward File (SAF) or if there are SAF records in progress, an

event message is sent indicating the number of records in the SAF and how many are in progress. All of these event messages are described in the BAHISO log message documentation file.

SAFPACING Command

A general use operator command that modifies the pacing interval between store-and-forward (SAF) transactions.

Syntax

DELIVER PROCESS *object-name,* “SAFPACING<*t*>DPC<dpc>”

object-name — The symbolic name assigned to the ISO Host Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

t — The number of hundredths of a second to wait between store-and-forward records. The minimum value is 0 and the maximum value is 9999. If *t* is set to a value of 0, pacing is turned off. A value must be specified in this command.

dpc — Identifies the data processing center (DPC) for which the SAF Pacing interval is being set.

Processing

The ISO Host Interface process performs the following steps when it receives a SAFPACING command:

1. Checks to make sure an interval is specified. If it is not specified, an event message is sent, and the ISO Host Interface process drops the message.
2. Checks to make sure a specific DPC is specified. If it is not, an event message is sent, and the ISO Host Interface process drops the message.
3. Checks to see whether SAF pacing is already in progress. If it is, the pacing timer is deleted, and an event message is sent to indicate the old value is being replaced with the new value.
4. Verifies whether the new interval value is between 0 and 9999 hundredths of a second. If it is not, an event message is sent, and the ISO Host Interface process drops the message.
5. Starts the SAF pacing interval by inserting a new SAF pacing timer for the new interval value. Sends an event message indicating that SAF pacing has been turned on and indicates the new interval value.

STARTSAF Command

A general use operator command that starts a store-and-forward cycle.

Syntax

DELIVER PROCESS *object-name,* "STARTSAF *dpc"*

DELIVER PROCESS *object-name,* "SAFSTART *dpc"*

object-name — The symbolic name assigned to the ISO Host Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

dpc — Identifies the data processing center (DPC) for which the store-and-forward cycle is being started.

Processing

The STARTSAF command is intended to restart a store-and-forward cycle stopped by previous entry of a STOPSAF command. Note that, since the STOPSAF command only stops the store-and-forward cycle temporarily, this command has limited value.

When the ISO Host Interface process receives the STARTSAF command, it performs the following steps:

1. Checks to make sure the DPC is not logged off or marked down.

   If the DPC is logged off or marked down, the ISO Host Interface process drops the message.

2. Checks to make sure that a Store-and-Forward File (SAF) extract is not currently in progress for the DPC. If a SAF extract is in progress for the DPC, the ISO Host Interface process drops the message.

3. Checks to make sure that a store-and-forward cycle, which could have been initiated by another station, is not already in progress for the DPC. If a store-and-forward cycle already is in progress, the ISO Host Interface process drops the message.

4. Starts the store-and-forward cycle and updates the DPC status to indicate that a store-and-forward cycle is in progress.

Example

The following are examples of STARTSAF commands. In these examples, the ISO Host Interface process is P1A^HISO5 and the DPC is 6751.

DELIVER PROCESS P1A^HISO5, “STARTSAF 6751”

DELIVER PROCESS P1A^HISO5, “SAFSTART 6751”

STA Command

A general use operator diagnostic command that requests the current input and output sequence numbers for each station associated with a data processing center (DPC).

Syntax

DELIVER PROCESS *object-name,* “STA *dpc”*

object-name — The symbolic name assigned to the ISO Host Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

dpc — Identifies the DPC for which station diagnostics are being run.

Processing

The STA command displays the input and output sequence numbers for each station associated with the specified DPC. These sequence numbers are presented in event messages 0998 and 0999, as described in the BAHISO log message documentation file.

STATUS Commands

General use operator commands that request status information for one or all of the data processing centers (DPCs) that the ISO Host Interface process supports.

Syntax

DELIVER PROCESS *object-name,* "STATUSALL"

DELIVER PROCESS *object-name*, "STATUS *dpc"*

object-name — The symbolic name assigned to the ISO Host Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

dpc — Identifies the DPC for which status information is requested. If the command form is STATUSALL, the status of all DPCs that the ISO Host Interface process supports is requested.

Processing

The ISO Host Interface process sends the requested status information in a series of event messages numbered 0460 through 0472. These event messages are described in the BAHISO log message documentation file.

STOPSAF Command

A general use operator command that requests the temporary termination of a store-and-forward cycle.

Syntax

DELIVER PROCESS *object-name,* "STOPSAF *dpc"*

DELIVER PROCESS *object-name*, "SAFSTOP *dpc"*

object-name — The symbolic name assigned to the ISO Host Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

dpc — Identifies the data processing center (DPC) for which the store-and-forward cycle is being terminated.

Processing

The STOPSAF command can be used to stop store-and-forward processing if a Store-and-Forward File (SAF) extract is required. Since a SAF extract cannot be started while a store-and-forward cycle is in progress, an operator must stop the cycle before starting the extract.

The STOPSAF command only stops store-and-forward processing temporarily. It will start again automatically under the normal conditions for starting a store-and-forward cycle. Refer to the STARTSAF command description provided earlier in this subsection for more information on starting the store-and-forward cycle.

Example

The following are examples of STOPSAF commands. In these examples, the ISO Host Interface process is P1A^HISO5 and the DPC is 8510.

DELIVER PROCESS P1A^HISO5, "STOPSAF 8510"

DELIVER PROCESS P1A^HISO5, "SAFSTOP 8510"

SWL Command

A general use operator command that changes the primary event message logging facility for the ISO Host Interface process from the current log destination to the destination specified in the command.

Syntax

DELIVER PROCESS *object-name,* "SWL *logging-facility*"

object-name — The symbolic name assigned to the ISO Host Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

logging-facility — The facility to which logging is being changed.

Processing

The ISO Host Interface process begins logging event messages to the logging facility specified in the SWL command.

Note: This command is not the same as the command used to switch log files for the XPNET process. For more information on network commands, refer to the ***NET24-XPNET Network Control Command Reference Manual***.

Example

The following is an example of a SWL command. In this example, the ISO Host Interface process is P1A^HISO3 and the logging facility is being switched from the current log destination to the HP NonStop primary Event Management Service (EMS) collector process $0.

DELIVER PROCESS P1A^HISO3, "SWL $0"

TRACE ON and TRACE OFF Commands

Restricted use operator commands that activate or deactivate a trace function within the ISO Host Interface process. When activated, the trace function generates an event message each time the ISO Host Interface process enters a new procedure. The event message displays the name of the procedure from which it was written.

Syntax

DELIVER PROCESS *object-name,* "TRACE { ON | OFF } *bit*"

object-name — The symbolic name assigned to the ISO Host Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

{ ON | OFF } — An identifier specifying whether trace is activated or deactivated. Valid values are as follows:

ON = Trace has been activated for the specified process.
OFF = Trace has been deactivated for the specified process.

bit — Identifies the product for which the trace function has been activated or deactivated. The bits corresponding to BASE24, BASE24-atm, BASE24-pos, BASE24-telebanking, and BASE24-teller are as follows:

bit 0 = BASE24
bit 1 = BASE24-atm
bit 2 = BASE24-pos
bit 3 = BASE24-teller
bit 14 = BASE24-telebanking

Processing

When the trace function is activated, event messages are sent to the current logging facility. The trace function should be activated only in a test environment.

The TRACE ON form of the message is used to turn on the trace function for all products. To turn on the trace function for only one product, the TRACE ON *bit* form is used.

The trace off message deactivates the trace function within the ISO Host Interface process. The TRACE OFF form of the message is used to turn off the trace function for all products. To turn off the trace function for only one product, the TRACE OFF *bit* form is used.

The ISO Host Interface process performs the following processing steps when it receives a TRACE command:

1. Reads the command.
2. Activates or deactivates the trace function within the ISO Host Interface process.

Example

The following are examples of TRACE commands. The ISO Host Interface process for which the commands are being entered is P1A^HISO2. In the TRACE OFF command example, the trace function is being deactivated for the BASE24-teller product.

```
DELIVER PROCESS P1A^HISO2, "TRACE ON"
DELIVER PROCESS P1A^HISO2, "TRACE OFF 3"
```

WARMBOOT Command

A general use operator command that causes the ISO Host Interface process to reinitialize.

Syntax

DELIVER PROCESS *object-name,* “9501”

DELIVER PROCESS *object-name*, “WARMBOOT”

object-name — The symbolic name assigned to the ISO Host Interface process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

This command warm boots the ISO Host Interface process. Warmboot processing is similar to process initialization, except that a warmboot only includes those steps required to update the file information carried by the ISO Host Interface process in memory.

During warmboot processing, the ISO Host Interface process opens the following files:

- Logical Network Configuration File (LCONF)
- Log Message Configuration File (LMCF)
- Host Configuration File (HCF)
- External Message File (EMF)
- Key File (KEYF)

The process retains all of its current timers, processing flags, status indicators, and performance accumulators so that current processing is not affected. It does not, however, reallocate extended memory or initiate new echo-tests to any of its data processing centers (DPCs). Nor does it open the Store-and-Forward File (SAF) and count the messages pending for each DPC.

During warmboot processing, the ISO Host Interface process retrieves the names of its security modules from the LCONF and attempts to open those modules. If the security modules named cannot be opened, the ISO Host Interface process continues using those that are already open.

Following successful warmboot processing, the ISO Host Interface process sends event message 0499 to the EMS collector process. This event message is described in the BAHISO log message documentation file.

Example

The following are examples of the WARMBOOT command. In these examples, the ISO Host Interface process is P1A^HISO7.

DELIVER PROCESS P1A^HISO7, “9501”

DELIVER PROCESS P1A^HISO7, “WARMBOOT”

Refresh Process Commands

This section describes the processing performed by the BASE24 Refresh processes and includes descriptions of the BASE24 commands recognized by each Refresh process. Four different BASE24 Refresh processes exist, each of which is described below.

- The Base National Negative Refresh process is an Enscribe process that is used to refresh the Base national negative files used by the BASE24-pos product.
- The INAS National Negative Refresh process is an Enscribe process that is used to refresh the INAS national negative files used by the BASE24-pos product.
- The Enscribe Refresh process is used to refresh Enscribe files other than national negative files.
- The SQL Refresh process is used to refresh SQL tables.

Each Refresh process can receive a number of BASE24 commands that cause it to perform specialized types of processing. These commands can be separated into three categories: process-generated commands, general use operator commands, and restricted use operator commands. All BASE24 general and restricted use operator commands that can be received by the Refresh processes are documented in this subsection. Process-generated commands are documented in appendix B.

Refresh processes allow full- or partial-file refreshes to be performed on BASE24. Using disks or computer tapes generated by a host computer, Refresh processes update particular BASE24 files or tables. Transactions can still be processed while performing full- or partial-file refreshes of BASE24 files or partial-file refreshes of BASE24 tables. A host computer is any computer other than the HP NonStop processor on which BASE24 is installed.

Caution: The HP NonStop Kernel and NonStop SQL/MP currently require system downtime in order to perform the full-file refresh of an SQL table. This is because the name of an SQL table cannot be changed without recompiling the processes that use the table. Therefore, a full-file refresh is recommended only at system installation for SQL tables that can be refreshed.

All of the commands that Refresh processes can receive are listed below by category. Each command is used by all Refresh processes unless indicated otherwise in parentheses.

Process-Generated Commands	**General Use Operator Commands**	**Restricted Use Operator Commands**
PAUSEACK (Enscribe only) PAUSEEND (Enscribe only)	CONTINUE (Enscribe only) REFRESH RESUME (Enscribe only) STATUS STOP SUSPEND (Enscribe only)	None

CONTINUE Command

A general use operator command that instructs the Enscribe Refresh process to continue processing, even though at least one parity error has been encountered.

Syntax

DELIVER PROCESS *object-name,* "CONTINUE"

object-name — The symbolic name assigned to the Enscribe Refresh process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

An operator sends this command using an XPNET network control facility. If the Enscribe Refresh process encounters a parity error, refresh processing stops. Entering the CONTINUE command causes the Enscribe Refresh process to begin processing again at the point where the parity error caused refresh processing to stop. The CONTINUE command must be entered any time the Enscribe Refresh process encounters a parity error in order for refresh processing to continue.

RESUME Command

A general use operator command that instructs the Enscribe Refresh process to begin processing again after processing has been suspended using a SUSPEND command. Upon entry of the RESUME command, refresh processing is taken out of the suspended state and restarted. The SUSPEND command is described later in this subsection.

Syntax

DELIVER PROCESS *object-name,* "RESUME *tape-date*, *tape-time*, *refresh-group*, *application-id*, [*partition-number*]"

object-name — The symbolic name assigned to the Enscribe Refresh process in the XPNET node. For more information on entering the object name, refer to section 1.

The following fields (from file header fields) comprise the Refresh Group Configuration File (RGCF) key:

tape-date — Indicates the date the refresh tape was created (YYYYMMDD).

The tape date, along with the other parameters in this command comprise the key into the Refresh Group Configuration File (RGCF).

tape-time — Indicates the time the refresh tape was created (hhmm). The tape time, along with the other parameters in this command comprise the key into the Refresh Group Configuration File (RGCF).

refresh-group — Identifies the refresh group used as a key into the Refresh Group Configuration File (RGCF) and Institution Definition File (IDF).

application-id — Identifies the application ID used as a key into the Refresh Group Configuration File (RGCF). Valid values are as follows:

CC	=	Positive Balance File (PBF) for credit accounts only
CF	=	Cardholder Authorization File (CAF)
CH	=	Card History File (CHF)
CI	=	Customer/Card Information File (CCIF)
CM	=	Customer/Card Memo File (CCMF)
CO	=	Corporate Check File (CCF)
CS	=	Check Status File (CSF)
DA	=	PBF for checking or NOW accounts only
MB	=	Mail Box File (MBF)
NB	=	No Book File (NBF)
NF	=	Negative Card File (NEG)

PF = PBF for all account types
SD = Switch Dispute File (SDF)
SP = Stop Payment File (SPF)
SV = PBF for savings accounts only
TH = Transaction History File (THF)
WH = Warning/Hold/Float File (WHFF)

[*partition-number*] — Identifies the number of the partition when refreshing a single partition. If a partition number is not specified, the Enscribe Refresh process refreshes the entire file. A partition number should not be specified during partial-file refreshes. Valid values are 0 through 15. This field is not taken from the file header fields. The partition number, along with the other parameters in this command comprise the key into the Refresh Group Configuration File (RGCF).

Processing

The Enscribe Refresh process performs the following processing steps when it receives a RESUME command:

1. Removes processing from a suspended state and places it into a resume state.
2. Begins processing at the point where processing was suspended.

Example

The following is an example of a RESUME command. In this command, refresh processing is resumed for refresh group GRP1 of the Cardholder Authorization File (CAF).

```
DELIVER PROCESS P1A^REFR6, “RESUME 19980901, 2300, GRP1,
    CF”
```

REFRESH Command

A general use operator command that starts a Base National Negative Refresh process, Enscribe Refresh process, INAS National Negative Refresh process, or SQL Refresh process.

Note: The *label-type* and *character-set* fields are required.

Syntax

DELIVER PROCESS *object-name,* "REFRESH *label-type*, *input-file-name*,
[*partition-number*], [*bypass-number*], [*mail-name*], [*mail-ID*],
[*mail-group*], [*atm-impact-start-date/time*],
[*pos-impact-start-date/time*], [*teller-impact-start-date/time*],
[*telebanking-impact-start-date/time*], [*character-set*],
[*mount-message*], [*file-ID*], [[*volume-ID*],]..."

object-name — The symbolic name assigned to the Refresh process in the XPNET node. For more information on entering the object name, refer to section 1.

label-type — Specifies the tape label type for the refresh tape. If this field and the actual tape label do not match, the Refresh process stops processing. Valid values are as follows:

IBM = IBM OS
DOS = IBM DOS
BLS = Burroughs Large Scale
BUR = Burroughs 5500
ANS = ANSI
NON = No label processing

input-file-name — The HP NonStop file name of the tape device on which the refresh tape is mounted.

This field can also contain the name of a disk file, if disk file input is being used. Disk file input is used primarily when disk files can be transferred from the host to the HP NonStop processor.

[*partition-number*] — The specific partition of a partitioned file to be refreshed during this operation. Valid values are 0 through 15. A value of spaces or AL signify all partitions.

[*bypass-number*] — The number of tape parity errors that can be encountered without causing the Refresh process to abort. The default is 0. Valid values are as follows:

-1	=	Bypass all tape parity errors.
0–32767	=	Bypass the number of tape parity errors specified.

An entry in this field overrides the value in the REF-BYPASS-PARITY-ERRORS param in the Logical Network Configuration File (LCONF). This field is only used for tape input.

[*mail-name*] — The name of the Mail process, for MBF refreshes. During an MBF refresh, the Refresh process creates mail messages from the information on the refresh tape, and sends the messages to the process named in this field. The Mail process writes the messages to the MBF.

[*mail-ID*] — The mail FIID. This field is not used.

[*mail-group*] — The Mail refresh group. This field is only used for previous-release MBF refreshes.

[*atm-impact-start-date/time*] — The impact starting date and time for BASE24-atm transactions, in the format YYYYMMDDhhmmssmmmmmm (milliseconds are optional). An entry in this field overrides the values in the ATM.IMP-STRT-DAT and ATM.IMP-STRT-TIM fields in the Refresh File Header (RFH).

[*pos-impact-start-date/time*] — The impact starting date and time for BASE24-pos transactions, in the format YYYYMMDDhhmmssmmmmmm (milliseconds are optional). An entry in this field overrides the values in the POS.IMP-STRT-DAT and POS.IMP-STRT-TIM fields in the Refresh File Header (RFH).

[*teller-impact-start-date/time*] — The impact starting date and time for BASE24-teller transactions, in the format YYYYMMDDhhmmssmmmmmm (milliseconds are optional). An entry in this field overrides the values in the TLR.IMP-STRT-DAT and TLR.IMP-STRT-TIM fields in the Refresh File Header (RFH).

[*telebanking-impact-start-date/time*] — The impact starting date and time for BASE24-telebanking transactions, in the format YYYYMMDD hhmmssmmmmmm (milliseconds are optional). An entry in this field overrides the values in the TB.IMP-STRT-DAT and TB.IMP-STRT-TIM fields in the Refresh File Header (RFH).

[*character-set*] — Indicates whether the form of the data on the refresh tape is ASCII or EBCDIC. Valid values are as follows:

A = ASCII
E = EBCDIC (default)

[*mount-message*] — A message to be displayed along with the system mount message when the Refresh process requests use of a tape drive. This field can be any alphanumeric message of 25 characters or less surrounded by single quotes. The default message is "MOUNT REFRESH TAPE." This field is not used for refreshes from disk.

[*file-ID*] — The name of the file on the refresh tape. This field is not used for refreshes from disk.

This field contains the value of the Data Set ID field in IBM header labels. This field is only used when the tape label type is IBM and the BASE24-TAPE-LBL-PROC-USED param in the LCONF is set to N (Not used with disk input, TAPECOM is processing the labels).

[[*volume-ID*],]... — The volume identifiers for each reel to be used in the refresh. Volume identifiers are required when the tape label type is IBM or ANSI. Up to 35 volume identifiers can be entered, each separated from the previous volume ID by a comma. This field is not used for refreshes from disk.

Note: The bypass number, mount message, file ID, and volume IDs are only used with tape input. The file ID and volume IDs are only used with TAPECOM processing the tape labels.

When optional items are included in the REFRESH command, the command may be too long to be entered directly in the NCS COMMAND field on the Network Control Supervisor (NCS) screen. If the REFRESH command exceeds 139 characters, it can be placed in an Obey file and then issued to the Refresh process using the OBEY command or from the Refresh screen. For more information on the OBEY command, refer to the command-specific documentation in the ***NET24-XPNET Network Control Command Reference Manual***.

Processing

An operator sends the REFRESH command from the Refresh (REFR) screens or an XPNET network control facility. Each Refresh process performs several processing steps when it receives a REFRESH command. For information on the processing performed by each type of Refresh process, refer to the ***BASE24 Refresh and Extract Operators Manual***.

Example

The following is an example of a REFRESH command.

```
DELIVER PROCESS P1A^REFR6, “REFRESH IBM, $TAPE, 15, 10, , , ,
    20010923150500000000, , , , E, , , 32”
```

In this example, the following variable values are used:

- Symbolic name of the refresh process is P1A^REFR6
- Label type is IBM
- Input file name is $TAPE
- Partition number is 15
- Bypass number is 10
- Impact BASE24-atm start date and time is September 23, 2001 at 3:05 p.m. (15:05)
- Character set is EBCDIC (E)
- Tape volume ID is 32

STATUS Command

A general use operator command that requests the current status of a Base National Negative Refresh process, Enscribe Refresh process, INAS National Negative Refresh process, or SQL Refresh process.

Syntax

DELIVER PROCESS *object-name,* "STATUS"

object-name — The symbolic name assigned to the Refresh process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

An operator sends this command from the Refresh (REFR) screens or an XPNET network control facility. STATUS commands can be sent at any time to monitor activity occurring in the system. When a Refresh process receives the STATUS command, it generates various event messages that are sent to the EMS collector process. The messages generated varies between the Enscribe-based Refresh processes (i.e., Base National Negative Refresh process, INAS National Negative Refresh process, and Enscribe Refresh process) and the SQL Refresh process.

Event messages generated for the STATUS command by Enscribe-based Refresh processes include events 0312, 0313, 0314, 0317, 0318, and 0387. These messages are described in the BAREFR log message documentation file.

Event messages generated for the STATUS command by the SQL Refresh process include events 0710, 0730, 0740, 0810, and 0820. These messages are described in the ISCOREFR log message documentation file.

STOP Command

A general use operator command that instructs a Base National Negative Refresh process, Enscribe Refresh process, INAS National Negative Refresh process, or SQL Refresh process to abort.

Syntax

DELIVER PROCESS *object-name,* “STOP”

object-name — The symbolic name assigned to the Refresh process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

Refresh processing can be terminated by issuing the STOP command at any time during processing. When a Refresh process receives the STOP command, it logs an event message indicating that refresh processing has been terminated.

SUSPEND Command

A general use operator command that instructs the Enscribe Refresh process to cease processing, but remain in a suspended state. This means the process actually stops, but the current information is saved in the Refresh Group Configuration File (RGCF) so that when the process is resumed, it can begin where it left off. The SUSPEND command is used in conjunction with the RESUME command. Refresh processing can remain suspended until an operator enters a RESUME command. When a RESUME command is issued, refresh processing continues. The RESUME command is described earlier in this subsection.

Syntax

DELIVER PROCESS *object-name,* "SUSPEND"

object-name — The symbolic name assigned to the Enscribe Refresh process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

An operator sends this command using an XPNET network control facility or the Refresh (REFR) screens. Refresh processing can be placed on hold by issuing the SUSPEND command at any time during processing. When the Enscribe Refresh process receives the SUSPEND command, it ceases processing when it reaches the next logical stopping point and logs one or more of the following event messages indicating that refresh processing has been suspended: 0332, 0347, and 0348. These messages are described in the BAREFR log message documentation file. The Enscribe Refresh process also logs the event messages described earlier in the STATUS command. Refresh processing can be restarted at the point at which it was suspended by entering the RESUME command.

Super Extract Process Commands

This section describes the BASE24 commands the Super Extract process recognizes. BASE24 processes can receive a number of BASE24 commands that cause them to perform specialized types of processing. These commands can be separated into three categories: process-generated commands, general use operator commands, and restricted use operator commands. All BASE24 general and restricted use operator commands that can be received by the Super Extract process are documented in this subsection. Process-generated commands are documented in appendix B.

The Super Extract process allows files on the HP NonStop computer to be placed on tapes or disk files so that hosts can use that information for processing. The process of placing files on tape or disk is known as extraction. One or more files can be extracted to a single tape or disk file.

All of the commands that can be received by the Super Extract process are listed below by category.

Process-Generated Commands	**General Use Operator Commands**	**Restricted Use Operator Commands**
SAFEXTRESP	EXTR REEX TLFRSET TLFSTAT TRACEOFF TRACEON WARMBOOT	None

EXTR Command

A general use operator command that causes the files indicated in the command or in the Extract Configuration File (ECF) to be extracted to tape or disk.

Syntax

DELIVER PROCESS *object-name,* “EXTR*ecf-key* [, *override-flag*] [, *file-map*] [, *volume-id*]”

object-name — The symbolic name assigned to the Super Extract process in the XPNET node. For more information on entering the object name, refer to section 1.

ecf-key — The tag to use when reading the ECF record for this extract. The value in this field must match the value in a TAG field in the ECF.

[*override-flag*] — Indicates whether information in the *file-map* variable is to override values in the ECF. Valid values are as follows:

Y = Yes, override the values in the ECF with the values in the *file-map* variable.
N = No, do not override the values in the ECF.

This variable is optional. The Super Extract process allows you to either extract a file already specified in the ECF or not extract a file already specified in the ECF.

[*file-map*] — Identifies the BASE24 files to be extracted if the *override-flag* variable is set to the value Y. There is a flag for each file supported by the Super Extract process (e.g., ILF, OMF, SAF, etc.). Valid values are as follows:

Y = Yes, extract this file.
N = No, do not extract this file.

[*volume-id*] — The volume identifier for the first tape the Super Extract process uses.

Processing

An operator sends this command from an XPNET network control facility or the Extract (EXTR) screen to start a manual extract. Using the file map from the Network Control Supervisor (NCS) screen does not work for all files, because the whole file map will not fit on the command line. The files to be extracted are controlled by the ECF record and the optional *override-flag* and *file-map* variables in the command. The *override-flag* variable defaults to a value of N, indicating that the ECF fields should determine which files are to be extracted. If a value of Y is placed in the *override-flag* variable, the *file-map* variable is compared byte by byte with the FILE-MAP field in the ECF, using the following tests:

- If the FILE-MAP field in the ECF and in the command contains the value Y, the Super Extract process uses a value of Y.
- If the FILE-MAP field contains the value Y in the ECF and contains the value N or blank in the command, the Super Extract process uses a value of N.
- If the FILE-MAP field contains the value N in the ECF and the value Y in the command, the Super Extract process uses a value of N and logs an event message indicating that the ECF cannot be overridden in this circumstance.
- If the FILE-MAP field contains the value N in the ECF and in the command, the Super Extract process uses a value of N.

If the above tests result in the Super Extract process using a value of Y for a file, the Super Extract process extracts the file. If the above tests result in the Super Extract process using a value of N for a file, the Super Extract process does not extract the file.

The byte positions in the *file-map* variable are identified in the following table.

Byte Position	File Acronym	Byte Position	File Acronym
1	ILF	22	not assigned
2	OMF	23	not assigned
3	SAF	24	not assigned
4	ICF	25	not assigned
5	IDF	26	not assigned
6	TLF	27	not assigned
7	ITLF	28	MBF
8	ICFE	29	HMBF
9	not assigned	30	ULF
10	not assigned	31	not assigned

Byte Position	File Acronym	Byte Position	File Acronym
11	PTLF	32	not assigned
12	PRDF	33	not assigned
13	TTLF	34	not assigned
14	TTF	35	HSF
15	not assigned	36	not assigned
16	not assigned	37	not assigned
17	not assigned	38	not assigned
18	not assigned	39	not assigned
19	not assigned	40	not assigned
20	not assigned	41–42	user field
21	not assigned		

Example

The following is an example of the EXTR command. In this example, the Super Extract process is P1A^EXTR1, the *ecf-key* is PTLFEXTR, the *override-flag* is set to a value of N, no *file-map* value is entered, and the *volume-id* is 000001.

```
DELIVER PROCESS P1A^EXTR1, “EXTR PTLFEXTR, N, , 000001”
```

Note: When optional items are included in the EXTR command, the command may be too long to be entered directly in the NCS COMMAND field on the Network Control Supervisor (NCS) screen. If the EXTR command exceeds 139 characters, it can be placed in an Obey file and then issued to the Super Extract process using the OBEY command. For more information on the OBEY command, refer to the ***NET24-XPNET Network Control Command Reference Manual***.

REEX Command

A general use operator command that directs the Super Extract process to reextract the files indicated in the command or in the ECF record to tape or disk. A reextract can only be performed after an initial extract has been performed.

Syntax

DELIVER PROCESS *object-name,* "REEX*ecf-key* [, *override-flag*] [, *file-map*] [, *volume-id*] [, *start-partial*] [, *end-partial*]"

object-name — The symbolic name assigned to the Super Extract process in the XPNET node. For more information on entering the object name, refer to section 1.

ecf-key — The tag to use when reading the ECF record for this reextract. The value in this field must match the value in a TAG field in the ECF.

[*override-flag*] — Indicates whether information in the *file-map* variable is to override values in the ECF. Valid values are as follows:

Y = Yes, override the values in the ECF with the values in the *file-map* field.
N = No, do not override the values in the ECF.

This field is optional. If this field is not included in the message, the Super Extract process reextracts the files identified in the ECF record.

[*file-map*] — Identifies the BASE24 files to be reextracted if the *override-flag* variable is set to the value Y. Valid values are as follows:

Y = Yes, reextract this file.
N = No, do not reextract this file.

[*volume-id*] — The volume identifier for the first tape the Super Extract process will use.

[*start-partial*] — The number of the partial extract at which to begin when reextracting a transaction log file. Valid values are 1 through 100. The value in the *end-partial* variable must be greater than or equal to the value in the *start-partial* variable.

[*end-partial*] — The number of the partial extract at which to end when reextracting a transaction log file. Valid values are 1 through 100. The value in the *end-partial* variable must be greater than or equal to the value in the *start-partial* variable.

Processing

An operator sends this command from an XPNET network control facility or the Extract (EXTR) screen. The files to be reextracted are controlled by the ECF record and the optional *override-flag* and *file-map* variables in the command. The *override-flag* variable defaults to a value of N, indicating that the ECF fields should determine which files are to be extracted. If a value of Y is placed in the *override-flag* variable, the *file-map* variable is compared byte by byte with the FILE-MAP field in the ECF, using the following tests:

- If the FILE-MAP field in the ECF and in the command contain the value Y, the Super Extract process uses a value of Y.
- If the FILE-MAP field contains the value Y in the ECF and contains the value N or blank in the command, the Super Extract process uses a value of N.
- If the FILE-MAP field contains the value N in the ECF and the value Y in the command, the Super Extract process uses a value of N and logs an event message indicating that the ECF cannot be overridden in this circumstance.
- If the FILE-MAP field contains the value N in the ECF and in the command, the Super Extract process uses a value of N.

If the above tests result in the Super Extract process using a value of Y for a file, the Super Extract process extracts the file. If the above tests result in the Super Extract process using a value of N for a file, the Super Extract process does not reextract the file.

The byte positions in the *file-map* variable are identified in the following table.

Byte Position	File Acronym	Byte Position	File Acronym
1	ILF	22	not assigned
2	OMF	23	not assigned
3	SAF	24	not assigned
4	ICF	25	not assigned
5	IDF	26	not assigned
6	TLF	27	not assigned
7	ITLF	28	MBF

Byte Position	File Acronym	Byte Position	File Acronym
8	ICFE	29	HMBF
9	not assigned	30	ULF
10	not assigned	31	not assigned
11	PTLF	32	not assigned
12	PRDF	33	not assigned
13	TTLF	34	not assigned
14	TTF	35	HSF
15	not assigned	36	not assigned
16	not assigned	37	not assigned
17	not assigned	38	not assigned
18	not assigned	39	not assigned
19	not assigned	40	not assigned
20	not assigned	41–42	user field
21	not assigned		

Example

The following is an example of the REEX command. In this example, the Super Extract process is P1A^EXTR1, the *ecf-key* is PTLFEXTR, the *override-flag* is set to a value of N, no *file-map* value is entered, the *volume-id* is 000001, the *start-partial* variable is 5, and the *end-partial* variable is 7.

DELIVER PROCESS P1A^EXTR1, “REEX PTLFEXTR, N, , 000001, 5, 7”

Note: When optional items are included in the REEX command, the command may be too long to be entered directly in the NCS COMMAND field on the Network Control Supervisor (NCS) screen. If the REEX command exceeds 139 characters, it can be placed in an Obey file and then issued to the Super Extract process using the OBEY command. For more information on the OBEY command, refer to the ***NET24-XPNET Network Control Command Reference Manual***.

TLFRSET Command

A general use operator command that causes the pointers in the specified transaction log file to be reset.

Syntax

DELIVER PROCESS *object-name,* “*file-id*TLFRSET *date* [, *start-partial*]”

object-name — The symbolic name assigned to the Super Extract process in the XPNET node. For more information on entering the object name, refer to section 1.

file-id — A letter identifying the transaction log file for which the extract position markers are being reset. Valid values are as follows:

A = BASE24-atm Transaction Log File (TLF)
B = BASE24-telebanking ITS Transaction Log File (ITLF)
P = BASE24-pos Transaction Log File (PTLF)
T = BASE24-teller Transaction Log File (TTLF)

For example, to reset the position markers in the BASE24-pos Transaction Log File (PTLF), the string would be PTLFRSET.

date — The date (YYMMDD) of the transaction log file in which to reset the extract position markers. The date is part of the transaction log file name.

[*start-partial*] — The number of the partial extract at which to begin resetting extract position marker information.

Processing

Caution should be used with this command because it impacts future extracts and reextracts performed on the file. When this command is executed, the extract history is deleted and the next extract begins after the number of the partial extract specified by the *start-partial* variable

Therefore, if the entire transaction log file is to be extracted for a second time, the end-of-file pointers must be reset and the EXTR command must be sent. Refer to the ***BASE24 Refresh and Extract Operators Manual*** for information on end-of-file pointers. The extract position markers can be reset by issuing a TLFRSET command from an XPNET network control facility. The command allows you to reset all of the extract markers, or only those from a specific extract through the remainder of the markers.

For example, the following command resets the extract position markers for all partial extracts *after* the seventh partial extract in the BASE24-pos Transaction Log File (PTLF) dated October 10, 1998. Extract position information for the first seven partial extracts are kept.

DELIVER PROCESS P1A^EXTR1 PTLFRSET 981022, 7

After initiating a TLFRSET command from an XPNET network control facility, check the network log to determine whether the command was processed successfully. If the TLFRSET command was successful, event messages 0065 and 0068 are displayed on the network log. If the command was unsuccessful, event messages 0008, 0011, and 0179 are displayed on the network log. Descriptions of these event messages are provided in the BASUPERX log message documentation file.

After the position markers have been reset, the EXTR command must be used to extract the file. If a reextract is attempted after resetting the position markers, an error occurs because the position markers indicate that an extract has not yet been performed.

Example

The following is an example of the TLFRSET command. In this example, the Super Extract process is P1A^EXTR1, the *file-id* is PTLFEXTR, the *date* of the PTLF whose markers are being reset is October 10, 1998, and the *start-partial* value is 7.

DELIVER PROCESS P1A^EXTR1, “PTLFRSET 981022, 7”

TLFSTAT Command

A general use operator command that directs the Super Extract process to log an event message indicating how many partial extracts have been performed against a transaction log file. Note that partial extracts are numbered from 0 to 99. There is a separate form of this command for each transaction log file that can be extracted.

Syntax

DELIVER PROCESS *object-name,* "*file-id*TLFSTAT *date"*

object-name — The symbolic name assigned to the Super Extract process in the XPNET node. For more information on entering the object name, refer to section 1.

file-id — A letter identifying the transaction log file for which the extract position markers are being reset. Valid values are as follows:

A = BASE24-atm Transaction Log File (TLF)
B = BASE24-telebanking ITS Transaction Log File (ITLF)
P = BASE24-pos Transaction Log File (PTLF)
T = BASE24-teller Transaction Log File (TTLF)

For example, to reset the position markers in the BASE24-pos Transaction Log File (PTLF), the string would be PTLFSTAT.

date — The date (YYMMDD) of the transaction log file in which to reset the extract position markers. The date is part of the transaction log file name.

Processing

The Super Extract process performs the following steps when it receives a TLFSTAT command:

1. Reads the command.
2. Checks the specified transaction log file and determines how many partial extracts have been performed against the specified transaction log file.
3. Logs an event message indicating the number of partial extracts that have been performed against the specified transaction log file.

Example

The following is an example of the TLFSTAT command. In this example, the Super Extract process is P1A^EXTR1, the *file-id* is PTLFSTAT, and the *date* of the PTLF whose status is being requested is October 22, 1998.

```
DELIVER PROCESS P1A^EXTR1, "PTLFSTAT 981022"
```

TRACEON and TRACEOFF Commands

Restricted use operator commands that activate or deactivate the trace function for the specified Super Extract process.

Syntax

DELIVER PROCESS *object-name,* "TRACEON"

DELIVER PROCESS *object-name*, "TRACEOFF"

object-name — The symbolic name assigned to the Super Extract process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

The Super Extract process uses the trace function to track token records during transaction log file extracts. The Super Extract process logs event messages 0196 and 0197 when the trace function is activated or deactivated, respectively. These event messages are described in the BASUPERX log message documentation file.

Each time the extract process encounters a token record, it logs the following message:

`TKN GROUP` *`group`*`, TKN TYPE` *`type,`* `TKN SUBTYPE` *`subtype`*

If no token records are encountered during the TLF extract, the Extract process logs the following message:

```
!!! *** NO TKN RECORD FOUND *** !!!
```

WARMBOOT Command

A general use operator command that causes the Super Extract process to reinitialize.

Syntax

DELIVER PROCESS *object-name,* "9501"

object-name — The symbolic name assigned to the Super Extract process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

The WARMBOOT command causes the Super Extract process to reinitialize. When the Super Extract process reinitializes, it deletes all the timers set in its timer table and then goes through all of its initialization steps. The Super Extract process performs the following processing steps when it receives a WARMBOOT command:

1. Closes and reopens the Logical Network Configuration File (LCONF) and reads the required assigns and params.
2. Reads the Log Message Configuration File (LMCF) to retrieve its modified event messages.

   The LMCF contains a record for each modified event message in the system.
3. Closes and reopens the Extract Configuration File (ECF) with read/write access. Any other process can read the ECF while the Super Extract process has the ECF open.
4. Reads the ECF for records with a matching symbolic name in the ALTKEY. SYM-NAME field in the Base segment. For each record found, the Super Extract process checks the EXTRACT-DAT and the RESTRT-FLG fields as follows:
   a. If the RESTRT-FLG field is set to a value of Y and the value of the EXTRACT-DAT field is less than the current day, the Super Extract process logs a warning message that shows the EXTRACT-DAT and TAG fields in the ECF and then changes the value of the EXTRACT-DAT field to the current date. The Super Extract process then starts a timer that expires at the time specified by the EXTRACT-DAT and EXTRACT-TIM fields.

b. If the RESTRT-FLG is set to a value of Y and the value of the EXTRACT-DAT field is equal to the current date or a future date, the Super Extract process starts a timer that expires at the time specified by the EXTRACT-DAT and EXTRACT-TIM fields. When this timer expires, an extract is performed.

The RESTRT-FLG field typically is set to a value of N when extracts are to be completed manually on a general basis. Although a manual extract can be performed when the RESTRT-FLG is set to a value of Y, the Super Extract process sets a timer and performs an extract when the timer expires. Thus, two extracts would be performed on the same files.

5. If records were found, the Super Extract process issues a netread that times out when the first timer expires.

   If no records are found, the Super Extract process logs an event message indicating this and abends.

3: BASE24-atm Process Commands

This section contains the general and restricted operator use commands received by the following BASE24-atm processes:

- Authorization
- Device Handler
- Settlement Initiator
- Transaction Context Manager

The processes are presented in alphabetical order. Furthermore, the specific commands are documented in alphabetical order within the processes. The documentation for each command includes the following information:

- A description of the command
- The syntax of the command
- The processing steps performed when the command is received
- An example of the command (when necessary)

Authorization Process Commands

This section describes the processing performed by the BASE24-atm Authorization process. The BASE24-atm Authorization process is the authorizing entity in the BASE24-atm system. It authorizes transactions according to an institution's authorization parameters and logs transactions to a Transaction Log File (TLF), which can then be extracted for affecting settlement immediately without waiting for transaction receipts.

The BASE24-atm Authorization process can receive a number of BASE24 commands that cause it to perform specialized processing. These commands can be separated into three categories: process-generated commands, general use operator commands, and restricted use operator commands. All BASE24 general and restricted use operator commands that can be received by the BASE24-atm Authorization process are documented in this subsection. Process-generated commands are documented in appendix B.

All of the commands that can be received by the BASE24-atm Authorization process are listed below by category.

Process-Generated Commands	General Use Operator Commands	Restricted Use Operator Commands
CUTOVER REFRESH	ALTERATTRIBUTE INITDNT INITIALIZE CCF STATUS WARMBOOT ALL WARMBOOT FIID	DEBUG TRACE OFF TRACE ON

ALTERATTRIBUTE Command

A general use operator command that causes the Authorization process to warmboot (reallocate) a specified shared extended memory table after it has been rebuilt. Whenever a shared extended memory table is modified, all processes that read the table must reallocate the table before the modified data is used in transaction processing. The Process Control Terminal (PCT) Server process automatically sends this command to each process specified in an EMT warmboot notify list param in the Logical Network Configuration File (LCONF) when a network operator enters an ALTERATTRIBUTE command from the EMT Control Command screen (screen 5) in the Device Control Terminal (DCT). You can also enter the ALTERATTRIBUTE command manually from a network control facility to warmboot an extended memory table for an individual process. The EMT warmboot notify list params are only used when issuing the command from the DCT. For more information on the EMT Control Command screen, refer to the ***BASE24 Device Control Manual***.

Syntax

DELIVER PROCESS *object-name,* “ALTERATTR[IBUTE]*file-id*EMT*”*

object-name — The symbolic name assigned to the Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

file-id — The acronym of the file from which the shared extended memory table is built. Valid values are as follows:

ERF = Exchange Rate File
IPCF = Issuer Processing Code File

Processing

This command causes the Authorization process to reallocate a specified shared extended memory table. If the command completes successfully, a message stating that the EMT warmboot completed successfully is displayed on the network log.

Examples

The following are examples of ALTERATTRIBUTE commands sent to an Authorization process from an XPNET network control facility. In these examples, the symbolic name of the Authorization process is P1A^AUTH1 and the shared extended memory table to be warmbooted is the Issuer Processing Code File (IPCF) extended memory table.

DELIVER PROCESS P1A^AUTH1, “ALTERATTRIBUTEIPCFEMT”

DELIVER PROCESS P1A^AUTH1, “ALTERATTRIPCFEMT”

INITDNT Command

A general use operator command that causes the Authorization process to load or verify the Diebold number table (DNT) to the Thales e-Security Host Security Module (HSM) or the Atalla Network Security Processor (NSP).

Syntax

DELIVER PROCESS *object-name,* "INITDNT"

object-name — The symbolic name assigned to the Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

This command causes the Authorization process to load or verify the DNT to the hardware module.

With the Atalla NSP, the Authorization process uses the clear key for encrypting the DNT and sends the encrypted version of the key to the NSP along with the encrypted DNT to allow the NSP to decrypt the the DNT.

With the Thales e-Security HSM, the Authorization process calls a utility and passes the DNT from the Key Authorization File (KEYA) to the HSM. If the DNT from the KEYA contains all zeros, the utility asks the HSM to verify that the HSM already has a valid DNT. If the DNT in the KEYA contains values other than zero, the utility loads the KEYA DNT to the HSM.

The DNT is not loaded if the security device type indicates the Transaction Security Services process is being used.

INITCCF Command

A general use operator command that causes the Corporate Check File (CCF) information included in the extended memory segment of the Authorization process to be reinitialized.

Note: The INITCCF command is only applicable if the Self-Service Banking (SSB) Check Application is supported.

Syntax

DELIVER PROCESS *object-name,* "INITCCF"

object-name — The symbolic name assigned to the Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

This command is entered by an operator from an XPNET network control facility if the CCF requires reinitialization. The Authorization process performs the following steps when it receives an INITCCF command:

1. Reads the Logical Network Configuration File (LCONF) to retrieve the SSB-CCF assign. If the SSB-CCF assign is not present or if the Self-Service Banking (SSB) Check Application is not supported, the Authorization process logs an event message indicating the condition and processing is terminated.
2. Opens the CCF.
3. Calculates the amount of extended memory required for the CCF.
4. Allocates the new extended memory segment for the CCF.
5. Keypositions to the beginning of the CCF.
6. Reads the CCF records and loads them into memory. Logs event message 0103. This event message is described in the ATAUTH log message documentation file.
7. Deallocates the existing extended memory segment for the CCF if no CCF records are read.

DEBUG Command

A restricted use operator command that places the Authorization process under control of the HP NonStop Inspect product and displays an Inspect prompt on the home terminal (the terminal specified for the process by the HOMETERM attribute or the home terminal of the XPNET process). Inspect is an HP NonStop program debugging tool.

Syntax

DELIVER PROCESS *object-name,* "DEBUG"

object-name — The symbolic name assigned to the Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

An operator enters this command from an XPNET network control facility. For more information on the DEBUG command, refer to appendix A.

STATUS Command

A general use operator command that causes the Authorization process to send miscellaneous information on its current processing parameters to the EMS collector process.

Syntax

DELIVER PROCESS *object-name,* "STATUS"

object-name — The symbolic name assigned to the Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

The Authorization process performs the following processing steps when it receives a STATUS command:

1. Reads the command.
2. Searches its current processing parameters for information including the current and next processing dates for the Authorization process, the number of IDF and KEYA records in extended memory, the current status of the IDF, and the security module type and name if one is being used.
3. Sends the processing parameters in a series of event messages numbered from 0300 through 0304 to the EMS collector process. These event messages are described in the ATAUTH log message documentation file.

TRACE OFF Command

A restricted use operator command that deactivates the trace function within the Authorization process.

Syntax

DELIVER PROCESS *object-name,* “TRACE OFF”

object-name — The symbolic name assigned to the Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing:

An operator sends this command from an XPNET network control facility. For more information on the TRACE OFF command, refer to appendix A.

TRACE ON Command

A restricted use operator command that activates a trace function within the Authorization process.

Syntax

DELIVER PROCESS *object-name,* “TRACE ON”

object-name — The symbolic name assigned to the Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

An operator sends this command from an XPNET network control facility. For more information on the TRACE ON command, refer to appendix A.

WARMBOOT ALL Command

A general use operator command that reinitializes the Authorization process.

Syntax

DELIVER PROCESS *object-name,* “9501****”

object-name — The symbolic name assigned to the Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

When the Authorization process receives a WARMBOOT ALL command, it performs the same processing steps as it does during initialization, except all files are closed and reopened and the Authorization process does not retrieve its PPD name. When warmboot processing is completed, event message 0029 is displayed. For a detailed description of this event message, refer to the ATAUTH log message documentation file.

An operator sends this command from an XPNET network control facility.

WARMBOOT FIID Command

A general use operator command that causes the Authorization process to update the current information that it has in memory for an institution.

Syntax

DELIVER PROCESS *object-name,* "9501*FIID*"

object-name — The symbolic name assigned to the Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

FIID — The institution for which current information is updated by the Authorization process. The Authorization process updates the information in its memory for the identified institution.

Processing

Note: A WARMBOOT FIID command does not cause the Authorization process to reread the CPF. To update this information, a WARMBOOT ALL command must be entered or the process must be stopped and restarted. Also, a WARMBOOT FIID command cannot be used to add an IDF record to memory or to add KEYA data to memory where the IDF record did not have a KEYA record associated with it previously. Again, this must be done either by using a WARMBOOT ALL command or by stopping and restarting the Authorization process.

The Authorization process performs the following processing steps when it receives a WARMBOOT FIID command:

1. Locates the IDF record for the FIID in extended memory.
2. Reads the corresponding IDF record from the disk file.

   If a record cannot be found, the Authorization process generates an event message indicating the message cannot be processed.
3. Closes any data files open for the institution.

   Using the indexes in the FI table, the Authorization process identifies the cardholder files (CAF, PBF, NEG, UAF) open for the institution. If any of the files are opened exclusively for the institution, the Authorization process closes the files.

   The Authorization process builds and attaches an FI table to each internal IDF record when it is initialized. This FI table allows for faster access to certain pieces of information used internally for processing an institution.

4. Overlays the IDF record in extended memory with the new IDF record from disk, field-by-field.
5. Initializes all fields in the FI table portion of the IDF record (setting the FATAL ERROR field to indicate no error).
6. Opens the cardholder files required for processing if they are not already open and updates the indexes in the FI table.
7. Edits the IDF record fields. To do this, the Authorization process follows the steps taken in initialization.
8. Generates event message 0030. For a detailed description of this event message, refer to the ATAUTH log message documentation file.

WARMBOOT SURCHARGE Command

A general use operator command that causes the Authorization process to update the current surcharging information that it has in memory from the Surcharge File (SURF).

Syntax

DELIVER PROCESS *object-name,* “9501SURCHARGE”

object-name — The symbolic name assigned to the Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

A WARMBOOT SURCHARGE command causes the Authorization process to reread the SURF into extended memory. If the SURF is successfully reread, the process logs event message 0140. If the SURF is not successfully reread, the process logs event message 0145 and continues to process using the old SURF data. For detailed descriptions of these event messages, refer to the ATAUTH log message documentation file.

Device Handler Process Commands

This section describes the commonly used BASE24 commands the BASE24-atm Device Handler processes recognize. BASE24-atm Device Handler processes are the primary interfaces between ATMs and BASE24. They translate BASE24-atm internal messages to and from the native mode messages of the ATMs, enforcing each ATM's transaction sequence, updating accumulator fields and status flags in each ATM's BASE24-atm Terminal Data files record, and forcing cutover of terminals during daily settlement (in limited cases).

BASE24-atm Device Handler processes can receive a number of BASE24 commands that cause them to perform specialized processing. These commands can be separated into three categories: process-generated commands, general use operator commands, and restricted use operator commands. These common BASE24 general and restricted use operator commands that can be received by a Device Handler process are documented in this subsection. Process-generated commands are documented in appendix B.

Most of the commands described in this subsection can also be also be executed from a Device Control Terminal (DCT) screen. In addition, some commands that can be executed from a DCT screen, cannot be executed using the DELIVER PROCESS command. For more information on DCT screens, refer to the ***BASE24 Device Control Manual***. All of the commands that can be received by one or more BASE24-atm Device Handler processes are listed below by category. A table listing the commands that can be used by each type of Device Handler process is shown on the following pages.

.

Process-Generated Commands	General Use Operator Commands	Restricted Use Operator Commands
DEVICE STATE MESSAGES	9529ATD ALTERATTRIBUTE BJP CCK CDM CIM CLEARQ CLOSE[A] COD COPY BLOCK COPY CELL COPY FONT COPY ICON CPH CPLC DAS DCO DCS DEL DELPUBLIC DLS DMD DMS DVD DOWNLOAD DPUBLIC GETCNFG GETCNTRS LAFITS LAGROUP LALLKEYS LASCREENS LASTATES LCSTATE LDATE LFIT *continued on next page*	RESET TRACE OFF/TOF TRACE ON/TON

Process-Generated Commands	General Use Operator Commands	Restricted Use Operator Commands
	continued LISTQ LKEYMODE LKEYTYPE LMACKEY LMASTER LMK LMSTRCOM LOAD LOADALL LOADALLKEYS LOADGROUP LOADGROUPSCREEN LOADGROUPSTATE LOADKEY LOADKEYMODE LOADMACKEY LOADMASTER LOADMSTRCOM LOADPRIPUB LOADPUBLIC LOADSCREEN LOADSTATE LPRIPUB LPUBLIC LSCREEN MACINGOFF MACINGON NAM OPEN[A] REL RESTORE RVR SDT SENDSTATUS SENDSUPPLY SER SHUTDOWN STARTUP STATUS STATUSEXTMEM SUS	
	continued on next page	

Process-Generated Commands	General Use Operator Commands	Restricted Use Operator Commands
	continued UDC UPLOAD[!] WARMBOOT ZEROSUPPLY	

The following table lists the text commands supported by BASE24-atm Device Handler processes. Commands are listed down the lefthand side of the table. BASE24-atm Device Handler processes are listed across the top of the table. If a check mark is present where a specific command intersects with a specific Device Handler process, that Device Handler process supports the command. The absence of the check mark at the intersection indicates the Device Handler process does not support the command. Device Handler processes are identified in the table as follows:

D910	=	Diebold 910/911/912 Device Handler process
1000	=	Diebold 10XX/478X Device Handler process
3624	=	IBM 3624 Device Handler process
5XXX	=	NCR 5XXX Device Handler process
Fujitsu	=	Fujitsu Device Handler process
473X	=	IBM 473X-series Device Handler process
Fujitsu7	=	Fujitsu 7000 Device Handler process
Tidel	=	Tidel Anycard Device Handler process
Triton	=	Triton MiniATM Device Handler process
D-Cash	=	Diebold CashSource Plus Device Handler process
N-Cash	=	NCR “Cash” Device Handler process

BASE24-atm Device Handler Process Text Commands											
Command	**BASE24-atm Device Handler Process**										
	D910	**1000**	**3624**	**5XXX**	**Fujitsu**	**473X**	**Fujitsu7**	**Tidel**	**Triton**	**D-Cash**	**N-Cash**
9501 (WARMBOOT)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
9529ATD (WARMBOOT ATD)		✓		✓							
ALTERATTRIBUTE		✓		✓							
BJP			✓								
CCK			✓			✓					
CDM			✓								
CIM			✓			✓					
CLEARQ		✓		✓							
CLOSE[A]	✓	✓	✓	✓	✓	✓	✓				
COD			✓								
COPY BLOCK		✓									
COPY CELL		✓									
COPY FONT		✓									
COPY ICON		✓									
CPH						✓					
CPLC		✓		✓							
DAS			✓								
DCO			✓								
DCS			✓								
DEL						✓					

BASE24-atm Device Handler Process Text Commands

Command	BASE24-atm Device Handler Process										
	D910	1000	3624	5XXX	Fujitsu	473X	Fujitsu7	Tidel	Triton	D-Cash	N-Cash
DELPUBLIC				✓							
DLS			✓								
DMD			✓								
DMS						✓					
DPUBLIC				✓							
DVD			✓								
DOWNLOAD						✓					
GETCNFG				✓							
GETCNTRS				✓							
LAFITS				✓							
LAGROUP				✓							
LALLKEYS				✓							
LASCREENS				✓							
LASTATES				✓							
LCSTATE				✓							
LDATE				✓							
LFIT				✓							
LISTQ		✓		✓							
LKEYMODE				✓							
LKEYTYPE				✓							

BASE24-atm Device Handler Process Text Commands											
Command	BASE24-atm Device Handler Process										
	D910	1000	3624	5XXX	Fujitsu	473X	Fujitsu7	Tidel	Triton	D-Cash	N-Cash
LMACKEY				✓							
LMASTER				✓							
LMK			✓								
LMSTRCOM				✓							
LOAD	✓	✓	✓		✓	✓	✓			✓	✓
LOADALL				✓							
LOADALLKEYS		✓		✓				✓	✓		
LOADGROUP	✓	✓			✓		✓				
LOADGROUPSCREEN	✓	✓			✓		✓				
LOADGROUPSTATE	✓	✓			✓		✓				
LOADKEY	✓	✓			✓		✓				
LOADKEYMODE				✓							
LOADMACKEY	✓	✓					✓				
LOADMASTER		✓		✓							
LOADMSTRCOM				✓							
LOADPRIPUB				✓							
LOADPUBLIC		✓		✓							
LOADSCREEN	✓	✓			✓		✓				
LOADSTATE	✓	✓			✓		✓				
LPRIPUB				✓							

BASE24-atm Device Handler Process Text Commands											
Command	BASE24-atm Device Handler Process										
	D910	1000	3624	5XXX	Fujitsu	473X	Fujitsu7	Tidel	Triton	D-Cash	N-Cash
LPUBLIC				✓							
LSCREEN				✓							
MACINGOFF									✓		
MACINGON									✓		
NAM			✓			✓					
OPEN[A]	✓	✓	✓	✓	✓	✓	✓				
REL			✓								
RESET	✓	✓	✓	✓	✓		✓	✓	✓	✓	✓
RESTORE		✓									
RVR			✓								
SDT						✓					
SENDSTATUS					✓		✓				
SENDSUPPLY					✓		✓				
SER			✓								
SHUTDOWN	✓	✓		✓	✓						
STARTUP	✓	✓		✓	✓						
STATUS								✓	✓	✓	✓
STATUSEXTMEM		✓		✓							
SUS			✓								
TOF			✓								

BASE24-atm Device Handler Process Text Commands											
Command	BASE24-atm Device Handler Process										
	D910	1000	3624	5XXX	Fujitsu	473X	Fujitsu7	Tidel	Triton	D-Cash	N-Cash
TON			✓								
TRACE OFF		✓		✓		✓	✓	✓	✓	✓	✓
TRACE ON		✓		✓		✓	✓	✓	✓	✓	✓
UDC						✓					
UPLOAD[!]						✓					
ZEROSUPPLY							✓				

ALTERATTRIBUTE Command

A general use operator command that causes a Diebold 10XX/478X or NCR 5XXX Device Handler process to warmboot (reallocate) a specified shared extended memory table after it has been rebuilt. Whenever a shared extended memory table is modified, all processes that read the table must reallocate the table before the modified data is used in transaction processing. The Process Control Terminal (PCT) Server process automatically sends this command to each process specified in an EMT warmboot notify list param in the Logical Network Configuration File (LCONF) when a network operator enters an ALTERATTRIBUTE command from the EMT Control Command screen (screen 5) in the Device Control Terminal (DCT). You can also enter the ALTERATTRIBUTE command manually from a network control facility to warmboot an extended memory table for an individual process. The EMT warmboot notify list params are only used when issuing the command from the DCT. For more information on the EMT Control Command screen, refer to the ***BASE24 Device Control Manual***.

Syntax

DELIVER PROCESS *object-name,* “ALTERATTR[IBUTE]*file-id*EMT”

object-name — The symbolic name assigned to the Diebold 10XX/478X or NCR 5XXX Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

file-id — The acronym of the file from which the shared extended memory table is built. Valid values are as follows:

APCF = Acquirer Processing Code File
IDF = Institution Definition File
RCPT = Receipt File
TRF = Terminal Receipt File

Processing

This command causes a Diebold 10XX/478X or NCR 5XXX Device Handler process to reallocate a specified shared extended memory table. If the command completes successfully, a message stating that the EMT warmboot completed successfully is displayed on the network log.

Examples

The following are examples of ALTERATTRIBUTE commands sent to an NCR 5XXX Device Handler process from an XPNET network control facility. In these examples, the symbolic name of the NCR 5XXX Device Handler process is P1A^N50^3 and the shared extended memory table to be warmbooted is the Acquirer Processing Code File (APCF) extended memory table.

DELIVER PROCESS P1A^N50^3, “ALTERATTRIBUTEAPCFEMT”

DELIVER PROCESS P1A^N50^3, “ALTERATTRAPCFEMT”

BJP Command

A general use operator command that causes the IBM 3624 ATM to print a message on a blank receipt at the ATM and store that receipt in the escrow hopper of the ATM. You can use this command to leave messages for institution personnel who service the ATM.

Syntax

DELIVER PROCESS *object-name,* "9500BJP *terminal-name journal-message"*

object-name — The symbolic name assigned to the IBM 3624 Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the Terminal Data File. This ID must match a symbolic station name defined in the XPNET node.

journal-message — The message you want to be printed on the receipt. Only uppercase letters (A–Z), numbers (0–9), hyphens (**-**), apostrophes ('), and slashes (/) can be printed on the receipt printer. To enter a quotation mark ("), use two consecutive apostrophes. You can enter up to four lines of text, with up to 34 characters per line.

Processing

This command causes the IBM 3624 Device Handler process to send the journal text in a native-mode message to the specified ATM. The ATM then prints the message on a blank receipt and stores the receipt in the escrow hopper of the ATM.

CCK Command

A general use operator command that causes the IBM 3624 or 473X-series Device Handler process to download a new communications key to the terminal. This command changes the data encryption communications key to a random value and downloads the new value to the ATM. Sets of communications keys are stored by both BASE24-atm, IBM 36XX-series, and IBM 473X series terminals, each of which maintains its own encrypted version of the key.

Syntax

DELIVER PROCESS *object-name,* "9500CCK *terminal-name console"*

object-name — The symbolic name assigned to the IBM 3624 or 473X-series Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the Terminal Data File. This ID must match a symbolic station name defined in the XPNET node.

console — The number of the IBM 4730 ATM console unit on which to execute the CCK command. This variable does not apply to IBM 36XX-series terminals. Valid values are as follows:

1 = Console 1
2 = Console 2

Processing

This command causes the IBM 3624 Device Handler process to send the new communications key in a native-mode message to the specified ATM.

CDM Command

A general use operator command that causes the IBM 3624 Device Handler process to temporarily replace a display message at the ATM. This command replaces a display message in the memory of the ATM with a temporary display message such as a holiday or marketing message. When the ATM is reloaded, it displays the original display message.

Syntax

DELIVER PROCESS *object-name,* “9500CDM *terminal-name msg-num display-msg”*

object-name — The symbolic name assigned to the IBM 3624 Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the Terminal Data File. This ID must match a symbolic station name defined in the XPNET node.

msg-num — The number of the display message to be temporarily replaced.

display-msg — The text of the replacement message to be temporarily displayed. You can use any combination of upper and lower case alphanumeric characters. To enter a quotation mark ("), use two consecutive apostrophes. You can enter up to six lines of text, with up to 40 characters per line.

Processing

This command causes the IBM 3624 Device Handler process to send the display message text in a native-mode message to the specified ATM. The ATM then temporarily displays the message instead of the original message until it is reloaded.

CIM Command

A general use operator command that causes a specified IBM 3624 or 473X-series Device Handler process to close the specified terminal immediately, regardless of the terminal's current state.

Syntax

DELIVER PROCESS *object-name,* "9500CIM *terminal-name*"

object-name — The symbolic name assigned to the IBM 3624 or 473X-series Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the Terminal Data File. This ID must match a symbolic station name defined in the XPNET node.

console — The number of the IBM 4730 ATM console unit on which to execute the CIM command. This variable does not apply to IBM 36XX-series terminals. Valid values are as follows:

1 = Console 1
2 = Console 2

Processing

This command causes the IBM 3624 or 473X-series Device Handler process to send a native-mode message to the specified terminal, requesting it to close immediately.

CLEARQ Command

A general use operator command that causes a specified Diebold 10XX/478X or NCR 5XXX Device Handler process to clear (delete) all records from the ATM Dial-Up Command File (ADCF) for the specified terminal ID. The commands are listed in event messages on the network log before they are deleted. This command is valid only for Diebold 10XX/478X and NCR 5XXX terminals configured as dial-up terminals in the DIAL-UP field on Diebold 10XX/478X screen 7 or NCR 5XXX screen 7 of the BASE24-atm Terminal Data files (ATD), respectively.

Syntax

DELIVER PROCESS *object-name,* “9500CLEARQ *terminal-name”*

object-name — The symbolic name assigned to the Diebold 10XX/478X or NCR 5XXX Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the dial-up ATM in the BASE24-atm Terminal Data files.

Processing

This command causes the Diebold 10XX/478X or NCR 5XXX Device Handler process to read the ADCF using the terminal ID passed in the command, generate an event message for each command queued for the terminal, and then delete all records for the terminal ID from the ADCF.

Examples

The following is an examples of the CLEARQ command sent to an NCR 5XXX Device Handler process from an XPNET network control facility. In this example, the symbolic name of the NCR 5XXX Device Handler process is is P1A^N50^1 and the terminal ID is DIALUP50801.

DELIVER PROCESS P1A^N50^1, “9500CLEARQ DIALUP50801”

CLOSE, CLOSEA and SHUTDOWN Commands

General use operator commands that cause the Device Handler process to close the specified ATM. The CLOSE and SHUTDOWN commands close the specified ATM for all transactions, while the CLOSEA command closes the specified ATM for administrative transactions only.

Syntax

DELIVER PROCESS *object-name,* "9500CLOSE *terminal-name"*

DELIVER PROCESS *object-name,* "9500CLOSEA *terminal-name"*

DELIVER PROCESS *object-name,* "9500SHUTDOWN *terminal-name"*

object-name — The symbolic name assigned to the Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files. This ID must match a symbolic station name defined in the XPNET node.

Processing

The Device Handler process performs the following processing steps when it receives a CLOSE, CLOSEA, or SHUTDOWN command:

1. Translates the command into the appropriate native mode format for the specified ATM.
2. Sends a message to close the specified ATM.

COD Command

A general use operator command that causes the IBM 3624 Device Handler process to change the customized option data used by an ATM. Refer to the ATM's documentation from IBM for valid values. The new option data submitted to the ATM in this command replaces the current option data.

Syntax

DELIVER PROCESS *object-name,* "9500COD *terminal-name opt-name opt-data*"

object-name — The symbolic name assigned to the IBM 3624 Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the Terminal Data File. This ID must match a symbolic station name defined in the XPNET node.

opt-name — The name of the customized option data to be replaced.

opt-data — The customized option data. Up to 40 hexadecimal characters can be specified.

Processing

This command causes the IBM 3624 Device Handler process to send new customized option data to the specified ATM. If the command is successful, event message 0062 is generated. This event message is described in the AT3624 log message documentation file.

COPY BLOCK Commands

A general use operator command that provides support for Diebold 10XX/478X ATMs with Enhanced Monochrome Graphics (EMG). Terminals that support EMG contain an additional memory board dedicated to storing graphics data. EMG memory is divided into seven banks, six of which are supported by BASE24. Each of these banks holds eight libraries, and one library holds 128 cells. A cell is one screen element, such as a character in a font, or a part of a pictorial icon. A screen can display graphics from only one bank at a time. By moving cells between libraries, you can select any combination of graphics for display on a given screen. The Diebold ***Enhanced Monochrome Graphics (EMG) Programming Manual*** contains additional information about these concepts, and also provides a list of all fonts and icons that Diebold provides with the EMG module.

The following command allows you to copy a block of cells from one library to another.

Syntax

DELIVER PROCESS *object-name,* "9500COPY BLOCK *terminal-name num-cells source-library src-beg-cell target-library targ-beg-cell*"

object-name — The symbolic name assigned to the Diebold 10XX/478X Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files. This ID must match a symbolic station name defined in the XPNET node.

num-cells — The total number of cells to be copied. Valid values are 001–128.

source-library — The source library from which a block of cells is to be copied. Valid values for specifying library IDs are uppercase letters, numbers from 0 to 9, and the following special characters:

: ; < = > ? @ [\ ] ^ _

src-beg-cell — The first cell in the block of cells to be copied from the source library. Valid values are 001–128.

target-library — The target library to which the block of cells is to be copied. Valid values for specifying library IDs are uppercase letters, numbers from 0 to 9, and the following special characters:

: ; < = > ? @ [\ ] ^ _

targ-beg-cell — The first cell in the target library to which the first cell in the block is to be copied. Valid values are 001–128.

Processing

This command causes the Diebold 10XX/478X ATM to copy the specified block of cells from the source library to the target library.

COPY CELL Command

A general use operator command that provides support for Diebold 10XX/478X ATMs with Enhanced Monochrome Graphics (EMG). Terminals that support EMG contain an additional memory board dedicated to storing graphics data. EMG memory is divided into seven banks, six of which are supported by BASE24. Each of these banks holds eight libraries, and one library holds 128 cells. A cell is one screen element, such as a character in a font, or a part of a pictorial icon. A screen can display graphics from only one bank at a time. By moving cells between libraries, you can select any combination of graphics for display on a given screen. The Diebold ***Enhanced Monochrome Graphics (EMG) Programming Manual*** contains additional information about these concepts, and also provides a list of all fonts and icons that Diebold provides with the EMG module.

The following command allows you to copy all cells from one library to another.

Syntax

DELIVER PROCESS *object-name,* “9500COPY CELL *terminal-name source-library target-library”*

object-name — The symbolic name assigned to the Diebold 10XX/478X Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files. This ID must match a symbolic station name defined in the XPNET node.

source-library — The source library from which all cells are to be copied. Valid values for specifying library IDs are uppercase letters, numbers from 0 to 9, and the following special characters:

: ; < = > ? @ [\ ] ^ _

target-library — The target library that receives the cells. Valid values for specifying library IDs are uppercase letters, numbers from 0 to 9, and the following special characters:

: ; < = > ? @ [\ ] ^ _

Processing

This command causes the Diebold 10XX/478X ATM to copy all cells from the source library to the specified target library.

COPY FONT Command

A general use operator command that provides support for Diebold 10XX/478X ATMs with Enhanced Monochrome Graphics (EMG). Terminals that support EMG contain an additional memory board dedicated to storing graphics data. EMG memory is divided into seven banks, six of which are supported by BASE24. Each of these banks holds eight libraries, and one library holds 128 cells. A cell is one screen element, such as a character in a font, or a part of a pictorial icon. A screen can display graphics from only one bank at a time. By moving cells between libraries, you can select any combination of graphics for display on a given screen. The Diebold ***Enhanced Monochrome Graphics (EMG) Programming Manual*** contains additional information about these concepts, and also provides a list of all fonts and icons that Diebold provides with the EMG module.

The following command allows you to copy a font to a library.

Syntax

DELIVER PROCESS *object-name,* "9500COPY FONT *terminal-name font-id target-library"*

object-name — The symbolic name assigned to the Diebold 10XX/478X Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files. This ID must match a symbolic station name defined in the XPNET node.

font-id — The font that is to be copied into a library. Valid values are uppercase letters, numbers from 1 to 9, and the following special characters:

: ; < = > ? @ [\ ] ^ _

target-library — The target library that receives the font value. Valid values for specifying library IDs are uppercase letters, numbers from 0 to 9, and the following special characters:

: ; < = > ? @ [\ ] ^ _

Processing

This command causes the Diebold 10XX/478X ATM to store the specified font in the selected library.

COPY ICON Command

A general use operator command that provides support for Diebold 10XX/478X ATMs with Enhanced Monochrome Graphics (EMG). Terminals that support EMG contain an additional memory board dedicated to storing graphics data. EMG memory is divided into seven banks, six of which are supported by BASE24. Each of these banks holds eight libraries, and one library holds 128 cells. A cell is one screen element, such as a character in a font, or a part of a pictorial icon. A screen can display graphics from only one bank at a time. By moving cells between libraries, you can select any combination of graphics for display on a given screen. The Diebold ***Enhanced Monochrome Graphics (EMG) Programming Manual*** contains additional information about these concepts, and also provides a list of all fonts and icons that Diebold provides with the EMG module.

The following command allows you to copy a pictorial icon to a library.

Syntax

DELIVER PROCESS *object-name,* "9500COPY ICON *terminal-name icon-id target-library"*

object-name — The symbolic name assigned to the Diebold 10XX/478X Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files. This ID must match a symbolic station name defined in the XPNET node.

icon-id — Specifies the icon that is to be copied to a library. Valid values are 000–254. If this variable is not included in the command, it defaults to a value of 000.

target-library — The target library that receives the icon value. Valid values for specifying library IDs are uppercase letters, numbers from 0 to 9, and the following special characters:

: ; < = > ? @ [\ ] ^ _

Processing

This command causes the Diebold 10XX/478X ATM to store the specified icon in the selected library.

CPH Command

A general use operator command that causes the IBM 473X-series Device Handler process to add a new phrase or change an existing phrase on an ATM screen. A phrase is a word or a group of words used in screen and statement printer headings, prompts, or messages.

Syntax

DELIVER PROCESS *object-name,* "9500CPH *terminal-name console lang-num phrase-num phrase"*

object-name — The symbolic name assigned to the IBM 473X-series Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the Terminal Data File. This ID must match a symbolic station name defined in the XPNET node.

console — The number of the IBM 4730 ATM console unit on which to execute the CPH command. Valid values are as follows:

1 = Console 1
2 = Console 2

lang-num — The number of the language in which you want the phrase to be displayed. Valid values are as follows:

01 = Primary language
02 = Secondary language

If this variable is not specified in the command it defaults to 01.

phrase-num — The number of the phrase you want to change. Valid values are 000–999. If this variable is not specified in the command it defaults to 000.

phrase — The text of the phrase to be changed. The phrase can consist of up to 72 alphanumeric characters.

Processing

This command causes the IBM 473X-series Device Handler process to send the changed phrase to the specified ATM. If the command is successful, event message 0732 or 0733 is generated. These event messages are described in the AT4730 log message documentation file.

CPLC Command

A general use operator command that causes a specified Diebold 10XX/478X or NCR 5XXX terminal to print the check proof list. This command is only available with the BASE24-atm self-service banking (SSB) add-on product. The check proof listing feature enables BASE24-atm to track the checks cashed and deposited at each terminal and to provide this information back to terminals for printing proof lists. These proof listings can be printed on the ATM audit or receipt printer. Check proof listings are intended to facilitate balancing and reconciling check-related transactions by institution personnel. Check proof lists contain details of check activity by terminal bin and include the MICR information read from the check.

The terminal must be closed when this command is entered; otherwise, the command is rejected. After the check proof list is printed, the terminal should be reopened.

Syntax

DELIVER PROCESS *object-name,* "9500CPLC *terminal-name*"

object-name — The symbolic name assigned to the Diebold 10XX/478X or NCR 5XXX Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files. This ID must match a symbolic station name defined in the XPNET node.

Processing

This command causes the Diebold 10XX/478X or NCR 5XXX Device Handler process to send a native-mode message to the specified ATM, requesting that the check proof list be printed. If the command is successful, event message 0799 is generated. This event message is described in the AT1000 log message documentation file.

DAS Command

A general use operator command that causes the IBM 3624 Device Handler process to deactivate the Systems Network Architecture (SNA) session between itself and the specified ATM. This command deactivates the SNA session between the ATM and the IBM 3624 Device Handler process. You can use this command to reset SNA deadlock conditions. This command does not affect the external status of the ATM. If, for example, the ATM is open when this command is executed, the ATM appears to be open until the next attempt to use the ATM.

Syntax

DELIVER PROCESS *object-name,* "9500DAS *terminal-name"*

object-name — The symbolic name assigned to the IBM 3624 Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the Terminal Data File. This ID must match a symbolic station name defined in the XPNET node.

Processing

This command causes the IBM 3624 Device Handler process to send a native-mode message to the specified ATM, requesting that the current SNA session be deactivated.

DCO Command

A general use operator command that causes the IBM 3624 Device Handler process to request customized option data from the specified ATM and display the data on the network logging facility. The ATM transmits the customized option data to the Device Handler process, which then sends it to the log.

Syntax

DELIVER PROCESS *object-name,* "9500DCO *terminal-name"*

object-name — The symbolic name assigned to the IBM 3624 Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the Terminal Data File. This ID must match a symbolic station name defined in the XPNET node.

Processing

This command causes the IBM 3624 Device Handler process to request the customized data from the specified ATM, and display this data in event messages 0072 and 0073. These event messages are described in the AT3624 log message documentation file.

DCS Command

A general use operator command that causes the IBM 3624 Device Handler process to request command status information from the specified ATM. This command displays command status information for the ATM on the network logging facility. The ATM transmits the command status information to the Device Handler process, which then sends it to the log.

Syntax

DELIVER PROCESS *object-name,* “9500DCS *terminal-name”*

object-name — The symbolic name assigned to the IBM 3624 Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the Terminal Data File. This ID must match a symbolic station name defined in the XPNET node.

Processing

This command causes the IBM 3624 Device Handler process to request the customized data from the specified ATM, and display this data in event message 0069. This event message is described in the AT3624 log message documentation file.

DEL Command

A general use operator command that causes the IBM 473X-series Device Handler process to display event messages for the last six errors that have occurred at an ATM. These event messages are displayed on the network logging facility for the ATM.

Syntax

DELIVER PROCESS *object-name,* “9500DEL *terminal-name console”*

object-name — The symbolic name assigned to the IBM 473X-series Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the Terminal Data File. This ID must match a symbolic station name defined in the XPNET node.

console — The number of the IBM 4730 ATM console unit on which to execute the DEL command. Valid values are as follows:

1 = Console 1
2 = Console 2

Processing

This command causes the IBM 473X-series Device Handler process to display the last six event messages for the specified ATM with a severity of 2 (Error). All possible event messages for the IBM 473X-series Device Handler process are described in the AT4730 log message documentation file.

DELPUBLIC Command

A general use operator command that causes the NCR 5XXX Device Handler process to delete the BASE24 Primary Public key and optionally the BASE24 Root Public key if the ATM supports the NCR Enhanced AKDS scheme.

Syntax

DELIVER PROCESS *object-name,* “9500DELPUBLIC *terminal-name x*”

object-name — The symbolic name assigned to the ATM Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files.

x — A code indicating which keys to delete. Value values are as follows:

P = Delete the BASE24 Primary Public key
R = Delete the BASE24 Root Public key and the BASE24 Primary Public key

Processing

This command causes the Device Handler process to remove the BASE24 Primary Public key and/or the BASE24 Root Public key from the EPP. The Device Handler process sends an operational shutdown command to the ATM. When it receives a response from the ATM indicating that it has been shut down, the Device Handler process then sends a configuration request to the ATM. Information in the configuration response indicates whether the ATM has AKDS Basic or Enhanced capability. For ATMs with AKDS Enhanced protocol, the Device Handler requests a self signature from TSS of the key to be deleted. If the Root Public key is deleted, the Primary Public key is also deleted. If only the Primary Public key is deleted, the Root Public key remains in the EPP and can be used to load another Primary Public key. The ATM is not brought back into service upon completion of the command.

DLS Command

A general use operator command that causes the IBM 3624 Device Handler process to send the status of the last transaction data for the ATM to the network logging facility. This status information can be used during a transaction failure recovery.

Syntax

DELIVER PROCESS *object-name,* "9500DLS *terminal-name"*

object-name — The symbolic name assigned to the IBM 3624 Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the Terminal Data File. This ID must match a symbolic station name defined in the XPNET node.

Processing

This command causes the IBM 3624 Device Handler process to display the status of the last transaction processed from the specified ATM in event message 0070. This event message is described in the AT3624 log message documentation file.

DMD Command

A general use operator command that causes a specified IBM 3624 Device Handler process to send a native-mode message to a terminal to request maintenance data. The maintenance data for the ATM is sent to the network logging facility. Maintenance data includes machine status, error log, error statistics, bill counters, configuration data, transaction sequence numbers, feed calibrations, required intervention code, and machine control flags.

Syntax

DELIVER PROCESS *object-name,* "9500DMD *terminal-name"*

object-name — The symbolic name assigned to the IBM 3624 Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the Terminal Data File. This ID must match a symbolic station name defined in the XPNET node.

Processing

This command causes the IBM 3624 Device Handler process to perform the following:

1. Send a native-mode message to the specified terminal requesting maintenance data.
2. Receive the native-mode response from the terminal, reformat it, and generate event message 0064 and 0134 to display the maintenance data. These event messages are described in the AT3624 log message documentation file.

DMS Command

A general use operator command that causes the IBM 473X-series Device Handler process to display the status of all modules, or parts, that make up an IBM 4730 ATM. This status information is displayed on the network logging facility for the ATM.

Syntax

DELIVER PROCESS *object-name,* “9500DMS *terminal-name console”*

object-name — The symbolic name assigned to the IBM 473X-series Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the Terminal Data File. This ID must match a symbolic station name defined in the XPNET node.

console — The number of the IBM 4730 ATM console unit on which to execute the DMS command. Valid values are as follows:

1 = Console 1
2 = Console 2

Processing

This command causes the IBM 473X-series Device Handler process to display the status of all modules for the specified ATM in a series of event messages on the network logging facility. These event messages are described in the AT4730 log message documentation file.

DPUBLIC Command

A general use operator command that causes the NCR 5XXX Device Handler process to delete the BASE24 Primary Public key and optionally the BASE24 Root Public key if the ATM supports the NCR Enhanced AKDS scheme.

Syntax

DELIVER PROCESS *object-name,* “9500DPUBLIC *terminal-name x”*

object-name — The symbolic name assigned to the ATM Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files.

x — A code indicating which keys to delete. Value values are as follows:

P = Delete the BASE24 Primary Public key
R = Delete the BASE24 Root Public key and the BASE24 Primary Public key

Processing

This command causes the Device Handler process to remove the BASE24 Primary Public key and/or the BASE24 Root Public key from the EPP. The Device Handler process sends an operational shutdown command to the ATM. When it receives a response from the ATM indicating that it has been shut down, the Device Handler process then sends a configuration request to the ATM. Information in the configuration response indicates whether the ATM has AKDS Basic or Enhanced capability. For ATMs with AKDS Enhanced protocol, the Device Handler requests a self signature from TSS of the key to be deleted. If the Root Public key is deleted, the Primary Public key is also deleted. If only the Primary Public key is deleted, the Root Public key remains in the EPP and can be used to load another Primary Public key. The ATM is not brought back into service upon completion of the command.

DVD Command

A general use operator command that causes the IBM 3624 Device Handler process to send configuration version data for the ATM to the network logging facility. For more information on ATM configuration version data, refer to the ***IBM 3624 Programmer's Guide***.

Syntax

DELIVER PROCESS *object-name,* "9500DVD *terminal-name"*

object-name — The symbolic name assigned to the IBM 3624 Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the Terminal Data File. This ID must match a symbolic station name defined in the XPNET node.

Processing

This command causes the IBM 3624 Device Handler process to request configuration version data from the specified ATM and display this data in event message 0146. This event message is described in the AT3624 log message documentation file.

DOWNLOAD Command

A general use operator command that causes the IBM 473X-series Device Handler process to download one or more configuration files to the specified ATM. One or more data files of a particular type, microcode files, and microcode patch files can be downloaded with this command.

Syntax

DELIVER PROCESS *object-name,* "9500DOWNLOAD *terminal-name console config-file-name... data-file-type purge-existing"*

object-name — The symbolic name assigned to the IBM 473X-series Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the Terminal Data File. This ID must match a symbolic station name defined in the XPNET node.

console — The number of the IBM 4730 ATM console unit on which to execute the DOWNLOAD command. Valid values are as follows:

1 = Console 1
2 = Console 2

config-file-name — The HP NonStop file name for one or more data files, microcode files, or microcode patch files that you want to download to the ATM. Each file name must be a valid HP NonStop file name and can contain up to 34 characters.

data-file-type — A flag indicating the type of data files to be downloaded. Valid values are as follows:

C = Custom image file
P = Picture file
J = Journal file

purge-existing — A flag indicating whether existing files are deleted and replaced with the new files specified in the command. Valid values are as follows:

Y = Yes, delete and replace existing files with the new files specified in the command.
N = No, do not delete and replace existing files with the new files specified in the command.

Processing

This command causes the IBM 473X-series Device Handler process to download one or more configuration files to the specified ATM.

GETCNFG Command

A general use operator command that causes the NCR 5XXX Device Handler process to retrieve configuration information from the ATM and display the data in event messages on the network logging facility. If the ATM needs to be reconfigured, this command must be executed before the LOAD ALL command.

Syntax

DELIVER PROCESS *object-name,* "9500GETCNFG *terminal-name"*

object-name — The symbolic name assigned to the NCR 5XXX Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files. This ID must match a symbolic station name defined in the XPNET node.

Processing

This command causes the NCR 5XXX Device Handler process to request configuration information from the specified ATM and display this data in a series of event messages. If the command is successful, event message 0514 is generated. For a detailed description of this event message, refer to the AT5070 log message documentation file.

GETCNTRS Command

A general use operator command that causes the NCR 5XXX Device Handler process to retrieve supply counters from the ATM and display the data in a series of event messages on the network logging facility.

Syntax

DELIVER PROCESS *object-name,* "9500GETCNTRS *terminal-name"*

object-name — The symbolic name assigned to the NCR 5XXX Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files. This ID must match a symbolic station name defined in the XPNET node.

Processing

This command causes the NCR 5XXX Device Handler process to request supply counter information from the specified ATM and display this data in a series of event messages. If the command is successful, event message 0515 is generated. For a detailed description of this event message, refer to the AT5070 log message documentation file.

LAFITS Command

A general use operator command that causes the NCR 5XXX Device Handler process to download all financial institution tables (FITs) from load group 0000 and the customized load group specified in the TERMINAL GROUP field on BASE24-atm Terminal Data files (ATD) screen 7. The information to be downloaded is taken from the state values in the CNFG5070 file.

Syntax

DELIVER PROCESS *object-name,* “9500LAFITS *terminal-name”*

object-name — The symbolic name assigned to the NCR 5XXX Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files. This ID must match a symbolic station name defined in the XPNET node.

Processing

This command causes the NCR 5XXX Device Handler process to send all configured financial institution tables (FITs) in a native-mode message to the specified ATM.

LAGROUP Command

A general use operator command that causes the NCR 5XXX Device Handler process to download all configuration information (i.e., all customer, supervisor, and printer configuration screens, states, financial institution tables (FITs), and the configuration ID) for a specified load group. The configuration information for a load group is taken from the CNFG5070 file.

Syntax

DELIVER PROCESS *object-name,* "9500LAGROUP *terminal-name load-group"*

object-name — The symbolic name assigned to the NCR 5XXX Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files. This ID must match a symbolic station name defined in the XPNET node.

load-group — The number of the load group in the CNFG5070 file from which all configuration information is to be loaded.

Processing

This command causes the NCR 5XXX Device Handler process to send all the configuration information for the specified load group in native-mode messages to the specified ATM.

LALLKEYS Command

A general use operator command that causes the NCR 5XXX Device Handler process to download up to four keys to the encrypting PIN pad (EPP) as part of Automated Key Distribution System (AKDS) processing. For devices that support AKDS, the Device Handler process downloads the public key, master key, MAC encryption key, and PIN encryption key. For devices that do not support AKDS, the Device Handler process downloads the MAC encryption key and the PIN encryption key. For devices supporting the NCR Enhanced AKDS scheme, the BASE24 Root public and BASE24 Primary Public keys are deleted prior to loading the public keys, master key, and MAC encryption and PIN encryption keys. Refer to the ***BASE24-atm NCR 5XXX Device Support Manual*** for more information about the Automated Key Distribution add-on product.

Syntax

DELIVER PROCESS *object-name,* “9500LALLKEYS *terminal-name”*

object-name — The symbolic name assigned to the NCR 5XXX Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files.

Processing

This command causes the NCR 5XXX Device Handler process to download the public key, master key, MAC encryption key, and PIN encryption key to the EPP. The Device Handler process sends an operational shutdown command to the ATM. When the Device Handler receives a response from the ATM indicating that it has been shut down and is ready, the Device Handler process sends a configuration request to the ATM. Information in the configuration response indicates whether the ATM has AKDS capability. For ATMS with Enhanced AKDS protocol, the device handler deletes both the BASE24 Root and Primary public keys from the EPP first. For ATMs with AKDS capability, the Device Handler process downloads public, master, MAC, and PIN keys. For ATMs without AKDS capability, the Device Handler process downloads only the MAC and PIN keys. The Device Handler process then sends an operational startup command to the ATM.

LASCREENS Command

A general use operator command that causes the NCR 5XXX Device Handler process to download all screens from load group 0000 and the customized load group specified in the TERMINAL GROUP field on BASE24-atm Terminal Data files (ATD) screen 7. The information to be downloaded is taken from the state values in the CNFG5070 file.

Syntax

DELIVER PROCESS *object-name,* “9500LASCREENS *terminal-name”*

object-name — The symbolic name assigned to the NCR 5XXX Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files. This ID must match a symbolic station name defined in the XPNET node.

Processing

This command causes the NCR 5XXX Device Handler process to send all configured screens in a native-mode message to the specified ATM.

LASTATES Command

A general use operator command that causes the NCR 5XXX Device Handler process to download all states from load group 0000 and the customized load group specified in the TERMINAL GROUP field on BASE24-atm Terminal Data files (ATD) screen 7. Use caution when changing a state since changing state types changes the functionality of the screens associated with that state. The information to be downloaded is taken from the state values in the CNFG5070 file.

Syntax

DELIVER PROCESS *object-name,* "9500LASTATES *terminal-name"*

object-name — The symbolic name assigned to the NCR 5XXX Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files. This ID must match a symbolic station name defined in the XPNET node.

Processing

This command causes the NCR 5XXX Device Handler process to send all configured states in a native-mode message to the specified ATM.

LCSTATE Command

A general use operator command that causes the NCR 5XXX Device Handler process to download a specified state from the customized load group specified in the TERMINAL GROUP field on BASE24-atm Terminal Data files (ATD) screen 7. Use caution when changing a state since changing state types changes the functionality of the screens associated with that state. The information to be downloaded is taken from the state values for the customized load group in the CNFG5070 file.

Syntax

DELIVER PROCESS *object-name,* "9500LCSTATE *terminal-name state-number*"

object-name — The symbolic name assigned to the NCR 5XXX Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files. This ID must match a symbolic station name defined in the XPNET node.

state-number — The number of the state to be loaded from the device-specific configuration file.

Processing

This command causes the NCR 5XXX Device Handler process to send the specified state from the customized load group in a native-mode message to the specified ATM.

LDATE Command

A general use operator command that causes the NCR 5XXX Device Handler process to download the current date and time to the specified ATM. The new date and time are calculated from the current HP NonStop system time, plus the value from the TIME OFFSET field in the BASE24-atm Terminal Data files record for this terminal.

Syntax

DELIVER PROCESS *object-name,* “9500LDATE *terminal-name”*

object-name — The symbolic name assigned to the NCR 5XXX Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files. This ID must match a symbolic station name defined in the XPNET node.

Processing

This command causes the NCR 5XXX Device Handler process to calculate and send the current date and time in a native-mode message to the specified ATM.

LFIT Command

A general use operator command that causes the NCR 5XXX Device Handler process to download the specified financial institution table (FIT) from load group 0000 or the customized load group specified in the TERMINAL GROUP field on BASE24-atm Terminal Data files (ATD) screen 7. The information to be downloaded is taken from the FIT value in the CNFG5070 file.

Syntax

DELIVER PROCESS *object-name,* "9500LFIT *terminal-name fit-num"*

object-name — The symbolic name assigned to the NCR 5XXX Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files. This ID must match a symbolic station name defined in the XPNET node.

fit-num — The number of the financial institution table (FIT) you want downloaded to the ATM. If you do not include this variable in the command, it defaults to 000. Valid values are 000–999.

Processing

This command causes the NCR 5XXX Device Handler process to send the specified financial institution table (FIT) in a native-mode message to the specified ATM.

LISTQ Command

A general use operator command that causes a specified Diebold 10XX/478X or NCR 5XXX Device Handler process to list all records from the ATM Dial-Up Command File (ADCF) for the specified terminal ID. Each command is listed in an event message on the network log. This command is valid only for Diebold 10XX/478X and NCR 5XXX terminals configured as dial-up terminals in the DIAL-UP field on Diebold 10XX/478X screen 7 or NCR 5XXX screen 7 of the BASE24-atm Terminal Data files (ATD), respectively.

Syntax

DELIVER PROCESS *object-name,* “9500LISTQ *terminal-name”*

object-name — The symbolic name assigned to the Diebold 10XX/478X or NCR 5XXX Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the dial-up ATM in the BASE24-atm Terminal Data files.

Processing

This command causes the Diebold 10XX/478X or NCR 5XXX Device Handler process to read the ADCF using the terminal ID passed in the command and generate an event message for each command queued for the terminal.

Examples

The following is an examples of the CLEARQ command sent to a Diebold 10XX/ 478X Device Handler process from an XPNET network control facility. In this example, the symbolic name of the Diebold 10XX/478X Device Handler process is is P1A^D1000^1 and the terminal ID is DIALUPD10001.

DELIVER PROCESS P1A^N50^1, “9500CLEARQ DAILUPD10001”

LKEYMODE Command

A general use operator command that causes the NCR 5XXX Device Handler process to download the key entry mode to the Encrypting PIN Pad (EPP) of an ATM.

Syntax

DELIVER PROCESS *object-name,* “9500LKEYMODE *terminal-name x”*

object-name — The symbolic name assigned to the NCR 5XXX Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files. This ID must match a symbolic station name defined in the XPNET node.

x— A code indicating the key entry mode. Valid values are as follows:

1 = Single length without XOR
2 = Single length with XOR
3 = Double length with XOR
4 = Double length restricted

Processing

This command causes the NCR 5XXX Device Handler process to load the key entry mode to the EPP.

LKEYTYPE Command

A general use operator command that causes the NCR 5XXX Device Handler process to download a new communications key. The communications key must be downloaded to change the key currently being used.

Syntax

DELIVER PROCESS *object-name,* "9500LKEYTYPE *terminal-name key-type"*

object-name — The symbolic name assigned to the NCR 5XXX Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files. This ID must match a symbolic station name defined in the XPNET node.

key-type — The number of the decryption type you want used in downloading the communications key. Currently, the NCR 5XXX Device Handler process does not support changing the master key of the ATM. Valid values are as follows:

1 = Decrypt new master key using current master key (not currently supported).
2 = Decrypt new communications key using current master key.
3 = Decrypt new communications key using current communications key.
4 = Use power up communications key as current communications key (not currently supported).

Processing

This command causes the NCR 5XXX Device Handler process to decrypt the communications key as specified in the *key-type* variable and send the key in a native-mode message to the specified ATM.

LMACKEY Command

A general use operator command that causes the NCR 5XXX Device Handler process to download a new Message Authentication Code (MAC) key from the BASE24-atm Terminal Data files or a security module. A new MAC key must be downloaded to change the MAC key currently being used. For this command to execute, message authentication must be enabled for the ATM in the BASE24-atm Terminal Data files.

Syntax

DELIVER PROCESS *object-name,* "9500LMACKEY *terminal-name"*

object-name — The symbolic name assigned to the NCR 5XXX Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files. This ID must match a symbolic station name defined in the XPNET node.

Processing

This command causes the NCR 5XXX Device Handler process to send the new MAC key from the BASE24-atm Terminal Data files or a security module in a native-mode message to the specified ATM.

LMASTER Command

A general use operator command that causes the NCR 5XXX Device Handler process to download the master key to the Encrypting PIN Pad (EPP). This command is used with Automated Key Distribution System processing. Refer to the ***BASE24-atm NCR 5XXX Device Support Manual*** for more information about the Automated Key Distribution System add-on product.

Syntax

DELIVER PROCESS *object-name,* “9500LMASTER *terminal-name”*

object-name — The symbolic name assigned to the NCR 5XXX Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files. This ID must match a symbolic station name defined in the XPNET node.

Processing

This command causes the NCR 5XXX Device Handler process to load the master key to the EPP. For ATMs with AKDS Basic protocol, the Device Handler process causes the HSM to generate a new master key and downloads it to the ATM. The ATM is not brought back into service when the command is completed. For ATMs with AKDS Enhanced protocol, an additional step is required. The Device Handler requests a random number from the ATM before the host generates a new master key, then ensures both the key and random number are downloaded to the ATM.

LMK Command

A general use operator command that causes the IBM 3624 Device Handler process to download the PIN master key for the terminal from the Terminal Data File or a security module to the terminal.

Syntax

DELIVER PROCESS *object-name,* “9500LMK *terminal-name”*

object-name — The symbolic name assigned to the IBM 3624 Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the Terminal Data File. This ID must match a symbolic station name defined in the XPNET node.

Processing

This command causes the IBM 3624 Device Handler process to retrieve the PIN master key for the terminal from the Terminal Data File or a security module and send it in a native-mode message to the specified terminal.

LMSTRCOM Command

A general use operator command that causes the NCR 5XXX Device Handler process to download the new master, PIN, and MAC keys to the encrypting PIN pad (EPP) with a single command to change master keys if the ATM supports the NCR Enhanced AKDS scheme. Refer to the ***BASE24-atm NCR 5XXX Device Support Manual*** for more information about the Automated Key Distribution add-on product.

Syntax

DELIVER PROCESS *object-name,* “9500LMSTRCOM *terminal-name”*

object-name — The symbolic name assigned to the ATM Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files.

Processing

This command causes the Device Handler process to load the new Master key to the ATM, and then download the new PIN and MAC keys to the EPP. The Device Handler process sends an operational shutdown command to the ATM. When it receives a response from the ATM indicating that it has been shut down, the Device Handler process then sends a configuration request to the ATM. Information in the configuration response indicates if the ATM has AKDS capability, and if so, whether it is Basic or Enhanced. For ATMs with AKDS Enhanced protocol, the Device Handler downloads the new Master key and loads the PIN and MAC keys (if supported), allowing a single command to load a new master key to an ATM. The Device Handler process sends an operational startup command to the ATM.

LOAD Command

A general use operator command that causes the Device Handler process to download all information from the customized load group specified in the BASE24-atm Terminal Data files record of the terminal and all information from load group 0000. This information includes financial institution tables (FITs), screens, states, communications key information, and miscellaneous data. This command is generally used to load the configuration for a new ATM.

Syntax

DELIVER PROCESS *object-name,* "9500LOAD *terminal-name*"

object-name — The symbolic name assigned to the Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files. This ID must match a symbolic station name defined in the XPNET node.

Processing

The exact processing performed varies by the type of Device Handler process. In general, this command causes the Device Handler process to send all necessary configuration information in a native-mode message to the specified ATM.

LOADALL Command

A general use operator command that causes the NCR 5XXX Device Handler process to download all screens, states, and the financial institution table (FIT) to the ATM. Enhanced configuration parameter information from load group 0000 and the customized load group specified in the TERMINAL GROUP field on ATD screen 7 are also downloaded for this command. The information to be downloaded is taken from the CNFG5070 File.

If the ATM is to be reconfigured, the GETCNFG command must be executed before the LOADALL command.

Syntax

DELIVER PROCESS *object-name,* "9500LOADALL *terminal-name"*

object-name — The symbolic name assigned to the NCR 5XXX Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files. This ID must match a symbolic station name defined in the XPNET node.

Processing

Causes the NCR 5XXX Device Handler process to load all configuration data from load group 0000 and the customized load group in a native-mode message to the specified ATM.

LOADALLKEYS Command

A general use operator command that causes the Diebold 10XX/478X, Tidel, Triton, or NCR 5XXX Device Handler process to download up to four keys to the encrypting PIN pad (EPP) as part of Automated Key Distribution System (AKDS) processing. For devices that support AKDS, the Device Handler process downloads the public key, master key, MAC encryption key, and PIN encryption key. For devices that do not support AKDS, the Device Handler process downloads the MAC encryption key and the PIN encryption key. Refer to the ***BASE24-atm Diebold 10XX/478X Device Support Manual*** or ***BASE24-atm NCR 5XXX Device Support Manual*** for more information about the Automated Key Distribution add-on product.

Syntax

DELIVER PROCESS *object-name,* "9500LOADALLKEYS *terminal-name"*

object-name — The symbolic name assigned to the ATM Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files.

Processing

This command causes the Device Handler process to load the public key, master key, MAC encryption key, and PIN encryption key to the EPP. The Device Handler process sends an operational shutdown command to the ATM. When the Device Handler receives a response from the ATM indicating that it has been shut down, the Device Handler process then sends a configuration request to the ATM. Information in the configuration response indicates if the ATM has AKDS capability, and if so, whether it is Basic or Enhanced. For ATMs with AKDS Basic protocol, the Device Handler process downloads public, master, MAC, and PIN keys. For ATMs with AKDS Enhanced protocol, the device handler deletes both the BASE24 Root Public and Primary Public keys before the LOADPUBLIC step is performed to download the public keys, master key, and MAC and PIN keys. An alternative is to use the LOADMSTRCOM command in place of LOADALLKEYS when changing master keys at an ATM (once the initial load public has been performed). LOADMSTRCOM generates a new master key and loads the PIN key and Mac key as appropriate without exchanging the public keys. For ATMs without AKDS capability, the Device Handler process downloads only the MAC and PIN keys. For all protocols, the Device Handler process then sends an operational startup command to the ATM.

LOADGROUP Command

A general use operator command that causes the Device Handler process to download all configuration information (i.e., configuration identifier, screens, states, financial institution tables, communications keys, and miscellaneous data) for a specified customized load group. Miscellaneous data and print configuration text screens from load group 0000 are also downloaded with this command. The configuration information for a load group is taken from the device-specific configuration file (e.g., the CNFG5070 file for NCR 5XXX terminals) and the BASE24-atm Terminal Data files.

Syntax

DELIVER PROCESS *object-name,* "9500LOADGROUP *terminal-name load-group"*

object-name — The symbolic name assigned to the Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files. This ID must match a symbolic station name defined in the XPNET node.

load-group — The number of the load group in the device-specific configuration file from which all configuration information is to be loaded.

Processing

The exact processing performed varies by the type of Device Handler process. In general, this command causes the Device Handler process to send all the configuration information listed above in native-mode messages to the specified ATM.

LOADGROUPSCREEN Command

A general use operator command that causes the Device Handler process to download a specified screen from the specified load group. The screen to be downloaded is taken from the screen values for the specified group in the device-specific configuration file (e.g., the CNFG5070 file for NCR 5XXX terminals).

Syntax

DELIVER PROCESS *object-name,* "9500LOADGROUPSCREEN *terminal-name screen-number load-group"*

object-name — The symbolic name assigned to the Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files. This ID must match a symbolic station name defined in the XPNET node.

screen-number — The number of the screen to be loaded from the device-specific configuration file.

load-group — The number of the load group in the device-specific configuration file from which the specified screen is to be loaded.

Processing

The exact processing performed varies by the type of Device Handler process. In general, this command causes the Device Handler process to send the specified screen from the specified group in a native-mode message to the specified ATM.

LOADGROUPSTATE Command

A general use operator command that causes the Device Handler process to download a specified state from the specified load group. Use caution when changing a state since changing state types changes the functionality of the screens associated with that state. The information to be downloaded is taken from the state values for the specified group in the device-specific configuration file (e.g., the CNFG5070 file for NCR 5XXX terminals).

Syntax

DELIVER PROCESS *object-name,* “9500LOADGROUPSTATE *terminal-name state-number load-group”*

object-name — The symbolic name assigned to the Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files. This ID must match a symbolic station name defined in the XPNET node.

state-number — The number of the state to be loaded from the device-specific configuration file.

load-group — The number of the load group in the device-specific configuration file from which the specified state is to be loaded.

Processing

The exact processing performed varies by the type of Device Handler process. In general, this command causes the Device Handler process to send the specified state from the specified group in a native-mode message to the specified ATM.

LOADKEY Command

A general use operator command that causes the Device Handler process to download a new communications key from the BASE24-atm Terminal Data files or a security module. The communications key must be downloaded to change the key currently being used.

Syntax

DELIVER PROCESS *object-name,* "9500LOADKEY *terminal-name decrypt-type"*

object-name — The symbolic name assigned to the Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files. This ID must match a symbolic station name defined in the XPNET node.

decrypt-type — The number of the decryption type you want used in downloading the communications key. Currently, the NCR 5XXX Device Handler process does not support changing the master key of the ATM. Valid values are as follows:

1 = Decrypt new master key using current master key (not currently supported).
2 = Decrypt new communications key using current master key.
3 = Decrypt new communications key using current communications key.
4 = Use power up communications key as current communications key (not currently supported).

Processing

The exact processing performed varies by the type of Device Handler process. In general, this command causes the Device Handler process to send the new communications key in a native-mode message to the specified ATM.

LOADKEYMODE Command

A general use operator command that causes the NCR 5XXX Device Handler process to download the key entry mode to the Encrypting PIN Pad (EPP) of an ATM.

Syntax

DELIVER PROCESS *object-name,* “9500LOADKEYMODE *terminal-name x*”

object-name — The symbolic name assigned to the NCR 5XXX Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files. This ID must match a symbolic station name defined in the XPNET node.

x— A code indicating the key entry mode. Valid values are as follows:

1	=	Single length without XOR
2	=	Single length with XOR
3	=	Double length with XOR
4	=	Double length restricted

Processing

This command causes the NCR 5XXX Device Handler process to load the key entry mode to the EPP.

LOADMACKEY Command

A general use operator command that causes the Device Handler process to download a new message authentication code (MAC) key from the BASE24-atm Terminal Data files or a security module. A new MAC key must be downloaded to change the MAC key currently being used. For this command to execute, message authentication must be enabled for the ATM in the BASE24-atm Terminal Data files.

Syntax

DELIVER PROCESS *object-name,* "9500LOADMACKEY *terminal-name"*

object-name — The symbolic name assigned to the Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files. This ID must match a symbolic station name defined in the XPNET node.

Processing

The exact processing performed varies by the type of Device Handler process. In general, this command causes the Device Handler process to send the new MAC key from the BASE24-atm Terminal Data files or a security module in a native-mode message to the specified ATM.

LOADMASTER Command

A general use operator command that causes the Diebold 10XX/478X Device Handler process or NCR 5XXX Device Handler process to download the master key to the encrypting PIN pad (EPP). This command is issued as the final phase of Automated Key Distribution System processing. Refer to the ***BASE24-atm Diebold 10XX/478X Device Support Manual*** or ***BASE24-atm NCR 5XXX Device Support Manual*** for more information about the Automated Key Distribution add-on product.

Syntax

DELIVER PROCESS *object-name,* “9500LOADMASTER *terminal-name”*

object-name — The symbolic name assigned to the ATM Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files.

Processing

This command causes the Device Handler process to download the master key to the EPP. The Device Handler process sends an operational shutdown command to the ATM. When it receives a response from the ATM indicating that it has been shut down, the Device Handler process then sends a configuration request to the ATM. Information in the configuration response indicates if the ATM has AKDS capability, and if so, whether it is Basic or Enhanced. For ATMs with AKDS Basic protocol, the Device Handler process causes the HSM to generate a new master key and downloads it to the ATM. The ATM is not brought back into service when the command is completed. For ATMs with AKDS Enhanced protocol, an additional step is required. The Device Handler requests a random number from the ATM before the host generates a new master key, then ensures both the key and random number are downloaded to the ATM. For ATMs without AKDS capability, the Device Handler process sends an operational startup command to the ATM without first downloading the requested key.

LOADMSTRCOM Command

A general use operator command that causes the NCR 5XXX Device Handler process to download the new master, PIN, and MAC keys to the encrypting PIN pad (EPP) with a single command to change master keys if the ATM supports the NCR Enhanced AKDS scheme. Refer to the ***BASE24-atm NCR 5XXX Device Support Manual*** for more information about the Automated Key Distribution add-on product.

Syntax

DELIVER PROCESS *object-name,* "9500LOADMSTRCOM *terminal-name"*

object-name — The symbolic name assigned to the ATM Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files.

Processing

This command causes the Device Handler process to load the new Master key to the ATM, and then download the new PIN and MAC keys to the EPP. The Device Handler process sends an operational shutdown command to the ATM. When it receives a response from the ATM indicating that it has been shut down, the Device Handler process then sends a configuration request to the ATM. Information in the configuration response indicates if the ATM has AKDS capability, and if so, whether it is Basic or Enhanced. For ATMs with AKDS Enhanced protocol, the Device Handler downloads the new Master key and loads the PIN and MAC keys (if supported), allowing a single command to load a new master key to an ATM. The Device Handler process sends an operational startup command to the ATM.

LOADPRIPUB Command

A general use operator command that causes the NCR 5XXX Device Handler process to download the BASE24 Primary public key to the encrypting PIN pad (EPP). Refer to the ***BASE24-atm Diebold 10XX/478X Device Support Manual*** or ***BASE24-atm NCR 5XXX Device Support Manual*** for more information about the Automated Key Distribution add-on product.

Syntax

DELIVER PROCESS *object-name,* “9500LOADPRIPUB *terminal-name”*

object-name — The symbolic name assigned to the ATM Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files.

Processing

This command causes the Device Handler process to download the BASE24 Primary public key only to the EPP. The Device Handler process sends an operational shutdown command to the ATM. When it receives a response from the ATM indicating that it has been shut down, the Device Handler process then sends a configuration request to the ATM. Information in the configuration response indicates whether the ATM has AKDS Basic or Enhanced capability. For ATMs with AKDS Enhanced protocol, the BASE24 Primary public key is downloaded to the ATM. This command can be used to load a new BASE24 Primary public key after a DELPUBLIC command has deleted the old one.

LOADPUBLIC Command

A general use operator command that causes the Diebold 10XX/478X Device Handler process or NCR 5XXX Device Handler process to download the BASE24 public key to the encrypting PIN pad (EPP). This command is issued as the third phase of Automated Key Distribution System processing. Refer to the ***BASE24-atm Diebold 10XX/478X Device Support Manual*** or ***BASE24-atm NCR 5XXX Device Support Manual*** for more information about the Automated Key Distribution add-on product.

Syntax

DELIVER PROCESS *object-name,* “9500LOADPUBLIC *terminal-name”*

object-name — The symbolic name assigned to the ATM Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files.

Processing

This command causes the Device Handler process to download the BASE24 public key to the EPP. During this processing, BASE24 and the EPP authenticate each other before exchanging their public keys. Diebold ATM devices send two keys—public signature verification key and public encipherment key—to BASE24 for storage in the Transaction Security Services database. NCR ATM devices send one key —EPP public key—to BASE24 for storage in the Transaction Security Services database.

The Device Handler process sends an operational shutdown command to the ATM. When it receives a response from the ATM indicating that it has been shut down, the Device Handler process then sends a configuration request to the ATM. Information in the configuration response indicates whether the ATM has AKDS Basic or Enhanced capability. For ATMs with AKDS Basic capability, the Device Handler process downloads the BASE24 public key. For ATMs with AKDS Enhanced capability, the Device Handler process loads the BASE24 Root public key and signature to the ATM before downloading the primary public key and signature. For ATMs without AKDS capability, the Device Handler process sends an operational startup command to the ATM without first downloading the requested key or keys.

LOADSCREEN Command

A general use operator command that causes the Device Handler process to download a specified screen from load group 0000. The screen to be downloaded is taken from screen values in the device-specific configuration file (e.g., the CNFG5070 file for NCR 5XXX terminals).

Syntax

DELIVER PROCESS *object-name,* “9500LOADSTATE *terminal-name screen-number”*

object-name — The symbolic name assigned to the Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files. This ID must match a symbolic station name defined in the XPNET node.

screen-number — The number of the screen to be loaded from the device-specific configuration file.

Processing

The exact processing performed varies by the type of Device Handler process. In general, this command causes the Device Handler process to send the specified screen in a native-mode message to the specified ATM.

LOADSTATE Command

A general use operator command that causes the Device Handler process to download a specified state from load group 0000. Use caution when changing a state since changing state types changes the functionality of the screens associated with that state. The information to be downloaded is taken from the state values in the device-specific configuration file (e.g., the CNFG5070 file for NCR 5XXX terminals).

Syntax

DELIVER PROCESS *object-name,* "9500LOADSTATE *terminal-name state-number"*

object-name — The symbolic name assigned to the Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files. This ID must match a symbolic station name defined in the XPNET node.

state-number — The number of the state to be loaded from the device-specific configuration file.

Processing

The exact processing performed varies by the type of Device Handler process. In general, this command causes the Device Handler process to send the specified state in a native-mode message to the specified ATM.

LPRIPUB Command

A general use operator command that causes the NCR 5XXX Device Handler process to download the BASE24 Primary public key to the encrypting PIN pad (EPP). Refer to the ***BASE24-atm Diebold 10XX/478X Device Support Manual*** or ***BASE24-atm NCR 5XXX Device Support Manual*** for more information about the Automated Key Distribution add-on product.

Syntax

DELIVER PROCESS *object-name,* "9500LPRIPUB *terminal-name"*

object-name — The symbolic name assigned to the ATM Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files.

Processing

This command causes the Device Handler process to download the BASE24 Primary public key only to the EPP. The Device Handler process sends an operational shutdown command to the ATM. When it receives a response from the ATM indicating that it has been shut down, the Device Handler process then sends a configuration request to the ATM. Information in the configuration response indicates whether the ATM has AKDS Basic or Enhanced capability. For ATMs with AKDS Enhanced protocol, the BASE24 Primary public key is downloaded to the ATM. This command can be used to load a new BASE24 Primary public key after a DELPUBLIC command has deleted the old one.

LPUBLIC Command

A general use operator command that causes the NCR 5XXX Device Handler process to exchange public keys with the Encrypting PIN Pad (EPP). This command is used with Automated Key Distribution System processing. Refer to the ***BASE24-atm NCR 5XXX Device Support Manual*** for more information about the Automated Key Distribution System add-on product.

Syntax

DELIVER PROCESS *object-name,* "9500LPUBLIC *terminal-name"*

object-name — The symbolic name assigned to the NCR 5XXX Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files. This ID must match a symbolic station name defined in the XPNET node.

Processing

This command causes the Device Handler process to download the BASE24 public key to the EPP. During this processing, BASE24 and the EPP authenticate each other before exchanging their public keys. Diebold ATM devices send two keys—public signature verification key and public encipherment key—to BASE24 for storage in the Transaction Security Services database. NCR ATM devices send one key —EPP public key—to BASE24 for storage in the Transaction Security Services database.

The Device Handler process sends an operational shutdown command to the ATM. When it receives a response from the ATM indicating that it has been shut down, the Device Handler process then sends a configuration request to the ATM. Information in the configuration response indicates whether the ATM has AKDS Basic or Enhanced capability. For ATMs with AKDS Basic capability, the Device Handler process downloads the BASE24 public key. For ATMs with AKDS Enhanced capability, the Device Handler process loads the BASE24 Root public key and signature to the ATM before downloading the primary public key and signature. For ATMs without AKDS capability, the Device Handler process sends an operational startup command to the ATM without first downloading the requested key or keys.

LSCREEN Command

A general use operator command that causes the NCR 5XXX Device Handler process to download a specified screen from the customized load group specified in the TERMINAL GROUP field on BASE24-atm Terminal Data files (ATD) screen 7. The screen to be downloaded is taken from the screen values for the customized load group in the CNFG5070 file.

Syntax

DELIVER PROCESS *object-name,* "9500LSCREEN *terminal-name screen-number language-number*"

object-name — The symbolic name assigned to the NCR 5XXX Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files. This ID must match a symbolic station name defined in the XPNET node.

screen-number — The number of the screen to be loaded from the CNFG5070 file.

language-number — A code indicating the language of the customer screen to be loaded from the CNFG5070 file. Valid values are 0–9.

Processing

This command causes the NCR 5XXX Device Handler process to send the specified screen from the customized load group in a native-mode message to the specified ATM.

MACINGOFF Command

A general use operator command that causes the Triton Device Handler process to send an ASCII 0 in the field ID code (FID) "/" (solidus) in a response message to a specified Triton device. This FID value instructs the device to turn off message authentication in requests sent to BASE24.

Syntax

DELIVER PROCESS *object-name,* "9500MACINGOFF *terminal-name"*

object-name — The symbolic name assigned to the Triton Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files. This ID must match a symbolic station name defined in the XPNET node.

Processing

This command causes the Triton Device Handler process to send an ASCII 0 in the field ID code (FID) "/" (solidus) in a response message to the specified Triton ATM device.

MACINGON Command

A general use operator command that causes the Triton Device Handler process to send an ASCII 1 in the field ID code (FID) “/” (solidus) in a response message to a specified Triton device. This FID value instructs the device to turn on message authentication in requests sent to BASE24.

Syntax

DELIVER PROCESS *object-name,* “9500MACINGON *terminal-name”*

object-name — The symbolic name assigned to the Triton Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files. This ID must match a symbolic station name defined in the XPNET node.

Processing

This command causes the Triton Device Handler process to send an ASCII 1 in the field ID code (FID) “/” (solidus) in a response message to the specified Triton ATM device.

NAM Command

A restricted use operator command that causes the IBM 3624 or 473X-series Device Handler process to rename the file or device to which trace messages from the ATM are sent. This name must be a valid HP NonStop file or device name.

Syntax

DELIVER PROCESS *object-name,* "9500NAM *terminal-name console trace-facility"*

object-name — The symbolic name assigned to the IBM 3624 or 473X-series Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

console — The number of the IBM 4730 ATM console unit on which to execute the DOWNLOAD command. This variable does not apply to IBM 3624 terminals. Valid values are as follows:

1 = Console 1
2 = Console 2

terminal-name — The terminal ID defined for the ATM in the Terminal Data File. This ID must match a symbolic station name defined in the XPNET node.

trace-facility — The name of the trace file or device.

Processing

This command causes the IBM 3624 or 473X-series Device Handler process to direct trace messages from the specified ATM to a new trace facility location.

OPEN, OPENA, and STARTUP Commands

General use operator commands that cause the Device Handler process to send the appropriate native-mode message to open the ATM. The OPEN and STARTUP commands open the specified ATM for all transactions, while the OPENA command opens the specified ATM for administrative transactions only.

Syntax

DELIVER PROCESS *object-name,* “9500OPEN *terminal-name”*

DELIVER PROCESS *object-name,* “9500OPENA *terminal-name”*

DELIVER PROCESS *object-name,* “9500STARTUP *terminal-name”*

object-name — The symbolic name assigned to the Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files. This ID must match a symbolic station name defined in the XPNET node.

Processing

The Device Handler process performs the following processing steps when it receives an OPEN, OPENA, or STARTUP command:

1. Translates the command into the appropriate native mode format for the specified ATM.
2. Sends a message to open the specified ATM.

REL Command

A general use operator command that causes the IBM 3624 Device Handler process to open or release an ATM after it has been suspended using the SUS command. The REL command allows the ATM to return to normal operation.

Syntax

DELIVER PROCESS *object-name,* "9500REL *terminal-name"*

object-name — The symbolic name assigned to the IBM 3624 Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the Terminal Data File. This ID must match a symbolic station name defined in the XPNET node.

Processing

This command causes the IBM 3624 Device Handler process to release the specified ATM from a suspended state.

RESET Command

A restricted use operator command that causes the Device Handler process to interrupt any current transaction or load in progress, reset the transaction status, and clear any device faults for the specified ATM. For the IBM 3624 Device Handler process, this command causes the process to activate or deactivate a SNA session.

Syntax

DELIVER PROCESS *object-name,* “9500RESET *symbolic-station-name”*

object-name — The symbolic name assigned to the Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

symbolic-station-name — The station name for the ATM as defined in the XPNET node.

Processing

The transaction status for the ATM is reset so that the ATM is able to process transactions. Any device faults recorded by the Device Handler process prior to the reset are cleared. The Device Handler process performs the following steps when it receives a RESET command:

1. Interrupts any transaction or load currently in progress for the specified ATM.
2. Resets the transaction status for the ATM so that the ATM is able to process transactions.
3. Clears any device faults recorded by the Device Handler process prior to the reset.

RESTORE Command

A general use operator command that causes the Diebold 10XX/478X Device Handler process to restore the initial graphics configuration from load group 0000. This information includes default screens, states, communications key information, and miscellaneous data.

Syntax

DELIVER PROCESS *object-name,* "9500RESTORE *terminal-name*"

object-name — The symbolic name assigned to the Diebold 10XX/478X Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the BASE24-atm Terminal Data files. This ID must match a symbolic station name defined in the XPNET node.

Processing

This command causes the Diebold 10XX/478X Device Handler process to restore the initial graphics configuration from load group 0000 for the specified ATM.

RVR Command

A general use operator command that causes the IBM 3624 terminal to revert to using the base communications key. This command allows the IBM 3624 Device Handler process to resynchronize transaction sequence numbering.

Syntax

DELIVER PROCESS *object-name,* “9500RVR *terminal-name”*

object-name — The symbolic name assigned to the IBM 3624 Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the Terminal Data File. This ID must match a symbolic station name defined in the XPNET node.

Processing

This command causes the IBM 3624 Device Handler process to send a native-mode message to the specified terminal, requesting the terminal to revert to using the base communications key.

SDT Command

A general use operator command that causes the IBM 473X-series Device Handler process to download the current date and time to the specified ATM. If the IBM 473X-series terminal specified has two console units, this command sets the date and time on both consoles. The new date and time are calculated from the current HP NonStop system time, plus the value from the TIME OFFSET field in the Terminal Data File record for this terminal.

Syntax

DELIVER PROCESS *object-name,* "9500SDT *terminal-name"*

object-name — The symbolic name assigned to the 473X-series Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the Terminal Data File. This ID must match a symbolic station name defined in the XPNET node.

Processing

This command causes the IBM 473X-series Device Handler process to calculate and send the current date and time in a native-mode message to the specified ATM.

SENDSTATUS Command

A general use operator command that causes the Fujitsu Device Handler process to request hardware status information from the specified ATM.

Syntax

DELIVER PROCESS *object-name,* "9500SENDSTATUS *terminal-name"*

object-name — The symbolic name assigned to the Fujitsu Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the Terminal Data File. This ID must match a symbolic station name defined in the XPNET node.

Processing

This command causes the Fujitsu Device Handler process to send a native-mode message to the specified ATM, requesting hardware status information. The process displays this status information on the network logging facility.

SENDSUPPLY Command

A general use operator command that causes the Fujitsu Device Handler process to request supplies status information from the specified ATM.

Syntax

DELIVER PROCESS *object-name,* "9500SENDSUPPLY *terminal-name"*

object-name — The symbolic name assigned to the Fujitsu Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the Terminal Data File. This ID must match a symbolic station name defined in the XPNET node.

Processing

This command causes the Fujitsu Device Handler process to send a native-mode message to the specified ATM, requesting supplies status information. The process displays this information on the network logging facility.

SER Command

A general use operator command that causes the IBM 3624 Device Handler to place the ATM in service mode. Use this command when an ATM does not accept administrative cards. This command can be executed when an ATM is open, closed, or inactive.

Syntax

DELIVER PROCESS *object-name,* "9500SER *terminal-name"*

object-name — The symbolic name assigned to the IBM 3624 Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the Terminal Data File. This ID must match a symbolic station name defined in the XPNET node.

Processing

This command causes the IBM 3624 Device Handler process to place the specified ATM in service mode.

STATUS Command

A general use operator command that causes the Tidel or Triton Device Handler process to generate an event message indicating whether a transaction is currently in progress for the specified ATM.

Syntax

DELIVER PROCESS *object-name,* “9500STATUS *terminal-name”*

object-name — The symbolic name assigned to the Tidel or Triton Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the Terminal Data File. This ID must match a symbolic station name defined in the XPNET node.

Processing

This command causes the Tidel or Triton Device Handler process to generate an event message indicating whether a transaction is in currently in progress for the specified ATM.

STATUSEXTMEM Command

A general use operator command that causes a Diebold 10XX/478X or NCR 5XXX Device Handler process to display statistics about its BASE24-atm Terminal Data files (ATD) extended memory segment on the network log. This command allows network operators to monitor extended memory usage and determine whether adjustments are needed.

Syntax

DELIVER PROCESS *object-name,* “STATUSEXTMEM”

object-name — The symbolic name assigned to the Diebold 10XX/478X or NCR 5XXX Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

This command causes a Diebold 10XX/478X or NCR 5XXX Device Handler process to generate several event messages detailing its current ATD extended memory segment usage. Information in these messages includes the following:

- The current number of terminals in the Terminal table.
- The current number of terminals in the Dynamic Data table.
- The maximum number of terminals to be managed in the Terminal table.
- The maximum number of terminals allowed in the Dynamic Data table, including the last message data.
- The maximum number of terminals allowed in the Static Data table.
- The length of each record in the Dynamic Data table.
- The length of each record in the Static Data table.
- The length of each record in the last message data.

SUS Command

A general use operator command that causes the IBM 3624 Device Handler process to temporarily suspend an ATM. While operation is suspended, the ATM remains open but cannot process customer transactions. When a card is inserted while the ATM is suspended, the ATM screen displays the message "PLEASE WAIT A FEW MOMENTS." The REL command allows you to release the ATM from a suspended state and return it to normal operation.

Syntax

DELIVER PROCESS *object-name,* "9500SUS *terminal-name"*

object-name — The symbolic name assigned to the IBM 3624 Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the Terminal Data File. This ID must match a symbolic station name defined in the XPNET node.

Processing

This command causes the IBM 3624 Device Handler process to send a native-mode message to the specified ATM, requesting the ATM to temporarily suspend normal operations.

TOF Command

A restricted use operator command that causes the IBM 3624 Device Handler process to stop the BASE24 trace facility, which terminates the message flow to the currently designated trace file or device.

Syntax

DELIVER PROCESS *object-name,* "9500TOF *terminal-name"*

object-name — The symbolic name assigned to the IBM 3624 Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the Terminal Data File. This ID must match a symbolic station name defined in the XPNET node.

Processing

This command causes the IBM 3624 Device Handler process to stop the BASE24 trace facility. If the command is successful, event message 0207 is generated. This event message is described in the AT3624 log message documentation file.

TON Command

A restricted use operator command that causes the IBM 3624 Device Handler process to start the BASE24 trace facility, which sends a copy of all messages exchanged between the ATM and the HP NonStop processor to the currently designated trace file or device.

Syntax

DELIVER PROCESS *object-name,* "9500TON *terminal-name"*

object-name — The symbolic name assigned to the IBM 3624 Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the Terminal Data File. This ID must match a symbolic station name defined in the XPNET node.

Processing

This command causes the IBM 3624 Device Handler process to start the BASE24 trace facility. If the command is successful, event message 0206 is generated. This event message is described in the AT3624 log message documentation file.

TRACE OFF Command

A restricted use operator command that deactivates the trace function within a Device Handler process.

Syntax

DELIVER PROCESS *object-name,* “TRACE OFF”

object-name — The symbolic name assigned to the Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing:

An operator sends this command from an XPNET network control facility. For more information on the TRACE OFF command, refer to appendix A.

TRACE ON Command

A restricted use operator command that activates a trace function within a Device Handler process.

Syntax

DELIVER PROCESS *object-name,* "TRACE ON"

object-name — The symbolic name assigned to the Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

An operator sends this command from an XPNET network control facility. For more information on the TRACE ON command, refer to appendix A.

UDC Command

A general use operator command that causes the IBM 473X-series Device Handler process to update configuration of the modules of a specified ATM. Specifically, this command allows you to enable or disable one or more specific modules of an ATM.

Syntax

DELIVER PROCESS *object-name,* "9500UDC *terminal-name console statement-printer envelope-depository check-depository document-feed... coin-feed... journal-disk camera protective-door*"

object-name — The symbolic name assigned to the IBM 473X-series Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the Terminal Data File. This ID must match a symbolic station name defined in the XPNET node.

console — The number of the IBM 4730 ATM console unit on which to execute the UDC command. Valid values are as follows:

1 = Console 1
2 = Console 2

statement-printer — A flag disabling (deconfigure) or enabling (configure) the statement printer of the ATM. If this variable is not entered, it defaults to a value of 0. Valid values are as follows:

0 = No change
1 = Deconfigure
2 = Configure

envelope-depository — A flag disabling (deconfigure) or enabling (configure) the envelope depository of the ATM. If this variable is not entered, it defaults to a value of 0. Valid values are as follows:

0 = No change
1 = Deconfigure
2 = Configure

check-depository — A flag disabling (deconfigure) or enabling (configure) the check depository of the ATM. If this variable is not entered, it defaults to a value of 0. Valid values are as follows:

0 = no change
1 = Deconfigure
2 = Configure

document-feed — Flags disabling (deconfigure) or enabling (configure) each of the document feeders (1 through 5) of the ATM. If this variable is not entered for a particular feeder, it defaults to a value of 0. Valid values are as follows:

0 = No change
1 = Deconfigure
2 = Configure

coin-feed — Flags disabling (deconfigure) or enabling (configure) each of the coin feeders (1 through 4) of the ATM. If this variable is not entered for a particular feeder, it defaults to a value of 0. Valid values are as follows:

0 = No change
1 = Deconfigure
2 = Configure

journal-disk — A flag disabling (deconfigure) or enabling (configure) the journal disk of the ATM. This field is not valid for the IBM 4730 model. It is optional on other IBM 473X-series models. If this variable is not entered for a particular feeder, it defaults to a value of 0. Valid values are as follows:

0 = No change
1 = Deconfigure
2 = Configure

camera — A flag disabling (deconfigure) or enabling (configure) the camera of the ATM. This field is not valid for the IBM 4730 model. It is optional on other IBM 473X-series models. If this variable is not entered for a particular feeder, it defaults to a value of 0. Valid values are as follows:

0 = No change
1 = Deconfigure
2 = Configure

protective-door — A flag disabling (deconfigure) or enabling (configure) the protective door of the ATM. This field is not valid for the IBM 4730 model. It is optional on other IBM 473X-series models. If this variable is not entered for a particular feeder, it defaults to a value of 0. Valid values are as follows:

0 = No change
1 = Deconfigure
2 = Configure

Processing

This command causes the IBM 473X-series Device Handler process to send the module configuration information in a native-mode message to the specified ATM. The ATM then enables or disables each module as instructed.

UPLOAD[!] Command

A general use operator command that causes the IBM 473X-series Device Handler process to upload one or more configuration files from the specified ATM to BASE24. One or more data files of a particular type, microcode files, and microcode patch files can be uploaded with this command.

Syntax

DELIVER PROCESS *object-name,* "9500UPLOAD[!] *terminal-name console config-file-name... data-file-type"*

object-name — The symbolic name assigned to the IBM 473X-series Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

[!] — When this character is appended to the UPLOAD keyword, any existing files on the HP NonStop processor are purged and replaced with the new files specified in the command. Otherwise, an error message is generated if one or more of the files to be uploaded already exist on the HP NonStop processor.

terminal-name — The terminal ID defined for the ATM in the Terminal Data File. This ID must match a symbolic station name defined in the XPNET node.

console — The number of the IBM 4730 ATM console unit on which to execute the UPLOAD command. Valid values are as follows:

1 = Console 1
2 = Console 2

config-file-name — The HP NonStop file name for one or more data files, microcode files, or microcode patch files that you want to upload from the ATM to BASE24. Each file name must be a valid HP NonStop file name and can contain up to 34 characters.

data-file-type — A flag indicating the type of data files to be uploaded. Valid values are as follows:

C = Custom image file
P = Picture file
J = Journal file

Processing

This command causes the IBM 473X-series Device Handler process to upload configuration files from the specified ATM to BASE24.

WARMBOOT Command

A general use operator command that causes the Device Handler process to reinitialize.

Syntax

DELIVER PROCESS *object-name,* “9501”

object-name — The symbolic name assigned to the Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

When Device Handler processes are warmbooted, they do not go through their entire initialization routine. Instead, they perform the following steps:

1. Reread the Logical Network Configuration File (LCONF)
2. Close and reopen the files they use
3. Release the current extended memory, including any extended memory tables.
4. Close and reopen the security module
5. Reinitialize the Log Message Configuration File (LMCF)
6. Reallocate new extended memory and any extended memory tables based on the current settings in the LCONF.

WARMBOOT ATD Command

A general use operator command that causes a Diebold 10XX/478X or NCR 5XXX Device Handler process to warmboot static terminal data held in extended memory. This command should be issued any time an operator performs file maintenance on a BASE24-atm Terminal Data files (ATD) field that is stored in the ATM Terminal Data Static File—general data (ATDS1). Static data that is modified in the ATD will not be used in processing until the ATD is warmbooted or the Device Handler process itself is reinitialized.

Syntax

DELIVER PROCESS *object-name,* "9529ATD"

object-name — The symbolic name assigned to the Diebold 10XX/478X or NCR 5XXX Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

This command causes a Diebold 10XX/478X or NCR 5XXX Device Handler process to re-read its ATDS1 records into extended memory. If the command completes successfully, the following message is displayed on the network log:

```
'TDF Warmboot' command completed.
```

ZEROSUPPLY Command

A general use operator command that causes a Fujitsu 7000 Device Handler process to send a message to the specified terminal, instructing it to zero out (or clear) its subdevice supplies information.

Syntax

DELIVER PROCESS *object-name,* “9500ZEROSUPPLY *terminal-name”*

object-name — The symbolic name assigned to the Fujitsu 7000 Device Handler process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the ATM in the Terminal Data File. This ID must match a symbolic station name defined in the XPNET node.

Processing

This command causes a Fujitsu 7000 Device Handler process to send a native-mode Zero Supply Counts command message to the specified ATM. If the message is successfully sent, an informational message is displayed on the network log indicating that a zero supply count operation was requested for the specified terminal.

Settlement Initiator Process Commands

The BASE24-atm Settlement Initiator process is the prime controller of BASE24-atm's daily cutover and settlement. It acts as a timekeeper by taking actions at prescribed times each day to carry out cutover and settlement for each institution in the logical network. The BASE24-atm Settlement Initiator process creates reports for the processing day just ended. This subsection describes the processing performed by the BASE24-atm Settlement Initiator process. It includes descriptions of the BASE24 commands the BASE24-atm Settlement Initiator process recognizes.

BASE24 processes can receive a number of commands that cause them to perform specialized processing. These commands can be separated into three categories: process-generated commands, general use operator commands, and restricted use operator commands. All BASE24 general and restricted use operator commands that can be received by the BASE24-atm Settlement Initiator process are documented in this subsection. Process-generated commands are documented in appendix B.

All of the commands that can be received by the BASE24-atm Settlement Initiator process are listed below by category.

Process-Generated Commands	General Use Operator Commands	Restricted Use Operator Commands
None	WARMBOOT	DEBUG MIDNIGHT RESET DATES

DEBUG Command

A restricted use operator command that places the Settlement Initiator process under control of the HP NonStop Inspect product and displays an Inspect prompt on the home terminal (the terminal specified for the process by the HOMETERM attribute or the home terminal of the XPNET process). Inspect is an HP NonStop program debugging tool.

Syntax

DELIVER PROCESS *object-name,* "DEBUG"

object-name — The symbolic name assigned to the Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

An operator enters this command from an XPNET network control facility. For more information on the DEBUG command, refer to appendix A.

MIDNIGHT Command

A restricted use command that causes the Settlement Initiator process to immediately go through the processing it would at midnight when the midnight cutover timer expires.

Syntax

DELIVER PROCESS *object-name,* "MIDNIGHT"

object-name — The symbolic name assigned to the Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

Any action, such as purging the Usage Accumulation File (UAF), that is to take place at midnight using the midnight cutover timer is taken when this command is initiated.

Note: This command does not delete a midnight cutover timer that is currently running. Therefore, if a midnight timer is running, the same action is taken again at midnight.

When the MIDNIGHT command is entered from an XPNET network control facility, the Settlement Initiator process purges the data in the Usage Accumulation File (UAF) if required, clears the usage accumulation fields in the Cardholder Authorization File (CAF), and updates the date in the CUSTOMER PROCESSING DATE field in the BASE24-atm segment of the IDF.

When the Settlement Initiator process receives a MIDNIGHT command, it reads the first record in the IDF and performs the following steps:

1. Determines whether midnight processing is required.
   a. Verifies whether the UAF name is present in the IDF. If so, institution uses the Negative Authorization with Usage Accumulation method.
   b. Checks the FIELD CUTOVER field in the Base segment of the IDF. If that field contains a value of 2 (purge UAF at midnight; begin clearing CAF at midnight) or 5 (do not purge UAF at midnight; begin clearing CAF at midnight) and the date in the CUSTOMER PROCESSING DATE field in the BASE24-atm segment of the IDF is earlier than or equal to the date in the CURRENT BUSINESS DATE field in the

BASE24-atm segment of the IDF, the Settlement Initiator process continues with step 2. Otherwise, the Settlement Initiator process continues with step 4.

2. Updates the CUSTOMER PROCESSING DATE in the IDF. To do this, the Settlement Initiator process performs the following steps:

    a. Computes a new customer business date by checking the WORK DAY CODE field in the Base segment of the IDF and performing according to the following situations:

        1) If the value in the WORK DAY CODE field is 0, the Settlement Initiator process increases the date in the CUSTOMER PROCESSING DATE field in the IDF by one day.

        2) If the value in the WORK DAY CODE field is not 0, the Settlement Initiator process calculates the customer processing date based on the value in the WORK DAY CODE field and the holidays listed in the HOLIDAYS fields in the Base segment of the IDF. For example, if the institution is not set up to process on weekends or holidays, the customer processing date is set to the next non-weekend, non-holiday date.

    b. Moves the newly computed customer processing date to the CUSTOMER PROCESSING DATE field in the BASE24-atm segment of the IDF.

3. Establishes a new withdrawal period if necessary.

    If the date in the CUSTOMER PROCESSING DATE field in the BASE24-atm segment of the IDF is greater than or equal to the date in the NEXT BEGINNING DATE field in the Base segment of the IDF, the institution has reached the end of its current usage accumulation period. In this case, the Settlement Initiator process must compute a new withdrawal period based on the WORK DAY CODE field in the Base segment of the IDF.

    a. If the value in the WORK DAY CODE field is 0, the Settlement Initiator process performs the following steps:

        1) Computes a new next begin date based on the length of the institution’s withdrawal period set in the PERIOD LENGTH field in the Base segment of the IDF.

        2) Compares the new next begin date with the date currently in the NEXT BEGINNING DATE field in the Base segment of the IDF. If these dates are the same, the Settlement Initiator process moves to step 4).

        3) Moves the date in the NEXT BEGINNING DATE field to the BEGINNING DATE field in the Base segment of the IDF.

4) Sets the date in the NEXT BEGINNING DATE field to the new next begin date the Settlement Initiator process calculated earlier.

b. If the value in the WORK DAY CODE field is not 0, the Settlement Initiator process performs the following steps:

1) Moves the date in the NEXT BEGINNING DATE field to the BEGINNING DATE field in the Base segment of the IDF.

2) Computes a new next beginning date using the value in the WORK DAY CODE field, the date in the BEGINNING DATE field, and the holidays the institution established in the HOLIDAYS fields in the Base segment of the IDF.

3) Sets the date in the NEXT BEGINNING DATE field to the new next beginning date the Settlement Initiator process calculated earlier.

4. Determines whether the data in the UAF must be purged for the institution. The Settlement Initiator process performs this step only if the value in the FIELD CUTOVER field in the Base segment of the IDF is 2. If the value in the CUTOVER field is 5, the Settlement Initiator process continues processing with step 5.

To purge data in the UAF, the Settlement Initiator process performs as follows:

a. Checks the UAF FILE NAME field in the Base segment of the IDF to determine whether the institution is using the Negative Authorization with Usage Accumulation method. If the name of a UAF is not located in this field, the institution is not using the Negative Authorization with Usage Accumulation method and no UAF exists. In that case, the Settlement Initiator process continues processing with step 5.

b. Retrieves the file name of the UAF from the UAF FILE NAME field in the Base segment of the IDF if a file name exists.

c. Opens the UAF and purges the data in the file.

5. Broadcasts a cutover (9502) message to the processes specified in the ATM-SETL-MIDNIGHT-NOTIFY params in the LCONF.

   The Settlement Initiator process reads the LCONF at initialization and maintains the midnight broadcast list in memory.

   The cutover message contains a date of zeros. The Settlement Initiator process uses this message to notify each authorization process to reinitialize internal memory (i.e., set new IDF dates).

6. Writes the IDF record back to the file.

RESET DATES Commands

Restricted use operator commands that cause the Settlement Initiator process to reset dates in the BASE24-atm Terminal Data files and Institution Definition File (IDF).

Syntax

DELIVER PROCESS *object-name,* "RESETATDDATES*yymmdd*"

DELIVER PROCESS *object-name,* "RESETIDFDATES*yymmdd*"

DELIVER PROCESS *object-name,* "RESETALLDATES*yymmdd*"

object-name — The symbolic name assigned to the Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

yymmdd — The year, month, and day to which the dates in the BASE24-atm Terminal Data files, IDF, or both are set.

Processing

The processing performed by the RESET DATES commands varies according to each command entered. The following table describes the processing performed for each command.

Command	Processing
RESETATDDATES	Changes the POST-DAT field for all records in the BASE24-atm Terminal Data files to the date in the command.
RESETIDFDATES	Changes the CUR-BUS-DAT field and the CUS-BUS-DAT field in the BASE24-atm segment of the IDF to the date in the command. This command also changes the NXT-BUS-DAT field in the BASE24-atm segment of the IDF to the date following the date in the command and the RPT-BUS-DAT field in the BASE24-atm segment of the IDF to the date prior to the date in the command.

Continued on the following page

Command	Processing
RESETIDFDATES *continued*	Additionally, the RESETIDFDATES command resets the usage accumulation period dates, if the date in the command is not in the current usage accumulation period. In this case, the date in the command is used as the begin date of the usage accumulation period.
RESETIDFDATES	Additionally, the RESETIDFDATES command resets the usage accumulation period dates, if the date in the command is not in the current usage accumulation period. In this case, the date in the command is used as the begin date of the usage accumulation period. If the date in the command is within the current usage accumulation period, the Settlement Initiator process does not recalculate the usage accumulation dates. The fields that are changed if the usage accumulation period is recalculated are the BEG-DAT field and the NXT-BEG-DAT field in the Base segment of the IDF.
RESETALLDATES	Changes all of the dates affected by the RESETATDDATES and RESETIDFDATES commands as noted above.

WARMBOOT Command

A general use operator command that causes the Settlement Initiator process to reinitialize.

Syntax

DELIVER PROCESS *object-name,* “9501”

object-name — The symbolic name assigned to the Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

The WARMBOOT command is the only general use operator command accepted by the Settlement Initiator process.

When the Settlement Initiator process is warmbooted it performs the same processing steps as it does during initialization, except that it closes and reopens all files.

Transaction Context Manager Process Commands

The BASE24-atm Transaction Context Manager (TCM) module can receive a number of BASE24 commands that cause it to perform specialized processing. The commands used by the BASE24-atm Transaction Context Manager module can be separated into three categories: process-generated commands, general use operator commands, and restricted use operator commands. All BASE24 general and restricted use operator commands that can be received by the BASE24-atm TCM module are documented in this subsection. Process-generated commands are documented in appendix B.

All of the commands that can be received by the BASE24-atm Transaction Context Manager module are listed below by category.

Process-Generated Commands	General Use Operator Commands	Restricted Use Operator Commands
CUTOVER	ALTERATTRIBUTE WARMBOOT	TRACE OFF TRACE ON

ALTERATTRIBUTE Command

A general use operator command that causes the Transaction Context Manager process to warmboot (reallocate) a specified shared extended memory table after it has been rebuilt. Whenever a shared extended memory table is modified, all processes that read the table must reallocate the table before the modified data is used in transaction processing. The Process Control Terminal (PCT) Server process automatically sends this command to each process specified in an EMT warmboot notify list param in the Logical Network Configuration File (LCONF) when a network operator enters an ALTERATTRIBUTE command from the EMT Control Command screen (screen 5) in the Device Control Terminal (DCT). You can also enter the ALTERATTRIBUTE command manually from a network control facility to warmboot an extended memory table for an individual process. The EMT warmboot notify list params are only used when issuing the command from the DCT. For more information on the EMT Control Command screen, refer to the ***BASE24 Device Control Manual***.

Syntax

DELIVER PROCESS *object-name,* “ALTERATTR[IBUTE]*file-id*EMT*”*

object-name — The symbolic name assigned to the Transaction Context Manager process in the XPNET node. For more information on entering the object name, refer to section 1.

file-id — The acronym of the file from which the shared extended memory table is built. Valid values are as follows:

STRF = Split Transaction Routing File

Processing

This command causes the Transaction Context Manager process to reallocate a specified shared extended memory table. If the command completes successfully, a message stating that the EMT warmboot completed successfully is displayed on the network log.

Examples

The following are examples of ALTERATTRIBUTE commands sent to a Transaction Context Manager process from an XPNET network control facility. In these examples, the symbolic name of the Transaction Context Manager process is P1A^TCB1 and the shared extended memory table to be warmbooted is the Split Transaction Routing File (STRF) extended memory table.

```
DELIVER PROCESS P1A^TCB1, “ALTERATTRIBUTESTRFEMT”
DELIVER PROCESS P1A^TCB1, “ALTERATTRSTRFEMT”
```

TRACE OFF Command

A restricted use operator command that deactivates the trace function within a Transaction Context Manager process.

Syntax

DELIVER PROCESS *object-name,* "TRACE OFF"

object-name — The symbolic name assigned to the Transaction Context Manager process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing:

An operator sends this command from an XPNET network control facility. For more information on the TRACE OFF command, refer to appendix A.

TRACE ON Command

A restricted use operator command that activates a trace function within a Transaction Context Manager process.

Syntax

DELIVER PROCESS *object-name,* "TRACE ON"

object-name — The symbolic name assigned to the Transaction Context Manager process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

An operator sends this command from an XPNET network control facility. For more information on the TRACE ON command, refer to appendix A.

WARMBOOT Command

A general use operator command that causes the Transaction Context Manager process to reinitialize.

Syntax

DELIVER PROCESS *object-name,* “9501****”

object-name — The symbolic name assigned to the Transaction Context Manager process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

Upon receiving this command, the Transaction Context Manager process opens, reads, and closes all files it read during initialization. Any file information retained in extended memory at initialization is written over in extended memory when the particular files are reread during the warmboot. For definitions-based mobile top-up, this also includes the Mobile Operator File (MOF).

When the warmboot processing is complete, an event message is displayed indicating the warmboot is complete.

An operator sends this command from an XPNET network control facility

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4: BASE24-billpay Process Commands

This section contains the general and restricted operator use commands received by the following BASE24-billpay processes:

- Remittance Generator process
- Scheduled Transaction Initiator process

The processes are presented in alphabetical order. Furthermore, the specific commands are documented in alphabetical order within the processes. The documentation for each command includes the following information:

- A description of the command
- The syntax of the command
- The processing steps taken when the command is received
- An example of the command (when necessary)

Remittance Generator Process Commands

This section describes the processing performed by the Remittance Generator process. The Remittance Generator process identifies successfully completed BASE24-billpay payment transactions in the History Table (HIST) and writes them to vendor remittance output files. A transaction must reside in the HIST for a specified period of time before it is written to a vendor remittance output file. This allows time for an approved transaction entered at a customer service representative (CSR) terminal to be canceled before the vendor remittance is generated. Separate remittance output files are created based on the business date and remittance method.

The Remittance Generator process can receive a number of BASE24 commands that cause it to perform specialized processing. These commands can be separated into three categories: process-generated commands, general use operator commands, and restricted use operator commands. All BASE24 general and restricted use operator commands that can be received by the Remittance Generator process are documented in this subsection. Process-generated commands are documented in appendix B.

All of the commands that can be received by the Scheduled Transaction Initiator process are listed below by category.

Process-Generated Commands	General Use Operator Commands	Restricted Use Operator Commands
None	ACTIVATE HELP STATUS SUSPEND	None

ACTIVATE Command

A general use operator command that instructs the Remittance Generator process to resume processing after processing has been suspended using a SUSPEND command. Upon entry of the ACTIVATE command, remittance processing is taken out of the suspended state and restarted. The SUSPEND command is described later in this section.

Syntax

DELIVER PROCESS *object-name,* “ACTIVATE”

object-name — The symbolic name assigned to the Remittance Generator process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

The Remittance Generator process performs the following processing steps when it receives an ACTIVATE command:

1. Removes processing from a suspended state and places it into a active state (i.e., sets the state to a value of ACTIVE).
2. Generates event message 0012 to indicate that the process is activated. This message is described in the TPBPRMTG log message documentation file.
3. Resumes processing at the point where processing was suspended.

HELP Command

A general use operator command that causes the Remittance Generator process to send help information on its currently supported commands to the EMS collector process.

Syntax

DELIVER PROCESS *object-name,* “HELP”

object-name — The symbolic name assigned to the Remittance Generator process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

When the Remittance Generator process receives the HELP command, it generates event message 0013, which identifies the currently supported commands. This message is described in the TPBPRMTG log message documentation file.

STATUS Command

A general use operator command that causes the Remittance Generator process to send miscellaneous information on its current processing parameters and state to the EMS collector process.

Syntax

DELIVER PROCESS *object-name,* "STATUS"

object-name — The symbolic name assigned to the Remittance Generator process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

When the Remittance Generator process receives the STATUS command, it generates an event message that displays the current settings of all its LCONF assigns and params, the current sleep interval, and the current state of the process. The Remittance Generator process performs the following processing steps when it receives a STATUS command:

1. Reads the command.
2. Searches its current processing parameters for information including the Remittance Generator process's current processing values, current sleep interval, and internal state.
3. Sends the above information in event message 0016 to the EMS collector process. This message is described in the TPBPRMTG log message documentation file.

SUSPEND Command

A general use operator command that instructs the Remittance Generator process to finish processing the transaction it is currently processing and then cease processing, but remain in a suspended state. This means the process actually stops, but the current information is saved so that when the process is resumed, it can begin processing where it left off. The SUSPEND command is used in conjunction with the ACTIVATE command. Remittance processing can remain suspended until an operator enters an ACTIVATE command. When an ACTIVATE command is issued, remittance processing continues. The ACTIVATE command is described earlier in this section.

Syntax

DELIVER PROCESS *object-name,* “SUSPEND”

object-name — The symbolic name assigned to the Remittance Generator process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

When the Remittance Generator process receives the SUSPEND command, it finishes processing the transaction it is currently processing, sets its state to a value of SUSPENDED, and generates event message 0015 to indicate that remittance processing has been suspended. This message is described in the TPBPRMTG log message documentation file.

Remittance processing can be restarted at the point at which it was suspended by entering the ACTIVATE command.

Scheduled Transaction Initiator Process Commands

This section describes the processing performed by the BASE24-billpay Scheduled Transaction Initiator process. The Scheduled Transaction Initiator process initiates payments and transfers that were previously scheduled.

The Scheduled Transaction Initiator process can receive a number of BASE24 commands that cause it to perform specialized types of activities. These commands can be separated into three categories: process-generated commands, general use operator commands, and restricted use operator commands. All BASE24 general and restricted use operator commands that can be received by the Scheduled Transaction Initiator process are documented in this subsection. Process-generated commands are documented in appendix B.

All of the commands that can be received by the Scheduled Transaction Initiator process are listed below by category.

Process-Generated Commands	General Use Operator Commands	Restricted Use Operator Commands
None	ACTIVATE HELP STATUS SUSPEND	None

ACTIVATE Command

A general use operator command that instructs the Scheduled Transaction Initiator process to resume processing after processing has been suspended using a SUSPEND command. Upon entry of the ACTIVATE command, scheduled transaction processing is taken out of the suspended state and restarted. The SUSPEND command is described later in this section.

Syntax

DELIVER PROCESS *object-name,* "ACTIVATE"

object-name — The symbolic name assigned to the Scheduled Transaction Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

The Scheduled Transaction Initiator process performs the following processing steps when it receives an ACTIVATE command:

1. Removes processing from a suspended state and places it into a active state (i.e., it sets a suspended flag to a value of FALSE).
2. Generates event message 0014 to indicate that the process is activated. This event message is described in the TPBPSTI log message documentation file.
3. Resumes processing at the point where processing was suspended.

HELP Command

A general use operator command that causes the Scheduled Transaction Initiator process to send help information on its currently supported commands to the EMS collector process.

Syntax

DELIVER PROCESS *object-name,* “HELP”

object-name — The symbolic name assigned to the Scheduled Transaction Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

When the Scheduled Transaction Initiator process receives the HELP command, it generates a series of event messages (0015, 0017–0020) that identify the currently supported commands and briefly describe the processing performed by each command. These event messages are described in the TPBPSTI log message documentation file.

STATUS Command

A general use operator command that causes the Scheduled Transaction Initiator process to send miscellaneous information on its current processing parameters and state to the EMS collector process.

Syntax

DELIVER PROCESS *object-name,* “STATUS”

object-name — The symbolic name assigned to the Scheduled Transaction Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

When the Scheduled Transaction Initiator process receives the STATUS command, it generates a series of event messages that display the current settings of all its LCONF assigns and params, the current internal state of the process, and values indicating whether the process is currently suspended or stopped. If the process has an authorization request pending when this command is received, it also returns identification information for the Future Table (FUTR) row with the pending authorization. The Scheduled Transaction Initiator process performs the following processing steps when it receives a STATUS command:

1. Reads the command.
2. Searches its current processing parameters for information including the Scheduled Transaction Initiator process’s current processing values, internal state, and FUTR row information for any transaction with a pending authorization request.
3. Sends the above information in a series of event messages (0021–0040) to the EMS collector process. These event messages are described in the TPBPSTI log message documentation file.

SUSPEND Command

A general use operator command that instructs the Scheduled Transaction Initiator process to finish processing the transaction it is currently processing and then cease processing, but remain in a suspended state. This means the process actually stops, but the current information is saved so that when the process is resumed, it can begin where it left off. The SUSPEND command is used in conjunction with the ACTIVATE command. Scheduled transaction processing can remain suspended until an operator enters an ACTIVATE command. When an ACTIVATE command is issued, scheduled transaction processing continues. The ACTIVATE command is described earlier in this section.

Syntax

DELIVER PROCESS *object-name,* “SUSPEND”

object-name — The symbolic name assigned to the Scheduled Transaction Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

When the Scheduled Transaction Initiator process receives the SUSPEND command, it finishes processing the transaction it is currently processing, sets a suspended flag to a value of TRUE, and generates an event message indicating that scheduled transaction processing has been suspended. If the process is already suspended when this command is received, it generates event message 0044 to indicate that the process is already suspended. This event message is described in the TPBPSTI log message documentation file.

Scheduled transaction processing can be restarted at the point at which it was suspended by entering the ACTIVATE command.

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5: BASE24-pos Process Commands

This section contains the general operator use and restricted operator use commands received by the following BASE24-pos modules or processes:

- Device Handler
- Router/Authorization
- Main and Secondary Settlement Initiator
- Transaction Context Manager

The processes are presented in alphabetical order. Furthermore, specific commands are documented in alphabetical order within the command categories. The documentation for each command includes the following information:

- A description of the command
- The syntax of the command
- The processing steps taken when the command is received
- An example of the command (when necessary)

Note: Operators are required to enter the commands described in this section using uppercase letters. In most cases, lowercase letters are not acceptable.

Device Handler Module Commands

The BASE24-pos Device Handler module can receive a number of BASE24 commands that causes it to perform specialized processing. The commands used by the BASE24-pos Device Handler module can be separated into three categories: process-generated commands, general use operator commands, and restricted use operator commands. All BASE24 general and restricted use operator commands that can be received by the BASE24-pos Device Handler module are documented in this subsection. Process-generated commands are documented in appendix B.

Note: In a typical BASE24-pos environment, the BASE24-pos Device Handler module is bound together with a Router/Authorization module to create the Device Handler/Router/Authorization process. Several different types of Device Handler modules are available for use with BASE24-pos. This subsection describes the commands used only by the specific Device Handler modules. For information on the commands used by Router/Authorization modules, refer to the BASE24-pos Router/Authorization discussion in this section.

All of the commands that can be received by one or more BASE24-pos Device Handler modules are listed below by category. A table listing the commands that can be used by each type of Device Handler module is shown on the following page.

Process-Generated Commands	**General Use Operator Commands**	**Restricted Use Operator Commands**
None	ALTERATTRIBUTE CLOSEBATCH CLOSEFLAGON CLOSEFLAGOFF DEACTIVATE FLAG FACILITY KEYSTATUS LOAD LOADFLAGOFF LOADFLAGON LOADKEY LOADWK PERFORMANCE READPTDF STATUS STATUSEXTMEM SYSTEM TERM TESTFLAGON TESTFLAGOFF TOTALS WARMBOOT ALL WARMBOOT PTD	COMMKEY RESETSTATUS TRACEOFF TRACEON TRACE1 TRACE2 TRACE3 TRACE4

The following table lists the text commands supported by BASE24-pos Device Handler modules. Commands are listed down the lefthand side of the table. BASE24-pos Device Handler modules are listed across the top of the table. If a check mark is present where a specific command intersects with a specific Device Handler module, that Device Handler module supports the command. The absence of the check mark at the intersection indicates the Device Handler module does not support the command. A check mark with an asterisk (✓*) indicates that the command is not yet supported by the Device Handler module. Device Handler modules are identified in the table as follows:

VISA I = Visa I Device Handler module
SPDH = Standard POS Device Handler module
HPDH = Hypercom Device Handler module
MHI = Merchant Host Interface module
NCRNDP = NCR/NDP Device Handler module

BASE24-pos Device Handler Module Text Commands

Command	BASE24-pos Device Handler Module				
	VISA I	SPDH	HPDH	MHI	NCRNDP
9501 (WARMBOOT)	✓	✓	✓	✓	✓
9525 (DEACTIVATE FLAG)	✓	✓	✓	✓	✓
9529 (WARMBOOT PTD)		✓	✓	✓	
9549 TESTFLAGON			✓		
9549 TESTFLAGOFF			✓		
ALTERATTRIBUTE		✓	✓	✓	
CLOSEBATCH			✓		
CLOSEFLAGOFF			✓		
CLOSEFLAGON			✓		
COMMKEY			✓		
FACILITY					✓

BASE24-pos Device Handler Module Text Commands					
Command	**BASE24-pos Device Handler Module**				
	VISA I	**SPDH**	**HPDH**	**MHI**	**NCRNDP**
KEYSTATUS					✓
LOAD		✓*		✓*	
LOADFLAGOFF		✓	✓	✓	
LOADFLAGON		✓	✓	✓	
LOADKEY		✓*		✓*	
LOADWK					✓
PERFORMANCE				✓	
READPTDF				✓	
RESETSTATUS	✓	✓	✓	✓	✓
STATUS				✓	
STATUSEXTMEM		✓	✓	✓	
SYSTEM					✓
TERM					✓
TOTALS					✓
TRACE1		✓		✓	
TRACE2		✓		✓	
TRACE3		✓		✓	
TRACE4		✓		✓	
TRACEOFF		✓		✓	
TRACEON		✓		✓	

ALTERATTRIBUTE Command

A general use operator command that causes the Device Handler/Router/Authorization process to warmboot (reallocate) a specified shared extended memory table after it has been rebuilt. This command is used by SPDH, Merchant Host Interface, or Hypercom Device Handler modules. Whenever a shared extended memory table is modified, all processes that read the table must reallocate the table before the modified data is used in transaction processing. The Process Control Terminal (PCT) Server process automatically sends this command to each process specified in an EMT warmboot notify list param in the Logical Network Configuration File (LCONF) when a network operator enters an ALTERATTRIBUTE command from the EMT Control Command screen (screen 5) in the Device Control Terminal (DCT). You can also enter the ALTERATTRIBUTE command manually from a network control facility to warmboot an extended memory table for an individual process. The EMT warmboot notify list params are only used when issuing the command from the DCT. For more information on the EMT Control Command screen, refer to the ***BASE24 Device Control Manual***.

Syntax

DELIVER PROCESS *object-name,* "ALTERATTR[IBUTE]*file-id*EMT"

object-name — The symbolic name assigned to the Device Handler/Router/Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

file-id — The acronym of the file from which the shared extended memory table is built. Valid values are as follows:

APCF = Acquirer Processing Code File
RCDF = Response Code Description File

Processing

This command causes the Device Handler/Router/Authorization process to reallocate a specified shared extended memory table. If the command completes successfully, a message stating that the EMT warmboot completed successfully is displayed on the network log.

Examples

The following are examples of ALTERATTRIBUTE commands sent to a Device Handler/Router/Authorization process from an XPNET network control facility. In these examples, the symbolic name of the Device Handler/Router/Authorization process is P1A^RTAU1 and the shared extended memory table to be warmbooted is the Acquirer Processing Code File (APCF) extended memory table.

DELIVER PROCESS P1A^RTAU1, “ALTERATTRIBUTEAPCFEMT”

DELIVER PROCESS P1A^RTAU1, “ALTERATTRAPCFEMT”

COMMKEY Command

A restricted use operator command that causes the BASE24-pos Terminal Data files to be updated with a new PIN key. The COMMKEY command is supported by the Hypercom Device Handler (HPDH) module only.

Syntax

DELIVER PROCESS *object-name,* “9500 *terminal-name* COMMKEY *commkey”*

object-name — The symbolic name assigned to the Device Handler/Router/Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the POS device in the BASE24-pos Terminal Data files.

commkey — Identifies the communication key to be placed into the terminal record. The communication key can either be in the clear (software encryption) or encrypted (hardware encryption).

Processing

The HPDH performs the following processing steps when it receives a COMMKEY command from an XPNET network control facility:

1. Checks the BASE24-pos Terminal Data files and obtains PIN information.
2. Places the specified communications key into the specified terminal record.
3. Generates an event message indicating the processing results. For detailed descriptions of event messages generated by the HPDH module, refer to the PSHYPRCM log message documentation file.

For more information on PINs, refer to the ***BASE24 Transaction Security Manual***.

Example

The following is an example of the COMMKEY command. In the example, the Device Handler/Router/Authorization process name is P1A^HPDH11, the terminal name is PSTERM3, and the communication key being placed into the BASE24-pos Terminal Data files record is 1112223338880000.

DELIVER PROCESS P1A^HPDH11, “9500 PSTERM3 COMMKEY 1112223338880000”

CLOSE Commands

General use operator commands that cause the Device Handler module to request a batch close transaction. These commands are used only with Hypercom Device Handler modules.

Syntax

DELIVER PROCESS *object-name,* "9500 *terminal-name command-text*"

object-name — The symbolic name assigned to the Device Handler/Router/Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the POS device in the BASE24-pos Terminal Data files.

command-text — Directs the Device Handler module to perform some action relevant to the POS devices whose symbolic station name follows in the command.

Processing

The HPDH performs the following processing when it receives a CLOSE command. There are three versions of the CLOSE command described below.

9500 TERM1 CLOSEFLAGON — A batch close flag is set in all responses to terminal TERM1 to indicate that a batch close transaction is required. This function is supported only by Hypercom T4, T5, T6, and T7 terminals. A flag is set in all responses to the terminal until it is cleared by a settlement transaction or by a CLOSEFLAGOFF command.

9500 TERM1 CLOSEFLAGOFF — A batch close flag is deactivated in all responses to terminal TERM1 to indicate that a batch close transaction is no longer required. This function is supported only by Hypercom T4, T5, T6, and T7 terminals.

9500 TERM2 CLOSEBATCH — A close batch transaction is required for terminal TERM2, but has not been received. The close batch transaction is manually initiated from an XPNET network control facility to prevent an out-of-balance condition from occurring.

DEACTIVATE FLAG Command

A restricted use operator command that sets the status of the terminal in the BASE24-pos Terminal Data files to activated or deactivated. This command can be used to deactivate a single terminal, a range of terminals, or all terminals.

Syntax

DELIVER PROCESS *object-name,* "9525 *terminal-name code"*

DELIVER PROCESS *object-name*, "9525 *low-term-id*/*high-term-id code"*

DELIVER PROCESS *object-name*, "9525 ALL *code"*

object-name — The symbolic name assigned to the Device Handler/Router/Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the POS device in the BASE24-pos Terminal Data files.

low-term-id — The first BASE24-pos Terminal Data files record in a range of records to be activated or deactivated. The Device Handler module activates or deactivates each BASE24-pos Terminal Data files record beginning with the low terminal ID and continuing until it reaches the high terminal ID or the end of the file.

high-term-id — The last BASE24-pos Terminal Data files record in a range of records to be activated or deactivated. The Device Handler module activates or deactivates each BASE24-pos Terminal Data files record until the high terminal ID or the end of the file is reached.

code — A code specifying whether the deactivation flag for the specified terminal is on, or off. The valid values are as follows:

0 = Activate the terminal. This means the terminal is able to perform transactions.
1 = Deactivate the terminal. This means the terminal is not be able to perform transactions.

Processing

The Device Handler module performs the following processing when it receives a DEACTIVATE FLAG command. If the terminal is deactivated, the Device Handler module declines transactions originated by that terminal. The default (initialized value) is deactivated. If the deactivate flag in the command message is set to 1 (deactivated) and all totals (batch, shift, day, and network) are zero, the terminal is deactivated. If any totals are not zero, the terminal is identified as pending deactivation and only balancing transactions are accepted from the terminal. If the deactivate flag in the command message is not set to 1, the terminal is activated.

FACILITY Command

A general use operator command used by the NCR NDP Device Handler module to identify the retail facility where the card was accepted (the card acceptor) or BASE24 (for unsolicited requests). This command is only used by the NCR NDP Device Handler module.

Syntax

DELIVER PROCESS *object-name,* "9500 *terminal-id* FACILITY f*acility-id"*

object-name — The symbolic name assigned to the Device Handler/Router/Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-id — The terminal ID defined for the POS device in the POS Terminal Data File.

facility-id — Identifies the facility ID associated with the specified terminal. The facility ID is carried in the EFT header type 2000 for the specified terminal.

Processing

The NCR NDP Device Handler module performs the following steps when it receives a FACILITY command:

1. Reads the FACILITY command.
2. Sets the facility ID for the specified terminal in the device dependent data contained in the EFT header type 2000 of the NCR external message.
3. Generates an event message indicating the processing results. For detailed descriptions of event messages generated by the NCR NDP Device Handler module, refer to the PSNCRNDP log message documentation file.

KEYSTATUS Command

A general use operator command that allows an operator to request the status of the Key Exchange Key (KEK), Working Key (WK), or both from the terminal.

An operator initiates a key status request message from an XPNET network control facility by entering the KEYSTATUS command. The NCR NDP Device Handler module receives the message, formats the 0800 key status request, sets a timer, and sends the key status request to the NCR 2127 terminal.

The NCR 2127 terminal receives the key status request, formats a 0800 key status response, and sends the response to the NCR NDP Device Handler module. The NCR NDP Device Handler module receives the response, logs an event message indicating the key status, and deletes the network management timer set earlier.

If the network management timer expires before the terminal responds to the 0800 message, the NCR NDP Device Handler module verifies that the timeout is for the previous transaction by checking the sequence number. If there is an error at this point, the Device Handler module logs an event message. If there are no errors, the Device Handler module logs an event message indicating the 0800 message timed out. The NCR terminal sends the late key status response to the NCR NDP Device Handler module. The NCR NDP determines the response is late, logs an event message indicating this condition, and drops the response.

The KEYSTATUS command is only supported by NCR 2127 terminals and the NCR NDP Device Handler module.

Syntax

DELIVER PROCESS *object-name,* “9500 *terminal-id* KEYSTATUS *station* [*key-type*]”

object-name — The symbolic name assigned to the Device Handler/Router/Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-id — The terminal ID defined for the POS device in the POS Terminal Data File.

station — The symbolic name of the station associated with the terminal in the XPNET node.

[*key-type*] — The type of key whose status is being requested. Valid values are as follows:

01 = Key Exchange Key (KEK)
02 = Working Key (WK)
03 = KEK and WK

If the *key-type* is not present in the command, this field defaults to 03 (KEK and WK).

Processing

The NCR NDP Device Handler module performs the following steps when it receives a KEYSTATUS command:

1. Reads the KEYSTATUS command.
2. Determines if the terminal is an NCR 2127. If it is, the Device Handler module moves to the next step. If it is not, the Device Handler module stops processing the command and logs an event message.
3. Checks the key type and returns the key status for the specified terminal.
4. Generates an event message indicating the processing result. For detailed descriptions of event messages generated by the NCR NDP Device Handler module, refer to the PSNCRNDP log message documentation file.

LOADFLAGON and LOADFLAGOFF Commands

A general use operator command that directs the Device Handler module to initiate a downline load containing some form of configuration information to the terminal.

Syntax

DELIVER PROCESS *object-name,* "9500 *terminal-name command-text optional-data"*

object-name — The symbolic name assigned to the Device Handler/Router/Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal ID defined for the POS device in the BASE24-pos Terminal Data files.

command-text — Directs the Device Handler module to perform some action relevant to the type of POS device specified in the command.

optional-data — Varies in content and format by Device Handler module.

Processing

The information is taken from a configuration file specific to the type of POS device, or can be generated within the Device Handler module (e.g., communication keys, configuration parameters, etc.). These commands take varied forms depending on the Device Handler. For example:

9500 TERM1 LOADFLAGON — Downline load information is waiting for terminal TERM1. Upon receipt of the LOADFLAGON command, the SPDH module sets the REQUEST-LOAD flag in the device-dependent area of the BASE24-pos Terminal Data files to on. When the next response is being formatted to the terminal, the SPDH module sets the PROCESSING FLAG 2 field in the standard message header to indicate that a downline load is waiting.

9500 TERM1 LOADFLAGOFF — Downline load information is not waiting for terminal TERM1. The LOADFLAGOFF command sets the REQUEST-LOAD flag in the BASE24-pos Terminal Data files to off. When the next response is being formatted to the terminal, the SPDH module sets the PROCESSING 2 field in the standard message header to indicate that a downline load is not waiting.

LOADWK Command

A general use operator command that directs the NCR NDP Device Handler module to initiate a downline load of the working key to the NCR terminal. This command is only supported by the NCR NDP Device Handler module.

Syntax

DELIVER PROCESS *object-name,* "*terminal-id* LOADWK *station-id*"

object-name — The symbolic name assigned to the Device Handler/Router/Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-id — The terminal ID defined for the POS device in the POS Terminal Data File.

station-id — The station ID associated with the terminal ID for which the working key is being downline loaded. The station is defined in the XPNET node.

Processing

From an XPNET network control facility, the operator initiates a load key message requesting the downline load of the working key to the terminal. Upon entry of the LOADWK command, the NCR NDP Device Handler module formats an 0800 network management request message, sets a key management timer, and sends the request to the NCR terminal. The 0800 message contains the working key value taken from the device-dependent data in the POS Terminal Data File.

The NCR terminal receives the request and sends the NCR NDP Device Handler module an 0810 network management response message indicating whether the working key was downline loaded successfully. If the downline load was successful, the terminal uses the new working key and logs an event message indicating this condition. If the downline load was unsuccessful, the terminal continues to use the old working key and logs an event message indicating this condition.

If the key management timer expires before the terminal can respond to the 0800 message, the NCR NDP Device Handler module verifies that the timeout is for the previous transaction by checking the sequence number. If an error occurs at this point, the Device Handler module logs an event message. If no errors occur, the Device Handler module logs an event message indicating that the 0800 message timed out, sets the transaction status to "TRANSACTION IN PROGRESS", continues to use the old working key, and updates the POS Terminal Data File with the transaction in progress status.

PERFORMANCE Command

A general use operator command that notifies the Merchant Host Interface (MHI) process to send miscellaneous information on request processing activity for a specific DPC or all DPCs. This command is used only by the Merchant Host Interface (MHI) process.

Syntax

DELIVER PROCESS *object-name,* “9500 PERFORMANCE *dpc-number”*

object-name — The symbolic name assigned to the MHI process in the XPNET node. For more information on entering the object name, refer to section 1.

dpc-number — The number of the DPC whose processing activity is being requested. DPC numbers are established in the Merchant Host Configuration File (MHCF). If ALL is entered in this field, the performance statistics for all DPCs associated with the MHI process are displayed.

Processing

When the MHI process receives a PERFORMANCE command, it performs the following steps:

1. Searches its processing activity parameters for performance information including the total number of requests that have been received in the current processing period, the total number of requests that timed out in the current processing period, and the total response time. The following event message is generated when a PERFORMANCE command is received:

   ```
   DPC NUMBER:         MINUTES IN PERIOD:
   RQSTS    TIMED OUT    PERCENT    RESP TIME
   ```

   The DPC NUMBER field contains the DPC number taken from the Merchant Host Interface Configuration File (MHCF). This field identifies the DPC for which the performance data is reported. The MINUTES IN PERIOD field identifies the number of minutes in a processing period and is also taken from the MHCF.

   The RQSTS field indicates the number of financial requests or force posts that have been received during the specified processing period. The TIMED OUT field indicates the number of requests that have timed out during the specified processing period. The PERCENT field indicates the percentage of the total number of requests received that timed out. The RESP TIME field indicates the average response time from the authorizing entity per request. A time out is considered to have a response time equal to the value of the authorization request limit taken from the associated Merchant Host Interface Terminal Configuration File (MHCF) record for the originating terminal.

2. Sends the processing parameters to the EMS collector process.

Example

The following is an example of a PERFORMANCE command. The symbolic name of the MHI process is P1A^MHIDH1. In the example, performance statistics for all DPCs associated with the MHI process are being requested.

```
DELIVER PROCESS P1A^MHIDH1, "9500 PERFORMANCE ALL"
```

READPTDF Command

A general use operator command used to read a particular record held in memory. This command is only used with the Merchant Host Interface (MHI) process when the customer has requested the host control mode. It cannot be used if a customer is using terminal control mode.

Syntax

DELIVER PROCESS *object-name,* “9500 *terminal-id* READPTDF”

object-name — The symbolic name assigned to the MHI process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-id — The terminal ID whose associated POS Terminal Data File record is being read.

Processing

When the MHI process receives a READPTDF command, it performs the following steps:

1. Reads the command.
2. Determines if host control mode is being used. If it is, then the process continues with step 3. If terminal control mode is being used, the READPTDF command cannot be used.
3. Reads the POS Terminal Data File record for the terminal ID indicated in the command.

RESETSTATUS Command

A restricted use operator command that causes the Device Handler module to reset the transaction status to “no transaction in progress” in theBASE24-pos Terminal Data files. This command is used in cases where a process abends or another unforeseen event causes the transaction status to be corrupted in the BASE24-pos Terminal Data files.

Syntax

DELIVER PROCESS *object-name,* “9500 *terminal-name* RESETSTATUS”

object-name — The symbolic name assigned to the Device Handler/Router/Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-name — The terminal name of POS device as defined in the XPNET node.

Processing

The Device Handler module resets the transaction status in the TRAN-STATUS field of the BASE24-pos Terminal Data files to NO-TRAN-IN-PROGRESS. This command is intended for use if a transaction abends or some other uncontrollable event leaves the transaction status corrupted in the BASE24-pos Terminal Data files.

STATUS Command

A general use operator command notifying the specified process to send status information on its current processing parameters in event messages to its designated logging destination. This command is used only by the Merchant Host Interface (MHI) process.

Syntax

DELIVER PROCESS *object-name,* "9500 STATUS"

object-name — The symbolic name assigned to the MHI process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

When the MHI process receives a STATUS command, it performs the following steps:

1. Reads the command.
2. Searches its current processing parameters for information including a report on the DPCs associated with the MHI. The following event message is generated when a STATUS command is issued.

```
MHI STATUS SUMMARY -- BASE24 RELEASE 5.0 YY/MM/DD HH:MM:SS
REQUESTS -- PENDING:      ALLOWED:
```

Pending requests indicate the number of current outstanding requests. Allowed requests indicate the maximum number of outstanding requests allowed based on the POS-MHI-MAX-REQUESTS param in the Logical Network Configuration File (LCONF).

```
***IDX***NUM***STAT***INT***STA***SUP***SUS
```

The IDX field identifies the index for the indicated DPC in the MHI DPC table. The NUM field identifies the DPC number taken from the Merchant Host Interface Configuration File (MHCF). The STAT field indicates the status of the DPC (i.e., whether the DPC is UP or DOWN).

The INT field indicates the Store-and-Forward (SAF) method used by the MHI process taken from the MHCF. The valid values for this field are as follows:

0 = Interspersed from the DPC
1 = SAF sent from the DPC before a real time transaction.

The STA field identifies the number of stations defined for the DPC. The SUP field identifies the number of stations marked up for the indicated DPC.

The SUS field indicates the number of outstanding requests for the indicated DPC that are waiting for authorization.

3. Sends the processing parameters to the EMS collector process.

STATUSEXTMEM Command

A general use operator command that causes a SPDH or Hypercom Device Handler module to display statistics about its BASE24-pos Terminal Data files (PTD) extended memory segment on the network log. This command allows network operators to monitor extended memory usage and determine whether adjustments are needed.

Syntax

DELIVER PROCESS *object-name,* “STATUSEXTMEM*”*

object-name — The symbolic name assigned to the Device Handler/Router/Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

This command causes a SPDH or Hypercom Device Handler process to generate several event messages detailing its current PTD extended memory segment usage. Information in these messages includes the following:

- The current number of terminals in the Terminal Table.
- The current number of terminals in the Dynamic Data Table.
- The maximum number of terminals to be managed in the Terminal Table.
- The maximum number of terminals allowed in the Dynamic Data Table, including the last message data. More than one dynamic data record can exist for a terminal.
- The maximum number of terminals allowed in the Static Data Table. More than one static data record can exist for a terminal.
- The length of each record in the Dynamic Data Table.
- The length of each record in the Static Data Table.
- The length of each record in the last message data.

SYSTEM Command

A general use operator command used by the NCR NDP Device Handler module to set the system ID associated with the specified terminal in the device dependent data contained in the EFT header type 2000 of the NCR external message.

Syntax

DELIVER PROCESS *object-name,* "9500 *terminal-id* SYSTEM *system-id*"

object-name — The symbolic name assigned to the Device Handler/Router/Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-id — Identifies the terminal for which the system ID is being set. The system ID is carried in the EFT header type 2000 for the specified terminal.

system-id — Identifies the system ID associated with the specified terminal. The system ID is carried in the EFT header type 2000 for the specified terminal.

Processing

The system ID is a 4-character code required to format the EFT header type 2000 and is only used by the NCR NDP Device Handler module.

TERM Command

A general use operator command used by the NCR NDP Device Handler module to set the terminal ID for the specified terminal in the device dependent data contained in the EFT header type 2000 of the NCR external message. This command is only used by the NCR NDP Device Handler.

Syntax

DELIVER PROCESS *object-name,* "9500 *terminal-id-1* TERM *terminal-id-2*"

object-name — The symbolic name assigned to the Device Handler/Router/ Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-id-1 — Identifies the terminal for which the terminal ID is being set. The terminal ID is carried in the EFT header type 2000 for the specified terminal.

terminal-id-2 — Identifies the terminal ID associated with the specified terminal. The terminal ID is carried in the EFT header type 2000 for the specified terminal.

Processing

The terminal ID is a 4-character code required to format the EFT header type 2000 and is used to determine accumulation and accounting totals at the terminal level.

TESTFLAGON and TESTFLAGOFF Commands

General use operator commands directing the Device Handler module to determine if downline load information is waiting for the terminal. These commands are used by the Hypercom Device Handler module when downloading information to VISA II terminals.

Syntax

DELIVER PROCESS *object-name,* "9549TESTFLAGON *terminal-id*"

DELIVER PROCESS *object-name,* "9549TESTFLAGOFF *terminal-id*"

object-name — The symbolic name assigned to the Device Handler/Router/Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-id — The terminal ID defined for the POS device in the BASE24-pos Terminal Data files.

Processing

Upon receipt of the TESTFLAGON command, the HPDH sets the REQUEST-LOAD flag in the device-dependent area of the BASE24-pos Terminal Data files to on. When the next response is being formatted to the terminal, the HPDH sets the PROCESSING FLAG 2 field in the standard message header to indicate that a downline load is waiting.

Upon receipt of the TESTFLAGOFF command, the HPDH sets the REQUEST-LOAD flag in the device-dependent area of the BASE24-pos Terminal Data files to off. When the next response is being formatted to the terminal, the HPDH sets the PROCESSING FLAG 2 field in the standard message header to indicate that a downline load is not waiting.

TOTALS Command

A general use operator command used by the NCR NDP Device Handler module to generate totals for the specified terminal ID.

Syntax

DELIVER PROCESS *object-name,* "9500 *terminal-id* TOTALS *station-id*"

object-name — The symbolic name assigned to the Device Handler/Router/Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

terminal-id — Identifies the terminal ID for which totals are being generated. The totals are carried in the EFT header type 2000 for the specified terminal ID.

station-id — Identifies the station associated with the specified terminal for which totals are being generated. The station ID is defined in the XPNET node for the specified terminal.

Processing

The totals generated depend on how the External Message File (EMF) is set up for the terminal. At a minimum, terminal totals include the net value of all gross amounts for approved transactions. In addition, terminal totals may include the total dollar amount of all approved 0210 and 0230 transactions that were processed and the total dollar amount of all reversals that were processed for approved 0210 and 0230 transactions.

From an XPNET network control facility, an operator initiates a reconciliation request message by entering the TOTALS command. The NCR NDP Device Handler module receives the TOTALS command and formats a 0500 reconciliation request, sets a timer, and sends the 0500 request to the NCR terminal.

The NCR terminal formats its totals and sends a 0510 reconciliation response to the NCR NDP Device Handler module.

At this point, the NCR NDP Device Handler module receives the 0510 message from the terminal and sends an 0220 administrative request to the Authorization process to determine whether cutover should be performed for the terminal.

The Authorization process sends an 0210 administrative response to the NCR NDP Device Handler module to indicate whether the terminal is within the cutover window.

If the terminal is within the cutover window, cutover is performed. If the terminal is not within the cutover window, cutover is not performed. If cutover is performed, the BASE24 totals are logged by card type (service totals) to the POS Transaction Log File (PTLF). BASE24 batch totals are logged to the PTLF with the BALANCE FLAG field set to either in balance or out-of-balance. Network and NCR terminal totals are also logged to the PTLF at this time. In addition, the posting date in the BASE24-pos Terminal Data files is increased by one.

Note: The next 0210 response sent to the NCR terminal contains the new BASE24-pos Terminal Data files posting date. The new BASE24-pos Terminal Data files posting date causes the NCR terminal to move active totals to inactive and clear the active totals. If the 0210 response contains an approved transaction, this amount is added to the new set of active totals.

If the reconciliation timer expires before the terminal can respond to the 0500 message, the NCR NDP Device Handler module verifies that the timeout is for the previous transaction by checking the sequence number. If an error occurs at this point, the Device Handler module logs an event message. If no errors occur, the Device Handler module logs an event message indicating that the 0500 message timed out. The NCR terminal sends the late reconciliation response to the NCR NDP Device Handler module. The NCR NDP determines that the response is late, logs an event message indicating this condition, and drops the response.

The TOTALS command is only used by the NCR NDP Device Handler module and is not supported by NCR 2126 terminals.

TRACE Commands

Restricted use operator commands that activate or deactivate a trace function within the Device Handler/Router/Authorization process.

Syntax:

DELIVER PROCESS *object-name,* "9500 *term-id* { TRACEON | TRACEOFF | TRACE1 | TRACE2 | TRACE3 | TRACE4 *num* }"

object-name — The symbolic name assigned to the Device Handler/Router/Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

term-id — The terminal ID of the terminal to be traced or an asterisk (*) if all terminals are to be traced.

{ TRACEON | TRACEOFF | TRACE1 | TRACE2 | TRACE3 | TRACE4 *num* } — Specifies whether the trace is activated or deactivated. Valid values are as follows:

TRACEON	=	Turns the trace on for all trace types
TRACEOFF	=	Turns the trace off for all trace types
TRACE1	=	Activates the message trace only
TRACE2	=	Activates the transaction information trace only
TRACE3	=	Activates the key trace only
TRACE4 *num*	=	Activates the response time trace only, where *num* is the number of transactions that are accumulated before the average response time is displayed

Processing

When activated, the trace function generates an event message each time a different type of trace is performed. The SPDH and MHI support all of the TRACE commands.

The trace function is intended for use in trouble-shooting the Device Handler/Router/Authorization process. It can be activated for all processes or for a specific module. The TRACEON command activates the trace function for all of the trace types. To activate the trace function for only one type of trace, use one of the following commands: TRACE1, TRACE2, TRACE3, or TRACE4 *num*. When the trace function is set to on, event messages are sent to the current logging facility.

The TRACE OFF command deactivates all trace functions.

The Device Handler module performs the following processing steps when it receives a TRACE command:

1. Reads the command.
2. Determines whether the TRACE command activates all traces, activates a specific type of trace, or deactivates all traces.
3. Activates or deactivates the trace function within the Device Handler module.

Examples

The following are examples of TRACE commands. The Device Handler/Router/Authorization process for which the commands are entered is P1A^PSVISA2. In the first example, the message trace function is activated (i.e., the TRACE1 command is used) for terminal 37000001. In the second example, all traces are deactivated for all terminals.

DELIVER PROCESS P1A^PSVISA2, “9500 37000001 TRACE1”

DELIVER PROCESS P1A^PSVISA2, “9500 * TRACEOFF”

WARMBOOT ALL Command

A general use operator command that causes the Device Handler/Router/Authorization process to reinitialize.

Syntax

DELIVER PROCESS *object-name,* "9501****"

object-name — The symbolic name assigned to the Device Handler/Router/Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

Upon receiving this command, the Device Handler/Router/Authorization process opens, reads, and closes all files it read during initialization. Any file information retained in extended memory at initialization is written over in extended memory when the particular files are reread during the warmboot.

WARMBOOT PTD Command

A general use operator command that causes a SPDH or Hypercom Device Handler module to warmboot static terminal data held in extended memory. This command should be issued any time an operator performs file maintenance on a BASE24-pos Terminal Data files (PTD) field that is stored in the BASE24-pos Terminal Data - Static File (PTDS1). Static data that is modified in the PTD will not be used in processing until the PTD is warmbooted or the Device Handler module is reinitialized.

Syntax

DELIVER PROCESS *object-name,* “9529PTD”

object-name — The symbolic name assigned to the Device Handler/Router/Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

This command causes a SPDH or Hypercom Device Handler module to re-read its PTDS1 records into extended memory. If the command completes successfully, the following message is displayed on the network log:

```
'TDF Warmboot' command completed.
```

Router/Authorization Module Commands

This section describes the processing performed by the BASE24-pos Router/Authorization modules. The Router/Authorization modules are the authorizing and routing entities in the BASE24-pos system. They authorize transactions according to an institution's authorization parameters and log transactions to a POS Transaction Log File (PTLF), which can then be extracted for affecting settlement immediately without waiting for transaction receipts.

The BASE24-pos Router/Authorization modules can receive a number of BASE24 commands that cause them to perform specialized processing. The commands used by the Router/Authorization modules can be separated into three categories: process-generated commands, general use operator commands, and restricted use operator commands. All BASE24 general and restricted use operator commands that can be received by the Router/Authorization process are documented in this subsection. Process-generated commands are documented in appendix B.

Note: In a typical BASE24-pos acquirer environment, the Router/Authorization modules are bound together with a Device Handler module to create the Device Handler/Router/Authorization process. In a typical BASE24-pos issuer environment, the Router/Authorization modules are bound together to create the Router/Authorization process.

Several different types of Device Handler modules are available for use with BASE24-pos. This subsection describes the commands used only by the Router/Authorization modules. For information on the commands used by specific Device Handler modules, refer to the BASE24-pos Device Handler module discussion provided earlier in this section. All of the commands that can be received by the Router/Authorization process are listed in the following table by category.

Process-Generated Commands	General Use Operator Commands	Restricted Use Operator Commands
LOGICAL NETWORK CUTOVER REFRESH CAF, PBF, NEG REFRESH INR, BNR, INEG, BNEG, or ANEG	ALTERATTRIBUTE INITIALIZE DNT WARMBOOT ALL WARMBOOT AST WARMBOOT CART WARMBOOT FIID WARMBOOT INR, BNR, or CAPF WARMBOOT PRF WARMBOOT PTLF WARMBOOT RDF	ADX DEBUG INA ANA, ANB ROUTERTRACE ROUTERTRACE PAN ROUTERTRACE PREFIX SWA

ADX Command

A restricted use operator command that changes the extent size for the audit file named with the ANA (Name Audit File) command described later. The ADX command originates from an XPNET network control facility.

Syntax

DELIVER PROCESS *object-name,* "ADX *file-name size"*

object-name — The symbolic name assigned to the Device Handler/Router/ Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

file-name — The name of the audit file for which the extent size is being changed.

size — The size to which the extents for the audit file should be changed.

Processing

Upon entry of this command, the file extents for the specified audit file are changed to the specified size. This command must be entered before the audit has started. Refer to the SWA command description for an explanation of starting an audit.

Note: This command is not the same as the system command to change audit file extent sizes. For more information on system commands, refer to the ***NET24-XPNET Network Control Command Reference Manual.***

Example

The following is an example of an ADX command. The Device Handler/Router/ Authorization process in the example is P1A^RTAU1. The file name is $DATA1. AUDIT.AUDITA and the extent size is 150.

DELIVER PROCESS P1A^RTAU1, "ADX $DATA1.AUDIT.AUDITA 150"

ALTERATTRIBUTE Command

A general use operator command that causes the Device Handler/Router/Authorization process to warmboot (reallocate) a specified shared extended memory table after it has been rebuilt. Whenever a shared extended memory table is modified, all processes that read the table must reallocate the table before the modified data is used in transaction processing. The Process Control Terminal (PCT) Server process automatically sends this command to each process specified in an EMT warmboot notify list param in the Logical Network Configuration File (LCONF) when a network operator enters an ALTERATTRIBUTE command from the EMT Control Command screen (screen 5) in the Device Control Terminal (DCT). You can also enter the ALTERATTRIBUTE command manually from a network control facility to warmboot an extended memory table for an individual process. The EMT warmboot notify list params are only used when issuing the command from the DCT. For more information on the EMT Control Command screen, refer to the ***BASE24 Device Control Manual***.

Syntax

DELIVER PROCESS *object-name,* “ALTERATTR[IBUTE]*file-id*EMT*”*

object-name — The symbolic name assigned to the Device Handler/Router/Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

file-id — The acronym of the file from which the shared extended memory table is built. Valid values are as follows:

APCF	=	Acquirer Processing Code File
ERF	=	Exchange Rate File
IPCF	=	Issuer Processing Code File

Processing

This command causes the Device Handler/Router/Authorization process to reallocate a specified shared extended memory table. If the command completes successfully, a message stating that the EMT warmboot completed successfully is displayed on the network log.

Examples

The following are examples of ALTERATTRIBUTE commands sent to a Device Handler/Router/Authorization process from an XPNET network control facility. In these examples, the symbolic name of the Device Handler/Router/Authorization process is P1A^RTAU1 and the shared extended memory table to be warmbooted is the Issuer Processing Code File (IPCF) extended memory table.

DELIVER PROCESS P1A^RTAU1, “ALTERATTRIBUTEIPCFEMT”

DELIVER PROCESS P1A^RTAU1, “ALTERATTRIPCFEMT”

ANA and ANB Commands

Restricted use operator commands that name the audit file to be used during an audit.

Syntax

DELIVER PROCESS *object-name,* “ANA *file-name”*

DELIVER PROCESS *object-name*, “ANB *file-name”*

object-name — The symbolic name assigned to the Device Handler/Router/Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

file-name — The name of the audit file to be used during an audit.

Processing

Upon entry of these commands, the specified audit file is named for the specified Device Handler/Router/Authorization process. One audit file can be named with the ANA command and another audit file can be named with the ANB command.

Note: These commands are not the same as the system command to name audit files. For more information on system commands, refer to the ***NET24-XPNET Network Control Command Reference Manual.***

Example

The following is an example of an ANA command. In the example, the Device Handler/Router/Authorization process is P1A^RTAU1.

DELIVER PROCESS P1A^RTAU1, “ANA $DATA.AUDIT.AUDITA”

DEBUG Command

A restricted use operator command that places the Device Handler/Router/Authorization process into the debug state. The debug message places the process into the HP NonStop Inspect and displays an Inspect prompt on the home terminal (the terminal specified for the process by the HOMETERM attribute or the home terminal of the XPNET process). Inspect is an HP NonStop programming tool. For more information on the DEBUG command, refer to appendix A.

INA Command

A restricted use operator command that displays information about audits.

Syntax

DELIVER PROCESS *object-name,* “INA”

object-name — The symbolic name assigned to the Device Handler/Router/Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

When the Device Handler/Router/Authorization process receives an INA command, the information displayed includes the name of the current audit file, names of other audit files, extent sizes, and percentage of the current audit file that is full.

Note: This command is not the same as the system command to inquire into an audit. For more information on system commands, refer to the ***NET24-XPNET Network Control Command Reference Manual.***

INITIALIZE DNT Command

A general use operator command that causes the Device Handler/Router/Authorization process to load or verify the Diebold number table (DNT) to the Thales e-Security Host Security Module (HSM) or the Atalla Network Security Processor (NSP).

Syntax

DELIVER PROCESS *object-name,* “9531DNT”

object-name — The symbolic name assigned to the Device Handler/Router/Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

This command causes the Device Handler/Router/Authorization process to load or verify the DNT to the hardware module.

With the Atalla NSP, the Device Handler/Router/Authorization process uses the clear key for encrypting the DNT and sends the encrypted version of the key to the NSP along with the encrypted DNT to allow the NSP to decrypt the the DNT.

With the Thales e-Security HSM, the Device Handler/Router/Authorization process calls a utility and passes the DNT from the Key Authorization File (KEYA) to the HSM. If the DNT from the KEYA contains all zeros, the utility asks the HSM to verify that the HSM already has a valid DNT. If the DNT in the KEYA contains values other than zero, the utility loads the KEYA DNT to the HSM.

The DNT is not loaded if the security device type indicates the Transaction Security Services process is being used.

ROUTERTRACE PREFIX Command

A restricted use operator command that activates a trace function for a particular prefix. The indicated prefix can be up to 11 characters in length.

Syntax

DELIVER PROCESS *object-name,* "ROUTERTRACE PREFIX *prefix#"*

object-name — The symbolic name assigned to the Device Handler/Router/Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

prefix# — The prefix number within the Device Handler/Router/Authorization process for which the ROUTETRACE PREFIX command is being entered.

Processing

The Device Handler/Router/Authorization process performs the following processing steps when it receives a ROUTERTRACE PREFIX command:

1. Reads the command.
2. Activates the router trace function within the Device Handler/Router/Authorization process for the specified prefix number.

Example

The following is an example of a command activating the specified router trace prefix number. In the example, the Device Handler/Router/Authorization process is P1A^RTAU1 and the prefix number is 4360.

DELIVER PROCESS P1A^RTAU1, "ROUTERTRACE PREFIX 4360"

ROUTERTRACE PAN Command

A restricted use operator command that activates a trace function for a particular PAN within the Device Handler/Router/Authorization process. The indicated PAN can be up to 19 characters in length.

Syntax

DELIVER PROCESS *object-name,* "ROUTERTRACE PAN *pan#*"

object-name — The symbolic name assigned to the Device Handler/Router/Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

pan# — The Primary Account Number (PAN) within the specified Device Handler/Router/Authorization process for which the ROUTERTRACE PAN command is being entered.

Processing

The Device Handler/Router/Authorization process performs the following processing steps when it receives a ROUTERTRACE PAN command:

1. Reads the command.
2. Activates the router trace function within the Device Handler/Router/Authorization process for the specified PAN.

Example

The following is an example of a command activating the specified router trace PAN. In the example, the Device Handler/Router/Authorization process is P1A^RTAU1 and the PAN is 4420000345000786543.

DELIVER PROCESS P1A^RTAU1, "ROUTERTRACE PAN 4420000345000786543"

ROUTERTRACE Command

A restricted use operator command that activates a trace function within the Authorization module. The Authorization module keeps a record of the POS Standard Internal Message (PSTM) as it passes in and out of the Authorization module and logs this information to an audit file if an audit file has been specified and opened using an ANA or ANB text command.

When the trace function is set to on, event messages are sent to the current audit file. In addition, trace can be activated to include the detailed steps performed by the Router module in the information written to the network log. This is accomplished by entering ROUTERTRACE 1 instead of ROUTERTRACE ON in the command syntax.

The ROUTERTRACE OFF command simply deactivates the trace function within the Authorization module.

Syntax

DELIVER PROCESS *object-name,* "ROUTERTRACE { ON | 1 | OFF }"

object-name — The symbolic name assigned to the Device Handler/Router/Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

{ ON | 1 | OFF } — An identifier specifying whether trace is activated or deactivated. Valid values are as follows:

ON	=	Trace has been activated for the specified process. In this case, 0200 and 0210 PSTM messages are written to the audit file.
1	=	Trace has been activated for the specified process and includes the detailed steps the Router module performs. In this case, 0200 and 0210 PSTM messages, and the Router steps are written to the audit file.
OFF	=	Trace has been deactivated for the specified process.

Processing

The Device Handler/Router/Authorization process performs the following processing steps when it receives a ROUTERTRACE command:

1. The Authorization module reads the command. In addition, the Router module sends the trace on confirmation to the log.
2. The Authorization module activates or deactivates the trace function within the Authorization module.

Examples

The following are examples of ROUTERTRACE commands. The Device Handler/Router/Authorization process for which the command is being entered is P1A^RTAU2. In the first example, the trace is being activated to include in the information it logs to the network log the steps the Router module performs. The second example deactivates the trace function.

```
DELIVER PROCESS P1A^RTAU2, “ROUTERTRACE 1”
DELIVER PROCESS P1A^RTAU2, “ROUTERTRACE OFF”
```

SWA Command

A restricted use operator command that changes logging from the current audit file to the next audit file.

Syntax

DELIVER PROCESS *object-name,* "SWA"

object-name — The symbolic name assigned to the Device Handler/Router/Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

When the Device Handler/Router/Authorization process receives an SWA command, the Authorization module closes the current audit file and opens the next audit file.

The name of the next audit file must be specified using the ANA or ANB command prior to issuing the SWA command. If audit file A has not been named, the current audit file will be closed and audit information stops logging to a file. If the audit file A has been named, the audit A file becomes the current audit file. The audit B file then becomes the new audit file A when the SWA command is issued. The audit B file name is then set to 0 (no file). An ANB command can be issued after the SWA command to name a new audit B file.

Before issuing the SWA command, the audit disk extent size can be changed using the ADX command. If the ADX command is not issued before the SWA command, the next audit file assumes the characteristics of the current audit file.

Note: This command is not the same as the system command to switch audit files. For more information on system commands, refer to the ***NET24-XPNET Network Control Command Reference Manual.***

WARMBOOT ALL Command

A general use operator command that causes the Device Handler/Router/Authorization process to reinitialize.

Syntax

DELIVER PROCESS *object-name,* "9501****"

object-name — The symbolic name assigned to the Device Handler/Router/Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

Upon receiving this command, the Device Handler/Router/Authorization process opens, reads, and closes all files it read during initialization. Any file information retained in extended memory at initialization is written over in extended memory when the particular files are reread during the warmboot.

The Router module accesses the Check Authorization Routing Table File (CART), Card Prefix File (CPF), Authorization Selection Table File (AST), POS Retailer Definition File (PRDF), and SPROUTE Files. The processing performed for the Router module is the same as that for initialization.

The processing performed for the Authorization module is the same as that for initialization with the following exceptions:

- The Authorization module closes all of the files it currently has open prior to starting the reinitialization.
- The Authorization module generates the following event message when it completes its processing: "Warmboot completed for this process."

WARMBOOT AST Command

A general use operator command that causes the Router module to reinitialize the Authorization Selection Table File (AST).

Syntax

DELIVER PROCESS *object-name,* "9527AST"

object-name — The symbolic name assigned to the Device Handler/Router/Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

When the Router module receives a 9527AST command, it reinitializes the AST. This includes opening the file, allocating space in extended memory, and closing the file.

WARMBOOT CART Command

A general use operator command that causes the Router module to reinitialize the Check Authorization Routing Table File (CART).

Syntax

DELIVER PROCESS *object-name,* “9530CART”

object-name — The symbolic name assigned to the Device Handler/Router/Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

When the Device Handler/Router/Authorization process receives a WARMBOOT CART command, it performs the following steps:

1. The Router module retrieves the name of the CART from the POS-CART assign in the LCONF and rereads the CART.
2. The Router module opens the CART, allocates space in extended memory for the CART, and then closes the CART.

Example

The following is an example of a WARMBOOT CART command. In the example, the Router/Authorization module is P1A^RTAU8.

DELIVER PROCESS P1A^RTAU8, “9530CART”

WARMBOOT FIID Command

A general use operator command that causes the Authorization module to update the current information that it has in memory for an institution.

Note: The WARMBOOT FIID command does not cause the Authorization module to reread the Card Prefix File (CPF), add an Institution Definition File (IDF) record to memory, or add Key Authorization File (KEYA) data to memory if the IDF record did not have a KEYA record associated with it previously. To update this information, the WARMBOOT ALL command (9500****) must be issued or the process must be stopped and restarted.

Syntax

DELIVER PROCESS *object-name,* "9501*fiid*"

object-name — The symbolic name assigned to the Device Handler/Router/Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

fiid — The FIID of the financial institution being warmbooted.

Processing

When the Authorization module receives the WARMBOOT FIID command, it performs the following steps:

1. Locates the IDF record for the FIID in extended memory.
2. Reads the corresponding IDF record from the disk file.

   If a record cannot be found, the Authorization module generates an event message indicating the message cannot be processed and no further steps are performed.
3. Closes any data files open for the institution.

   The Authorization module identifies the cardholder files (e.g., CAF, PBF, NEG, or UAF) open for the institution. If any of the files are opened exclusively for the institution, the Authorization module closes the files.
4. Overlays the IDF record in extended memory with the new IDF record from disk, field-by-field.
5. Sets the fatal errors to indicate no error for the institution.
6. Opens the cardholder files required for processing if they are not already open.

7. Edits the IDF record fields. To do this, the Authorization module performs the following steps:

   a. Checks each RT-TBL.SYM-NAME field in the BASE24-pos segment of the IDF for leading and trailing asterisks, replacing the asterisks with corresponding characters from its own name.

   b. Generates the event message “Warmboot complete for bank *fiid*.”

Example

The following is an example of a WARMBOOT FIID command. The Router/Authorization process being warmbooted in the command is P1A^PSRTAU1. In the example, the FIID being warmbooted is BNK1.

```
DELIVER PROCESS P1A^PSRTAU1, “9501BNK1”
```

WARMBOOT INR, BNR, and CAPF Commands

General use operator commands that cause the Router or Authorization module to reinitialize the file specified in the command. The files that can be warmbooted with these commands are the INAS National Range File (INR), Base National Range File (BNR), and Card Authorization Parameters File (CAPF).

Syntax

DELIVER PROCESS *object-name,* "9520INR"

DELIVER PROCESS *object-name,* "9520BNR"

DELIVER PROCESS *object-name*, "9520CAPF"

object-name — The symbolic name assigned to the Device Handler/Router/Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

When the Router or Authorization module receives a 9520 command, it opens the file associated with the command, loads extended memory, and closes the file.

Example

The following is an example of a WARMBOOT INR command. In the example, the Router/Authorization module is P1A^RTAU6.

DELIVER PROCESS P1A^RTAU6, "9520INR"

WARMBOOT PRF Command

A general use operator command that causes the Router/Authorization module to reinitialize the POS Referral File (PRF). The PRF is used with the BASE24-pos CRT Authorization product.

Syntax

DELIVER PROCESS *object-name,* “9530PRF”

object-name — The symbolic name assigned to the Device Handler/Router/Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

When the Device Handler/Router/Authorization process receives a WARMBOOT PRF command, it performs the following steps:

1. The Router module retrieves the name of the PRF from the CRT-AUTH-PRF assign in the LCONF and rereads the PRF.
2. The Authorization module then closes the current and next day’s PRF to which it is currently logging and opens the current and next day’s PRF named in the CRT-AUTH-PRF assign.

   The PRF is used by the CRT Authorization Control System. When transactions are referred to the CRT Authorization Control System, the Router module must know the location of the PRF in order to send transactions to the appropriate location.

Example

The following is an example of a WARMBOOT PRF command. In the example, the Router/Authorization module is P1A^RTAU6.

DELIVER PROCESS P1A^RTAU6, “9530PRF”

WARMBOOT PTLF Command

A general use operator command that causes the Router/Authorization module to reinitialize the POS Transaction Log File (PTLF).

Syntax

DELIVER PROCESS *object-name,* "9530PTLF"

object-name — The symbolic name assigned to the Device Handler/Router/Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

When the Device Handler/Router/Authorization process receives a WARMBOOT PTLF command, it performs the following steps:

1. The Router module retrieves the name of the PTLF from the POS-PTLF assign in the LCONF and rereads the PTLF.
2. The Authorization module then closes the current and next day's PTLF to which it is currently logging and opens the current and next day's PTLF named in the POS-PTLF assign.

Example

The following is an example of a WARMBOOT PTLF command. In the example, the Router/Authorization module is P1A^RTAU7.

DELIVER PROCESS P1A^RTAU7, "9530PTLF"

WARMBOOT PRDF Command

A general use operator command that causes the Router module to reinitialize the POS Retailer Definition File (PRDF).

Syntax

DELIVER PROCESS *object-name,* "9527RDF"

object-name — The symbolic name assigned to the Device Handler/Router/Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

When the Device Handler/Router/Authorization process receives a WARMBOOT PRDF command, it performs the following steps:

1. The Router module retrieves the name of the PRDF from the POS-PRDF assign in the LCONF and rereads the PRDF.
2. The Router module opens the PRDF, allocates space in extended memory for the PRDF, and then closes the PRDF.

Settlement Initiator Process Commands

This section describes the BASE24 commands recognized by the Settlement Initiator processes. The Settlement Initiator processes can receive a number of BASE24 commands that cause it to perform specialized processing. These commands can be separated into three categories: process-generated commands, general use operator commands, and restricted use operator commands. All BASE24 general and restricted use operator commands that the Settlement Initiator process can receive are documented in this subsection. Process-generated commands are documented in appendix B.

BASE24-pos Settlement Initiator processing is divided among a Main Settlement Initiator process and a configurable number of Secondary Settlement Initiator processes. The Main Settlement Initiator process performs some settlement functions while the Secondary Settlement Initiator process performs other aspects of Settlement Initiator processing.

The Main Settlement Initiator processes perform the following tasks:

- Opening the Secondary Settlement Initiator processes for communication, if Secondary Settlement Initiator processes are defined in the system
- Creating the report obey files for running reports, if reports are configured to be run
- Creating and purging POS Transaction Log Files (PTLFs) and POS Referral Files (PRFs)
- Cutting over financial institutions
- Cutting over the logical network

The Secondary Settlement Initiator processes perform the following tasks:

- Opening and closing the PTLF
- Cutting over terminals
- Cutting over retailers

For more information on BASE24-pos settlement, refer to the ***BASE24-pos Settlement and Reporting Manual***.

All of the commands that can be received by the Settlement Initiator processes are listed below by category. For more information on BASE24 command categories, refer to section 1.

Process-Generated Commands	General Use Operator Commands	Restricted Use Operator Commands
ALIVEMSG	CREATEPTLF	DEBUG
CHECKMASTER	CREATEPRF	DUMP
DUMPIDFTBL	DUMPIDF	MIDNIGHT
DUMPRDF	DUMPIDFTBL	RESETIDFDATES
DUMPRDFTBL	DUMPIDFTBLALL	RESETRDFDATES
DUMPTIMERTBL	DUMPIDFTBL	RESETPTDDATES
DUMPTIMES	SECONDARY	RESETALLDATES
DUMPPTD	DUMP NOTIFY	RESETRDF TIMESALL
DUMPPTDTBL	DUMPRDFALL	RESETRDFTIMES
FIID	DUMPRDF SECONDARY	SECONDARY
LOGICAL NETWORK	DUMPRDFTBLALL	RESYNC
CUTOVER	DUMPRDFTBL SECONDARY	TRACEOFF
NOTIFY MHI	DUMPPTDALL	TRACEON
RESETRDFDATES	DUMPPTD SECONDARY	
RESETPTDDATES	DUMPPTDTBLALL	
RESETALLDATES	DUMPPTDTBL SECONDARY	
RESETRDFTIMES	DUMPSLAVETBL	
START UP	DUMPTIMERTBL SECONDARY	
STARTUPCOMPLETE	DUMPTIMES SECONDARY	
TRACEOFF SECONDARY	DUMPTIMERTBL	
TRACEON SECONDARY	DUMPTIMERTBLALL	
WARMBOOT PRDF TABLE ENTRY	DUMPTIMES	
WB MASTER PPD	DUMPTIMESALL	
WB COMPLETE	WARMBOOT ALL	
	WARMBOOT MASTER	
	WARMBOOT NOTIFY	
	WARMBOOTRDFKEY	
	WARMBOOT SECONDARY	

CREATEPTLF and CREATEPRF Commands

A set of general use operator commands that cause the Main Settlement Initiator process to create the POS Transaction Log File (PTLF) or the POS Referral File (PRF) for the date specified.

Syntax

DELIVER PROCESS *object-name,* "CREATEPTLF *yymmdd*"

DELIVER PROCESS *object-name*, "CREATEPRF *yymmdd*"

object-name — The symbolic name assigned to the Main Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

yymmdd — Identifies the date (yymmdd) for which the Main Settlement Initiator process will create the file.

Processing

The Main Settlement Initiator process creates the POS Transaction Log Files (PTLFs) and POS Referral Files (PRFs) when a CREATEPTLF or CREATEPRF command is received. In addition to creating these files, the Main Settlement Initiator process also creates the alternate key files associated with these files. The Main Settlement Initiator process performs the following steps when it receives a CREATEPTLF or CREATEPRF command:

1. Allocates new PTLF extents.
2. Writes the extract positions record for the PTLF.
3. Creates the new PTLF or PRF and the corresponding alternate key files for the date (YYMMDD) specified in the command.

   When creating the PTLF, the Main Settlement Initiator process names the files POYYMMDD P0YYMMDD, P1YYMMDD, P2YYMMDD, P3YYMMDD, and P4YYMMDD as specified in the template of the POS-PTLF assign in the Logical Network Configuration File (LCONF). POYYMMDD is the name of the PTLF and P0YYMMDD, P1YYMMDD, P2YYMMDD, P3YYMMDD, and P4YYMMDD are the alternate key files for the PTLF. The Main Settlement Initiator process creates one of each of these files for the date specified in the command.

   When creating the PRF, the Main Settlement Initiator process names the files FFYYMMDD and F0YYMMDD, as specified in the template of the POS-PRF assign in the LCONF. FFYYMMDD is the name of the PRF and

F0YYMMDD is the alternate key file for the PRF. The Main Settlement Initiator process creates one of each of these files for the date specified in the command.

4. Closes the PTLF or PRF.

Example

The following are examples of CREATEPTLF and CREATEPRF commands. In the examples, the Main Settlement Initiator process is P1A^SETL1 and the date is December 22, 2001.

```
DELIVER PROCESS P1A^SETL1, “CREATEPTLF 011222”
DELIVER PROCESS P1A^SETL1, “CREATEPRF 011222”
```

DEBUG Command

A restricted use operator command that places a Settlement Initiator process into the HP NonStop Inspect and displays an Inspect prompt on the home terminal (the terminal specified for the process by the HOMETERM attribute or the home terminal of the XPNET process). Inspect is an HP NonStop programming tool. If the optional *secondary-name* variable is included in the command, the Main Settlement Initiator process sends an interprocess DEBUG command to the specified Secondary Settlement Initiator process, and the specified Secondary Settlement Initiator process is put into Debug. If the *secondary-name* variable is not included, the Main Settlement Initiator process is put into debug. For more information on the DEBUG command, refer to appendix A.

Syntax

DELIVER PROCESS *object-name,* “DEBUG[*secondary-name*]”

Note: There is no space between DEBUG and the variable *secondary-name*.

object-name — The symbolic name assigned to the Main Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

[*secondary-name*] — The symbolic name of the Secondary Settlement Initiator process to be put into debug.

DUMPIDF Command

A general use operator command that causes the Main Settlement Initiator process to read each of its Institution Definition File (IDF) record and provide the FIID, cutover end date, current business date, and customer business date.

Syntax

DELIVER PROCESS *object-name,* "DUMPIDF"

object-name — The symbolic name assigned to the Main Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

The Main Settlement Initiator process receives the DUMPIDF command from an XPNET network control facility; reads each of its associated IDF records; and displays the FIID, cutover end time, current business date, and customer business date in the following format:

```
FI-ID  CUTOVER-END   CUR-BUS-DATE  CUS-BUS-DATE
```

The IDF record information described above is placed here. The CUTOVER-END field contains the ending time for BASE24-pos cutover. The CUR-BUS-DATE field contains the current processing date (YYMMDD). The CUS-BUS-DATE field contains the current customer processing date (YYMMDD). When all IDF record information has been displayed, the following message is shown:

```
END OF IDF DISPLAY
```

DUMPIDFTBL Command

A general use operator command that causes the Main Settlement Initiator process to display the contents of its Institution Definition File (IDF) extended memory table in a series of event messages.

Note: The DUMPIDFTBL command can be considered both a process-generated command and a general use operator command, depending on where it originates. If a DUMPIDFTBLALL or DUMPIDFTBL SECONDARY command is sent to the Main Settlement Initiator process from an XPNET network control facility, the Main Settlement Initiator process sends a DUMPIDFTBL command as an interprocess command to Secondary Settlement Initiator processes. When a DUMPIDFTBLALL command is sent to the Main Settlement Initiator process, the Main Settlement Initiator process's IDF table is displayed as well as the Secondary Settlement Initiator process's IDF table. The Main Settlement Initiator process notifies the Secondary Settlement Initiator processes that IDF table information is being requested. Refer to the "Process-Generated Commands" topic discussed earlier for an explanation of process-generated dump IDF table commands.

If the command originates from an XPNET network control facility as described in this subsection, it is a general use operator command. The network control request is for Main Settlement Initiator process IDF table information. In this case, the Main Settlement Initiator process displays only its IDF table information. No interprocess commands are sent.

Syntax

DELIVER PROCESS *object-name,* "DUMPIDFTBL"

object-name — The symbolic name assigned to the Main Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

The Main Settlement Initiator process receives the DUMPIDFTBL command from an XPNET network control facility and displays the following information taken from the IDF extended memory table in a series of event messages:

- FIID
- Cutover time
- Unique flag
- Clear at midnight
- Need to query
- Cutover flag

The IDF extended memory table information shown below is displayed in a single line. It has been divided into two parts here due to space restrictions.

```
FI-ID CUTOVER TIME UNIQUE FLAG
```

The IDF extended memory table information is placed here. The CUTOVER TIME field identifies the cutover time for the FIID. The second CUTOVER FLAG field identifies the cutover flag (contains a value of Y, Yes, or N, No). The cutover flag indicates whether the FIID has cut over for the current day.

```
MIDNIGHT NEED QUERY CUTOVER FLAG
```

The UNIQUE FLAG field indicates (Y or N) whether the institution cutover time is unique among all institutions. The MIDNIGHT field is set according to the IDF FIELD CUTOVER field and indicates whether the Usage Accumulation File (UAF) is to be cleared at midnight for the FIID.

The NEED QUERY field is set to determine if the Main Settlement Initiator process is required to query each of its Secondary Settlement Initiator processes to determine whether they have finished cutting over all the retailers for a specified FIID. When all IDF table information has been displayed, the following message is shown:

```
END OF IDF TABLE DISPLAY
```

DUMPIDFTBLALL Command

A general use operator command that causes the Main Settlement Initiator process and all its associated Secondary Settlement Initiator processes to display the contents of their Institution Definition File (IDF) extended memory tables in a series of event messages. The Main Settlement Initiator process sends interprocess DUMPIDFTBL commands to all of its Secondary Settlement Initiator processes.

Syntax

DELIVER PROCESS *object-name,* "DUMPIDFTBLALL"

object-name — The symbolic name assigned to the Main Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

The Main Settlement Initiator process receives this command from an XPNET network control facility and sends interprocess DUMPIDFTBL messages to all of its Secondary Settlement Initiator processes. The IDF extended memory table is created at initialization by the Main and Secondary Settlement Initiator processes. When each Secondary Settlement Initiator process receives this command, it displays the following information taken from the IDF extended memory table in a series of event messages:

- FIID
- Cutover time
- Batch, shift, daily, service, and network totals
- Host Interface symbolic name

The IDF extended memory table information is formatted as follows:

```
FI-ID CUTOVER TIME   BATCH SHFT DLY SRVC NET HISF-NAME
```

The IDF extended memory table information is placed here. The FI-ID field identifies the FIID associated with the retailer. The CUTOVER TIME field identifies the cutover time for the retailer. The BATCH, SHIFT, DLY, SRVC, and NET fields indicate the batch, shift, daily, services, and network totals for the retailer. The HISF-NAME field identifies the symbolic name of the Host Interface process.

When all IDF table information has been displayed, the following message is shown:

```
END OF IDF TABLE DISPLAY
```

DUMPIDFTBL SECONDARY Command

A general use operator command that causes a specific Secondary Settlement Initiator process to display the contents of its Institution Definition File (IDF) extended memory table in a series of event messages. The Main Settlement Initiator process sends the interprocess DUMPIDFTBL command to the specified Secondary Settlement Initiator process.

Syntax

DELIVER PROCESS *object-name,* "DUMPIDFTBL*secondary-name"*

Note: There is no space between DUMPIDFTBL and the variable *secondary-name*.

object-name — The symbolic name assigned to the Main Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

secondary-name — The symbolic name of the Secondary Settlement Initiator process that is displaying its IDF extended memory table.

Processing

The Main Settlement Initiator process receives this command from an XPNET network control facility and sends an interprocess DUMPIDFTBL message to a specific Secondary Settlement Initiator process. The IDF extended memory table is created at initialization by the Main and Secondary Settlement Initiator processes. When the specified Secondary Settlement Initiator process receives this command, it displays the following information taken from the IDF extended memory table in a series of event messages.

- FIID
- Cutover time
- Send batch, shift, daily, service, and network totals
- Host Interface symbolic name

The IDF extended memory table information is formatted as follows:

```
FI-ID CUTOVER BATCH SHFT DLY SRVC NET HISF-NAME
```

The IDF extended memory table information is placed here. The FI-ID field identifies the FIID associated with the retailer. The CUTOVER TIME field identifies the cutover time for the retailer. The BATCH, SHIFT, DLY, SRVC, and

NET fields indicate the batch, shift, daily, services, and network totals for the retailer. The HISF-NAME field identifies the symbolic name of the Host Interface process. When all IDF table information has been displayed, the following message is shown:

```
END OF IDF TABLE DISPLAY
```

DUMPNOTIFY Command

A general use operator command that causes the Main Settlement Initiator process to display the contents of its internal cutover, midnight, and Merchant Host Interface (MHI) tables in a series of event messages. These tables contain the symbolic process names that the Main or Secondary Settlement Initiator processes must notify for cutover to be successful. The internal cutover table identifies the Device Handler/Router/Authorization processes that the Main Settlement Initiator process must notify at logical network cutover. The midnight table identifies the Device Handler/Router/Authorization processes that the Main Settlement Initiator process must notify at midnight cutover. The MHI table identifies the name of the Merchant Host Interface (MHI) the Secondary Settlement Initiator process must notify at retailer cutover (if MHI is being used).

Syntax

DELIVER PROCESS *object-name,* "DUMPNOTIFY"

object-name — The symbolic name assigned to the Main or Secondary Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

The DUMPNOTIFY command is sent from an XPNET network control facility to the Main Settlement Initiator process. In response to this command, the Main Settlement Initiator process displays the contents of its internal cutover tables in a series of event messages. The Main Settlement Initiator process creates these internal tables at initialization based on the following params identified in the Logical Network Configuration File (LCONF):

- POS-SETL-NOTIFY — Identifies the Device Handler/Router/Authorization processes the Main Settlement Initiator process must notify at logical network cutover.
- POS-SETL-MIDNIGHT-NOTIFY — Identifies the Device Handler/Router/Authorization processes the Main Settlement Initiator process must notify at midnight cutover.
- POS-SETL-MHI-NOTIFY — Identifies the MHI Device Handler/Router/Authorization processes the Secondary Settlement Initiator process must notify at retailer cutover.

The Main Settlement Initiator process also sends DUMPNOTIFY commands to its Secondary Settlement Initiator processes to retrieve the MHI notify tables from the secondary processes. Notify information is displayed in the following format. Each internal table can contain up to 150 process names.

```
SYMBOLIC NAME  - NOTIFY TABLE
```

Symbolic process names for all Device Handler/Router Authorization processes that must be notified by the Main Settlement Initiator process at logical network cutover are placed here. When all notify table information has been displayed, the following message is shown:

```
END OF NOTIFY TABLE DISPLAY
SYMBOLIC NAME  - MIDNIGHT TABLE
```

Symbolic process names for all Device Handler/Router Authorization processes that must be notified by the Main Settlement Initiator process at midnight cutover are placed here. When all midnight notify information has been displayed, the following message is shown:

```
END OF MIDNIGHT TABLE DISPLAY
SYMBOLIC NAME  - MHI TABLE
```

Symbolic process names for all MHI Device Handler/Router/Authorization processes that must be notified by Secondary Settlement Initiator processes at retailer cutover are placed here. When all MHI notify information has been displayed, the following message is shown:

```
END OF MHI TABLE DISPLAY
```

DUMPRDFALL Command

A general use operator command that causes all Secondary Settlement Initiator processes associated with the Main Settlement Initiator process identified in the command to display POS Retailer Definition File (PRDF) records according to the retailer range defined for a specific Secondary Settlement Initiator process in a series of event messages. DUMPRDFALL commands entered from an XPNET network control facility are sent directly to the Main Settlement Initiator process. The Main Settlement Initiator process sends interprocess DUMPRDF commands to the appropriate Secondary Settlement Initiator processes.

Syntax

DELIVER PROCESS *object-name,* "DUMPRDFALL"

object-name — The symbolic name assigned to the Main Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

When the Main Settlement Initiator process receives a DUMPRDFALL command from an XPNET network control facility, it sends the interprocess DUMPRDF message to all Secondary Settlement Initiator processes. When a Secondary Settlement Initiator process receives this command, it displays PRDF record information only for those retailer IDs included in the retailer range defined for the specific Secondary Settlement Initiator process. Retailer ID ranges for Secondary Settlement Initiator processes are identified in the POS-SLAVE-KEY param in the LCONF. The Secondary Settlement Initiator process displays the following PRDF record information for each retailer ID in its retailer ID range in a series of event messages:

- Retailer ID
- FIID
- End of retailer cutover window
- PRDF posting date

PRDF record information is displayed in the following format:

```
RETAILER-ID    FI-ID    BAL-CUTOVER-END    CUR-BUS-DATE
```

The PRDF record information described above is displayed here. The RETAILER-ID field identifies the retailer associated with the PRDF records. The FI-ID identifies the FIID associated with the retailer. The BAL-CUTOVER-END field identifies the end of the retailer cutover window. The CUR-BUS-DATE identifies the posting date in the PRDF. When all the information has been displayed, the following message is shown:

```
END OF RDF DISPLAY
```

DUMPRDF SECONDARY Command

A general use operator command that causes a specific Secondary Settlement Initiator process identified in the command to display POS Retailer Definition File (PRDF) records according to its retailer range in a series of event messages. The Main Settlement Initiator process sends interprocess DUMPRDF commands to the specified Secondary Settlement Initiator process. If the optional *secondary-name* variable is included in the command, the Main Settlement Initiator process sends an interprocess DUMPRDF command to the specified Secondary Settlement Initiator process, and the specified Secondary Settlement Initiator process displays the PRDF records. If the *secondary-name* variable is not included, the *object-name* variable must contain the name of the Secondary Settlement Initiator process, and the DUMPRDF command is sent directly to the Secondary Settlement Initiator process.

Syntax

DELIVER PROCESS *object-name,* “DUMPRDF[*secondary-name*]”

Note: There is no space between DUMPRDF and the variable *secondary-name*.

object-name — The symbolic name assigned to a Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

[*secondary-name*] — The symbolic name of the Secondary Settlement Initiator process whose PRDF records are being displayed.

Processing

When the Main Settlement Initiator process receives a DUMPRDF SECONDARY command from an XPNET network control facility, it sends the interprocess DUMPRDF message to the specified Secondary Settlement Initiator process. When a Secondary Settlement Initiator process receives this command, it displays PRDF record information only for those retailer IDs included in the retailer range defined for the specific Secondary Settlement Initiator process in a series of event messages. Retailer ID ranges for Secondary Settlement Initiator processes are identified in the POS-SLAVE-KEY param in the LCONF. The Secondary Settlement Initiator process displays the following PRDF record information for each retailer ID in its retailer ID range:

- Retailer ID
- FIID

- End of retailer cutover window
- PRDF posting date

PRDF record information is displayed in the following format:

```
RETAILER-ID    FI-ID    BAL-CUTOVER-END    CUR-BUS-DATE
```

The PRDF record information described above is displayed here. The RETAILER-ID field identifies the retailer associated with the PRDF records. The FI-ID identifies the FIID associated with the retailer. The BAL-CUTOVER-END field identifies the end of the retailer cutover window. The CUR-BUS-DATE identifies the posting date in the PRDF. When all the information has been displayed, the following message is shown:

```
END OF RDF DISPLAY
```

DUMPRDFTBLALL Command

A general use operator command that causes all of the Secondary Settlement Initiator processes associated with the Main Settlement Initiator process identified in the command to display its internal POS Retailer Definition File (PRDF) extended memory table in a series of event messages. The Main Settlement Initiator process sends interprocess DUMPRDFTBL commands to all of its Secondary Settlement Initiator processes.

Syntax

DELIVER PROCESS *object-name,* "DUMPRDFTBLALL"

object-name — The symbolic name assigned to the Main Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

When the Main Settlement Initiator process receives a DUMPRDFTBLALL command from an XPNET network control facility, it sends an interprocess DUMPRDFTBL message to all Secondary Settlement Initiator processes. When a Secondary Settlement Initiator process receives this command, it displays its internal PRDF extended memory table in a series of event messages. The Secondary Settlement Initiator process displays the following PRDF table information for each retailer ID in its retailer ID range:

- Retailer ID
- FIID
- Cutover time
- Unique flag
- Cutover flag

The PRDF extended memory table information is formatted as follows:

```
RETAILER-ID FI-ID CUTOVER TIME UNIQUE FLAG CUTOVER FLAG
```

The PRDF extended memory table information is placed here. The CUTOVER TIME field identifies the cutover time for the retailer. The second CUTOVER FLAG field identifies the cutover flag (contains a value of Y, Yes, or N, No). The cutover flag indicates whether the retailer has cut over for the current day. The

UNIQUE FLAG field indicates (Y or N) whether the retailer cutover time is unique among all retailers.When all PRDF table information has been displayed, the following message is shown:

```
END OF RDF TABLE DISPLAY
```

DUMPRDFTBL SECONDARY Command

A general use operator command that causes a specific Secondary Settlement Initiator process identified in the command to display its internal POS Retailer Definition File (PRDF) extended memory table in a series of event messages. The Main Settlement Initiator process sends an interprocess DUMPRDFTBL command to the specified Secondary Settlement Initiator process. If the optional *secondary-name* variable is included in the command, the Main Settlement Initiator process sends an interprocess DUMPRDFTBL command to the specified Secondary Settlement Initiator process, and the specified Secondary Settlement Initiator process displays its internal PRDF extended memory table. If the *secondary-name* variable is not included, the *object-name* variable must contain the name of the Secondary Settlement Initiator process, and the DUMPRDFTBL command is sent directly to the Secondary Settlement Initiator process.

Syntax

DELIVER PROCESS *object-name,* “DUMPRDFTBL*secondary-name”*

Note: There is no space between DUMPRDFTBL and the variable *secondary-name*.

object-name — The symbolic name assigned to a Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

secondary-name — The symbolic name of the Secondary Settlement Initiator process whose PRDF extended memory table is being displayed.

Processing

When the Main Settlement Initiator process receives a DUMPRDFTBL SECONDARY command from an XPNET network control facility, it sends an interprocess DUMPRDFTBL message to the specified Secondary Settlement Initiator process. When the Secondary Settlement Initiator process receives this command, it displays its internal PRDF extended memory table in a series of event messages. The Secondary Settlement Initiator process displays the following PRDF table information for each retailer ID in its retailer ID range:

- Retailer ID
- FIID
- Cutover time

- Unique flag
- Cutover flag

The PRDF extended memory table information is formatted as follows:

```
RETAILER-ID FI-ID CUTOVER TIME UNIQUE FLAG CUTOVER FLAG
```

The PRDF extended memory table information is placed here. The CUTOVER TIME field identifies the cutover time for the retailer. The second CUTOVER FLAG field identifies the cutover flag (contains a value of Y, Yes, or N, No). The cutover flag indicates whether the retailer has cut over for the current day. The UNIQUE FLAG field indicates (Y or N) whether the retailer cutover time is unique among all retailers. When all PRDF table information has been displayed, the following message is shown:

```
END OF RDF TABLE DISPLAY
```

DUMPPTDALL Command

A general use operator command that causes all Secondary Settlement Initiator processes to display BASE24-pos Terminal Data files record information for their retailer ranges in a series of event messages. The Main Settlement Initiator process notifies all of its associated Secondary Settlement Initiator processes to display BASE24-pos Terminal Data files record information.

Syntax

DELIVER PROCESS *object-name,* "DUMPPTDALL"

object-name — The symbolic name assigned to the Main Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

The Main Settlement Initiator process receives the DUMPPTDALL command from an XPNET network control facility and sends a DUMPPTD interprocess command to all of its associated Secondary Settlement Initiator processes. When a specific Secondary Settlement Initiator process receives this command, it displays BASE24-pos Terminal Data files record information only for those retailer IDs included in the retailer range defined for the specific Secondary Settlement Initiator process in a series of event messages. Retailer ID ranges for Secondary Settlement Initiator processes are identified in the POS-SLAVE-KEY param in the LCONF. The Secondary Settlement Initiator process displays the following BASE24-pos Terminal Data files record information for each terminal ID in its retailer ID range:

- Terminal ID
- Retailer ID
- Device Handler/Router/Authorization process name
- Post date
- Deactivate flag

BASE24-pos Terminal Data files record information is displayed in the following format:

```
TERM-ID    RETAILER-ID    DH-PRO-NAME    POST-DATE DF
```

The TERM-ID field identifies the terminal ID from the BASE24-pos Terminal Data files record. The RETAILER-ID indicates the retailer ID associated with the terminal ID. The DH-PRO-NAME indicates the symbolic name of the Device Handler/Router/Authorization process as identified in the BASE24-pos Terminal Data files. The POST-DATE field identifies the posting date from the BASE24-pos Terminal Data files record. The DF field indicates whether the terminal is P (pending), A (activated), or D (deactivated). When all the information has been displayed, the following message is shown:

```
END OF TDF DISPLAY
```

DUMPPTD SECONDARY Command

A general use operator command that causes a specific Secondary Settlement Initiator process to display BASE24-pos Terminal Data files record information for its retailer range in a series of event messages. The Main Settlement Initiator process notifies the Secondary Settlement Initiator process identified in the command to display BASE24-pos Terminal Data files record information. If the optional *secondary-name* variable is included in the command, the Main Settlement Initiator process sends an interprocess DUMPPTD command to the specified Secondary Settlement Initiator process, and the specified Secondary Settlement Initiator process displays the BASE24-pos Terminal Data files records. If the *secondary-name* variable is not included, the *object-name* variable must contain the name of the Secondary Settlement Initiator process, and the DUMPPTD command is sent directly to the Secondary Settlement Initiator process.

Syntax

DELIVER PROCESS *object-name,* "DUMPPTD*secondary-name*"

Note: There is no space between DUMPPTD and the variable *secondary-name*.

object-name — The symbolic name assigned to a Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

secondary-name — The symbolic name of a Secondary Settlement Initiator process for which BASE24-pos Terminal Data files records have been requested.

Processing

The Main Settlement Initiator process receives the DUMPPTD SECONDARY command from an XPNET network control facility and sends a DUMPPTD interprocess command to the Secondary Settlement Initiator processes identified in the command. When the specified Secondary Settlement Initiator process receives this command, it displays BASE24-pos Terminal Data files record information only for those retailer IDs included in the retailer range defined for the specific Secondary Settlement Initiator process in a series of event messages. Retailer ID ranges for Secondary Settlement Initiator processes are identified in the POS-SLAVE-KEY param in the LCONF.

The Secondary Settlement Initiator process displays the following BASE24-pos Terminal Data files record information for each terminal ID in its retailer ID range:

- Terminal ID
- Retailer ID
- Device Handler/Router/Authorization process name
- Post date
- Deactivate flag

BASE24-pos Terminal Data files record information is displayed in the following format:

```
TERM-ID    RETAILER-ID    DH-PRO-NAME    POST-DATE DF
```

The TERM-ID field identifies the terminal ID from the BASE24-pos Terminal Data files record. The RETAILER-ID indicates the retailer ID associated with the terminal ID. The DH-PRO-NAME indicates the symbolic name of the Device Handler/Router/Authorization process as identified in the BASE24-pos Terminal Data files. The POST-DATE field identifies the posting date from the BASE24-pos Terminal Data files record. The DF field indicates whether the terminal is P (pending), A (activated), or D (deactivated). When all the information has been displayed, the following message is shown:

```
END OF TDF DISPLAY
```

DUMPPTDTBLALL Command

A general use operator command that causes all Secondary Settlement Initiator processes associated with the Main Settlement Initiator process identified in the command to display the contents of their internal BASE24-pos Terminal Data files extended memory tables in a series of event messages. DUMPPTDTBLALL commands entered from an XPNET network control facility are sent directly to the Main Settlement Initiator process. The Main Settlement Initiator process sends interprocess DUMPPTDTBL commands to the appropriate Secondary Settlement Initiator processes.

Syntax

DELIVER PROCESS *object-name,* "DUMPPTDTBLALL"

object-name — The symbolic name assigned to the Main Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

The Main Settlement Initiator process receives the DUMPPTDTBLALL command from an XPNET network control facility and sends a DUMPPTDTBL interprocess command to all of its associated Secondary Settlement Initiator processes. The Secondary Settlement Initiator processes respond by displaying the contents of their internal BASE24-pos Terminal Data files extended memory table in a series of event messages. The internal BASE24-pos Terminal Data files extended memory table for Merchant Host Interfaces (MHI) in host control mode contains all of the retailer IDs and Device Handler/Router/Authorization process names associated with a particular Secondary Settlement Initiator process. If there are no BASE24-pos Terminal Data files table entries for a particular Secondary Settlement Initiator process, that Secondary Settlement Initiator process logs an event message indicating that the BASE24-pos Terminal Data files extended memory table is empty.

BASE24-pos Terminal Data files table information is displayed in the following format:

```
RETAILER-ID      DH-PRO-NAME
```

When all the information has been displayed, the following message is shown:

```
END OF TDF TABLE DISPLAY
```

DUMPPTDTBL SECONDARY Command

A general use operator command that causes a specific Secondary Settlement Initiator processes to display the contents of its internal BASE24-pos Terminal Data files extended memory table in a series of event messages. The Main Settlement Initiator process sends interprocess DUMPPTDTBL commands to the Secondary Settlement Initiator process identified in the command. If the optional *secondary-name* variable is included in the command, the Main Settlement Initiator process sends an interprocess DUMPPTDTBL command to the specified Secondary Settlement Initiator process, and the specified Secondary Settlement Initiator process displays the contents of its internal BASE24-pos Terminal Data files extended memory table. If the *secondary-name* variable is not included, the *object-name* variable must contain the name of the Secondary Settlement Initiator process, and the DUMPPTDTBL command is sent directly to the Secondary Settlement Initiator process.

Syntax

DELIVER PROCESS *object-name,* "DUMPPTDTBL*secondary-name"*

Note: There is no space between DUMPPTDTBL and the variable *secondary-name*.

object-name — The symbolic name assigned to a Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

secondary-name — The symbolic name of the Secondary Settlement Initiator process that is displaying its BASE24-pos Terminal Data files extended memory table.

Processing

The Main Settlement Initiator process receives the DUMPPTDTBL SECONDARY command from an XPNET network control facility and sends a DUMPPTDTBL interprocess command to the Secondary Settlement Initiator process identified in the command. The Secondary Settlement Initiator process responds by displaying the contents of its internal BASE24-pos Terminal Data files extended memory table in a series of event messages. The internal BASE24-pos Terminal Data files extended memory table contains all of the retailer IDs and Device Handler/Router/Authorization process names for Merchant Host Interface (MHI) processes running in host control mode that are associated with a particular Secondary Settlement Initiator process. If there are no BASE24-pos Terminal

Data files table entries for the specified Secondary Settlement Initiator process, the Secondary Settlement Initiator process logs an event message indicating that the BASE24-pos Terminal Data files extended memory table is empty.

BASE24-pos Terminal Data files table information is displayed in the following format:

```
RETAILER-ID      DH-PRO-NAME
```

When all the information has been displayed, the following message is shown:

```
END OF TDF TABLE DISPLAY
```

DUMPSLAVETBL Command

A general use operator command that causes the Main Settlement Initiator process to display the contents of its internal Secondary Settlement Initiator process table in a series of event messages. The Secondary Settlement Initiator process table identifies the Secondary Settlement Initiator processes associated with the Main Settlement Initiator process.

Syntax

DELIVER PROCESS *object-name,* “DUMPSLAVETBL”

object-name — The symbolic name assigned to the Main Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

The Main Settlement Initiator process receives a DUMPSLAVETBL command from an XPNET network control facility and displays the contents of its internal Secondary Settlement Initiator process table in a series of event messages. The Main Settlement Initiator process creates the Secondary Settlement table at initialization. The Secondary Settlement Initiator process table includes the symbolic name, PPD name, process number, status flag, low retailer key range, and high retailer key range. This information is displayed in the following format:

```
SYMBOLIC NAME PPD NAME FNUM STAT CUTOVER LOW KEY  HIGH KEY
```

The Secondary Settlement Initiator process information is placed here. The symbolic name field indicates the symbolic name of the Secondary Settlement Initiator process. The PPD name field identifies the PPD name of the Secondary Settlement Initiator process. The FNUM field indicates the file number associated with the Secondary Settlement Initiator process. The status field indicates whether the Secondary Settlement Initiator process responded to a STARTUP command with a STARTUPCOMPLETE command. A 0 in the status field indicates the Secondary Settlement Initiator process did not respond and a 1 indicates the Secondary Settlement Initiator process did respond. In addition, the cutover field indicates whether the Secondary Settlement Initiator process has responded to a 9502 message (0 = No, has not responded, 1 = Yes, has responded). The low key and high key fields identify the retailer ID range that corresponds to a particular Secondary Settlement Initiator process. The low key identifies where the retailer ID range begins and the high key identifies where the range ends. When all secondary table information has been displayed, the following message is shown:

```
END OF SLAVE TABLE DISPLAY
```

DUMPTIMERTBL SECONDARY Command

A general use operator command that causes the timer table for a specific Secondary Settlement Initiator process to be displayed in a series of event messages. The Main Settlement Initiator process notifies the Secondary Settlement Initiator process specified in the command to display its timer table information.

Syntax

DELIVER PROCESS *object-name,* "DUMPTIMERTBL*secondary-name"*

Note: There is no space between DUMPTIMERTBL and the variable *secondary-name*.

object-name — The symbolic name assigned to the Main Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

secondary-name — The symbolic name of a Secondary Settlement Initiator process for which timer table information has been requested.

Processing

The Main Settlement Initiator process receives the DUMPTIMERTBL SECONDARYsecondary command from an XPNET network control facility and sends a DUMPTIMERTBL interprocess command to the Secondary Settlement Initiator process identified in the command. The Secondary Settlement Initiator process responds by displaying information from its internal timer table in a series of event messages. This information includes the timer type, the date and time the timer is scheduled to expire, and the information contained in the timer's user data buffer. Timer table information for the Secondary Settlement Initiator process is displayed in the following format:

```
TIME-TYPE    EXPIRE DATE/TIME    USER-BUF
```

Timer information is displayed for each entry in the internal timer table. The Secondary Settlement Initiator process displays information for the following timers:

- Retailer cutover timer
- Alive timer

When all timer information has been displayed, the following message is shown:

```
END OF TIMER DISPLAY
```

DUMPTIMES SECONDARY Command

A general use operator command that causes certain important times for a specific Secondary Settlement Initiator process to be displayed in a series of event messages. The Main Settlement Initiator process instructs the Secondary Settlement Initiator process specified in the command to display time information.

Syntax

DELIVER PROCESS *object-name,* "DUMPTIMES*secondary-name"*

Note: There is no space between DUMPTIMES and the variable *secondary-name*.

object-name — The symbolic name assigned to the Main Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

secondary-name — The symbolic name of a Secondary Settlement Initiator process for which time information has been requested.

Processing

The Main Settlement Initiator process receives the DUMPTIMES SECONDARY command from an XPNET network control facility and sends a DUMPTIMES interprocess command to the Secondary Settlement Initiator process identified in the command. The Secondary Settlement Initiator process responds by displaying the following time information taken from params set up in the Logical Network Configuration File (LCONF) in a series of event messages. The corresponding LCONF param where the information was taken is noted in parentheses:

- Logical network cutover time (POS-LN-CUTOVER)
- Financial institution cutover lead time (POS-SETL-FI-CUTOVER-LEAD-TIME)
- Terminal cutover delay time (POS-SETL-TERM-CUTOVER-DELAY-TIME)
- Alive timeout time (POS-SETL-ALIVE-TIMEOUT)

In addition to the above times, the Secondary Settlement Initiator process also displays the current business date, the next business date, and the trace flag. When all time information has been displayed, the following message is shown:

```
END OF TIME DISPLAY
```

DUMPTIMERTBL Command

A general use operator command that causes the specified Settlement Initiator process to display information in its timer table in a series of event messages.

Note: The DUMPTIMERTBL command can be considered both a process-generated command and a general use operator command, depending on where it originates. If a DUMPTIMERTBLALL or DUMPTIMERTBL SECONDARY command is sent to the Main Settlement Initiator process from an XPNET network control facility, the Main Settlement Initiator process sends a DUMPTIMERTBL command as an interprocess command to Secondary Settlement Initiator processes. The Main Settlement Initiator process notifies the Secondary Settlement Initiator processes that timer information is being requested. Refer to the process-generated commands discussed in appendix B for an explanation of interprocess DUMPTIMERTBL commands.

If the command originates from an XPNET network control facility as described in this subsection, it is a general use operator command. The network control request is for the specified Settlement Initiator process timer table information. In this case, the specified Settlement Initiator process displays only its timer table information. No interprocess commands are sent.

Syntax

DELIVER PROCESS *object-name,* "DUMPTIMERTBL"

object-name — The symbolic name assigned to a Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

The Settlement Initiator process receives the DUMPTIMERTBL command from an XPNET network control facility and displays the timer type, the date and time the timer is scheduled to expire, and the information contained in the timer's user data buffer in a series of event messages. Timer information is displayed in the following format:

```
TIME-TYPE    EXPIRE DATE/TIME    USER-BUF
```

Timer information is displayed for each entry in the internal timer table. Main Settlement Initiator processes display information for the following timers:

- Financial institution cutover timer
- Logical network cutover timer

- Midnight timer
- Settlement reports timer
- Start Secondary Settlement Initiator processes timer
- Wait interval timer
- Warmboot timer
- 9502 message timer

The Secondary Settlement Initiator process displays information for the following timers:

- Retailer cutover timer
- Alive timer

When all timer information has been displayed, the following message is shown:

```
END OF TIMER DISPLAY
```

DUMPTIMERTBLALL Command

A general use operator command that causes the timer tables for the Main Settlement Initiator process and all its associated Secondary Settlement Initiator processes to be displayed. The Main Settlement Initiator process notifies all of its associated Secondary Settlement Initiator processes to display their timer table information in a series of event messages.

Syntax

DELIVER PROCESS *object-name,* "DUMPTIMERTBLALL"

object-name — The symbolic name assigned to the Main Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

The Main Settlement Initiator process receives the DUMPTIMERTBLALL command from an XPNET network control facility and sends a DUMPTIMERTBL interprocess command to all of its associated Secondary Settlement Initiator processes. The Secondary Settlement Initiator processes respond by displaying information from their internal timer tables in a series of event messages. This information includes the timer type, the date and time the timer is scheduled to expire, and the information contained in the timer's user data buffer. Timer table information for each Secondary Settlement Initiator process is displayed in the following format:

```
TIME-TYPE    EXPIRE DATE/TIME    USER-BUF
```

Timer information is displayed for each entry in the internal timer table. Each Secondary Settlement Initiator process displays information for the following timers:

- Retailer cutover timer (Secondary)
- Financial institution cutover timer (Main)
- Logical network cutover timer (Main)
- Midnight timer (Main)
- Start settlement reports timer (Main)
- Start Secondary Settlement Initiator processes timer (Main)
- Wait interval timer (Main)
- Alive timer (Secondary)

- Warmboot timer (Main)
- 9502 timer (Main)

When all timer information has been displayed, the following message is shown:

```
END OF TIMER DISPLAY
```

DUMPTIMES Command

A general use operator command that causes the specified Settlement Initiator process to display certain important processing times in a series of event messages. An operator enters this command from an XPNET network control facility to request Main Settlement Initiator process timer information.

Note: The DUMPTIMES command can be considered both a process-generated command and a general use operator command, depending on where it originates. If a DUMPIDFTBLALL or DUMPTIMES SECONDARY command is sent to the Main Settlement Initiator process from an XPNET network control facility, the Main Settlement Initiator process sends a DUMPTIMES command as an interprocess command to Secondary Settlement Initiator processes. The Main Settlement Initiator process notifies the Secondary Settlement Initiator processes that time information is being requested. Refer to the “Process-Generated Commands” topic discussed earlier for an explanation of general use operator DUMPTIMES commands.

If the command originates from an XPNET network control facility as described in this subsection, it is a general use operator command. The network control request is for the specified Settlement Initiator process time information. In this case, the specified Settlement Initiator process displays only its time information. No interprocess commands are sent.

Syntax

DELIVER PROCESS *object-name,* “DUMPTIMES”

object-name — The symbolic name assigned to a Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

The Settlement Initiator process receives the DUMPTIMES command from an XPNET network control facility. The Main Settlement Initiator process displays the following time information taken from params set up in the Logical Network Configuration File (LCONF) in a series of event messages. The corresponding LCONF params are noted in parentheses.

- Logical network cutover time (POS-LN-CUTOVER)
- Logical network cutover lead time (POS-SETL-LN-CUTOVER-LEAD-TIME)
- Start settlement report time (POS-SETL-RPTS)

- Wait interval time (POS-SETL-WAIT-INTERVAL)

The Secondary Settlement process displays the following time information:

- Logical network cutover time (POS-LN-CUTOVER)
- Financial institution cutover lead time (POS-SETL-FI-CUTOVER-LEAD-TIME)
- Terminal cutover delay time (POS-SETL-TERM-CUTOVER-DELAY-TIME)
- Alive timeout time (POS-SETL-ALIVE-TIMEOUT)

In addition to the above times, the Settlement Initiator process also displays the current business date, the next business date, and the trace flag. When all time information has been displayed, the following message is shown:

```
END OF TIME DISPLAY
```

DUMPTIMESALL Command

A general use operator command that causes certain important times for the Main Settlement Initiator process and all its associated Secondary Settlement Initiator processes to be displayed in a series of event messages. The Main Settlement Initiator process notifies all of its associated Secondary Settlement Initiator processes to display their time information.

Syntax

DELIVER PROCESS *object-name,* "DUMPTIMESALL"

object-name — The symbolic name assigned to the Main Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

The Main Settlement Initiator process receives the DUMPTIMESALL command from an XPNET network control facility and sends a DUMPTIMES interprocess command to all of its associated Secondary Settlement Initiator processes. The Secondary Settlement Initiator processes respond by displaying the following time information taken from params set up in the Logical Network Configuration File (LCONF) in a series of event messages. The corresponding LCONF param where the information was taken is noted in parentheses:

- Logical network cutover time (POS-LN-CUTOVER)
- Financial institution cutover lead time (POS-SETL-FI-CUTOVER-LEAD-TIME)
- Terminal cutover delay time (POS-SETL-TERM-CUTOVER-DELAY-TIME)
- Alive timeout time (POS-SETL-ALIVE-TIMEOUT)

In addition to the above times, the Secondary Settlement Initiator processes also display the current business date, the next business date, and the trace flag. When all time information has been displayed, the following message is shown:

```
END OF TIME DISPLAY
```

MIDNIGHT Command

A restricted use operator command that causes the Main Settlement Initiator process to perform the processing it would at midnight when the Midnight Timer expires. When the command is initiated, the Settlement Initiator takes any action that would normally be taken at midnight. Such actions could include purging the Usage Accumulation File (UAF). However, midnight timers are not deactivated when this command is sent. Therefore, the same action taken when this command is sent is taken again at midnight.

Syntax

DELIVER PROCESS *object-name,* "MIDNIGHT"

object-name — The symbolic name assigned to the Main Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

When the Main Settlement Initiator process receives a MIDNIGHT command from an XPNET network control facility, it purges the data in the Usage Accumulation File (UAF) if required and clears the usage accumulation fields in the Cardholder Authorization File (CAF). In addition, the Main Settlement Initiator process also reads the first record in the IDF and performs the following steps:

1. Determines whether midnight processing is necessary. To do this, the Main Settlement Initiator process performs the following steps:
   a. Verifies that the UAF name is present in the IDF. If so, the Main Settlement Initiator process assumes the institution uses the Negative Authorization with Usage Accumulation method.
   b. Checks the FIELD CUTOVER field in the Base segment of the IDF. If that field contains a value of 2 (Purge UAF at midnight; begin clearing CAF usage accumulation fields at midnight) or 5 (Do not purge UAF; begin clearing CAF usage accumulation fields at midnight) and the date in the CUSTOMER PROCESSING DATE field in the BASE24-pos segment of the IDF is earlier than or equal to the date in the CURRENT BUSINESS DATE field in the BASE24-pos segment of the IDF, the Main Settlement Initiator process continues with step 2. Otherwise, the Main Settlement Initiator process continues with step 4.

2. Updates the CUSTOMER PROCESSING DATE field in the IDF. To do this, the Main Settlement Initiator process performs the following steps:

   a. Computes a new customer business date by checking the WORK DAY CODE field in the Base segment of the IDF and performing according to the following situations:

      1) If the value in the WORK DAY CODE field is 0, the Main Settlement Initiator process increments the data in the CUSTOMER PROCESSING DATE field in the IDF by 1.

      2) If the value in the WORK DAY CODE field is not 0, the Main Settlement Initiator process calculates the customer processing date based on the value in the WORK DAY CODE field and the holidays listed in the HOLIDAYS fields in the Base segment of the IDF. For example, if the institution is not set up to process on holidays or weekends, the customer processing date is set to the next non-weekend, non-holiday date.

   b. Moves the newly computed customer processing date to the CUSTOMER PROCESSING DATE field on screen 17 of the IDF.

3. Determines whether the data in the UAF must be purged for the institution. To do this, the Main Settlement Initiator process performs the following steps:

   a. Verifies that the newly computed customer processing date is equal to or greater than the date in the NEXT BEGINNING DATE field in the Base segment of the IDF. If so, the Main Settlement Initiator process continues with step b. If not, the Main Settlement Initiator process continues with step 4.

   b. Opens the UAF using the file name retrieved in step 1 and purges the data in the file.

   c. Generates a new withdrawal period according to the following situations:

      1) If the value in the WORK DAY CODE field is 0, the Main Settlement Initiator process computes the new beginning date based on the length of the institution's withdrawal period set in the PERIOD LENGTH field in the Base segment of the IDF. It also checks the new next beginning date against the date currently in the NEXT BEGINNING DATE field in the IDF. If these dates are the same, the Main Settlement Initiator process proceeds with step 4.

         If these dates are not the same, the Main Settlement Initiator process moves the date in the NEXT BEGINNING DATE field to the BEGINNING DATE field. It also places the new next beginning date in the NEXT BEGINNING DATE field.

2) If the value in the WORK DAY CODE field is not 0, the Main Settlement Initiator process moves the date in the NEXT BEGINNING DATE field to the BEGINNING DATE field. The new date for the NEXT BEGINNING DATE field is calculated based on the value in the WORK DAY CODE field and the holidays listed in the HOLIDAY fields in the IDF. For example, if the institution is not set up to process on holidays or weekends, the customer processing date is set to the next non-weekend, non-holiday date.

4. Reads the next IDF.

RESET DATES Commands

Restricted use operator commands that cause the Main Settlement Initiator process to reset the posting date in the Institution Definition File (IDF), and the Secondary Settlement Initiator processes to reset the posting dates in the BASE24-pos Terminal Data files and POS Retailer Definition File (PRDF). All RESET DATES commands are sent directly from an XPNET network control facility to the Main Settlement Initiator process. The Main Settlement Initiator process sends interprocess RESET DATES commands to its Secondary Settlement Initiator processes, depending on the file being reset. The file being reset is identified in the syntax of the command.

Syntax

DELIVER PROCESS *object-name,* "RESETIDFDATES *yymmdd"*

DELIVER PROCESS *object-name*, "RESETRDFDATES *yymmdd"*

DELIVER PROCESS *object-name*, "RESETPTDDATES *yymmdd"*

DELIVER PROCESS *object-name,* "RESETALLDATES *yymmdd"*

object-name — The symbolic name assigned to the Main Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

yymmdd — The date (YYMMDD) to which the file posting date is reset.

Processing

All RESET DATES commands are sent directly to the Main Settlement Initiator process. The Main Settlement Initiator process then determines if it must send interprocess commands to its associated Secondary Settlement Initiator processes based on the file being reset.

The Main Settlement Initiator process resets the posting dates in the IDF when it receives a RESETIDFDATES command or a RESETALLDATES command. Since the Main Settlement Initiator process resets the IDF itself, no interprocess commands are required if a RESETIDFDATES command is received.

The Secondary Settlement Initiator processes reset posting dates in the PRDF and BASE24-pos Terminal Data files. When the Main Settlement Initiator process receives a RESETRDFDATES, RESETPTDDATES, or RESETALLDATES command, it sends interprocess RESET DATES commands to all Secondary Settlement Initiator processes. When a Secondary Settlement Initiator process receives one of the RESET DATES commands, it resets the appropriate files.

The following table describes the processing performed by Main and Secondary Settlement Initiator processes when they receive RESET DATES commands.

Command	Processing
RESETIDFDATES	Causes the Main Settlement Initiator process to reset the posting date in the IDF. This message changes the CUR-BUS-DAT field and the CUS-BUS-DAT field in the BASE24-pos segment of the IDF to the date in the command. This message also changes the NXT-BUS-DAT field in the BASE24-pos segment of the IDF to the date following the date in the command, and changes the RPT-BUS-DAT field in the BASE24-pos segment of the IDF to the date prior to the date in the command. If the date in the command is not in the current usage accumulation period, the usage accumulation period dates are reset. In this case, the Main Settlement Initiator process uses the date in the command as the begin date of the usage accumulation period. If the date in the command is within the current usage accumulation period, the Main Settlement Initiator process does not recalculate the usage accumulation period dates. The fields that change when the usage accumulation period is recalculated are the BEG-DAT field and the NXT-BEG-DAT field in the Base segment of the IDF.
RESETALLDATES	Causes the Secondary Settlement Initiator processes to reset the posting dates in the PRDF and BASE24-pos Terminal Data files. In addition, the RESETALLDATES command causes the Main Settlement Initiator process to reset the posting date in the Institution Definition File (IDF).
RESETRDFDATES	Causes the Secondary Settlement Initiator processes to reset the posting dates in the PRDF. This message changes the CUR-POST-DAT field in the PRDF to the date in the command. This message also changes the NXT-POST-DAT field in the PRDF to the posting date following the date in the command.

Command	Processing
RESETPTDDATES	Causes the Secondary Settlement Initiator processes to reset the posting dates in the BASE24-pos Terminal Data files. The process changes the POST-DAT field in the BASE24-pos Terminal Data files to the date in the command.

Example

The following is an example of a RESETPTDDATES command sent to the Main Settlement Initiator process from an XPNET network control facility. In the example, the symbolic name of the Main Settlement Initiator process is P1A^ MAIN1 and the BASE24-pos Terminal Data files posting date is being changed to February 22, 2001.

```
DELIVER PROCESS P1A^MAIN1, “RESETPTDDATES 010222”
```

RESET TIMES Commands

A restricted use operator command that causes the Secondary Settlement Initiator processes to reset the retailer cutover end time in the POS Retailer Definition File (PRDF). All RESET TIMES commands are sent directly from an XPNET network control facility to the Main Settlement Initiator process. The Main Settlement Initiator process sends interprocess RESET TIMES commands to a specific Secondary Settlement Initiator process or to all of its Secondary Settlement Initiator processes, depending on the command issued. The RESET TIMES commands originate from the Main Settlement Initiator process. The new cutover times are identified in the syntax of the command.

Syntax:

DELIVER PROCESS *object-name,* "RESETRDFTIMESALL *old-time new-time"*

DELIVER PROCESS *object-name,* "RESETRDFTIMES *old-time new-time secondary-name"*

object-name — The symbolic name assigned to the Main Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

old-time — The cutover end time currently set in the RETAILER CUTOVER/ BALANCE END field in the PRDF.

new-time — The new cutover end time set in the RETAILER CUTOVER/ BALANCE END field in the PRDF as a result of processing this command.

secondary-name — The symbolic name of the Secondary Settlement Initiator process whose PRDF cutover end time is being changed.

Processing

When the Main Settlement Initiator process receives a RESET TIMES command from an XPNET network control facility, it determines from the command syntax whether the command is to reset all PRDF times (RESETRDFTIMESALL) or whether the PRDF times for a specific Secondary Settlement Initiator process should be reset (RESETRDFTIMES *secondary-name*). The Main Settlement Initiator process sends interprocess RESET TIMES commands to a specific or all Secondary Settlement Initiator processes.

When a Secondary Settlement Initiator process receives a RESET TIMES command, it performs the following steps:

1. Reads each PRDF record within its retailer ID range. If the cutover end time value currently set in the RETAILER CUTOVER/BALANCE END field in the PRDF matches the old time sent in the interprocess command, the Secondary Settlement Initiator process resets the value in the RETAILER CUTOVER/BALANCE END field to the new time contained in the interprocess command.
2. Reinitializes the PRDF extended memory.
    a. Deallocates the existing PRDF extended memory segment.
    b. Determines the segment size of the PRDF, BASE24-pos Terminal Data files, and IDF (if the MHI notify list contains at least one entry, BASE24-pos Terminal Data files extended memory must be rebuilt).
    c. Initializes the PRDF, BASE24-pos Terminal Data files, and IDF extended memory tables.
3. Initializes the timer table.
4. If the current time is after logical network cutover, the Secondary Settlement Initiator process sets the internal *up after cutover* flag to true. This flag causes retailers who have cut over after logical network cutover to have their timers reset to the current day as well as for tomorrow.
5. Sets the retailer timers.
6. Sets the Alive Timer. This step is only performed by the Senior Secondary process as identified in the POS-SETL-SENIOR-SLAVE param in the Logical Network Configuration File (LCONF).
7. Sets the internal *up after cutover* flag to false.

Example

The following is an example of a RESET TIMES command sent to a specific Secondary Settlement Initiator process by the Main Settlement Initiator process. In the example, the symbolic name of the Main Settlement Initiator process is P1A^MAIN1 and the symbolic name of the Secondary Settlement Initiator process is P1A^SEC1. The PRDF cutover end time is being changed from 11:00 a.m. to 2:00 p.m.

```
DELIVER PROCESS P1A^MAIN1, “RESETRDFTIMES 1100 1400
    P1A^SEC1”
```

RESYNC Command

A restricted use operator command that causes the Settlement Initiator processes to cut over all Institution Definition File (IDF), BASE24-pos Terminal Data files, and POS Retailer Definition File (PRDF) records to today's date. The Main Settlement Initiator process cuts over IDF (institution) records and the Secondary Settlement Initiator processes cut over the BASE24-pos Terminal Data files (terminals) and PRDF (retailer) records.

The RESYNC command performs the normal cutover processing necessary to synchronize all records with today's date. This command is used in situations where Settlement has been down for days and is being restarted before POS cutover time. This command works only if cutover has not yet taken place today. If cutover has already taken place, the command will be denied and a message is logged.

The RESYNC command is sent from the Main Settlement Initiator process to all Secondary Settlement Initiator processes.

Syntax

RESYNC

Processing

When the Main Settlement Initiator process receives a RESYNC command from an XPNET network control facility, it cuts over IDF records, changing the date to today's date. The Main Settlement Initiator process sends an interprocess RESYNC command to each of its Secondary Settlement Initiator processes to cut over the BASE24-pos Terminal Data files and PRDF, changing the dates to today's date. This causes all terminals and retailers that are in the retailer ID range for a specific Secondary Settlement Initiator process to reflect today's date. The Secondary Settlement Initiator processes respond when they have completed resynchronizing BASE24-pos Terminal Data files and PRDF records, and the Main Settlement Initiator process logs an event message indicating that all Secondary Settlement Initiator processes have completed resynchronization.

TRACEOFF Command

A restricted use operator command that deactivates the trace function within a specified Settlement Initiator process. If the optional *secondary-name* variable is included in the command, the Main Settlement Initiator process sends an interprocess TRACEOFF command to the specified Secondary Settlement Initiator process, and the trace function is deactivated for the Secondary Settlement Initiator process. If the *secondary-name* variable is not included, the trace function for the Main Settlement Initiator process is deactivated. For more information on the TRACEOFF command, refer to appendix A.

Syntax

DELIVER PROCESS *object-name,* "TRACEOFF[*secondary-name*]"

Note: There is no space between TRACEOFF and the variable *secondary-name*.

object-name — The symbolic name assigned to the Main or Secondary Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

[*secondary-name*] — The symbolic name of the Secondary Settlement Initiator process for which the trace function is being deactivated.

TRACEON Command

A restricted use operator command that activates the trace function within a specified Settlement Initiator process. If the optional *secondary-name* variable is included in the command, the Main Settlement Initiator process sends an interprocess TRACEON command to the specified Secondary Settlement Initiator process, and the trace function is activated for the Secondary Settlement Initiator process. If the *secondary-name* variable is not included, the trace function for the Main Settlement Initiator process is activated. For more information on the TRACEON command, refer to appendix A.

Syntax

DELIVER PROCESS *object-name,* “TRACEON[*secondary-name*]”

Note: There is no space between TRACEON and the variable *secondary-name*.

object-name — The symbolic name assigned to the Main or Secondary Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

[*secondary-name*] — The symbolic name of the Secondary Settlement Initiator process for which the trace function is being activated.

WARMBOOT ALL Command

A general use operator command that causes the Main and Secondary Settlement Initiator process to be warmbooted. Both the Main and Secondary Settlement Initiator processes must be warmbooted when the Institution Definition File (IDF) has been modified, an LCONF assign or param used by both the Main and Secondary Settlement Initiator processes has been modified, or the Secondary Settlement Initiator process table must be rebuilt due to a change in the high or low retailer key values for a Secondary Settlement Initiator process partition. The Main Settlement Initiator process then sends interprocess WARMBOOT commands to all of its associated Secondary Settlement Initiator processes.

Syntax

DELIVER PROCESS *object-name,* “9501ALL”

DELIVER PROCESS *object-name*, “9501****”

object-name — The symbolic name assigned to the Main Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

When both the Main and Secondary Settlement Initiator processes are warmbooted, the processes perform as they would during initialization, except they do not retrieve a new Logical Network Configuration File (LCONF) name and they close and open any files that were opened during initialization.

Example

The following is an example of a WARMBOOT ALL command sent from an XPNET network control facility to a Main Settlement Initiator process. In the example, the symbolic name of the Main Settlement Initiator process is P1A^ MAIN3.

DELIVER PROCESS P1A^MAIN3, “9501ALL”

WARMBOOT MAIN Command

A general use operator command that causes only the Main Settlement Initiator process to be warmbooted. The Main Settlement Initiator process must be warmbooted when an LCONF assign or param that only the Main Settlement Initiator process uses has been modified. For example, when the POS-SETL-PTLF-FILE-RETENTION or POS-LN-CUTOVER-LEAD-TIME param is changed, the Main Settlement Initiator process must be warmbooted.

Syntax

DELIVER PROCESS *object-name,* “9501MASTER”

object-name — The symbolic name assigned to the Main Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

When only the Main Settlement Initiator process is being warmbooted, the Main Settlement Initiator process performs as it would during initialization, with the following exceptions:

- It does not send interprocess warmboot messages to the Secondary Settlement Initiator processes and, therefore, the Secondary Settlement Initiator processes do not go through a warmboot procedure.
- It does not clear or rebuild the Secondary Settlement Initiator process table.
- It does not retrieve a new LCONF name.
- It closes and reopens any files it opened during initialization.

Example

The following is an example of a WARMBOOT MAIN command. In the example, the Main Settlement Initiator process being warmbooted is P1A^MAIN1.

DELIVER PROCESS P1A^MAIN1, “9501MASTER”

WARMBOOT NOTIFY Command

A general use operator command that directs the Main Settlement Initiator process to rebuild the settlement notify and settlement midnight tables. The Settlement Notify table identifies the names of the Device Handler/Router/Authorization processes the Main Settlement Initiator process must notify at logical network cutover. The Midnight Cutover table identifies the names of the Device Handler/Router/Authorization processes the Main Settlement Initiator process must notify at midnight network cutover.

The settlement and midnight notify tables are built by the Main Settlement Initiator process at initialization based on the POS-SETL-NOTIFY and POS-SETL-MIDNIGHT params in the LCONF. When setting these params, all applicable Device Handler/Router/Authorization processes must be contained in this list. The BASE24 system does not verify that all applicable processes are included in the list. If these LCONF params are modified, the tables must be rebuilt to reflect the changes. An operator enters a WARMBOOT NOTIFY command from an XPNET network control facility to update the Settlement and Midnight Notify tables.

Syntax

DELIVER PROCESS *object-name,* "9501NOTIFY"

object-name — The symbolic name assigned to the Main Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

When the Main Settlement Initiator process receives a WARMBOOT NOTIFY command, it warmboots only the Settlement Notify and Midnight Notify tables. No other processing associated with warmbooting is performed (i.e., reinitializing extended memory, closing and reopening files, etc.). The Main Settlement Initiator process performs the following steps when it receives a WARMBOOT NOTIFY command:

1. Retrieves all entries for the POS-SETL-NOTIFY and POS-SETL-MIDNIGHT params from the LCONF.
2. Checks to ensure the maximum number of process entries allowed is not surpassed. The maximum number of processes that can be specified for each param is 150.

   If this number is surpassed, the Main Settlement Initiator process logs an event message indicating the condition and abends.

3. Rebuilds the Settlement and Midnight Notify tables based on the POS-SETL-NOTIFY and POS-SETL-MIDNIGHT param information in the LCONF.

WARMBOOTRDFKEY Command

A general use command that causes the Main Settlement Initiator process to send an interprocess command notifying Secondary Settlement Initiator processes to update their internal PRDF extended memory tables for the retailer ID specified in the command.

Syntax

DELIVER PROCESS *object-name,* “9501RDFKEY *retailer-key”*

object-name — The symbolic name assigned to the Main Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

retailer-key — The ID of the retailer whose PRDF table entry will be updated.

Processing

When the Main Settlement Initiator process receives a WARMBOOTRDFKEY command from an XPNET network control facility, it sends an interprocess command to its Secondary Settlement Initiator processes indicating that the PRDF table entry for the specified retailer ID requires updating. This command is sent only when a PRDF table entry requires updating.

The Main Settlement Initiator process receives this command from an XPNET network control facility and searches through its internal Secondary Settlement Initiator process table for a match on the retailer key identified in the command. When a match is found, the Main Settlement Initiator process sends the interprocess WARMBOOTRDFKEY command to the Secondary Settlement Initiator process. When the Secondary Settlement Initiator process receives this interprocess command, it compares the cutover time and cutover flag in the PRDF record to the cutover time and cutover flag in the internal PRDF extended memory table for the specified retailer. Depending on the result of these comparisons, the Secondary Settlement Initiator process then performs the following processing:

- If the cutover times are not the same in the PRDF record and PRDF extended memory table, and the cutover flag is set to indicate retailer cutover has not occurred for the current day, the Secondary Settlement Initiator process logs an event message and does not warm boot its internal PRDF extended memory entry for the specified retailer ID.

- If the cutover times in the PRDF record and PRDF extended memory table are the same or different and the cutover flag is set to indicate that retailer cutover has occurred for the current day, the Secondary Settlement Initiator process warmboots its internal PRDF extended memory table entry with the new information from the PRDF record.

Example

The following is an example of a WARMBOOTRDFKEY command sent to the Main Settlement Initiator process for a Secondary Settlement Initiator process. In the example, the symbolic name of the Main Settlement Initiator process is P1A^ MAIN3 and the retailer key is PSRTLR11.

DELIVER PROCESS P1A^MAIN3, “9501RDFKEYPSRTLR11”

WARMBOOT SECONDARY Command

A general use operator command that causes a specific Secondary Settlement Initiator process to perform warmboot processing. A specific Secondary Settlement Initiator process must be warmbooted when a single POS Retailer Definition File (PRDF) or POS Terminal Data File (if the terminal is an MHI device in host mode) record has been modified from a file maintenance screen. Only the Secondary Settlement Initiator process that uses the partition that contains the modified record must be warmbooted. If the optional *secondary-name* variable is included in the command, the Main Settlement Initiator process sends an interprocess WARMBOOT command to the specified Secondary Settlement Initiator process, and the specified Secondary Settlement Initiator process performs warmboot processing. If the *secondary-name* variable is not included, the *object-name* variable must contain the name of the Secondary Settlement Initiator process, and the WARMBOOT command is sent directly to the Secondary Settlement Initiator process.

Syntax

DELIVER PROCESS *object-name,* “9501[*secondary-name*]”

Note: There is no space between 9501 and the variable *secondary-name*.

object-name — The symbolic name assigned to the Main or Secondary Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

[*secondary-name*] — The symbolic process name of the Secondary Settlement Initiator process being warmbooted.

Processing

When a specific Secondary Settlement Initiator process is warmbooted, the Main Settlement Initiator process sends an interprocess WARMBOOT command to the specified Secondary Settlement Initiator process. The Secondary Settlement Initiator process then performs as it would during initialization, except that it does not retrieve a new LCONF name and it closes and reopens any files it opened during initialization. The Main Settlement Initiator process does not perform any of its initialization or warmboot processing besides sending the warmboot message to the specified Secondary Settlement Initiator process.

Transaction Context Manager Process Commands

The BASE24-pos Transaction Context Manager (TCM) module can receive a number of BASE24 commands that cause it to perform specialized processing. The commands used by the BASE24-pos Transaction Context Manager module can be separated into three categories: process-generated commands, general use operator commands, and restricted use operator commands. All BASE24 general and restricted use operator commands that can be received by the BASE24-pos TCM module are documented in this subsection. Process-generated commands are documented in appendix B.

All of the commands that can be received by the BASE24-pos Transaction Context Manager module are listed below by category.

Process-Generated Commands	General Use Operator Commands	Restricted Use Operator Commands
LOGICAL NETWORK CUTOVER	ALTERATTRIBUTE WARMBOOT WARMBOOT MOF WARMBOOT PTLF	DEBUG DUMP

ALTERATTRIBUTE Command

A general use operator command that causes the Transaction Context Manager process to warmboot (reallocate) a specified shared extended memory table after it has been rebuilt. This command is used by SPDH Device Handler module. Whenever a shared extended memory table is modified, all processes that read the table must reallocate the table before the modified data is used in transaction processing. The Process Control Terminal (PCT) Server process automatically sends this command to each process specified in an EMT warmboot notify list param in the Logical Network Configuration File (LCONF) when a network operator enters an ALTERATTRIBUTE command from the EMT Control Command screen (screen 5) in the Device Control Terminal (DCT). You can also enter the ALTERATTRIBUTE command manually from a network control facility to warmboot an extended memory table for an individual process. The EMT warmboot notify list params are only used when issuing the command from the DCT. For more information on the EMT Control Command screen, refer to the ***BASE24 Device Control Manual***.

Syntax

DELIVER PROCESS *object-name,* "ALTERATTR[IBUTE]*file-id*EMT"

object-name — The symbolic name assigned to the Transaction Context Manager process in the XPNET node. For more information on entering the object name, refer to section 1.

file-id — The acronym of the file from which the shared extended memory table is built. Valid values are as follows:

STRF = Split Transaction Routing File

Processing

This command causes the Transaction Context Manager process to reallocate a specified shared extended memory table. If the command completes successfully, an event message stating that the EMT warmboot completed successfully is displayed on the network log.

Examples

The following are examples of ALTERATTRIBUTE commands sent to a Transaction Context Manager process from an XPNET network control facility. In these examples, the symbolic name of the Transaction Context Manager process is P1A^PTCM1 and the shared extended memory table to be warmbooted is the Split Transaction Routing File (STRF) extended memory table.

DELIVER PROCESS P1A^PTCM1, “ALTERATTRIBUTESTRFEMT”

DELIVER PROCESS P1A^PTCM1, “ALTERATTRSTRFEMT”

DEBUG Command

A restricted use operator command that places the Transaction Context Manager process under control of the HP NonStop Inspect product and displays an Inspect prompt on the home terminal (the terminal specified for the process by the HOMETERM attribute or the home terminal of the XPNET process). Inspect is an HP NonStop program debugging tool.

Syntax

DELIVER PROCESS *object-name,* "DEBUG"

object-name — The symbolic name assigned to the Transaction Context Manager process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

An operator enters this command from an XPNET network control facility. For more information on the DEBUG command, refer to appendix A.

DUMP Command

The DUMP command directs the Transaction Context Manager process to request a programmatic dump.

Syntax

DELIVER PROCESS *object-name,* "DUMP"

object-name — The symbolic name assigned to the process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

A programmatic dump allows a programmer to automatically capture an unusual situation for evaluation. The Transaction Context Manager can continue running after the dump is generated.

The Transaction Context Manager process performs the following processing steps when it receives a DUMP command from an XPNET network control facility.

1. Creates a dump file containing an image of the process' environment.
2. Logs messages to indicate the status of the programmatic dump.
3. At this point, an operator can read and analyze the dump file. The dump file contains results of the programmatic dump. Information from the dump file can be printed for further analysis.

WARMBOOT Command

A general use operator command that causes the Transaction Context Manager process to reinitialize.

Syntax

DELIVER PROCESS *object-name,* “9501****”

object-name — The symbolic name assigned to the Transaction Context Manager process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

Upon receiving this command, the Transaction Context Manager process opens, reads, and closes all files it read during initialization. Any file information retained in extended memory at initialization is written over in extended memory when the particular files are reread during the warmboot.

When the warmboot processing is complete, an event message is displayed on the network log.

WARMBOOT MOF Command

A general use operator command that causes the Transaction Context Manager process to reinitialize the Mobile Operator File (MOF).

Syntax

DELIVER PROCESS *object-name,* “9529MOF”

object-name — The symbolic name assigned to the Transaction Context Manager process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

When the Transaction Context Manager module receives a 9529MOF command, it reinitializes the MOF. This includes opening the file, allocating space in extended memory, and closing the file.

When warmboot processing is complete, an event message is displayed on the nwtwork log.

An operator sends this command from an XPNET network control facility.

WARMBOOT PTLF Command

A general use operator command that causes the Transaction Context Manager process to reinitialize the POS Transaction Log File (PTLF).

Syntax

DELIVER PROCESS *object-name,* "9530PTLF"

object-name — The symbolic name assigned to the Transaction Context Manager process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

When the Transaction Context Manager module receives a 9530PTLF command, it reinitializes the PTLF. This includes opening the file, allocating space in extended memory, and closing the file.

When warmboot processing is complete and event message is displayed on the network log.

An operator sends this command from an XPNET network control facility.

6: BASE24-teller Process Commands

This section contains the general and restricted operator use commands received by the following BASE24-teller processes:

- Authorization
- Settlement Initiator

The processes are presented in alphabetical order. Furthermore, specific commands are also documented in alphabetical order. The documentation for each command includes the following information:

- A description of the command
- The syntax of the command
- The processing steps taken when the command is received
- An example of the command (when necessary)

Authorization Process Commands

The Authorization process authorizes the transactions it receives or routes them to the appropriate BASE24-teller Host Interface process. The Authorization process logs all transactions it receives to the Teller Transaction Log File (TTLF), and generates completion messages to notify hosts regarding transactions approved or declined by BASE24-teller.

This section describes the processing performed by the BASE24-teller Authorization process. The BASE24-teller Authorization process can receive a number of BASE24 commands that cause it to perform specialized processing. These commands can be separated into three categories: process-generated, general use operator, and restricted use operator commands. All BASE24 general and restricted use operator commands that can be received by the BASE24-teller Authorization process are documented in this subsection. Process-generated commands are documented in appendix B.

The table below lists all of the commands that can be received by the BASE24-teller Authorization process and identifies the category under which the command is described.

Process-Generated Commands	General Use Operator Commands	Restricted Use Operator Commands
CUTOVER EXTRACT FIID CUTOVER REFRESH	ASSIGN FILES IDF TABLES PARAM STATUS WARMBOOT ALL WARMBOOT FIID	DEBUG TRACE OFF TRACE ON TRACE OVERRIDE TRACE TOKEN RECORD

DEBUG Command

A restricted use operator command that places the Authorization process into the HP NonStop Inspect and displays an Inspect prompt on the home terminal (the terminal specified for the process by the HOMETERM attribute or the home terminal of the XPNET process). Inspect is an HP NonStop programming tool. An operator enters this command from an XPNET network control facility. For more information on the DEBUG command, refer to appendix A.

FILES Command

A general use operator command that identifies the files and corresponding file numbers from the file table stored in the extended memory of the Authorization process. This information is displayed in a series of event messages.

Syntax

DELIVER PROCESS *object-name,* “FILES”

object-name — The symbolic name assigned to the Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

The Authorization process logs the following messages for each file name and corresponding file number from the files table when it receives a FILES command. The messages identify the file name and number.

```
9008  0  auth-process ---> file-name
9004  0  auth-process ---> file-number
```

These event messages are described in the TRAUTH log message documentation file.

IDFTBL Command

A general use operator command that causes the Authorization process to log miscellaneous information by FIID on its current processing parameters in the Institution Definition File (IDF) to the EMS collector process.

Syntax

DELIVER PROCESS *object-name,* “IDFTBL *fiid”*

object-name — The symbolic name assigned to the Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

fiid — The FIID of the institution for which the IDF status is being performed.

Processing

Information included when this command is sent is the Authorization process’s current and next processing dates, the current status of the IDF, and the values of the specific BASE24-teller fields set up in the BASE24-teller segment of the IDF. The Authorization process performs the following processing steps when it receives an IDFTBL command:

1. Searches its IDF extended memory segment by FIID for current processing parameters for the FIID. The IDF table includes the current status of the IDF including specific BASE24-teller parameters.
2. Logs the following event message concerning the IDF status for each FIID:

   `9003 0 IDF table entries for` *`FIID`* `are as follows:`

   The IDF status information for the specified FIID is displayed in event messages as follows:

   `9005 0 Refresh group` *`refresh-group-ID`*

   `9005 0 Institution ID` *`FIID`*

   `9005 0 Currency code` *`currency-code`*

   `9005 0 Exp date chk` *`expiration-date-check-value`*

   `9005 0 PIN chk` *`PIN-check-code`*

   `9005 0 Card hold select` *`cardholder-select-value`*

   `9005 0 Bad PIN disp` *`PIN-disposition-code`*

   `9005 0 Exp check disp` *`expiration-check-disposition-code`*

   `9005 0 DDA refresh flag` *`dda-refresh-flag`*

```
9005 0 SAV refresh flag sav-refresh-flag
9005 0 CCD refresh flag ccd-refresh-flag
9004 0 Fatal error error
9006 0 ORF address address
9004 0 ORF count count
9006 0 TTF address address
9004 0 TTF count count
9004 0 CAF number number
9004 0 DDA number number
9004 0 SAV number number
9004 0 CCD number number
9004 0 SPF number number
9004 0 NBF number number
9004 0 WHFF number number
9004 0 Cur TTLF number number
9004 0 Next TTLF number number
```

If the specified FIID is invalid, the Authorization process logs the following event message:

```
9007 2 Unable to find FIID FIID in IDF table
```

All of the above event messages are described in the TRAUTH log message documentation file.

ASSIGN Command

A general use operator command that causes assign information from the Logical Network Configuration File (LCONF) to be displayed in a series of event messages. The assign information displayed is the physical file name that corresponds to the applicable assign in the LCONF.

Syntax

DELIVER PROCESS *object-name,* "ASSIGN"

object-name — The symbolic name assigned to the Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

The Authorization process performs the following steps when it receives the ASSIGN command:

1. Accesses LCONF assign table information. The assign table contains LCONF assign information for the following files:
   - Account Routing File (ARF)
   - Card Prefix File (CPF)
   - Institution Definition File (IDF)
   - Key Authorization File (KEYA)
   - Log Message Configuration File (LMCF)
   - Override Response File (ORF)
   - Token File (TKN)
   - Teller Transaction File (TTF)
   - Teller Transaction Log File (TTLF)
2. Retrieves the physical names of the files corresponding to the LCONF assigns.

3. Logs the following event message indicating the assigns and corresponding file name, as defined in the LCONF:

   `9000  0  LCONF Assign` *`assign-name`* `---> Physical File:` *`file-name`*

PARAM Command

A general use operator command that causes param information from the Logical Network Configuration File (LCONF) to be displayed in a series of event messages. The PARAM command displays the param value that corresponds to the applicable param in the LCONF.

Syntax

DELIVER PROCESS *object-name,* "PARAM"

object-name — The symbolic name assigned to the Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

The Authorization process performs the following steps when it receives the PARAM command:

1. Accesses LCONF param table information. The param table contains LCONF param values for the following params:
   - CPF-REC-SIZE
   - LOGICAL-NET
   - TLR-PROCESS-CRD-PIN
   - TLR-MAX-NBF-REC-CNT
   - TLR-TTLF-NOTIFY-INTERVAL
   - TLR-TTLF-NOTIFY-PERCENT
   - TLR-WRITE-HLDS-DEP-FLOATS
2. Retrieves the param values corresponding to the LCONF params.
3. Logs the following event messages indicating the param and its corresponding param value, as defined in the LCONF.

   ```
   9001  0  LCONF Param param-name --->  param-value
   9002  0  LCONF Param param-name --->  param-value
   ```

STATUS Command

A general use operator command that causes the Authorization process to send miscellaneous information on its current processing parameters to the EMS collector process.

Syntax

DELIVER PROCESS *object-name,* "STATUS"

object-name — The symbolic name assigned to the Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

The Authorization process performs the following processing steps when it receives this command:

1. Reads the command.
2. Accesses its current processing parameters for information including the Authorization process number; PPD name; routing code; last transaction ID; segment IDs for the ARF, CPF, IDF, KEYA, ORF, TKN, and TTF; and record counts for the ARF, IDF, KEYA, and TKN files.
3. Logs the following event message describing the field and the actual value:

   ```
   9004  0  field-name ---> value
   ```

   ```
   9005  0  auth-process ---> PPD-name
   ```

   The 9005 event message is logged only to identify the PPD name. All other status event messages are logged as 9004 event messages.

TRACE ON and TRACE OFF Commands

Restricted use operator commands that activate or deactivate a trace function within the Authorization process. When activated, the trace function generates an event message each time the Authorization process enters a new procedure. The event message displays the name of the procedure from which it was written. An operator enters this command from an XPNET network control facility. For more information on the TRACE ON and TRACE OFF commands, refer to appendix A.

TRACE OVRRD Command

A restricted use operator command that activates or deactivates a trace function to monitor override processing.

Syntax

DELIVER PROCESS *object-name,* "TRACE OVRRD { ON | OFF }"

object-name — The symbolic name assigned to the Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

{ ON | OFF } — Indicates whether the trace function to monitor override processing is activated (ON) or deactivated (OFF).

TRACE TKNREC Command

A restricted use operator command that activates or deactivates a trace function related to the Token File (TKN) record used for logging.

Syntax

DELIVER PROCESS *object-name,* “TRACE TKNREC { ON | OFF }”

object-name — The symbolic name assigned to the Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

{ ON | OFF } — Indicates whether the trace function related to the Token File (TKN) record used for logging is activated (ON) or deactivated (OFF).

WARMBOOT ALL Command

A general use operator command that reinitializes the Authorization process.

Syntax

DELIVER PROCESS *object-name,* “9501****”

object-name — The symbolic name assigned to the Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

When Authorization processes are warmbooted, they do not go through their entire initialization routine. Instead, they perform the following steps:

1. Reread the Logical Network Configuration File (LCONF).
2. Close and reopen the files they use.
3. Release the current extended memory.
4. Close and reopen the security module.
5. Reinitialize the Log Message Configuration File (LMCF).
6. Reallocate new extended memory based on the current settings in the LCONF.

WARMBOOT FIID Command

A general use operator command that causes the Authorization process to update the current information that it has in memory for an institution.

Syntax

DELIVER PROCESS *object-name,* "9501*fiid*"

object-name — The symbolic name assigned to the Authorization process in the XPNET node. For more information on entering the object name, refer to section 1.

fiid — The the institution for which current information is updated by the Authorization process. The Authorization process updates the information in its memory for the identified institution.

Processing

When the Authorization process receives a WARMBOOT FIID command, the processing steps it performs are similar to the steps it performs during initialization. Therefore, only the differences are discussed here.

Note: The WARMBOOT FIID command does not cause the Authorization process to reread any files into memory except for the IDF record for the institution. To update this information, the WARMBOOT ALL command (9501****) must be performed or the process must be stopped and restarted. Also, the WARMBOOT FIID command cannot be used to add an IDF record to memory. Again, this must be done using the WARMBOOT ALL command or by stopping and restarting the Authorization process.

The Authorization process performs the following processing steps when it receives a WARMBOOT FIID command:

1. Locates the IDF record for the FIID in extended memory.
2. Reads the corresponding IDF record from the disk file.

   If a record cannot be found or if the institution is marked with a fatal error, the Authorization process generates an event message indicating that the command cannot be processed.

3. Closes any data files open for the institution.

   Using the indexes in the IDF table, the Authorization process identifies the files (CAF, PBF, SPF, NBF, and WHFF) open for the institution. If any of the files are opened exclusively for the institution, the Authorization process closes the files.

4. Closes the TTLFs for this institution.

5. Overlays the IDF record in extended memory with the new IDF record from disk field-by-field and sets the FATAL ERROR field in the IDF to indicate no error occurred.

6. Opens the files required for processing if they are not already open and updates the indexes in the IDF table.

7. Edits the IDF record fields. To do this, the Authorization process follows the steps taken at initialization.

8. Updates the KEYA pointer in the IDF table entry.

   This information is updated only if there is a KEYA currently in memory for the institution and the PIN-VRFY-TYP field in the Base segment of the IDF is not 0. If a KEYA record does not exist in memory for the institution, the Authorization process uses the old KEYA address from the IDF table entry. If an error is found, the Authorization process generates an event message and sets a fatal error for the institution.

9. Generates the following event message:

   ```
   6009  0  Proc auth-process Warmboot for FIID FIID completed.
   ```

Settlement Initiator Process Commands

The Settlement Initiator process initiates processing at prescribed times each day to carry out cutover and settlement for each institution in the logical network. This section describes the processing performed by the BASE24-teller Settlement Initiator process. It includes descriptions of the BASE24 commands the Settlement Initiator process recognizes.

The BASE24-teller Settlement Initiator process can receive a number of BASE24 commands that cause it to perform specialized processing. These commands can be separated into three categories: process-generated, general use operator commands, and restricted use operator commands. All BASE24 general and restricted use operator commands that can be received by the BASE24-teller Settlement Initiator process are documented in this subsection. Process-generated commands are documented in appendix B.

The table below lists all of the commands that can be received by the BASE24-teller Settlement Initiator process and identifies the category under which the command is described.

Process-Generated Commands	General Use Operator Commands	Restricted Use Operator Commands
None	ASSIGN PARAM STATUS WARMBOOT ALL	DEBUG TRACE OFF TRACE ON

ASSIGN Command

A general use operator command that causes assign information from the Logical Network Configuration File (LCONF) to be displayed. The assign information displayed is the physical file name that corresponds to the applicable assign in the LCONF.

Syntax

DELIVER PROCESS *object-name,* "ASSIGN"

object-name — The symbolic name assigned to the Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

The Settlement Initiator process performs the following steps when it receives an ASSIGN command:

1. Accesses the internal LCONF assign table containing LCONF assign information for the following files:
   - Institution Definition File (IDF)
   - Log Message Configuration File (LMCF)
   - Teller Transaction Log File (TTLF)
2. Provides the fully-expanded physical names of all the files corresponding to the LCONF assigns.
3. Logs the following event message indicating the assign and corresponding file name, as defined in the LCONF.

   `0061  LCONF Assign` *`assign-name`* `===> Physical file` *`file-name`*

   If the LCONF assign cannot be found or is invalid, the following message is logged:

   `9998  3  PROGRAM ERROR: No processing routine for` *`process-name`* `table entry` *`number`*

   The above event messages are described in the TRSETL log message documentation file.

DEBUG Command

A restricted use operator command that places the Settlement Initiator into the HP NonStop Inspect and displays an Inspect prompt on the home terminal (the terminal specified for the process by the HOMETERM attribute or the home terminal of the XPNET process). Inspect is an HP NonStop programming tool. An operator enters this command from an XPNET network control facility. For more information on the DEBUG command, refer to appendix A.

PARAM Command

A general use operator command that causes param information from the Logical Network Configuration File (LCONF) to be displayed in a series of event messages. The PARAM command displays the param value that corresponds to the applicable param in the LCONF.

Syntax

DELIVER PROCESS *object-name,* "PARAM"

object-name — The symbolic name assigned to the Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

The Settlement Initiator process performs the following steps when it receives a PARAM command:

1. Accesses the internal LCONF param table that contains values for the following params:
   - TLR-SETL-MASTER
   - TLR-LN-CUTOVER
   - TLR-SETL-TTLF-FILE-RETENTION
2. Provides the param values corresponding to all of the LCONF params.
3. Logs the following event messages indicating the param and its corresponding param value, as defined in the LCONF.

   ```
   0032  0  LCONF Param param-name --->  param-name
   ```

   ```
   0062  0  LCONF Param param-name --->  param-value
   ```

   If the LCONF param cannot be found or is invalid, the following message is logged:

   ```
   9998  3  PROGRAM ERROR: No processing routine for process-name
   table entry number
   ```

   The above event messages are described in the TRSETL log message documentation file.

TRACE ON and TRACE OFF Commands

Restricted use operator commands that activate or deactivate a trace function within the Settlement Initiator process. When the trace function is activated, it generates an event message each time the Settlement Initiator process enters a new procedure. The event message displays the name of the procedure from which it was written. For more information on the TRACE ON and TRACE OFF commands, refer to appendix A.

WARMBOOT ALL Command

A general use operator command that reinitializes the Settlement Initiator process. An operator sends this command from an XPNET network control facility.

Syntax

DELIVER PROCESS *object-name,* “9501****”

object-name — The symbolic name assigned to the Settlement Initiator process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

The processing steps taken by the Settlement Initiator process during a WARMBOOT ALL command are similar to the steps taken during initialization. However, there are some differences. These differences are described here. The Settlement Initiator process performs the following processing steps for a WARMBOOT ALL command:

1. Closes the IDF.
2. Opens the Logical Network Configuration File (LCONF) and retrieves the assigns and params it requires. Some assigns and params are required for processing, and the Settlement Initiator process abends if they are not present.
3. Reinitializes the logging facility. If the Settlement Initiator process is unable to reinitialize the logging facility, the Settlement Initiator process abends.
4. Opens the Institution Definition File (IDF) with Read/Write access. If the Settlement Initiator process is unable to open the IDF, the Settlement Initiator process abends.
5. Determines whether the logical network cutover has passed for the current day. To do this, the Settlement Initiator process checks the current time against the TLR-LN-CUTOVER param from the LCONF and performs according to the following situation.

   If the logical network cutover time has passed, the Settlement Initiator process increments its own dates for the current day and the next day by one. The Settlement Initiator process then verifies that all IDF records are operating on these new dates. If cutover has passed and an IDF is not operating, the Settlement Initiator process performs institution cutover for these IDFs.
6. Reinitializes its timer tables in internal memory. This is the memory area in which the Settlement Initiator process places its timers.

7. Sets its timers for processing.
8. Generates the following event message:

```
0035  0  Warmboot Completed.
```

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A: Restricted Use Operator Commands

This appendix contains descriptions of restricted use operator commands common to several processes across the BASE24 product lines. These universal restricted use operator commands are identified within the appropriate process section and readers are directed to this appendix for more information on the command. Each restricted use operator command documented in this appendix includes a description of the command, the command syntax, and a command example.

Operators should not enter these commands unless requested to do so by ACI personnel. In addition, care must be taken when using these commands since these commands allow programmers to test functionality and determine how a BASE24 process is running when changes are made to the code. The following restricted use operator commands are included in this appendix:

- DEBUG
- DUMP
- TRACE OFF
- TRACE ON

DEBUG Command

A restricted use operator command that places the BASE24 process into the HP NonStopHP Inspect product and displays an Inspect prompt on the home terminal (the terminal specified for the process by the HOMETERM attribute or the home terminal of the XPNET process). Inspect is an HP NonStop programming tool.

Syntax

DELIVER PROCESS *object-name,* “*id-list* DEBUG”

object-name — The symbolic name assigned to the process in the XPNET node. For more information on entering the object name, refer to section 1.

id-list — Identifies one or more processes for which the Debug status is set.

Processing

When the BASE24 process receives the DEBUG command, it performs the following processing steps:

1. Reads the message.
2. Places the process under control of the HP NonStop Inspect product and returns an Inspect prompt to the home terminal.

   For more information on Inspect, refer to the appropriate HP NonStop documentation.

Example

The following example sets the Debug status for BASE24-atm Authorization process P1A^AUTH2 to true.

DELIVER PROCESS P1A^AUTH2, “DEBUG”

DUMP Command

The DUMP command directs the BASE24 process to request a programmatic dump.

Syntax

DELIVER PROCESS *object-name,* "DUMP"

object-name — The symbolic name assigned to the process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

A programmatic dump allows a programmer to automatically capture an unusual situation for evaluation. The BASE24 process can continue running after the dump is generated.

The BASE24 process performs the following processing steps when it receives a DUMP command from an XPNET network control facility.

1. Creates a dump file containing an image of the process' environment.
2. Logs messages to indicate the status of the programmatic dump.
3. At this point, an operator can read and analyze the dump file. The dump file contains results of the programmatic dump. Information from the dump file can be printed for further analysis.

Example

The following is an example of a DUMP command. In the example, the Enscribe Refresh process P1A^REFR1 is being dumped.

DELIVER PROCESS P1A^REFR1, "DUMP"

TRACE OFF Command

A restricted use operator command that turns off the trace function within a BASE24 process. For a description of the trace function, refer to the TRACE ON command description.

Syntax

DELIVER PROCESS *object-name,* "TRACE OFF"

object-name — The symbolic name assigned to the process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

The BASE24 process performs the following processing steps when it receives a TRACE OFF command:

1. Reads the command.
2. Deactivates the trace function within the process.

Example

The following is an example of a TRACE OFF command. The BASE24-pos Device Handler/Router/Authorization process for which the command is being entered is P1A^RTAU1.

DELIVER PROCESS P1A^RTAU1, "TRACE OFF"

TRACE ON Command

A restricted use operator command that activates a trace function within a BASE24 process.

Syntax

DELIVER PROCESS *object-name,* “TRACE ON”

object-name — The symbolic name assigned to the process in the XPNET node. For more information on entering the object name, refer to section 1.

Processing

Generally, the trace function is intended for use in trouble-shooting. The TRACE command is used differently by each process. For example, some processes cause the trace function to generate an event message each time the BASE24 process sets a response code, indicating the octal address within the program itself where the response code was set. Other processes cause the trace function to identify the procedures that are called from the code.

The BASE24 process performs the following processing steps when it receives a TRACE ON command:

1. Reads the command.
2. Activates the trace function within the BASE24 process.

Example

The following is an example of a TRACE ON command. The BASE24-pos Device Handler/Router/Authorization process for which the command is being entered is P1A^RTAU1.

DELIVER PROCESS P1A^RTAU1, “TRACE ON”

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B: Process-Generated Commands

Process-generated commands are generated by processes running in the system and sent to other processes running in the system. They are used to notify the process receiving the command of actions occurring, or that have already occurred, in other processes in the system. In general, process-generated commands notify a specific process about events occurring elsewhere in the system.

This section contains the process-generated commands generated for the following BASE24 products:

- BASE24 Base and Shared
- BASE24-atm
- BASE24 Customer Service
- BASE24-pos
- BASE24-telebanking
- BASE24-teller

Within each product, the commands are organized by process. All processes associated with a particular BASE24 product are documented in alphabetical order within that BASE24 product. In addition, BASE24 process-generated commands are documented in alphabetical order within the process.

Each process-generated command is organized using three main elements. The first element is a description of the command. The description explains the purpose of the command and its intended use. The description is followed by the syntax of the command. The command syntax indicates the format of the command when it is received by a process. Next, the processing steps describe the actions the process takes when it receives the command.

Base and Shared Process Commands

This subsection contains the process-generated commands received by the following Base or shared processes:

- ANSI Host Interface
- BIC ANSI Interface
- BIC ISO Interface
- From Host Maintenance
- ISO Host Interface
- Enscribe Refresh
- Super Extract

ANSI Host Interface Process Commands

This subsection describes the BASE24 process-generated commands the ANSI Host Interface process recognizes. The ANSI Host Interface process receives the following process-generated commands:

- CUTOVER
- DEVICE STATE
- SAFEXTCOMP
- SAFEXTRQST

CUTOVER Command

A process-generated command that notifies the ANSI Host Interface process that logical network cutover and midnight cutover have taken place.

Syntax

9502 *yymmdd product-identifier*

yymmdd — The new posting date sent to the ANSI Host Interface process by the Settlement Initiator process.

product-identifier — Identifies the product for which the CUTOVER command was sent. Valid values are as follows:

ATM = BASE24-atm
POS = BASE24-pos

Processing

This command initiates logical network cutover and originates from the BASE24-atm or BASE24-pos Settlement Initiator processes. The CUTOVER command updates the BASE24-atm or BASE24-pos posting date maintained by the ANSI Host Interface process. The Settlement Initiator process sends a new posting date (yymmdd) each day at logical network cutover.

When the ANSI Host Interface process receives a CUTOVER command, it updates the posting date in the Host Configuration File (HCF) to the next day and continues processing. For more information on ANSI Host Interface process, refer to the ***BASE24 ANSI Host Interface Manual***.

DEVICE STATE Commands

Process-generated commands that inform the ANSI Host Interface process of the current state of communications between the network and the station to the host. This message originates from the XPNET process.

Syntax:

9510*symbolic-station-name*

9511*symbolic-station-name*

9512*symbolic-station-name*

9513*symbolic-station-name*

9516*symbolic-station-name*

9517*symbolic-station-name*

9518*symbolic-station-name*

9519*symbolic-station-name*

symbolic-station-name — The name associated with a particular station to the host as defined in the XPNET node.

Processing

The meanings of the commands and the actions performed by the ANSI Host Interface process for each command are as follows:

Command	Action
9510	Communications with the station to the host whose symbolic station name is indicated are down. The ANSI Host Interface process marks the station's status as DOWN.
9511	Communications with the station to the host whose symbolic station name is indicated are up. The ANSI Host Interface process marks the station's status as UP.
9512	Communications with the station to the host whose symbolic station name is indicated have been suspended. The ANSI Host Interface process marks the station's status as DOWN.
9513	Communications with the station to the host whose symbolic station name is indicated have been reinstated. The ANSI Host Interface process marks the station's status as UP.

Command	Action
9516	The station to the host whose symbolic station name is indicated is not being polled. The ANSI Host Interface process marks the station’s status as DOWN.
9517	The station to the host whose symbolic station name is indicated is being polled. The ANSI Host Interface process marks the station’s status as UP.
9518	The station to the host whose symbolic station name is indicated is connected. The ANSI Host Interface process marks the station’s status as UP.
9519	The station to the host whose symbolic station name is indicated is disconnected. The ANSI Host Interface process marks the station’s status as DOWN.

SAFEXTCOMP Command

A process-generated command that notifies the ANSI Host Interface process that a Store-and-Forward (SAF) extract has been completed for the specified data processing center (DPC) and of the number of SAF records extracted for that DPC. The Super Extract process sends this notification to the ANSI Host Interface process.

Syntax

SAFEXTCOMP *dpc records*

dpc — Identifies the DPC for which the SAF extract has been completed.

records — Indicates the number of SAF records that were extracted.

Processing

Upon receipt of a SAFEXTCOMP command, the ANSI Host Interface process performs the following steps:

1. Checks the DPC number.

   If the DPC number is invalid, the ANSI Host Interface process logs an event message indicating that a SAF extract was completed for an unknown DPC.

   If the DPC number is valid, the ANSI Host Interface process logs an event message indicating that a SAF extract was completed for the indicated DPC.

2. Recalculates the number of SAF records it has in its internal table as pending for the DPC.

   To do this, the ANSI Host Interface process subtracts the number sent by the Super Extract process from the number in its table.

3. If there are more SAF records to be sent to the DPC, the ANSI Host Interface process initiates a store-and-forward cycle (provided the DPC is not logged off and there is a send/receive station available).

SAFEXTRQST Command

A process-generated command that notifies the ANSI Host Interface process that a Store-and-Forward (SAF) extract is being started for the specified data processing center (DPC). The Super Extract process sends this notification to the ANSI Host Interface process.

Syntax

SAFEXTRQST *dpc*

dpc — Identifies the DPC for which the SAF extract is being started.

Processing

This command requires a response from the ANSI Host Interface process. Upon receipt of a SAFEXTRQST command, the ANSI Host Interface process performs the following steps:

1. Checks the DPC number.

   If the DPC number is not in its table, the ANSI Host Interface process sends a response to the Super Extract process indicating not to start the extract.

2. Checks the DPC status to determine whether a store-and-forward cycle or a SAF extract is currently in progress (the SAF extract might have been initiated at another terminal).

   If either of these conditions exists, the ANSI Host Interface process sends a response to the Super Extract process indicating not to start the extract.

3. Determines whether any records exist in the SAF for the DPC.

   The ANSI Host Interface process keeps track of this internally at all times. If no records exist in the file for the DPC, the ANSI Host Interface process sends a response to the Super Extract process indicating not to start the extract.

4. Sends a SAFEXTRESP command to the Super Extract process indicating the number of records in the SAF for the DPC.

5. Updates the DPC status to indicate that a SAF extract is in progress for the DPC.

BIC ANSI Interface Process Commands

This subsection describes the BASE24 process-generated commands the BIC ANSI Interface process recognizes. The BIC ANSI Interface process receives the following process-generated commands:

- CUTOVER
- DEVICE STATE

CUTOVER Command

A process-generated command that notifies the BIC ANSI Interface process that logical network cutover and midnight cutover have taken place. This command initiates interchange cutover and originates from the BASE24-atm Settlement Initiator process and the BASE24-pos Main Settlement Initiator process.

Syntax

9502 *yymmdd product-identifier*

yymmdd — The new posting date sent to the BIC ANSI Interface process by the Settlement Initiator process.

product-identifier — Identifies the product for which the CUTOVER command was sent. Valid values are as follows:

ATM = BASE24-atm
POS = BASE24-pos

Processing

The steps taken to cut over the logical network depend on whether the co-networks are equal partners or one network is the main network and the second network is the secondary network. For more information on BIC ANSI Interface process cutover processing and message flows, refer to the ***BASE24 BIC ANSI Interface Manual***.

DEVICE STATE Commands

Process-generated commands that inform the BIC ANSI Interface process of the current state of communications between the network and the co-network. This message originates and is sent from the XPNET process.

Syntax

9510*symbolic-station-name*

9511*symbolic-station-name*

9512*symbolic-station-name*

9513*symbolic-station-name*

9516*symbolic-station-name*

9517*symbolic-station-name*

9518*symbolic-station-name*

9519*symbolic-station-name*

symbolic-station-name — The name associated with a particular station to the co-network as defined in the XPNET node.

Processing

The meanings of the commands and the actions performed by the BIC ANSI Interface process for each command are as follows:

Command	Action
9510	Communications with the station to the co-network whose symbolic station name is indicated are down. The BIC ANSI Interface process marks the station's status as DOWN.
9511	Communications with the station to the co-network whose symbolic station name is indicated are up. The BIC ANSI Interface process marks the station's status as UP.
9512	Communications with the station to the co-network whose symbolic station name is indicated have been suspended. The BIC ANSI Interface process marks the station's status as DOWN.

Command	Action
9513	Communications with the station to the co-network whose symbolic station name is indicated have been reinstated. The BIC ANSI Interface process marks the station’s status as UP.
9516	The station to the co-network whose symbolic station name is indicated is not being polled. The BIC ANSI Interface process marks the station’s status as DOWN.
9517	The station to the co-network whose symbolic station name is indicated is being polled. The BIC ANSI Interface process marks the station’s status as UP.
9518	The station to the co-network whose symbolic station name is indicated is connected. The BIC ANSI Interface process marks the station’s status as UP.
9519	The station to the co-network whose symbolic station name is indicated is disconnected. The BIC ANSI Interface process marks the station’s status as DOWN.

BIC ISO Interface Process Commands

This subsection describes the BASE24 process-generated commands the BIC ISO Interface process recognizes. The BIC ISO Interface process receives the following process-generated commands:

- CUTOVER
- DEVICE STATE

CUTOVER Command

A process-generated command that notifies the BIC ISO Interface process that logical network cutover has taken place. This command initiates interchange cutover and originates from the BASE24-atm Settlement Initiator process and the BASE24-pos Main Settlement Initiator process.

Syntax

9502 *yymmdd product-identifier*

yymmdd — The new posting date sent to the BIC ISO Interface process by the Settlement Initiator process.

product-identifier — Identifies the product for which the CUTOVER command was sent. Valid values are as follows:

ATM = BASE24-atm
POS = BASE24-pos

Processing

The steps taken to cut over the logical network depend on whether the co-networks are equal partners or one network is the main network and the second network is the secondary network. For more information on BIC ISO Interface process cutover processing and message flows, refer to the ***BASE24 BIC ISO Interface Manual***.

DEVICE STATE Commands

Process-generated commands that inform the BIC ISO Interface process of the current state of communications between the network and the co-network. This message originates and is sent from the XPNET process.

Syntax

9510*symbolic-station-name*

9511*symbolic-station-name*

9512*symbolic-station-name*

9513*symbolic-station-name*

9516*symbolic-station-name*

9517*symbolic-station-name*

9518*symbolic-station-name*

9519*symbolic-station-name*

symbolic-station-name — The name associated with a particular station as defined in the XPNET node.

Processing

The meanings of the commands and the actions performed by the BIC ISO Interface process for each command are as follows:

Command	Action
9510	Communications with the station to the co-network whose symbolic station name is indicated are down. The BIC ISO Interface process marks the co-network's status as DOWN.
9511	Communications with the station to the co-network whose symbolic station name is indicated are up. The BIC ISO Interface process marks the co-network's status as UP.
9512	Communications with the station to the co-network whose symbolic station name is indicated have been suspended. The BIC ISO Interface process marks the co-network's status as DOWN.

Command	Action
9513	Communications with the station to the co-network whose symbolic station name is indicated have been reinstated. The BIC ISO Interface process marks the co-network's status as UP.
9516	The station to the co-network whose symbolic station name is indicated is not being polled. The BIC ISO Interface process marks the co-network's status as DOWN.
9517	The station to the co-network whose symbolic station name is indicated is being polled. The BIC ISO Interface process marks the co-network's status as UP.
9518	The station to the co-network whose symbolic station name is indicated is connected. The BIC ISO Interface process marks the co-network's status as UP.
9519	The station to the co-network whose symbolic station name is indicated is disconnected. The BIC ISO Interface process marks the co-network's status as DOWN.

From Host Maintenance Process Commands

This subsection describes the BASE24 process-generated commands the From Host Maintenance process recognizes. The From Host Maintenance process receives the following process-generated commands:

- CUTOVER
- REFRESH

CUTOVER Command

A process-generated command that notifies secondary From Host Maintenance processes to start using a new Update Log File (ULF). The main From Host Maintenance process sends this notification in the form of a 9502 message. The command originates from main From Host Maintenance process when the cutover timer expires (determined by the value of the FHM-LN-CUTOVER param in the LCONF).

REFRESH Command

A process-generated command that notifies the From Host Maintenance process of a file refresh. The Enscribe Refresh process sends this notification in the form of a 9503 message. This command originates from the Enscribe Refresh process.

Syntax

9503**** *file-name ref-group file-type refr-indicator*

file-name — The fully-qualified name of the refreshed file.

ref-group — An identifier used to group institutions together for file refreshes and extracts.

file-type — Indicates which file has been refreshed. Valid values for files used by the From Host Maintenance process are as follows:

1 = Positive Balance File (PBF) for DDA and NOW accounts if multiple PBFs are used or all accounts if PBFs are combined
2 = Positive Balance File (PBF) for savings accounts if multiple PBFs are used
3 = Positive Balance File (PBF) for credit card accounts if multiple PBFs are used
5 = Stop Payment File (SPF)
6 = No Book File (NBF) — BASE24-teller only
7 = Cardholder Authorization File (CAF)
9 = Warning/Hold/Float File (WHFF) — BASE24-teller only
A = Corporate Check File (CCF) — BASE24-atm self-service banking (SSB) Check Application only
B = Check Status File (CSF) — BASE24-atm self-service banking (SSB) Check Application only
D = Negative Card File (NEG)
E = Customer/Card Information File (CCIF)
F = Customer/Card Memo File (CCMF)

refr-indicator — Value used by Authorization processes to log future transactions to the transaction log files. The Enscribe Refresh process uses this value to control impacting.

Processing

When the From Host Maintenance process receives a 9503 message, it performs the following processing steps:

1. Checks the *file-type* sent in the 9503 message.

   If the *file-type* is one of the recognized values, processing continues with step 2.

   If the *file-type* is not one of the refresh types recognized by the From Host Maintenance process, processing skips to step 8.

2. If the *file-type* sent in the 9503 message is A (CCF) or B (CSF), the From Host Maintenance process closes and reopens the specified file and continues processing with step 8.

3. Locates the file in its file table using the *file-name* sent in the 9503 message.

   If the From Host Maintenance process cannot locate the file in its table, it generates an event message to note the situation and skips to step 6.

4. Opens the file specified in the 9503 message.

5. Closes the file previously held open.

6. Adds the new Guardian file number to its file table.

7. Each Institution Definition File (IDF) record in extended memory contains a refresh indicator field for each of the three PBFs. If the *ref-group* sent in the 9503 message matches the value in the REFR-GRP field of the IDF record and the refreshed file is a PBF, the From Host Maintenance process updates the corresponding refresh indicator field of the IDF record in memory with the *refr-indicator* from the 9503 message. The Enscribe Refresh process updates the refresh indicator fields in the IDF disk file.

8. Sends a PAUSEACK command to the Enscribe Refresh process to indicate that it has processed the 9503 message.

ISO Host Interface Process Commands

This subsection describes the BASE24 process-generated commands the ISO Host Interface process recognizes. The ISO Host Interface process receives the following process-generated commands:

- CUTOVER
- SAFEXTCOMP
- SAFEXTRQST

CUTOVER Command

A process-generated command that notifies the ISO Host Interface process that logical network cutover and midnight cutover have taken place. This command initiates interchange cutover and originates from the BASE24-atm, BASE24-pos, or BASE24-teller Settlement Initiator processes. The CUTOVER command updates the BASE24-atm, BASE24-pos, or BASE24-teller posting date maintained by the ISO Host Interface process. The Settlement Initiator process sends a new posting date (yymmdd) each day at logical network cutover.

Syntax

9502 *yymmdd product-identifier*

yymmdd — The new posting date sent to the ISO Host Interface process by the Settlement Initiator process.

product-identifier — Identifies the product for which the CUTOVER command was sent. Valid values are as follows:

ATM = BASE24-atm
POS = BASE24-pos
TLR = BASE24-teller

Processing

The steps taken to cut over the interchange depend on whether the co-networks are equal partners or one network is the main network and the second network is the secondary network. For more information on ISO Host Interface process cutover processing and message flows, refer to the ***BASE24 ISO Host Interface Manual***.

DEVICE STATE Commands

Process-generated commands that inform the ISO Host Interface process of the current state of communications between the network and the station to the host. This message originates and is sent from the XPNET process.

Syntax

9510*symbolic-station-name*

9511*symbolic-station-name*

9512s*ymbolic-station-name*

9513*symbolic-station-name*

9516*symbolic-station-name*

9517*symbolic-station-name*

9518*symbolic-station-name*

9519*symbolic-station-name*

symbolic-station-name — The name associated with a particular station to the host as defined in the XPNET node.

Processing

The meanings of the commands and the actions performed by the ISO Host Interface process for each command are as follows:

Command	Action
9510	Communications with the station to the host whose symbolic station name is indicated are down. The ISO Host Interface process marks the ATM's status as DOWN.
9511	Communications with the station to the host whose symbolic station name is indicated are up. The ISO Host Interface process marks the station's status as UP.
9512	Communications with the station to the host whose symbolic station name is indicated have been suspended. The ISO Host Interface process marks the station's status as DOWN.
9513	Communications with the station to the host whose symbolic station name is indicated have been reinstated. The ISO Host Interface process marks the station's status as UP.

Command	Action
9516	The station to the host whose symbolic station name is indicated is not being polled. The ISO Host Interface process marks the station’s status as DOWN.
9517	The station to the host whose symbolic station name is indicated is being polled. The ISO Host Interface process marks the station’s status as UP.
9518	The station to the host whose symbolic station name is indicated is connected. The ISO Host Interface process marks the station’s status as UP.
9519	The station whose symbolic station name is indicated is disconnected. The ISO Host Interface process marks the station’s status as DOWN.

SAFEXTCOMP Command

A process-generated command that notifies the ISO Host Interface process that a Store-and-Forward (SAF) extract has been completed for the specified data processing center (DPC) and of the number of SAF records extracted for that DPC. The Super Extract process sends this notification to the ISO Host Interface process. This command originates from the Super Extract process.

Syntax

SAFEXTCOMP *dpc records*

dpc — Identifies the DPC for which the SAF extract has been completed.

records — Indicates the number of SAF records that were extracted.

Processing

Upon receipt of a SAFEXTCOMP command, the ISO Host Interface process performs the following steps:

1. Checks the DPC number.

   If the DPC number is invalid, the ISO Host Interface process logs an event message indicating that a SAF extract was completed for an unknown DPC.

   If the DPC number is valid, the ISO Host Interface process logs an event message indicating that a SAF extract was completed for the indicated DPC.

2. Recalculates the number of SAF records it has in its internal table as pending for the DPC.

   To do this, the ISO Host Interface process subtracts the number sent by the Super Extract process from the number in its table.

3. If SAF records still exist to be sent to the DPC, the ISO Host Interface process initiates a store-and-forward cycle (provided the DPC is not logged off and there is a send/receive station available).

SAFEXTRQST Command

A process-generated command that notifies the ISO Host Interface process that a Store-and-Forward (SAF) extract is being started for the specified data processing center (DPC). The Super Extract process sends this notification to the ISO Host Interface process. This command originates from the Super Extract process.

Syntax

SAFEXTRQST *dpc*

dpc — Identifies the DPC for which the SAF extract is being started.

Processing

This command requires a response from the ISO Host Interface process. Upon receipt of a SAFEXTRQST command, the ISO Host Interface process performs the following steps:

1. Checks the DPC number.

   If the DPC number is not in its table, the ISO Host Interface process sends a response to the Super Extract process indicating not to start the extract.

2. Checks the DPC status to determine whether a store-and-forward cycle or a SAF extract is currently in progress (the SAF extract might have been initiated at another terminal).

   If either of these conditions exists, the ISO Host Interface process sends a response to the Super Extract process indicating not to start the extract.

3. Determines whether any records exist in the SAF for the DPC.
   The ISO Host Interface process keeps track of this internally at all times. If no records exist in the file for the DPC, the ISO Host Interface process sends a response to the Super Extract process indicating not to start the extract.

4. Sends a SAFEXTRESP response to the Super Extract process indicating the number of records in the SAF for the DPC.

5. Updates the DPC status to indicate that a SAF extract is in progress for the DPC.

Enscribe Refresh Process Commands

This subsection describes the BASE24 process-generated commands the Enscribe Refresh process recognizes. The Enscribe Refresh process receives the following process-generated commands:

- PAUSEACK
- PAUSEEND

PAUSEACK Command

A process-generated command that notifies the Enscribe Refresh process that the BASE24-atm Authorization, BASE24-teller Authorization, BASE24-from host maintenance From Host Maintenance process, and BASE24-pos Device Handler/Router/Authorization process has handled a 9503 REFRESH command successfully.

Syntax

PAUSEACK

Processing

The Enscribe Refresh process receives a PAUSEACK command after the Authorization process sending the message has processed the 9503 (PAUSE) command successfully. The PAUSEACK command serves as notification that a REFRESH command has been received and handled successfully.

PAUSEEND Command

A process-generated command that notifies the Enscribe Refresh process that the BASE24-atm Authorization, BASE24-teller Authorization, or BASE24-pos Device Handler/Router/Authorization processes cannot process a 9503 REFRESH command successfully. In this case, refresh processing continues to normal termination.

Syntax

PAUSEEND

Processing

Since refresh continues to process in a loop until all processes that were notified have sent acknowledgments, the PAUSEEND command allows an operator to cause the Enscribe Refresh process to continue until the normal End-of-Job (EOJ). After the Enscribe Refresh process receives a PAUSEEND command, it immediately continues processing to the EOJ regardless of the number of processes that have not responded.

Super Extract Process Commands

This subsection describes the BASE24 process-generated commands the Super Extract process recognizes. The Super Extract process receives the SAFEXTRESP process-generated command.

SAFEXTRESP Command

A process-generated command received in response to a SAFEXTRQST command. The SAFEXTRESP command notifies the Super Extract process on the status of the Store-and-Forward (SAF) extract being performed for the specified data processing center (DPC). The ISO Host Interface process sends this notification to the Super Extract process. This command originates from the ISO Host Interface process.

Syntax

SAFEXTRESP *dpc num-of-saf-records saf-name*

dpc — Identifies the DPC for which the SAF extract is being performed.

num-of-saf-records — The number of Store-and-Forward (SAF) records.

saf-name — The fully qualified name of the SAF.

Processing

This command is a response from the ISO Host Interface process. The ISO Host Interface process returns the DPC number, the number of SAF records for the indicated DPC, and the fully qualified SAF name.

BASE24-atm Process Commands

This subsection contains the process-generated commands received by the following BASE24-atm processes:

- Authorization
- Device Handler
- Transaction Context Manager

Authorization Process Commands

This subsection describes the BASE24 process-generated commands the Authorization process recognizes. The Authorization process is the authorizing entity in a BASE24-atm system. The Authorization process receives the following process-generated commands:

- CUTOVER
- REFRESH

CUTOVER Command

A process-generated command that notifies the Authorization process that logical network cutover and midnight cutover have taken place. The Settlement Initiator process sends this notification in the form of a 9502 message.

Syntax

9502*yymmdd*

yymmdd — The date associated with the processing day just ending.

Processing

Logical network cutover and midnight cutover notification is sent to the Authorization process in the form of a 9502 message. This message originates from the BASE24-atm Settlement Initiator process. When the Authorization process receives a 9502 message, it performs the following processing steps:

1. Reads the Institution Definition File (IDF) disk record for each institution in its IDF table. For each IDF record it reads, it overlays the IDF BEG-DAT, NXT-BEG-DAT, CUS-BUS-DAT in extended memory with the IDF record from disk.
2. Adjusts its current processing date.

   If the current processing date for the Authorization process is less than or equal to the date sent in the 9502 message, the Authorization process adds one day to the date sent in the 9502 message and uses that for the current processing date.

   If the current processing date for the Authorization process is greater than the date sent in the 9502 message, the Authorization process keeps its own current processing date.
3. Creates its next processing date.

   To get its next processing date, the Authorization process adds two calendar days to the date sent in the 9502 message.
4. Closes the TLFs it currently has open.
5. Opens the TLFs for the current and next day.

   The Authorization process opens the TLFs for its new current and next processing dates.
6. Generates the event message “Cutover completed for this process.”

REFRESH Command

A process-generated command that notifies the Authorization process of a file refresh. The Enscribe Refresh process sends this notification in the form of a 9503 message. The command originates from the Enscribe Refresh process.

Syntax

9503**** *file-name ref-group file-type refr-indicator*

file-name — The fully-qualified name of the refreshed file.

ref-group — An identifier used to group institutions together for file refreshes and extracts.

file-type — Indicates which file has been refreshed. Valid values for files used by the Authorization process are as follows:

- 1 = Positive Balance File (PBF) for DDA and NOW accounts if multiple PBFs are used or all accounts if PBFs are combined
- 2 = Positive Balance File (PBF) for savings accounts if multiple PBFs are used
- 3 = Positive Balance File (PBF) for credit card accounts if multiple PBFs are used
- 5 = Stop Payment File (SPF)
- 7 = Cardholder Authorization File (CAF)
- A = Corporate Check File (CCF) — BASE24-atm self-service banking (SSB) Check Application only
- B = Check Status File (CSF) — BASE24-atm self-service banking (SSB) Check Application only
- D = Negative Card File (NEG)
- E = Customer/Card Information File (CCIF)
- F = Customer/Card Memo File (CCMF)

refr-indicator — A value used by the Authorization process to log future transactions to the transaction log files. The Enscribe Refresh process uses this value to control impacting.

Processing

When the Authorization process receives a 9503 message, it performs the following processing steps:

1. Checks the *file-type* sent in the 9503 message.

   If the *file-type* is one of the recognized values, processing continues with step 2.

   If the *file-type* is not one of the refresh types recognized by the Authorization process, processing skips to step 8.

2. If the *file-type* sent in the 9503 message is A (CCF) or B (CSF), the Authorization process closes and reopens the specified file and continues processing with step 8.

3. Locates the file in its file table using the *file-name* sent in the 9503 message.

   If the Authorization process cannot locate the file in its table, it generates an event message to note the situation and skips to step 6.

4. Opens the file specified in the 9503 message.

5. Closes the file previously held open.

6. Adds the new Guardian file number to its file table.

7. Each Institution Definition File (IDF) record in extended memory contains a refresh indicator field for the CAF and each of the PBFs. If the *ref-group* sent in the 9503 message matches the value in the REFR-GRP field of the IDF record and the refreshed file is a CAF or PBF, the Authorization process updates the corresponding refresh indicator field of the IDF record in memory with the *refr-indicator* from the 9503 message. The Enscribe Refresh process updates the refresh indicator fields in the IDF disk file.

8. Sends a PAUSEACK command to the Enscribe Refresh process to indicate that it has processed the 9503 message.

Device Handler Process Commands

This subsection describes the BASE24 process-generated DEVICE STATE commands the BASE24-atm Device Handler processes recognize. Device Handler processes are the primary interfaces between ATMs and BASE24.

DEVICE STATE Command

Process-generated commands that inform the Device Handler process of the current state of communications between the network and the ATM. This message originates and is sent from the XPNET process.

Syntax

9510*symbolic-station-name*

9511*symbolic-station-name*

9512*symbolic-station-name*

9513*symbolic-station-name*

9514*symbolic-station-name*

9515*symbolic-station-name*

9518*symbolic-station-name*

9519*symbolic-station-name*

symbolic-station-name — The name associated with a particular ATM as defined in the XPNET node.

Processing

There are several device state messages that can be sent to a Device Handler process. The meanings of the commands and the actions performed by the Device Handler process for each command are as follows:

Command	Action
9510	Communications with the ATM whose symbolic station name is indicated are down. The Device Handler process marks the ATM's status as DOWN.
9511	Communications with the ATM whose symbolic station name is indicated are up. The Device Handler process marks the ATM's status as UP.
9512	Communications with the ATM whose symbolic station name is indicated have been suspended. The Device Handler process marks the ATM's status as DOWN.

Command	Action
9513	Communications with the ATM whose symbolic station name is indicated have been reinstated. The Device Handler process marks the ATM's status as UP.
9514	The ATM responded positively to an SNA activate LU message. The Device Handler process marks the ATM's status as UP. This command is valid for the ENVOY SNA only.
9515	The ATM responded positively to an SNA deactivate LU message. The Device Handler process marks the ATM's status as DOWN. This command is valid for the ENVOY SNA only
9518	The ATM responded positively to an X.25 activate LU message. The Device Handler process marks the ATM's status as UP.
9519	The ATM responded positively to an X.25 deactivate LU message. The Device Handler process marks the ATM's status as DOWN.

Note: Only Device Handler processes enhanced to support XPNET process Resident SSCP use the 9514 and 9515 messages.

Transaction Context Manager Commands

This subsection describes the BASE24 process-generated commands the Transaction Context Manager (TCM) process recognizes. The TCM process enables the BASE24-atm system to receive a single request from an endpoint and split the routing between two or more destinations. The TCM process receives the following process-generated command:

- CUTOVER

CUTOVER Command

A process-generated command that notifies the Transaction Context Manager (TCM) process that logical network cutover and midnight cutover have taken place. The Settlement Initiator process sends this notification in the form of a 9502 message.

Syntax

9502*yymmdd*

yymmdd — The date associated with the processing day just ending.

Processing

Logical network cutover and midnight cutover notification is sent to the Transaction Context Manager process in the form of a 9502 message. This message originates from the BASE24-atm Settlement Initiator process. When the Transaction Context Manager process receives a 9502 message, it performs the following processing steps:

1. Adjusts its current processing date.

   If the current processing date for the TCM process is less than or equal to the date sent in the 9502 message, TCM process adds one day to the date sent in the 9502 message and uses that for the current processing date.

   If the current processing date for the TCM process is greater than the date sent in the 9502 message, the TCM process keeps its own current processing date.

2. Closes the Transaction Log Files (TLFs) for the previous day.

3. Opens the TLFs for the next day.

   The TCM process opens the TLFs for its next processing dates.

   Generates the event message “Cutover completed for this process.”

BASE24-pos Process Commands

This subsection contains the process-generated commands received by the following BASE24-pos modules or processes:

- Device Handler
- Router/Authorization
- Settlement Initiator
- Transaction Context Manager

Device Handler Module Commands

This subsection describes the BASE24 process-generated commands the BASE24-pos Device Handler module recognizes. The Device Handler module receives only the CUTOVER process-generated command.

CUTOVER Command

A process-generated command that notifies the Device Handler module that logical network cutover and midnight cutover have taken place. The Settlement Initiator process sends this notification in the form of a 9502 message. This command originates from the BASE24-pos Settlement Initiator process.

Syntax

9502 *yymmdd*

yymmdd — The date (yymmdd) of the processing day just ending.

Processing

The 9502 command simply notifies the Device Handler process that cutover has taken place. Cutover processing is performed by the Settlement Initiator process and Authorization module.

Router/Authorization Module Commands

This subsection describes the BASE24 process-generated commands the BASE24-pos Router/Authorization module recognizes. The Router/Authorization module receives the following process-generated commands:

- CUTOVER EAF
- LOGICAL NETWORK CUTOVER
- MIDNIGHT CUTOVER
- REFRESH
- REFRESH INR, BNR, INEG, BNEG, OR ANEG

CUTOVER EAF

A process-generated command that notifies the Device Handler/Router/ Authorization process that E-Commerce Authentication File (EAF) cutover has taken place. The Settlement Initiator process sends this notification to the Router/ Authorization module. This command originates from the BASE24-pos Main Settlement Initiator process.

Syntax

9502EAF

Processing

When the Router/Authorization module receives a 9502EAF message, it closes the oldest EAF and opens the new EAF that has been created by the Settlement Initiator process. The EAF that had been the current file becomes the previous file. The Settlement Initiator process then purges the oldest EAF.

LOGICAL NETWORK CUTOVER Command

A process-generated command that notifies the Router/Authorization module that logical network cutover has taken place. The Settlement Initiator process sends this notification to the Router/Authorization module. This command originates from the BASE24-pos Main Settlement Initiator process.

Syntax

9502*yymmdd*

yymmdd — The date (yymmdd) of the processing day just ending.

Processing

When the Authorization module receives a 9502*yymmdd* message, it performs the following steps:

1. Reads the Institution Definition File (IDF) disk record for each institution whose IDF record it has in memory. For each IDF record it reads, it performs the following steps:
   a. Overlays the IDF record in memory with the IDF record from disk, field-by-field.
   b. Closes and reopens the authorization file (i.e., Cardholder Authorization File (CAF), Negative Card File (NEG), or Positive Balance File (PBF)).
2. Generates the event message “Cutover completed for this process.”
3. Adjusts its current processing date.

   If the current processing date is greater than the date sent in the 9502*yymmdd* message, the Authorization module takes no further steps.

   If the current processing date is less than or equal to the date sent in the 9502*yymmdd* message, the Authorization module performs the following steps:

   a. Creates its next processing date.

      To get its next processing date, the Authorization module adds one calendar day to the current processing date.

   b. Closes the POS Transaction Log Files (PTLFs) and POS Referral Files (PRFs) it currently has open.
   c. Opens the PTLFs, and PRFs if the referral processing flag is set, for its new current and next processing dates.

MIDNIGHT CUTOVER Command

A process-generated command that notifies the Router/Authorization module that midnight cutover has taken place. The Settlement Initiator process sends this notification to the Authorization module. This command originates from the BASE24-pos Main Settlement Initiator process.

Syntax

9502

Processing

When the Authorization module receives a 9502 message, it performs the following steps:

1. Reads the Institution Definition File (IDF) disk record for each institution whose IDF record it has in memory. For each IDF record it reads, it performs the following steps:
   a. Overlays the IDF record in memory with the IDF record from disk, field-by-field.
   b. Closes and reopens the Authorization file (i.e., Cardholder Authorization File (CAF), Negative Card File (NEG), or Positive Balance File (PBF)).
2. Generates the event message "Cutover completed for this process."

REFRESH Command

A process-generated command that notifies the Authorization module of a file refresh. The Enscribe Refresh process sends this notification in the form of a 9503 message. The command originates from the Enscribe Refresh process.

Syntax

9503**** *file-name ref-group file-type refr-indicator*

file-name — The fully-qualified name of the refreshed file.

ref-group — An identifier used to group institutions together for file refreshes and extracts.

file-type — Indicates which file has been refreshed. Valid values for files used by the Authorization module are as follows:

1 = Positive Balance File (PBF) for DDA and NOW accounts if multiple PBFs are used or all accounts if PBFs are combined
2 = Positive Balance File (PBF) for savings accounts if multiple PBFs are used
3 = Positive Balance File (PBF) for credit card accounts if multiple PBFs are used
7 = Cardholder Authorization File (CAF)
D = Negative Card File (NEG)
E = Customer/Card Information File (CCIF)
F = Customer/Card Memo File (CCMF)

refr-indicator — Value used by the Authorization module to log future transactions to the transaction log files. The Enscribe Refresh process uses this value to control impacting.

Processing

When the Authorization module receives a 9503 message, it performs the following processing steps:

1. Checks the *file-type* sent in the 9503 message.

   If the *file-type* is one of the recognized values, processing continues with step 2.

   If the *file-type* is not one of the refresh types recognized by the Authorization module, processing skips to step 7.

2. Locates the file in its file table using the *file-name* sent in the 9503 message.

   If the Authorization module cannot locate the file in its table, it generates an event message to note the situation and skips to step 5.

3. Closes the file previously held open.

4. Opens the file specified in the 9503 message.

5. Adds the new Guardian file number to its file table.

6. Each Institution Definition File (IDF) record in extended memory contains a refresh indicator field for each of the three PBFs. If the *ref-group* sent in the 9503 message matches the value in the REFR-GRP field of the IDF record and the refreshed file is a PBF, the Authorization module updates the corresponding refresh indicator field of the IDF record in memory with the *refr-indicator* from the 9503 message. The Enscribe Refresh process updates the refresh indicator fields in the IDF disk file.

7. Sends a PAUSEACK command to the Enscribe Refresh process to indicate that it has processed the 9503 message.

REFRESH INR, BNR, INEG, BNEG, and ANEG Command

A process-generated command that notifies the Device Handler/Router/Authorization process that an INAS National Range File (INR), Base National Range File (BNR), INAS National Negative Card File (INEG), Base National Negative Card File (BNEG), and American Express National Negative Card File (ANEG) refresh has started. The Base National Negative Refresh or INAS National Negative Refresh process sends this notification to the Authorization module in the form of a 9528 message. This command originates from the Refresh process.

Syntax

9528 *file-name*

Processing

When the Authorization module receives a 9528 message, it performs the following steps by group.

Group 1: (performs if *file-name* is the INEG)

1. Closes the INR.
2. Opens the INR and adds the new Guardian file number to its internal file table.
3. Reloads the extended memory with INR records.
4. Closes the INEG.
5. Opens the INEG and adds the new Guardian file number to its internal file table.

Group 2: (performs if *file-name* is the BNEG)

1. Closes the BNR.
2. Opens the BNR and adds the new Guardian file number to its internal file table.
3. Reloads the extended memory with BNR records.
4. Closes the BNEG.
5. Opens the BNEG and adds the new Guardian file number to its internal file table.

Group 3: (performs if *file-name* is the ANEG)

1. Closes the ANEG.
2. Opens the ANEG and adds the new Guardian file number to its internal table.

After a refresh is completed for any of the above groups, the Authorization module sends a PAUSEACK command back to the Refresh process to indicate that it has processed the refresh message.

Settlement Initiator Commands

This subsection describes the BASE24 process-generated commands recognized by the BASE24-pos Settlement Initiator process. The Settlement Initiator process receives the following process-generated commands:

- ALIVEMSG
- CHECKMASTER
- DUMPIDFTBL
- DUMPRDF
- DUMPRDFTBL
- DUMPTIMERTBL
- DUMPTIMES
- DUMPPTD
- DUMPPTDTBL
- FIID
- LOGICAL NETWORK CUTOVER
- NOTIFY MHI
- RESETRDFDATES
- RESETPTDDATES
- RESETALLDATES
- RESETRDFTIMES
- START UP
- STARTUPCOMPLETE
- TRACEOFF SECONDARY
- TRACEON SECONDARY
- WARMBOOT PRDF TABLE ENTRY
- WB MAIN PPD
- WB COMPLETE

ALIVEMSG Command

A command generated by a Senior Secondary Settlement Initiator process that causes it to periodically verify that the Main Settlement Initiator process is running.

Syntax

ALIVEMSG

Processing

The Senior Secondary Settlement Initiator process is identified in the POS-SENIOR-SLAVE-PROCESS param in the Logical Network Configuration File (LCONF). If a Senior Secondary Settlement Initiator process is not identified in this param, no Secondary Settlement Initiator process will verify that the Main Settlement Initiator process is up and running during processing. A Main Settlement Initiator process must be identified in the POS-SETL-MASTER param in the LCONF. If this param is not found at initialization, no further processing can occur.

The Senior Secondary Settlement Initiator process periodically checks the Main Settlement Initiator process to determine if it is up and running. The Senior Secondary Settlement Initiator process uses the Alive Timer to perform this function. The Senior Secondary Settlement Initiator process sets the Alive Timer at initialization and each time an Alive Timer expires. When this timer expires, the Senior Secondary Settlement Initiator process opens the Main Settlement Initiator process if it is not open already and sends an alive message to the Main Settlement Initiator process to verify it is running.

In addition, the POS-SETL-ALIVE-TIMEOUT param in the LCONF identifies how often, in minutes, the Senior Secondary Settlement Initiator process must query the Main Settlement Initiator process to verify that the Main Settlement Initiator process is up and running. If this param is not set and a Senior Secondary process is being used, the default value is every 120 minutes.

The ALIVEMSG command originates from a Senior Secondary Settlement Initiator process. For more information on the Alive Timer, refer to the ***BASE24-pos Settlement and Reporting Manual***.

When the Alive Timer expires, the Senior Secondary Settlement Initiator process performs the following steps:

1. Attempts to open the Main Settlement Initiator process if the Main Settlement Initiator process is not opened already.

2. Sends an ALIVEMSG command to the Main Settlement Initiator process.
3. Waits for a protocol-level response from the Main Settlement Initiator process and processes as follows:
   - If the Senior Secondary Settlement Initiator process receives a protocol-level response from the Main Settlement Initiator process, the Main Settlement Initiator process is alive. Processing continues with the next step.
   - If the Senior Secondary Settlement Initiator process does not receive a protocol-level response from the Main Settlement Initiator process, the Senior Secondary Settlement Initiator process closes the Main Settlement Initiator process and logs an event message indicating that the Senior Secondary Settlement Initiator process was unable to send an alive message to the Main Settlement Initiator process. Processing then continues with the next step.
4. Sets another Alive Timer based on the value of the POS-SETL-ALIVE-TIMEOUT param.

CHECKMASTER Command

A command generated by a Senior Secondary Settlement Initiator process and sent to the Main Settlement Initiator process.

Syntax

CHECKMASTER *symbolic-name*

symbolic-name — The symbolic name of the Senior Secondary Settlement Initiator process as defined in the POS-SENIOR-SLAVE-PROCESS param in the LCONF. If the Senior Secondary Settlement Initiator process already knows the symbolic name of the Main Settlement Initiator process, no further processing is necessary.

If the Senior Secondary Settlement Initiator process cannot determine the PPD name of the Main Settlement Initiator process, it uses the PPD name of the Main Settlement Initiator process as identified in the checkmaster response.

Processing

The Main Settlement Initiator process responds to the CHECKMASTER command with a message that contains the Main Settlement Initiator process's PPD name. If the Senior Secondary Settlement Initiator process does not already have the Main process PPD name, it uses the Main process PPD name in the CHECKMASTER command response to open the Main Settlement Initiator process. In addition, the Senior Secondary resets the Alive Timer based on the value of the POS-SETL-ALIVE-TIMEOUT param in the LCONF. This command is used as a recovery mechanism when the Senior Secondary Settlement Initiator process is stopped and restarted.

DUMPIDFTBL Command

A process-generated command that causes the Secondary Settlement Initiator processes to display the contents of their Institution Definition File (IDF) extended memory tables.

Syntax

DUMPIDFTBL

Processing

The DUMPIDFTBL command is sent from the Main Settlement Initiator process to the Secondary Settlement Initiator processes when a DUMPIDFTBL command (i.e., DUMPIDFTBLALL or DUMPIDFTBL SECONDARY command) is entered from an XPNET network control facility. DUMPIDFTBL commands entered from an XPNET network control facility are sent directly to the Main Settlement Initiator process. The Main Settlement Initiator process sends interprocess DUMPIDF commands to the appropriate Secondary Settlement Initiator processes. An operator can request IDF extended memory table information for all Secondary Settlement Initiator processes or for a specific Secondary Settlement Initiator process.

Note: The DUMPIDFTBL command can be considered to be both a process-generated command and a general use operator command, depending on where it originates. If a DUMPIDFTBLALL or DUMPIDFTBL SECONDARY command is sent to the Main Settlement Initiator process from an XPNET network control facility, the Main Settlement Initiator process sends a DUMPIDFTBL command as an interprocess command to Secondary Settlement Initiator processes. The Main Settlement Initiator process notifies Secondary Settlement Initiator processes that IDF table information is being requested.

If the command originates from an XPNET network control facility as a DUMPIDFTBL command, it is a general use operator command. The network control request is for Main Settlement Initiator process IDF table information. In this case, the Main Settlement Initiator process displays only its IDF table information. No interprocess commands are sent to the Secondary Settlement Initiator processes.

The Main Settlement Initiator process sends the interprocess DUMPIDFTBL command to Secondary Settlement Initiator processes. The IDF extended memory table is created at initialization by the Main and Secondary Settlement Initiator processes.

When the appropriate Secondary Settlement Initiator processes receive this command, they display the following information taken from the IDF extended memory table in a series of event messages:

- FIID
- Cutover time
- Batch, shift, daily, service, and network totals
- Host Interface symbolic name

The IDF extended memory table information is formatted as follows:

```
FI-ID CUTOVER TIME BATCH SHFT DLY SRVC NET HISF-NAME
```

The IDF extended memory table information is placed here. The FI-ID field identifies the FIID associated with the retailer. The CUTOVER TIME field identifies the cutover time for the retailer. The BATCH, SHIFT, DLY, SRVC, and NET fields indicate the batch, shift, daily, services, and network totals for the retailer. The HISF field identifies the symbolic name of the Host Interface process. When all IDF table information has been displayed, the following message is shown:

```
END OF IDF TABLE DISPLAY
```

DUMPRDF Command

A process-generated command that causes the Secondary Settlement Initiator processes to display POS Retailer Definition File (PRDF) records according to the retailer range defined for a specific Secondary Settlement Initiator process. This command is sent from the Main Settlement Initiator process to the Secondary Settlement Initiator processes when a DUMPRDF command (i.e., DUMPRDFALL or DUMPRDF SECONDARY command) is entered from an XPNET network control facility. DUMPRDF records commands entered from an XPNET network control facility are sent directly to the Main Settlement Initiator process. The Main Settlement Initiator process sends interprocess DUMPRDF records commands to the appropriate Secondary Settlement Initiator processes. An operator can request PRDF records for all Secondary Settlement Initiator processes or for a specific Secondary Settlement Initiator processes.

Syntax

DUMPRDF

Processing

The Main Settlement Initiator process sends the interprocess DUMPRDF command to Secondary Settlement Initiator processes. When a Secondary Settlement Initiator process receives this command, it displays PRDF record information only for those retailer IDs included in the retailer range defined for the specific Secondary Settlement Initiator process. Retailer ID ranges for Secondary Settlement Initiator processes are identified in the POS-SLAVE-KEY param in the LCONF. The Secondary Settlement Initiator process displays the following PRDF record information for each retailer ID in its retailer ID range in a series of event messages:

- Retailer ID
- FIID
- End of retailer cutover window
- PRDF posting date

PRDF record information is displayed in the following format:

```
RETAILER-ID    FI-ID    BAL-CUTOVER-END    CUR-BUS-DATE
```

The PRDF record information described above is displayed here. The RETAILER-ID field identifies the retailer associated with the PRDF records. The FI-ID identifies the FIID associated with the retailer. The BAL-CUTOVER-END field identifies the end of the retailer cutover window. The CUR-BUS-DATE field identifies the posting date in the PRDF. When all time information has been displayed, the following message is shown:

```
END OF RDF DISPLAY
```

DUMPRDFTBL Command

A process-generated command that causes the Secondary Settlement Initiator processes to display their internal POS Retailer Definition File (PRDF) extended memory tables. This command is sent from the Main Settlement Initiator process to the Secondary Settlement Initiator processes when a DUMPRDFTBL command (i.e., DUMPRDFTBLALL or DUMPRDFTBL SECONDARY command) is entered from an XPNET network control facility. DUMPRDFTBL commands entered from an XPNET network control facility are sent directly to the Main Settlement Initiator process. The Main Settlement Initiator process sends interprocess DUMPRDFTBL commands to the appropriate Secondary Settlement Initiator processes. An operator can request PRDF tables for all Secondary Settlement Initiator processes or for a specific Secondary Settlement Initiator processes.

Syntax

DUMPRDFTBL

Processing

The Main Settlement Initiator process sends the interprocess DUMPRDFTBL command to Secondary Settlement Initiator processes. When a Secondary Settlement Initiator process receives this command, it displays its internal PRDF extended memory table. The Secondary Settlement Initiator process displays the following PRDF table information for each retailer ID in its retailer ID range in a series of event messages:

- Retailer ID
- FIID
- Cutover time
- Unique flag
- Cutover flag

The PRDF extended memory table information is formatted as follows:

```
RETAILER-ID CUTOVER TIME   UNIQUE FLAG  CUTOVER FLAG
```

The PRDF extended memory table information is placed here. The CUTOVER TIME field identifies the cutover time for the retailer. The second CUTOVER FLAG field identifies the cutover flag (contains a value of Y (Yes) or N (No)). The cutover flag indicates whether the retailer has cut over for the current day. The UNIQUE FLAG field indicates (Y or N) whether the retailer cutover time is unique among all retailers processed by the secondary. When all PRDF table information has been displayed, the following message is shown:

```
END OF RDF TABLE DISPLAY
```

DUMPTIMERTBL Command

A process-generated command that causes the Secondary Settlement Initiator processes to display their timer tables. This command is sent from the Main Settlement Initiator process to the Secondary Settlement Initiator processes when a DUMPTIMERTBL command (i.e., DUMPTIMERTBLALL or DUMPTIMERTBL SECONDARY command) is entered from an XPNET network control facility. DUMPTIMERTBL commands entered from an XPNET network control facility are sent directly to the Main Settlement Initiator process. The Main Settlement Initiator process sends interprocess DUMPTIMERTBL commands to the appropriate Secondary Settlement Initiator processes. An operator can request timer tables for all Secondary Settlement Initiator processes or for a specific Secondary Settlement Initiator processes.

Note: The DUMPTIMERTBL command can be considered both a process-generated command and a general use operator command, depending on where it originates. If a DUMPTIMERTBLALL or DUMPTIMERTBL SECONDARY command is sent to the Main Settlement Initiator process from an XPNET network control facility, the Main Settlement Initiator process sends a DUMPTIMERTBL command as an interprocess command to Secondary Settlement Initiator processes. The Main Settlement Initiator process notifies Secondary Settlement Initiator processes that timer information is being requested.

If the command originates from an XPNET network control facility as a DUMPTIMERTBL command, it is a general use operator command. The network control request is for Main Settlement Initiator process timer table information. In this case, the Main Settlement Initiator process displays only its timer table information. No interprocess commands are sent.

Syntax

DUMPTIMERTBL

Processing

The Main Settlement Initiator process sends the interprocess DUMPTIMERTBL command to Secondary Settlement Initiator processes. When the appropriate Secondary Settlement Initiator processes receive this command, they display the timer type, the date and time the timer is scheduled to expire, and the information contained in the timer’s user data buffer in a series of event messages.

Timer information is displayed in the following format:

```
TIME-TYPE    EXPIRE DATE/TIME    USER-BUF
```

Timer information is displayed for each entry in the internal timer table. The TIME-TYPE field displays information for the following Secondary Settlement Initiator process timers:

- Retailer cutover timer
- Alive Timer (Senior Secondary Settlement Initiator process only)

The EXPIRE DATE/TIME field indicates the time and date the retailer cutover timer and Alive Timer will expire. The USER-BUF field indicates the information contained in the timer's user data buffer. When all timer information has been displayed, the following message is shown:

```
END OF TIMER DISPLAY
```

DUMPTIMES Command

A process-generated command that causes the Secondary Settlement Initiator processes to display certain important processing times. This command is sent from the Main Settlement Initiator process to the Secondary Settlement Initiator processes when a DUMPTIMES command (i.e., DUMPTIMESALL or DUMPTIMES SECONDARY command) is entered from an XPNET network control facility. DUMPTIMES commands entered from an XPNET network control facility are sent directly to the Main Settlement Initiator process. The Main Settlement Initiator process sends interprocess DUMPTIMES commands to the appropriate Secondary Settlement Initiator processes. An operator can request times for all Secondary Settlement Initiator processes or for a specific Secondary Settlement Initiator processes.

Note: The DUMPTIMES command can be considered both a process-generated command and a general use operator command, depending on where it originates. If a DUMPTIMESALL or DUMPTIMES SECONDARY command is sent to the Main Settlement Initiator process from an XPNET network control facility, the Main Settlement Initiator process sends a DUMPTIMES command as an interprocess command to Secondary Settlement Initiator processes. The Main Settlement Initiator process notifies Secondary Settlement Initiator processes that time information is being requested.

If the command originates from an XPNET network control facility as a DUMPTIMES command, it is a general use operator command. The network control request is for Main Settlement Initiator process time information. In this case, the Main Settlement Initiator process displays only its time information. No interprocess commands are sent.

Syntax

DUMPTIMES

Processing

The Main Settlement Initiator process sends the interprocess DUMPTIMES command to Secondary Settlement Initiator processes. When the appropriate Secondary Settlement Initiator processes receive this command, they display the following time information taken from params set up in the Logical Network Configuration File (LCONF) in a series of event messages. The corresponding LCONF param where the information displayed is taken is noted in parentheses:

- Financial institution cutover lead time (POS-SETL-FI-CUTOVER-LEAD-TIME)
- Terminal cutover delay time (POS-SETL-TERM-CUTOVER-DELAY-TIME)

- Alive timeout time (POS-SETL-ALIVE-TIMEOUT)

In addition to the above times, the Secondary Settlement Initiator processes also display the current business date, the next business date, and the trace flag. When all time information has been displayed, the following message is shown:

```
END OF TIME DISPLAY
```

DUMPPTD Command

A process-generated command that causes the Secondary Settlement Initiator processes to display BASE24-pos Terminal Data files records according to the retailer range defined for a specific Secondary Settlement Initiator process. This command is sent from the Main Settlement Initiator process to the Secondary Settlement Initiator processes when a DUMPPTD command (i.e., DUMPPTDALL or DUMPPTD SECONDARY command) is entered from an XPNET network control facility. DUMPPTD commands entered from an XPNET network control facility are sent directly to the Main Settlement Initiator process. The Main Settlement Initiator process sends interprocess DUMPPTD commands to the appropriate Secondary Settlement Initiator processes. An operator can request BASE24-pos Terminal Data files records for all Secondary Settlement Initiator processes or for a specific Secondary Settlement Initiator process.

Syntax

DUMPPTD

Processing

The Main Settlement Initiator process sends the interprocess DUMPPTD command to Secondary Settlement Initiator processes. When a Secondary Settlement Initiator process receives this command, it displays BASE24-pos Terminal Data files record information only for those terminal IDs included in the retailer range defined for the specific Secondary Settlement Initiator process. Retailer ID ranges for Secondary Settlement Initiator processes are identified in the POS-SLAVE-KEY param in the LCONF. If there are no BASE24-pos Terminal Data files table entries for a particular Secondary Settlement Initiator process, that Secondary Settlement Initiator process logs an event message indicating that the BASE24-pos Terminal Data files extended memory table is empty. The Secondary Settlement Initiator process displays the following BASE24-pos Terminal Data files record information for each terminal ID in its retailer ID range in a series of event messages:

- Terminal ID
- Retailer ID
- Device Handler process name
- Post date
- Deactivate flag

BASE24-pos Terminal Data files record information is displayed in the following format:

```
TERM-ID    RETAILER-ID    DH-PRO-NAME    POST-DATE DF
```

The TERM-ID field identifies the terminal ID from the BASE24-pos Terminal Data files record. The RETAILER-ID indicates the retailer ID associated with the terminal ID. The DH-PRO-NAME indicates the symbolic name of the Device Handler process as identified in the BASE24-pos Terminal Data files. The POST-DATE field identifies the posting date from the BASE24-pos Terminal Data files record. The DF field indicates whether the terminal is P (pending), A (activated), or D (deactivated). When all the information has been displayed, the following message is shown:

```
END OF TDF DISPLAY
```

DUMPPTDTBL Command

A process-generated command that causes the Secondary Settlement Initiator processes to display the contents of their internal BASE24-pos Terminal Data files extended memory tables. This command is sent from the Main Settlement Initiator process to the Secondary Settlement Initiator processes when a DUMPPTDTBL command (i.e., DUMPPTDTBLALL or DUMPPTDTBL SECONDARY command) is entered from an XPNET network control facility. DUMPPTDTBL commands entered from an XPNET network control facility are sent directly to the Main Settlement Initiator process. The Main Settlement Initiator process sends interprocess DUMPPTDTBL commands to the appropriate Secondary Settlement Initiator processes. An operator can request BASE24-pos Terminal Data files table information for all Secondary Settlement Initiator processes or for a specific Secondary Settlement Initiator process.

Syntax

DUMPPTDTBL

Processing

The Main Settlement Initiator process sends the interprocess DUMPPTDTBL command to Secondary Settlement Initiator processes. When a Secondary Settlement Initiator process receives this command, it displays the contents of its internal BASE24-pos Terminal Data files extended memory table in a series of event messages. The internal BASE24-pos Terminal Data files extended memory table contains all of the retailer IDs and Merchant Host Interface (MHI) Device Handler process names associated with a particular Secondary Settlement Initiator process. This table is maintained for MHI terminals in host control mode. If there are no BASE24-pos Terminal Data files table entries for a particular Secondary Settlement Initiator process, that Secondary Settlement Initiator process logs an event message indicating that the BASE24-pos Terminal Data files extended memory table is empty.

BASE24-pos Terminal Data files table information is displayed in the following format:

```
RETAILER-ID      DH-PRO-NAME
```

The RETAILER-ID field identifies all of the retailer IDs in the internal BASE24-pos Terminal Data files extended memory table associated with a particular Secondary Settlement Initiator process. The DH-PRO-NAME identifies the Merchant Host Interface (MHI) Device Handler process names in the internal BASE24-pos Terminal Data files extended memory table associated with a particular Secondary Settlement Initiator process. When all the information has been displayed, the following message is shown:

```
END OF TDF TABLE DISPLAY
```

FIID Command

A process-generated command that causes all Secondary Settlement Initiator processes to verify that their retailers and terminals associated with the FIID specified in the command are cut over. If they are not cut over, the Secondary Settlement Initiator processes must force cutover. The FIID command originates from the Main Settlement Initiator process. The Main Settlement Initiator process sends this command when it is attempting to perform institution cutover.

Syntax

FIID *fiid*

fiid — The FIID of the financial institution currently being cut over.

Processing

When the Financial Institution Cutover Timer expires, the Main Settlement Initiator process begins institution cutover. The Main Settlement Initiator process maintains a separate timer for each institution in the logical network. For more information on financial institution cutover, refer to the ***BASE24-pos Settlement and Reporting Manual***.

As a part of institution cutover, when the financial institution cutover timer expires, the Main Settlement Initiator process begins sending FIID commands to its Secondary Settlement Initiator processes to ensure that all retailers and their terminals have cut over. The Main process waits for responses from all its Secondary Settlement Initiator processes before cutting over the FIID specified in the command and moving to the next FIID.

The Main Settlement Initiator process uses information contained in its internal Institution Definition File (IDF) extended memory table to perform IDF cutover. The Main Settlement Initiator process searches the IDF extended memory table for FIID and the NEED TO QUERY FLAG field. The NEED TO QUERY FLAG field is set when the financial institution cutover timer expires. If this flag is set to a value of 1, the Main Settlement Initiator process sends an FIID command to each of its Secondary Settlement Initiator processes. When all Secondary Settlement Initiator processes have responded to the FIID query, the Main Settlement Initiator process sets the NEED TO QUERY FLAG field to a value of 0 and searches for the next FIID with a NEED TO QUERY FLAG field value of 1. The Main Settlement performs this processing until all of the necessary FIIDs have been queried.

In addition, the Main Settlement Initiator process performs the following processing steps when it sends an FIID command:

1. Sends an FIID command to all the Secondary Settlement Initiator processes to verify that all the retailers and their terminals have cut over.

   When the Secondary Settlement Initiator process receives the command, it performs the following steps:

   a. Uses the FIID of the retailer to read the PRDF. If the retailer has been cut over, processing continues with step 2. If the retailer has not been cut over, processing continues with step 1b.

   b. Uses the BASE24-pos Terminal Data files alternate key of the TERMINAL OWNER field to locate the retailer's terminal records. The Secondary Settlement Initiator process then checks the POST DATE field in the BASE24-pos Terminal Data files record for each terminal belonging to the retailer.

      1) If the terminal has been cut over to a new day, the Secondary Settlement Initiator process moves on to the next BASE24-pos Terminal Data files record or, if there are no more BASE24-pos Terminal Data files records, the Secondary Settlement Initiator process continues with step 1a until all retailers for the FIID have been processed.

      2) If the date in the POST DATE field indicates that the terminal has not been cut over to a new day, the Secondary Settlement Initiator process cuts the terminal over as it does when the retailer cutover timer expires. The Secondary Settlement Initiator process then checks the POST DATE field for the next terminal and repeats this step. If there are no more BASE24-pos Terminal Data files records, the Secondary Settlement Initiator process continues with step 2 until all retailers for the FIID have been processed.

      3) When all the terminals have been cut over for all of the Secondary Settlement Initiator process's retailers, the Secondary Settlement Initiator process sends a cutover complete message to the Main Settlement Initiator process.

2. When the Main Settlement Initiator process receives cutover complete messages from all the retailers belonging to the financial institution, the Main Settlement Initiator process performs institution cutover.

LOGICAL NETWORK CUTOVER Command

A process-generated command that notifies all Secondary Settlement Initiator processes and all processes specified in the POS-SETL-NOTIFY params in the LCONF that logical network cutover is to occur. When these processes receive this command, they perform their cutover procedures. This command originates from the Main Settlement Initiator process and is sent when the time specified in the POS-LN-CUTOVER param in the LCONF expires.

Syntax

9502 *yymmdd*

yymmdd — The date (YYMMDD) of the logical network cutover. For logical network cutover to occur, this date must be less than or equal to the current business date.

Processing

When the logical network cutover timer expires, the Main Settlement Initiator process begins logical network cutover.

The Main Settlement Initiator process sets its logical network cutover timer so that it expires at the time specified in the POS-LN-CUTOVER param in the LCONF. The Main Settlement Initiator process sets this timer at initialization and at logical network cutover. For more information on logical network cutover, refer to the ***BASE24-pos Settlement and Reporting Manual***.

As a part of logical network cutover, when the logical network cutover timer expires, the Main Settlement Initiator process begins broadcasting 9502 (CUTOVER) commands to all of its Secondary Settlement Initiator processes and to those processes specified in the POS-SETL-NOTIFY params in the LCONF. In addition, the Main Settlement Initiator process performs the following steps:

1. The Main Settlement Initiator process reads the LCONF at initialization and maintains the broadcast list and Secondary Settlement Initiator process table in memory.

   The cutover message contains the current date, which is the day just ending. The Main Settlement Initiator process uses this message to notify each process to perform its cutover procedures (e.g., close files, open files, reinitialize internal memory for the new day).

When each Secondary Settlement Initiator process receives a cutover message from the Main Settlement Initiator process, it performs the following steps:

a. Closes the current POS Transaction Log File (PTLF) and opens the next day's PTLF.

b. Increases the current and next business dates in memory.

c. Sets the retailer cutover timer for the current day.

d. Returns a message to the Main Settlement Initiator process indicating that the Secondary Settlement Initiator process has completed its logical network cutover procedures.

2. Sets a 9502 timer to three minutes. When the timer expires, the Main Settlement Initiator process sends new 9502 messages to all secondary processes that have not yet responded to the original 9502 message.

NOTIFY MHI Command

A process-generated command sent by the Main Settlement Initiator process in the form of a 9506 message to notify all Merchant Host Interface (MHI) processes with terminals running in host control mode that cutover has occurred. The MHIs are identified in the POS-SETL-MHI-NOTIFY param in the Logical Network Configuration File (LCONF). One POS-SETL-MHI-NOTIFY param exists for each MHI that requires notification when cutover occurs. The POS-SETL-MHI-NOTIFY params are read by the Main Settlement Initiator process at initialization.

Syntax

9506

Processing

The MHIs store POS Terminal Data File records in memory for terminals running in host control mode. The dates in these records must be updated when cutover occurs. Therefore, at terminal cutover, the Main Settlement Initiator process issues a 9506 notify message to each MHI that has terminals running in host control mode and that is identified in the POS-SETL-MHI-NOTIFY params. The MHI releases the POS Terminal Data File records from memory for the terminal being cut over when it receives the 9506 message. The next time a transaction is received by the MHI for the terminal, the MHI deletes the existing POS Terminal Data File record from memory for the terminal and re-reads the POS Terminal Data File record back into memory with the new date.

RESET DATES Commands

Process-generated commands that cause all Secondary Settlement Initiator processes to reset their posting dates in the POS Retailer Definition File (PRDF), BASE24-pos Terminal Data files, or both. These commands are sent to each Secondary Settlement Initiator process associated with the Main Settlement Initiator process. The RESET DATES commands originate from the Main Settlement Initiator process. The file to be reset is identified in the syntax of the command.

Syntax

RESETRDFDATES *yymmdd*

RESETPTDDATES *yymmdd*

RESETALLDATES *yymmdd*

yymmdd — The date (YYMMDD) to which the file posting date is reset.

Processing

The Main Settlement Initiator process sends interprocess RESET DATES commands to all Secondary Settlement Initiator processes. When a Secondary Settlement Initiator process receives one of the RESET DATES commands, it performs as follows:

- The RESETALLDATES command causes the Secondary Settlement Initiator processes to reset the posting dates in the PRDF and BASE24-pos Terminal Data files. In addition, the RESETALLDATES command causes the Main Settlement Initiator process to reset the posting date in the Institution Definition File (IDF).
- The RESETRDFDATES command causes the Secondary Settlement Initiator processes to reset the posting dates in the PRDF. This form of the message changes the CUR-POST-DAT field in the PRDF to the date in the command. This message form also changes the NXT-POST-DAT field in the PRDF to the posting date following the date in the command.
- The RESETPTDDATES command causes the Secondary Settlement Initiator processes to reset the posting dates in the BASE24-pos Terminal Data files. This form of the message changes the POST-DAT field in the BASE24-pos Terminal Data files to the date in the command.

RESETRDFTIMES Command

A process-generated command that causes Secondary Settlement Initiator processes to reset their retailer cutover end times in the POS Retailer Definition File (PRDF). This command can be sent by the Main Settlement Initiator process to a specific Secondary Settlement Initiator process or to all of its Secondary Settlement Initiator processes. The RESETRDFTIMES command originates from the Main Settlement Initiator process. The new cutover times are identified in the syntax of the command.

Syntax

RESETRDFTIMES[ALL] *old-time new-time secondary-symbolic-name*

old-time — The cutover end time currently set in the RETAILER CUTOVER/BALANCE END field on screen 1 of the PRDF (in hhmm format).

new-time — The new cutover end time set in the RETAILER CUTOVER/BALANCE END field on screen 1 of the PRDF as a result of processing this command (in hhmm format).

secondary-symbolic-name — The symbolic name of a specific Secondary Settlement Initiator process.

Processing

The Main Settlement Initiator process sends interprocess RESETRDFTIMES commands to a specific or all Secondary Settlement Initiator processes. When a Secondary Settlement Initiator process receives the RESETRDFTIMES command, it performs the following steps:

1. Reads each PRDF record within its retailer ID range. If the cutover end time value currently set in the RETAILER CUTOVER/BALANCE END field in the PRDF matches the old time sent in the interprocess command, the Secondary Settlement Initiator process resets the value in the RETAILER CUTOVER/BALANCE END field to the new time contained in the interprocess command.
2. Reinitializes the PRDF extended memory.
   a. Deallocates the existing PRDF extended memory segment.
   b. Determines the segment size of the PRDF, BASE24-pos Terminal Data files, and Institution Definition File (IDF). If the MHI notify list contains at least one entry, the BASE24-pos Terminal Data files extended memory needs to be rebuilt.

   c. Initializes the PRDF, BASE24-pos Terminal Data files, and IDF extended memory tables.

3. Initializes the timer table.
4. If the current time is after logical network cutover, the Secondary Settlement Initiator process sets the internal cutover flag to true. This flag causes retailers who have cut over after logical network cutover to have their timers reset to the current day as well as for the next day.
5. Sets the retailer timers.
6. Sets the Alive Timer. This step is only performed by the Senior Secondary process as identified in the POS-SETL-SENIOR-SLAVE param in the Logical Network Configuration File (LCONF).
7. Sets the internal cutover flag to false.

STARTUP Command

A process-generated command sent by the Main Settlement Initiator process to all Secondary Settlement Initiator processes during initialization. The STARTUP command causes all Secondary Settlement Initiator processes to initialize and return STARTUPCOMPLETE commands to the Main Settlement Initiator process. The STARTUP command originates from the Main Settlement Initiator process and is sent using the XPNET process to the Secondary Settlement Initiator processes.

Syntax

STARTUP *symbolic-name*

symbolic-name — The symbolic name of the Main Settlement Initiator process as defined in the POS-MASTER-SETL param in the LCONF.

Processing

During Main Settlement Initiator process initialization, it sends STARTUP commands to each Secondary Settlement Initiator process configured in the logical network. Up to 16 Secondary Settlement Initiator processes can be defined in a logical network. An issuer-only logical network does not require any Secondary Settlement Initiator processes since there are no terminals or retailers.

Note: The Main and Secondary Settlement Initiator processes communicate directly, except in the case of STARTUP and STARTUPCOMPLETE commands. In the case of STARTUP and STARTUPCOMPLETE commands, the Main and Secondary Settlement Initiator processes communicate using the XPNET process. When the Main or Secondary Settlement Initiator process sends messages to another process or to the host, the messages are sent using the XPNET process.

The Main Settlement Initiator process sends, using the XPNET process, start up messages to all Secondary Settlement Initiator processes, if they exist. The Main Settlement Initiator process also sets a 60 second timer during which all Secondary Settlement Initiator processes must perform their initialization steps and send STARTUPCOMPLETE commands back to the Main Settlement Initiator process. Each Secondary Settlement Initiator process performs the following steps when it receives a STARTUP command at initialization:

1. Reads the params and assigns it uses in processing from the LCONF and then closes the LCONF.

2. Determines the current and next business dates.

   To do this, the Secondary Settlement Initiator process uses the current date known to the system for the current date and adds one to this date to determine the next business date. If cutover has already occurred, the current date is used.

3. Opens the Institution Definition File (IDF).

   The Secondary Settlement Initiator process can read a locked IDF, allowing multiple Secondary Settlement Initiator processes to access the IDF simultaneously. If a Secondary Settlement Initiator process cannot access the IDF, the process abends.

4. Opens the POS Retailer Definition File (PRDF). If the Secondary Settlement Initiator process is unable to open the PRDF, the process abends.

5. Opens the BASE24-pos Terminal Data files. If the Secondary Settlement Initiator process is unable to open the BASE24-pos Terminal Data files, the process abends.

6. Opens the POS Transaction Log File (PTLF). If the Secondary Settlement Initiator process is unable to open the PTLF, the process abends.

7. Initializes its timer tables. The Secondary Settlement Initiator process does not set any timers at this point.

8. Reads the primary PRDF and retrieves the retailer range that identifies its partition. The Secondary Settlement Initiator process then closes the primary PRDF and opens its individual partition. If the Secondary Settlement Initiator process cannot open its partition, the process abends.

9. Calculates the extended memory required for its individual partition of the PRDF and for the BASE24-pos Terminal Data files if any Device Handler processes exist for a Merchant Host Interface (MHI) device in host control mode.

10. Closes its individual partition of the PRDF and opens the primary PRDF. If the Secondary Settlement Initiator process is unable to open the primary PRDF, the process abends.

11. Calculates the extended memory required for the IDF, PRDF and PBASE24-pos Terminal Data files and initializes its extended memory tables.

12. Sets its timers.

    The Secondary Settlement Initiator process sets the retailer cutover timer. If this is the Senior Secondary Settlement Initiator process, as determined by the POS-SETL-SENIOR-SLAVE param in the LCONF, it also sets the Alive Timer.

13. Sends a startupcomplete message containing its PPD name to the Main Settlement Initiator process using the XPNET process.

STARTUPCOMPLETE Command

A process-generated command sent by a Secondary Settlement Initiator process to the Main Settlement Initiator process. The STARTUPCOMPLETE command is sent in response to a STARTUP command from the Main Settlement Initiator process and is used to indicate that a specific Secondary Settlement Initiator process has initialized. The STARTUPCOMPLETE command originates from the Secondary Settlement Initiator process and is sent using the XPNET process to the Main Settlement Initiator processes.

Syntax

STARTUPCOMPLETE *symbolic-name*

symbolic-name — The symbolic name of the Secondary Settlement Initiator process for which initialization has completed.

Processing

The Main Settlement Initiator process sends STARTUP commands to initialize each Secondary Settlement Initiator process. Each Secondary Settlement Initiator process responds with a STARTUPCOMPLETE command. When the Main Settlement Initiator process receives a STARTUPCOMPLETE command, it opens the Secondary Settlement Initiator process with the PPD name it received in the STARTUPCOMPLETE command. After the Secondary Settlement Initiator process is open, the Main Settlement Initiator process and the Secondary Settlement Initiator process can communicate directly.

Secondary Settlement Initiator processes are defined in a Secondary Settlement Initiator process table built by the Main Settlement Initiator process at initialization. The Main Settlement Initiator process uses the POS-SLAVE-KEY param contained in the LCONF to build the Secondary Settlement Initiator process table. The POS-SLAVE-KEY param defines the range of retailer IDs assigned to an indicated Secondary Settlement Initiator process.

If the Main Settlement Initiator process does not receive a STARTUPCOMPLETE command from a Secondary Settlement Initiator processes before the 60 second startup timer set by the Main Settlement Initiator process expires, the Main Settlement Initiator process compares the number of times the 60 second startup timer has expired (the threshold counter) to the value in the POS-SETL-THRESHOLD param in the LCONF and performs according to the following:

- If the threshold counter is less than the value in the POS-SETL-THRESHOLD param, the Main Settlement Initiator process increases the threshold counter by 1 and sends another STARTUP command to the

Secondary Settlement Initiator process that did not send a STARTUPCOMPLETE command. The Main Settlement Initiator process then sets another 60 second timer.

- If the threshold counter is equal to the value in the POS-SETL-THRESHOLD param, the Main Settlement Initiator process sets its threshold counter to zero, logs a message indicating that the Main Settlement Initiator process is unable to start the Secondary Settlement Initiator process, and waits for the amount of time indicated in the POS-SETL-WAIT-INTERVAL param in the LCONF. After waiting the specified amount of time, the Main Settlement Initiator process sends another STARTUP command to the Secondary Settlement Initiator process and sets another 60 second timer.

TRACE OFF SECONDARY Command

A process-generated command sent by the Main Settlement Initiator process to a specified Secondary Settlement Initiator process to deactivate the trace function. The Main Settlement Initiator process sends an interprocess TRACE OFF command to the specified Secondary Settlement Initiator process. For more information on the TRACE OFF command, refer to appendix A.

Syntax

TRACE OFF *secondary-name*

secondary-name — The symbolic name of the Secondary Settlement Initiator process for which the trace function is being deactivated.

TRACE ON SECONDARY Command

A process-generated command sent by the Main Settlement Initiator process to a specified Secondary Settlement Initiator process to activate the trace function. The Main Settlement Initiator process sends an interprocess TRACE ON command to the specified Secondary Settlement Initiator process. For more information on the TRACE ON command, refer to appendix A.

Syntax

TRACE ON *secondary-name*

secondary-name — The symbolic name of the Secondary Settlement Initiator process for which the trace function is being activated.

WARMBOOT PRDF TABLE ENTRY Command

A process-generated command that causes the Main Settlement Initiator process to send an interprocess command that notifies Secondary Settlement Initiator processes to warmboot their internal PRDF extended memory table for the retailer ID specified in the command.

Syntax

9501RDFKEY *retailer-key*

retailer-key — The ID of the retailer whose PRDF records require warmbooting.

Processing

The Main Settlement Initiator process sends an interprocess command to its Secondary Settlement Initiator processes indicating the PRDF table entry for the specified retailer ID requiring a warmboot. This command is sent only when a PRDF table entry requires updating. The Main Settlement Initiator process receives this command from an XPNET network control facility and searches through its internal Secondary Settlement Initiator process table for a match on the retailer key identified in the command. When a match is found, the Main Settlement Initiator process sends the interprocess WARMBOOT PRDF TABLE ENTRY command to the Secondary Settlement Initiator process. The Secondary Settlement Initiator process receives the interprocess command and compares the cutover time and cutover flag in the PRDF record to the cutover time and cutover flag in the internal PRDF extended memory table for the specified retailer. The Secondary Settlement Initiator process performs the following processing:

- If the cutover times are not the same in the PRDF record and PRDF extended memory table, and the cutover flag is set to indicate retailer cutover has not occurred for the current day, the Secondary Settlement Initiator process logs an event message and does not warmboot its internal PRDF extended memory table entry for the specified retailer ID.
- If the cutover times in the PRDF record and PRDF extended memory table are the same or different and the cutover flag is set to indicate that retailer cutover has occurred for the current day, the Secondary Settlement Initiator process warmboots its internal PRDF extended memory table entry with the new information from the PRDF record.

WB MAIN PPD Command

A process-generated command sent by the Main Settlement Initiator process to all Secondary Settlement Initiator process when the Main Settlement Initiator process receives a 9501**** or 9501ALL command. When the Main Settlement Initiator process receives a 9501**** or 9501ALL command, it sends an interprocess WB MAIN PPD command, including the PPD name of the Main Settlement Initiator process, to each Secondary Settlement Initiator process contained in the Main Settlement Initiator process's Secondary table. This causes each Secondary Settlement Initiator process to perform its warmboot logic and to return a WBCOMPLETE command to the specified Main Settlement Initiator process when it is finished. The interprocess WB MAIN PPD command originates from the Main Settlement Initiator process.

Syntax

WB *main-ppd-name*

main-ppd-name — The PPD name of the Main Settlement Initiator process being warmbooted.

Processing

Both the Main and Secondary Settlement Initiator processes must be warmbooted when the Institution Definition File (IDF) has been modified, an assign or param used by both the Main and Secondary Settlement Initiator processes has been modified, or the Secondary table must be rebuilt due to a change in the high or low retail key values for a Secondary Settlement Initiator process partition.

When both the Main and Secondary Settlement Initiator processes are warmbooted, the processes perform as they would during initialization, except for the following:

- They do not receive a new LCONF name as they do during initialization.
- They close and open any files that were opened during initialization.

WBCOMPLETE Command

A process-generated command sent by each Secondary Settlement Initiator process to the Main Settlement Initiator process in response to an interprocess WARMBOOT command, including the PPD of the Main Settlement Initiator process. When the Main Settlement Initiator process receives a 9501**** or a 9501ALL command, it sends an interprocess WB MAIN PPD command, including the Main Settlement Initiator process PPD, to each Secondary Settlement Initiator process. This causes each Secondary Settlement Initiator process to perform its warmboot logic and to return a WBCOMPLETE command, including the PPD name of the Secondary Settlement Initiator process, to the Main Settlement Initiator process when it is finished. The Main Settlement Initiator process verifies all Secondary Settlement Initiator processes that have returned WBCOMPLETE commands. This command originates from the Secondary Settlement Initiator process.

Syntax

WBCOMPLETE *secondary-ppd-name*

secondary-ppd-name — The PPD name of the Secondary Settlement Initiator process that has just completed its warmboot logic.

Processing

The Main Settlement Initiator process sends interprocess WB MAIN PPD commands to warmboot each Secondary Settlement Initiator process. Each Secondary Settlement Initiator process responds with a WBCOMPLETE command. When the Main Settlement Initiator process receives a WBCOMPLETE command, it opens the Secondary Settlement Initiator process with the PPD name it received in the WBCOMPLETE command. After the Secondary Settlement Initiator process is open, the Main Settlement Initiator process and the Secondary Settlement Initiator process can communicate directly.

Secondary Settlement Initiator processes are defined in a Secondary Settlement Initiator process table built by the Main Settlement Initiator process at initialization. The Main Settlement Initiator process uses the POS-SLAVE-KEY param contained in the LCONF to build the Secondary Settlement Initiator process table. The POS-SLAVE-KEY param defines the range of retailer IDs assigned to an indicated Secondary Settlement Initiator process.

If the Main Settlement Initiator process does not receive a WBCOMPLETE command from a Secondary Settlement Initiator processes by the time the 60 second warmboot timer set by the Main Settlement Initiator process has expired, the Main Settlement Initiator process compares the number of times the 60 second warmboot timer has expired (the threshold counter) to the value in the POS-SETL-THRESHOLD param in the LCONF and performs according to the following:

- If the threshold counter is less than the value in the POS-SETL-THRESHOLD param, the Main Settlement Initiator process increases the threshold counter by 1 and sends another WB MAIN PPD command to the Secondary Settlement Initiator processes that did not send WBCOMPLETE commands. The Main Settlement Initiator process then sets another 60 second timer.
- If the threshold counter is equal to the value in the POS-SETL-THRESHOLD param, the Main Settlement Initiator process sets its threshold counter to zero, logs a message indicating that the Main Settlement Initiator process is unable to warmboot the Secondary Settlement Initiator process, and waits for the amount of time indicated in the POS-SETL-WAIT-INTERVAL param in the LCONF. After waiting the specified amount of time, the Main Settlement Initiator process sends another WB MAIN PPD command to the Secondary Settlement Initiator process and sets another 60 second timer.

Transaction Context Manager Commands

This subsection describes the BASE24 process-generated commands the Transaction Context Manager (TCM) process recognizes. The TCM process enables the BASE24-pos system to receive a single request from an endpoint and split the routing between two or more destinations. The TCM process receives the following process-generated command:

- LOGICAL NETWORK CUTOVER

LOGICAL NETWORK CUTOVER Command

A process-generated command that notifies the Transaction Context Manager (TCM) process that logical network cutover and midnight cutover have taken place. The Settlement Initiator process sends this notification to the TCM process. This command originates from the BASE24-pos Main Settlement Initiator process

Syntax

9502*yymmdd*

yymmdd — The date associated with the processing day just ending.

Processing

When the TCM process receives a 9502*yymmdd* message, it performs the following steps:

1. Adjusts its current processing date.

   If the current processing data is greater than the date sent in the 9502*yymmdd* message, the TCM process takes no further steps.

   If the current processing date is less than or equal to the date sent in the 9502*mmyydd* message, the TCM process performs the following steps.

   a. Creates its next processing date.

   To get its next processing date, the TCM process adds one calendar day to the current processing date.

   b. Closes the POS Transaction Log Files (PTLFs) for its previous day it currently has open.

   c. Opens the PTLFs for its next processing date.

2. Generates the event message “Cutover completed for this process.”

BASE24-teller Process Commands

This subsection contains the process-generated commands received by the BASE24-teller Authorization process.

Authorization Process

This subsection describes the BASE24 process-generated commands the Authorization process recognizes. The Authorization process receives the following process-generated commands:

- EXTRACT
- FIID CUTOVER
- LOGICAL NETWORK CUTOVER
- REFRESH

EXTRACT Command

A process-generated command that notifies the Authorization process that Teller Transaction Log File (TTLF) extract has begun.

Syntax

9505 *ref-group*

ref-group — The processing group for which the TTLF extract is being performed. The processing group is an identifier grouping institutions together for file refreshes and extracts. Refresh and extract groups are typically used in configurations where multiple hosts are used. File refreshes and extracts can use this identifier to group certain institutions together. The processing group is defined in the REFRESH GROUP field on Institution Definition File (IDF) screen 1. The REFRESH GROUP field can specify a value of ALL. In this case, all processing groups are refreshed or extracted.

Processing

The Super Extract process notifies the BASE24-teller Authorization process using a 9505 message that a TTLF extract has begun. This command originates from the Super Extract process. The Authorization process performs the following steps when it receives an EXTRACT command:

1. Checks the REFRESH GROUP field in the IDF to determine the institutions for which the TTLF extract is performed and logs the following event message indicating the processing group:

   ```
   6040 0  Proc: auth-process TTLF Extract NOTICE for group ref-
   group received
   ```

2. Rebuilds the BASE24-teller segment of the IDF stored in extended memory by setting the following fields to a value of N (No):

   - **DDA-CUR**. The DDA current indicator identifies whether the Positive Balance File (PBF) named in the PBF1-NAME (i.e., Demand Deposit Account) field in the IDF is current. The Authorization process sets this indicator to a value of N to indicate that the PBF1-NAME field is not current.

   - **SAV-CUR**. The SAV current indicator identifies whether the Positive Balance File (PBF) named in the PBF2-NAME (i.e., Savings Account) field in the IDF is current. The Authorization process sets this indicator to a value of N to indicate that the PBF2-NAME field is not current.

- **CCD-CUR**. The CCD current indicator identifies whether the Positive Balance File (PBF) named in the PBF3-NAME (i.e., Credit Card Account) field in the IDF is current. The Authorization process sets this indicator to a value of N to indicate that the PBF3-NAME field is not current.
- **SPF-CUR**. The SPF current indicator identifies whether the Stop Payment File (SPF) and PBF are current. The Extract process sets this indicator to a value of N to indicate that the PBF and SPF are not current.
- **NBF-CUR**. The NBF current indicator identifies whether the No Book File (NBF) and PBF are current. The Extract process sets this indicator to a value of N to indicate that the PBF and NBF are not current.
- **WHFF-CUR**. The WHFF current indicator identifies whether the Warning/Hold/Deposit Float File (WHFF) and PBF are current. The Extract process sets this indicator to a value of N to indicate that the PBF and WHFF are not current.

3. Logs event message 6041 if the IDF rebuild was successful or event message 6042 if the IDF rebuild was not successful. These event messages are described in the TRAUTH log message documentation file.

FIID CUTOVER Command

A process-generated command that notifies the Authorization process that FIID cutover has taken place. The Settlement Initiator process sends this notification to the Authorization process. This command originates from the BASE24-teller Settlement Initiator process.

Syntax

9504 *fiid yymmdd yymmdd*

fiid — The FIID of the financial institution being cut over.

yymmdd — The date (YYMMDD) of the new current processing day resulting from cutover. This is the cutover date for the specified FIID.

yymmdd — The date (YYMMDD) of the new next processing day resulting from cutover.

Processing

When the Authorization process receives an FIID CUTOVER command, it performs the following steps:

1. Searches the Institution Definition File (IDF) memory for the current FIID entry. If an entry is not found, the Authorization process logs the following event message:

   ```
   6030  1  Proc auth-process Cutover for unknown FIID FIID...
   rejected
   ```

   If the specified FIID is found with a fatal error, the Authorization process logs the following event message:

   ```
   6031  1  Proc auth-process Not processing FIID FIID...rejected
   ```

   If the command is successful, the Authorization process generates the following event message:

   ```
   6032  0  Proc auth-process Cutover for FIID FIID has completed
   ```

2. Adjusts its current processing date.

   If the current processing date is greater than the date sent in the FIID CUTOVER command, the Authorization process takes no further steps. If the current processing date is less than or equal to the date sent in the FIID CUTOVER command, the Authorization process performs the following steps:

   a. Creates its next processing date.

      To get its next processing date, the Authorization process uses the next date contained in the message.

   b. Closes the Teller Transaction Log Files (TTLFs) it currently has open.

   c. Opens the TTLFs for its new current processing date and next business date.

LOGICAL NETWORK CUTOVER Command

A process-generated command that notifies the Authorization process that logical network cutover has taken place. The Settlement Initiator sends this notification to the Authorization process. This command originates from the BASE24-teller Settlement Initiator process.

Syntax

9502*yymmdd*

yymmdd — The date (yymmdd) of the processing day just ending.

Processing

The Authorization process receives a LOGICAL NETWORK CUTOVER command from the Settlement Initiator process. This command contains the cutover date used by the Authorization process to find any FIIDs that are still processing on an old and invalid business date. All FIIDs must have a business date greater than the date sent in the 9502 command. Otherwise, the FIID is not processed by the Authorization process. When Authorization receives a LOGICAL NETWORK CUTOVER command, it performs the following steps:

1. Closes the current Teller Transaction Log Files (TTLFs) it currently has opened.
2. Checks the Institution Definition File (IDF) file table to determine if any institutions are still logging to the current TTLF.

   If so, the Authorization process sets a fatal error in the IDF and logs the following event message:

   ```
   6011  1  Proc auth-process FIID FIID still using TTLF-file-
   name at network cutover.  This FIID will not be processed.
   ```

   If cutover was successful, the Authorization process logs the following event message:

   ```
   6010  0  Proc auth-process Network cutover completed
   ```

REFRESH Command

A process-generated command that notifies the Authorization process of a file refresh. The Enscribe Refresh process sends this notification in the form of a 9503 message. The command originates from the Enscribe Refresh process.

Syntax

9503**** *file-name ref-group file-type refr-indicator*

file-name — The fully-qualified name of the refreshed file.

ref-group — An identifier used to group institutions together for file refreshes and extracts.

file-type — Indicates which file has been refreshed. Valid values for files used by the Authorization process are as follows:

1 = Positive Balance File (PBF) for DDA and NOW accounts if multiple PBFs are used or all accounts if PBFs are combined
2 = Positive Balance File (PBF) for savings accounts if multiple PBFs are used
3 = Positive Balance File (PBF) for credit card accounts if multiple PBFs are used
5 = Stop Payment File (SPF)
7 = Cardholder Authorization File (CAF)
6 = No Book File (NBF)
9 = Warning/Hold/Float File (WHFF)

refr-indicator — A value used by the Authorization process to log future transactions to the transaction log files. The Enscribe Refresh process uses this value to control impacting.

Processing

When the Authorization process receives a 9503 message, it performs the following processing steps:

1. Checks the *file-type* sent in the 9503 message.

   If the *file-type* is one of the recognized values, processing continues with step 2.

   If the *file-type* is not one of the refresh types recognized by the Authorization process, processing skips to step 8.

2. Locates the file in its file table using the *file-name* sent in the 9503 message.

   If the Authorization process cannot locate the file in its table, it generates an event message to note the situation and skips to step 8.

3. Closes the file previously held open.

4. Opens the file specified in the 9503 message.

5. Adds the new Guardian file number to its file table.

6. Each IDF record in extended memory and on disk contains a refresh indicator field for each of the files listed for the *file-type* variable. If the *ref-group* sent in the 9503 message matches the value in the REFR-GRP field of the IDF record, the Authorization process updates the corresponding refresh indicator field of the IDF record in memory with the *refr-indicator* from the 9503 message. The Enscribe Refresh process updates these fields in the IDF disk file.

7. Rebuilds the BASE24-teller segment of the IDF stored in extended memory by setting the following fields to a value of Y (yes) based on the *file-type* in the 9503 message:

file-type	**IDF Field Rebuilt in Extended Memory**
1	**DDA-CUR —** Identifies whether DDA and NOW account information in the PBF is current. This indicator is set to a value of N by the Super Extract process to indicate that the PBF is not current and to a value of Y by the Enscribe Refresh process to indicate that the PBF is current.
2	**SAV-CUR —** Identifies whether savings account information in the PBF is current. The indicator is set to a value of N by the Super Extract process to indicate that the PBF is not current and to a value of Y by the Enscribe Refresh process to indicate that the PBF is current.
3	**CCD-CUR —** Identifies whether credit card account information in the PBF is current. This indicator is set to a value of N by the Super Extract process to indicate that the PBF is not current and to a value of Y by the Enscribe Refresh process to indicate that the PBF is current.
5	**SPF-CUR —** Identifies whether the SPF is current. This indicator is set to a value of N by the Super Extract process to indicate that the SPF is not current and to a value of Y by the Enscribe Refresh process to indicate that the SPF is current.

file-type	**IDF Field Rebuilt in Extended Memory**
6	**NBF-CUR —** Identifies whether the NBF is current. This indicator is set to a value of N by the Super Extract process to indicate that the NBF is not current and to a value of Y by the Enscribe Refresh process to indicate that the NBF is current.
9	**WHFF-CUR —** Identifies whether the WHFF is current. This indicator is set to a value of N by the Super Extract process to indicate that the WHFF is not current and to a value of Y by the Enscribe Refresh process to indicate that the WHFF is current.

8. Sends a PAUSEACK command to the Enscribe Refresh process to indicate that it has processed the 9503 message.

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